FENGWEI FRIED RICE · STAGE 2 STUDY GUIDE

DAY EDITION

Fengwei 24-HOUR DINER

AI HARDWARE · NETWORKS · BEHAVIOURIST AI

30	6	115	60	74
MISSIONS	REGULARS	CODEX WORDS	EXAM QUESTIONS	DIAGRAMS

A playable cook-along study guide. Prep every mission, serve the customers who bring the questions, survive a rush hour in each of the diner's five rooms and win over the Professor, our VIP food critic, in the Final Exam. The interactive 3D version lives in **Fengwei_24H_Diner_Study_Guide.html**.

TEAM FENGWEI FRIED RICE · INTRODUCTION TO ARTIFICIAL INTELLIGENCE · NOVEMBER 2026

BEFORE YOUR FIRST SHIFT

Your chef licence

CHEF NAME _____

SHIFT STARTED _____

FAVOURITE ROOM _____

XP TRACKER · COLOUR ONE BOX PER 50 XP

Kitchen ranks

Dishwasher	Prep Cook	Line Cook	Wok Master	Sous Chef	Head Chef	Diner Legend
0 XP	150 XP	400 XP	800 XP	1400 XP	2200 XP	3200 XP

How to earn XP on paper

Mission tickets

Every mission starts with an order ticket. Tick each course (section) when you have read it: +10 XP.

At the end, serve the customers: each order is a question. Colour one ♡ for every happy customer. Half happy = ⭐ ⭐ and the ticket is cleared (+50 XP). All happy = ⭐ ⭐ ⭐.

Side quests

Mini-games are side quests: grab pen, paper and a calculator. Try the task, then check Mama's feedback in the Answer Vault at the end of the recipe: +25 XP.

Rush hours

Each room ends with a rush hour: 14 orders in a row, with a trouble customer at the end of the line. Colour a smiley box for every right dish. 9 or more happy customers = you survive: +150 XP and the recipe card.

Final exam

The Professor, our VIP food critic, orders all 60 Test Bank questions as a three-course tasting menu. Grade yourself honestly with the model answers: +300 XP.

Meet the kitchen

Mama Feng
Head chef and your teacher. Her tips are pink.

Papa Feng
Loves shortcuts: his "oops" boxes are common mistakes.

Leo
Asks the "but why?" questions.

Lily
Apprentice and number cruncher.

Dumpling
The kitchen pup, star of Recipe 03.

Grandma Feng
Founded the diner in 1986.

WHO COMES TO THE COUNTER

Meet the regulars

Six special customers visit the diner. In the interactive version their perks change the game; in this book they bring the orders next to each question. Tick each one when you meet them!

☐ The Professor

VIP

VIP food critic. He only comes for the Final Exam and grades every bite.

☐ The Big Eater

Orders two dishes in a row. The second one tips double.

☐ Blocky Guy

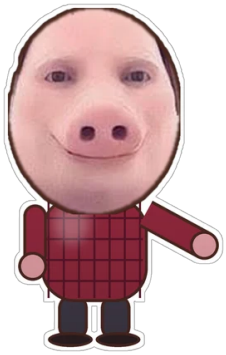
Orders the easy true/false dishes. Get his wrong and your streak is safe.

☐ The Night Owl

Night shift only. Serve him right and everyone's patience freezes for 10 seconds.

☐ The Hype Guy

Serve him right and the whole queue cheers up. Get it wrong and they all grumble.



☐ Mystery Pork

Rare! Orders a mystery dish from any chapter. Right = golden ×5 tip.

ONE DINER · 5 ROOMS + CENTRE STAGE

The diner map

Recipe 01 · The Kitchen Crew

MAIN KITCHEN · AI HARDWARE

- [ ] 1.1 What does AI cook? Definitions and fundame... 6
- [ ] 1.2 Meet the crew 14
- [ ] 1.3 Cooking in parallel 21
- [ ] 1.4 The pantry problem 27
- [ ] 1.5 Case study 33
- [ ] LAB Kitchen Lab 39
- [ ] Rush Hour 44
- [ ] Answer Vault 46

Recipe 02 · Order Up!

FRONT COUNTER · NETWORKS & AI INFRASTRUCTURE

- [ ] 2.1 How kitchens are connected (network archite... 50
- [ ] 2.2 The careful waiter and the shouting waiter (T... 57
- [ ] 2.3 Central kitchen, food truck or home kitchen? (... 65
- [ ] 2.4 Delivery time, road width and rush hour (late... 72
- [ ] 2.5 Case study 80
- [ ] LAB Kitchen Lab 87
- [ ] Rush Hour 91
- [ ] Answer Vault 93

Recipe 03 · Training the Pup

BBQ GARDEN · BEHAVIOURIST AI & RL

- [ ] 3.1 Treats and scoldings (definition and principles... 97
- [ ] 3.2 The pup and the kitchen (agent-environment ... 103
- [ ] 3.3 Choosing the right treat (reward systems and ... 110
- [ ] 3.4 New stall or favourite stall? (exploration vs ex... 115
- [ ] 3.5 Case study 122
- [ ] LAB Kitchen Lab 128
- [ ] Rush Hour 131
- [ ] Answer Vault 133

Recipe 04 · Chef's Senses

SUSHI BAR · AI SENSORS & FUSION

- [ ] 4.1 Types of sensors 137
- [ ] 4.2 Data acquisition and real-time processing 148
- [ ] 4.3 Sensor fusion and environmental perception 155
- [ ] 4.4 Integrating hardware with AI decision-making 164
- [ ] 4.5 Case study 171
- [ ] LAB Kitchen Lab 178
- [ ] Rush Hour 182
- [ ] Answer Vault 184

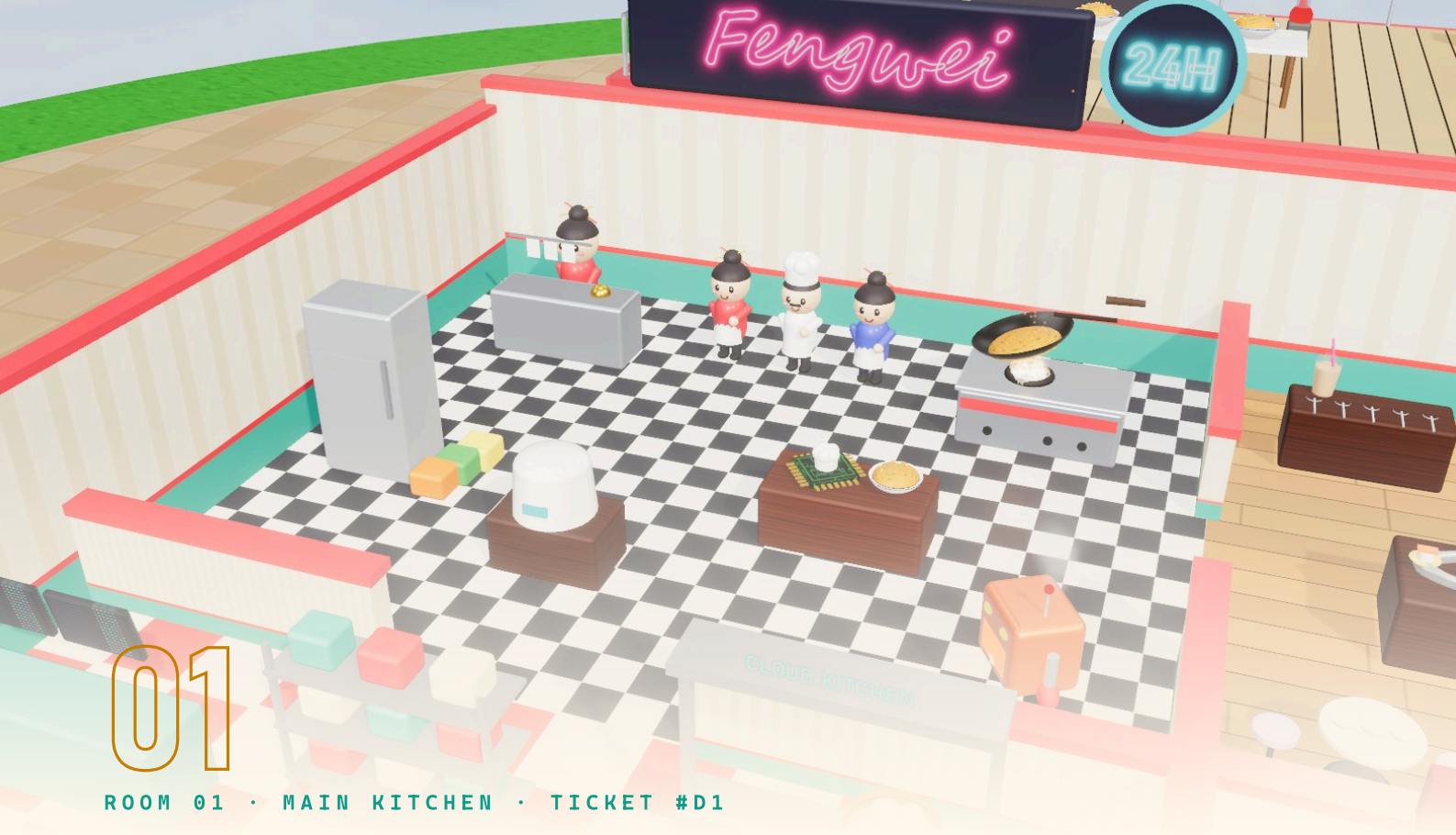
Recipe 05 · Running the Chain

ROOFTOP BANQUET · INTEGRATION & OPTIMISATION

- [ ] 5.1 Hardware-software interaction (the recipe car... 188
- [ ] 5.2 Distributed AI systems (many kitchens, one di... 194
- [ ] 5.3 Performance optimisation and resource alloca... 201
- [ ] 5.4 Challenges 207
- [ ] 5.5 Case study 215
- [ ] LAB Kitchen Lab 220
- [ ] Rush Hour 224
- [ ] Answer Vault 226

Centre stage & extras

- [ ] ★ The Professor's Final Exam 229
- [ ] ★ Final Exam answers 235
- [ ] ★ The Codex · kitchen words 237
- [ ] ★ Recipe book 240
- [ ] ★ Sources & credits 242



01

ROOM 01 · MAIN KITCHEN · TICKET #D1

The Kitchen Crew

AI HARDWARE

MAMA FENG

Mama's Problem: "1,000 plates of fried rice in one minute! Who should cook: one master chef, a thousand helpers, or a machine?" AI has the same problem: billions of tiny maths steps, and we must pick the right cook for the job.

By the end you will be able to

1. Define AI hardware and explain why neural networks are mostly MACs.
2. Count MACs and FLOPs for a layer and convert precisions to memory sizes.
3. Compare CPU, GPU, TPU and NPU and choose one for a workload.
4. Trace a 2 × 2 systolic array cycle by cycle.
5. Use Amdahl's law to predict the limit of parallel speed-up.
6. Explain the memory hierarchy and why moving data costs more than computing.
7. Use the roofline model to decide memory-bound vs compute-bound.
8. Analyse the Apple Neural Engine and Face ID as on-device AI.

TONIGHT'S TICKETS

- [ ] 1.1 What does AI cook? Definitions and … +170 XP
- [ ] 1.2 Meet the crew +140 XP
- [ ] 1.3 Cooking in parallel +120 XP
- [ ] 1.4 The pantry problem +110 XP
- [ ] 1.5 Case study +110 XP
- [ ] LAB Kitchen Lab +100 XP
- [ ] Rush Hour +150 XP

TICKET #1.1 · RECIPE 01 · MAIN KITCHEN

1.1 What does AI cook? Definitions and fundamentals

COURSES ☐☐☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 170 XP

BRIEFING

Mama's problem

+10 XP

MAMA FENG

Mama's Problem: "1,000 plates of fried rice in one minute! Who should cook: one master chef, a thousand helpers, or a machine?" AI has the same problem: billions of tiny maths steps, and we must pick the right cook for the job.

📋 Today's Adventure

1.1 What does AI cook? Definitions and fundamentals

1.2 Meet the crew: CPU vs GPU vs TPU (and NPU)

1.3 Cooking in parallel: AI acceleration

1.4 The pantry problem: memory and efficiency

1.5 Case study: the home rice cooker (Apple Neural Engine)

BRIEFING

Goals & what you need

+10 XP

✨ READY, SET, COOK!

Apron on! 👩‍🍳 By the end of this recipe you'll know why your phone has its own AI chip. Ready to level up?

🎯 By the end you will be able to

1. Define AI hardware and explain why neural networks are mostly MACs.
2. Count MACs and FLOPs for a layer and convert precisions to memory sizes.
3. Compare CPU, GPU, TPU and NPU and choose one for a workload.
4. Trace a 2 × 2 systolic array cycle by cycle.
5. Use Amdahl's law to predict the limit of parallel speed-up.
6. Explain the memory hierarchy and why moving data costs more than computing.
7. Use the roofline model to decide memory-bound vs compute-bound.
8. Analyse the Apple Neural Engine and Face ID as on-device AI.

Before You Jump In

Basic arithmetic (multiplying, adding, powers of ten) and the idea of a variable and a loop. No hardware background needed: every new word is explained when it first appears and collected in the glossary.

1.1

What Is AI Hardware?

+10 XP

✨ READY, SET, COOK!

Quick challenge! Your phone hides at least FIVE kinds of AI hardware. How many can you name before you read on? Count them!

The Big Idea

AI hardware is any processor, memory system or interconnect designed for, or well suited to, running machine-learning workloads efficiently.

Five categories:

- General-purpose CPUs
- GPUs (graphics processors)
- Domain-specific accelerators (TPU, NPU)
- Memory systems (caches, HBM)
- Interconnects (links between chips)

💡 MAMA'S SECRET TIP

A kitchen isn't only cooks. It's the cooks (processors), the fridge and pantry (memory) and the hallways between kitchens (interconnects). All of them decide how fast dinner arrives.

1.1

What Does AI Cook?

+10 XP

LEO

Predict first! A neural network "thinks" by doing... (a) logic puzzles, (b) one tiny maths move, billions of times, or (c) internet searches?

The Big Idea

Answer: (b). Almost every layer of a neural network multiplies its inputs by learned numbers called WEIGHTS, then adds the results up.

One "multiply, then add to the running total" step is called a MAC (multiply-accumulate). It is like adding one ingredient × its amount into the wok, again and again.

🔬 REAL-LIFE TECH

1 MAC = 2 FLOPs (one multiply + one add). Grouping the weights in a MATRIX lets hardware do many MACs at once. A TENSOR is a many-dimensional matrix: the "T" in TPU.

1.1

Worked Example: One Neuron

+10 XP

Stir-fry one neuron

Weights w = [0.5, −1.0, 2.0]

Inputs x = [4, 3, 1]

Start: total = 0

Step 1: + 0.5 × 4 = 2.0

Step 2: + (−1.0 × 3) = −1.0

Step 3: + 2.0 × 1 = 1.0

Output = 1.0, using 3 MACs.

KITCHEN TERMINAL · PYTHON

```python
w = [0.5, -1.0, 2.0]
x = [4, 3, 1]
total = 0
for wi, xi in zip(w, x):
    total += wi * xi      # one MAC
print(total)              # 1.0
```

Now scale it up

A layer with 1,024 inputs and 4,096 outputs needs 1,024 × 4,096 = 4,194,304 MACs for ONE input. Large language models need billions to trillions of MACs for every answer.

So the big question of Recipe 01: which cook can do MACs fastest, and cheapest?

1.1 · SEE IT

+10 XP

One Neuron, Step By Step

MAMA FENG

🔍 Spot it! Follow the running total from Start to the end. How many MACs does one neuron need?

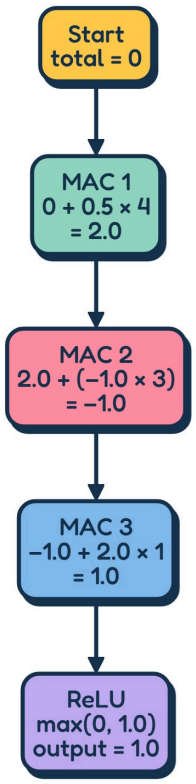


FIG Worked example: a single neuron computed one multiply-add (MAC) at a time.

1.1

Kitchen Words

+10 XP

✨ READY, SET, COOK!

Word hunt time! 🔎 Spot the words you already know, then learn the rest like a pro.

TERM	DEFINITION
MAC	Multiply two numbers and add the result to a running total
FLOP / FLOPS	One floating-point operation / FLOPs per second. 1 MAC = 2 FLOPs
TOPS	Tera (10^{12}) operations per second, often integer ops on AI chips
Throughput	How much work is finished per second
Latency	How long one task takes from start to finish
Precision	Bits per number: FP32 = 4 bytes, FP16 = 2, INT8 = 1
Accelerator	A chip working beside the CPU to speed up specific work
Bandwidth	Bytes per second moved between two parts of a system

⚠ PAPA'S OOPS!

"8 GB? 8 Gb? Same thing!" Not quite! 1 byte = 8 bits. FP32 takes 4 bytes, FP16 2 bytes, INT8 1 byte. Many hardware mistakes come from forgetting this conversion.

1.1

Prefixes & Precision

+10 XP

✨ READY, SET, COOK!

Kilo, mega, giga… tera?! 🤯 Guess: how many zeros are in one TERA? Check below!

PREFIX	SYMBOL	POWER	EXAMPLE
kilo	k	10^3	1 kB = 1,000 bytes
mega	M	10^6	8 MB of cache
giga	G	10^9	500 GB/s bandwidth
tera	T	10^{12}	92 TOPS
peta	P	10^{15}	Data-centre PFLOPS
milli / micro / nano / pico	m / μ / n / p	thousandth … trillionth	1 ms, 1 μs, 1 ns, 1 pJ

FORMAT	BITS	BYTES	TYPICAL USE
FP32	32	4	Classic training, "full" precision
BF16 / FP16	16	2	Training and inference on modern accelerators
FP8 / INT8	8	1	Newer hardware; phones, edge, TPUs
INT4	4	0.5	Aggressively compressed models

💡 MAMA'S SECRET TIP

Lower precision = cutting vegetables into fewer, bigger chunks: faster and lighter to carry, a little less detail. 1 MiB = 2^{20} = 1,048,576 bytes; the small difference doesn't change any conclusion here.

1.1

Why Not A Faster CPU?

+10 XP

✨ READY, SET, COOK!

Why not just build one super-fast CPU? 🤔 Make a guess, then see if you beat Mama!

Two reasons

Reason 1: the work is repetitive and independent. Millions of MACs don't depend on each other (neuron 17 doesn't change neuron 18), so a chip that does many simple things at once wins.

Reason 2: energy and memory limits. Chips can't simply be clocked faster without overheating. As transistors shrank they used to use less power (Dennard scaling), but that ended in the mid-2000s and clock speeds flattened at a few GHz.

Hennessy & Patterson (2019): big gains must now come from DOMAIN-SPECIFIC hardware.

🔑 REMEMBER THIS!

Gains now come from two directions this whole recipe returns to:

1. doing more things at once (PARALLELISM)
2. moving less data (MEMORY EFFICIENCY)

1.1

Writing the Recipe vs Cooking the Order

+10 XP

✨ READY, SET, COOK!

Writing a recipe vs cooking it 1,000 times a day: which one needs more speed? Pick a side! ⚖️

	TRAINING	INFERENCE
Goal	Learn weights from data	Use weights to answer a request
Frequency	Rare, long jobs (days-months)	Constant: millions of requests
Main limit	Raw compute, chip-to-chip communication	Latency, energy, cost per request
Place	Large data-centre clusters	Data centres and edge devices
Precision	Higher (FP32, BF16, FP16)	Lower (INT8, FP8, 4-bit)
Data movement	Forward + backward pass	Forward pass only
Batch size	Large (hundreds+)	Often 1 (one user)

💡 MAMA'S SECRET TIP

Training = Mama writing and testing a new recipe for weeks. Inference = cooking that recipe for every customer who orders it, quickly, all day long.

🔬 REAL-LIFE TECH

The first Google TPU was built for inference only (Jouppi et al., 2017); later generations added training, and Ironwood (2025) is again aimed at inference at scale.

1.1

Choosing the Crew: Six Metrics

+10 XP

✨ READY, SET, COOK!

You're the boss! 💼 Which SIX things would you check before hiring a chip? Let's find out.

METRIC	QUESTION	WHY IT MATTERS
Throughput (TOPS, FLOP/s)	Work per second?	Training speed
Latency	How long for one request?	User experience; real-time control
Performance per watt	Work per joule?	Battery life; power bill
Utilisation	Share of peak actually achieved?	Money paid for idle silicon

METRIC	QUESTION	WHY IT MATTERS
Performance per dollar	Work per unit of cost?	Budget decisions
Memory capacity	How big a model fits?	Which models can run at all

⚠ PAPA'S OOPS!

"Easy! The quoted peak speed is what I'll get." Not quite! Peak means EVERY unit busy EVERY cycle. Real programs reach much less: a chip quoted at 100 TOPS at 30% utilisation really gives 30 TOPS.

☐ 📖 SIDE QUEST · MINI-GAME 1.1 +25 XP

Stir-Fry Maths

✨ READY, SET, COOK!

Ready, set, STIR-FRY! 🔥 Can you beat Mama? Solve all five before your tea gets cold!

🎯 Your Mission

Compute a neuron output with MACs, count the work in a layer, and classify training vs inference.

🧺 Grab These First

Pen, paper, calculator. Re-read the worked neuron.

🔪 Let's Go! Your Turn

(1) w = [2, −1, 0.5], x = [1, 4, 6]: write each MAC step and the output.

(2) A layer has 512 inputs and 2,048 outputs: MACs per input? FLOPs?

(3) A chip quoted at 100 TOPS reaches 30% utilisation. Observed throughput?

(4) Training or inference? (a) Face ID unlocking (b) fine-tuning on 100 GPUs for 3 weeks (c) a chatbot answering 10,000 requests/s.

(5) A 7B-parameter model in FP16: weight memory? In INT8?

👀 What Did You Spot?

Write your running total after every MAC.

💭 Think It Over

Why would a chip that does MACs very fast speed up almost the whole model?

🔒 **Mama's feedback in the Answer Vault · V-1-1** · page 46 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #1.1

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.1) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. One MAC (multiply-accumulate) counts as two FLOPs. ♡ ♡ ♡

- (T) True
- (F) False

2. Weights [2, −1, 0.5], inputs [1, 4, 6]. What is the neuron output (no bias)? ♡ ♡ ♡

- (A) 1.0
- (B) 3
- (C) 5
- (D) −1

3. What does “Amdahl's law” mean? ♡ ♡ ♡

- (A) Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.
- (B) Converting numbers to lower precision (e.g. FP32 → INT8).
- (C) Apple framework that routes model parts to CPU, GPU or ANE.
- (D) Thousands of simple cores working in parallel.

4. Which kitchen word means: “Attainable = min(peak, intensity × bandwidth).”? ♡ ♡ ♡

- (A) Utilisation
- (B) GPU
- (C) NPU
- (D) Roofline model

TICKET #1.2 · RECIPE 01 · MAIN KITCHEN

1.2 Meet the crew

CPU vs GPU vs TPU (and NPU)

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 140 XP

1.2

Meet the Crew

+10 XP

👩‍🍳 CPU = the Head Chef

A few very skilled cores; can cook ANY dish, one step at a time each. Flexible, low latency. Best for mixed tasks, data loading, small models.

👩‍🍳 GPU = 1,000 Line Cooks

Thousands of simple cores all doing "chop these onions" together. Built for graphics, perfect for matrices: both are "same operation, lots of data". Best for training and serving large models.

🥟 TPU = the Dumpling Conveyor

Google's chip built around a grid of MAC units. Highest efficiency for neural-network maths, least flexible.

🍙 NPU = the Home Rice Cooker

A small, low-power AI engine inside a phone or laptop chip (Apple Neural Engine, Qualcomm Hexagon).

🔑 REMEMBER THIS!

The more specialised the cook, the more efficient at THAT dish, and the less useful for anything else.

1.2

The Crew Scorecard

+10 XP

✨ READY, SET, COOK!

Scorecard time! 🏆 Before reading, guess which chip wins on speed per watt.

FEATURE	CPU	GPU	TPU
Design goal	Fast, flexible, low latency	Throughput on parallel data	Max matrix throughput per watt
Cores	Few to ~100 complex	Thousands of simple	Grid of tens of thousands of MACs
Control logic	Complex: caches, out-of-order, branch prediction	Simpler; threads hide delays	Minimal; deterministic
Flexibility	Highest	High	Lowest

FEATURE	CPU	GPU	TPU
AI energy efficiency	Lowest	Medium-high	Highest (supported work)
AI role	Pre-processing, data loading, small models	Training and inference of big models	Large-scale training/inference in Google Cloud
Weakness	Too few parallel units	More power than an ASIC; bandwidth	Less flexible; tied to a stack
Programming	Any language	CUDA, OpenCL, PyTorch	TensorFlow/JAX via XLA

1.2

Where the Chip Area Goes

+10 XP

✨ READY, SET, COOK!

Where does all that chip space go? 🧩 Guess first: cleverness or raw muscle?

TERM	MEANING
Core	A unit that can execute instructions on its own
Cache	Small fast memory keeping copies of recently used data
Out-of-order execution	Running instructions in a safe different order to avoid waiting
Branch prediction	Guessing which way an "if" goes to keep working
Deterministic execution	Every run takes the same predictable number of cycles
Compiler	Translates a program into a chip's instructions (XLA for TPUs)

Silicon budget (qualitative sketch)

CPU: much area on caches and control logic that make ONE thread fast.

GPU: most area on arithmetic units.

TPU: nearly all area on the MAC grid and the memory that feeds it.

1.2

The Dumpling Conveyor

+10 XP

✨ READY, SET, COOK!

All aboard the dumpling conveyor! 🥟 Watch the numbers march through, step by step.

Systolic array

A grid of MAC cells. Each cell keeps ONE weight. Inputs and partial sums are passed hand-to-hand to the neighbour, like dumplings on a conveyor, instead of being fetched from memory again and again.

Worked example: 2 × 2 (weight-stationary)

W = [[1, 2], [3, 4]], x = [5, 6]

Cycle 1: x1 = 5 enters column 1 → 1 × 5 = 5 and 3 × 5 = 15

Cycle 2: x2 = 6 reaches column 2 → 2 × 6 + 5 = 17 and 4 × 6 + 15 = 39

Result y = [17, 39] with 4 MACs. Check: 1·5 + 2·6 = 17; 3·5 + 4·6 = 39.

Memory reads: systolic 4 weights + 2 inputs = 6; without passing = 8. The saving grows with array size.

KITCHEN TERMINAL · PYTHON

```python
reads_no_reuse = 2 * 2 * 2   # each MAC: 1 weight + 1 input
reads_systolic = 4 + 2
print(reads_no_reuse, reads_systolic)   # 8 6
```

1.2 · SEE IT

+10 XP

The 2 × 2 Conveyor

MAMA FENG

🔍 Spot it! Watch each input pass through two cooks. Which weight does each input meet first?

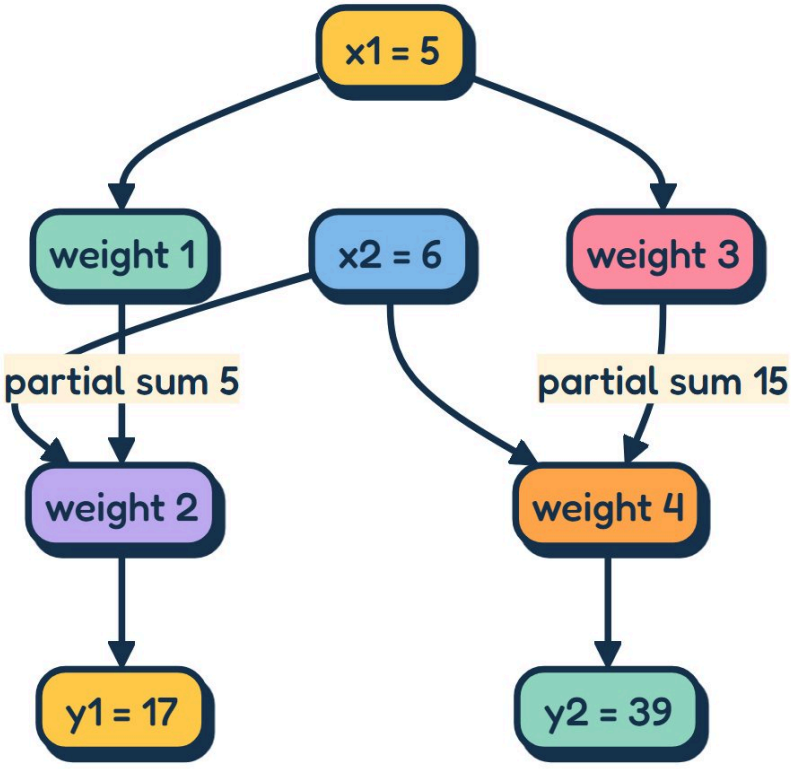


FIG A 2 × 2 systolic array: inputs flow right, partial sums flow down.

1.2 · SEE IT

+10 XP

The 2 × 2 Conveyor (2)

MAMA FENG

🔍 Spot it! Watch each input pass through two cooks. Which weight does each input meet first?

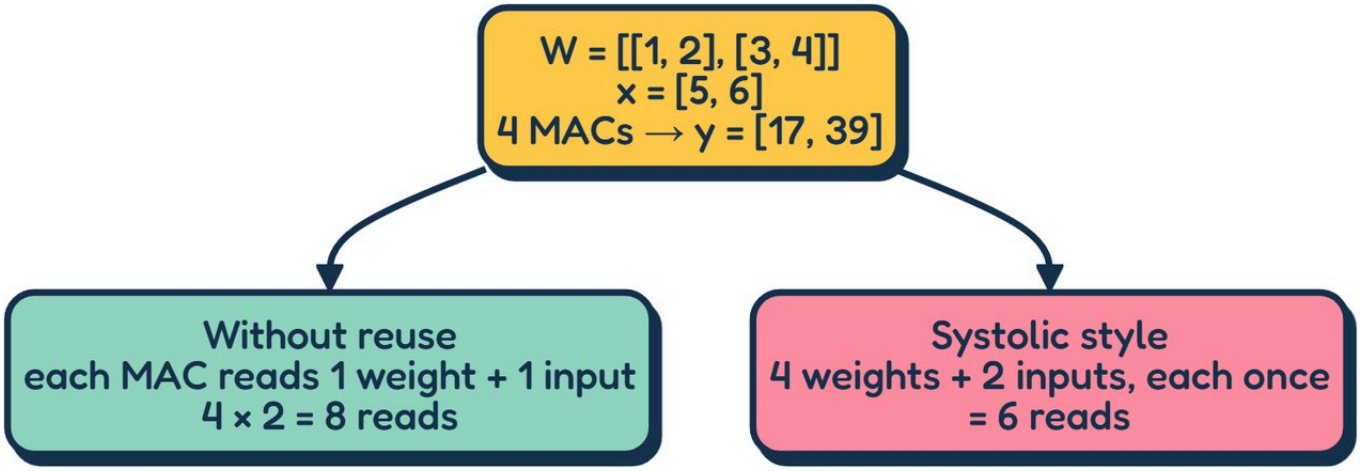


FIG The same example worked out: W = [[1, 2], [3, 4]], x = [5, 6].

1.2

+10 XP

What the First TPU Measured

✨ READY, SET, COOK!

Detective mode ON 🕵️ What did Google discover when they tested their very first TPU?

Jouppi et al. (2017)

- **Heart:** a 65,536-element 8-bit MAC matrix unit, 92 TOPS peak, 28 MiB on-chip memory.
- Versus a Haswell CPU and a K80 GPU in the same data centres: ~15-30× faster and ~30-80× more TOPS per watt.
- Deterministic execution helped the 99th-percentile response time user-facing services need.
- BUT 4 of 6 benchmarks were memory-bandwidth limited. Faster GDDR5 memory would have tripled achieved TOPS.

⚠ PAPA'S OOPS!

"So TPUs beat GPUs, end of story!" Not quite! This is 2015-2017 hardware, a design lesson, not today's ranking. GPUs now have their own matrix units (Tensor Cores), and the gap has narrowed.

🔑 REMEMBER THIS!

Even 65,536 MACs spent most of their time WAITING for data on four of six workloads. More arithmetic doesn't help if it can't be fed (see 1.4).

1.2

The Home Rice Cooker & Its Tools

+10 XP

✨ READY, SET, COOK!

Your phone has a tiny rice cooker inside! 🍚 Can you guess what it's for?

The NPU

A phone can't carry a 300-watt accelerator. An NPU uses the TPU ideas (MAC grid, low precision, on-chip memory) scaled down to a few watts inside a SYSTEM-ON-A-CHIP (SoC) with the CPU, GPU and shared memory.

Names: Apple Neural Engine • Google Tensor SoC • Qualcomm Hexagon • laptop "AI Engine".

CHIP	HOW PROGRAMMERS REACH IT
CPU	Any language; compiler targets it directly
GPU	CUDA, OpenCL, or PyTorch/TensorFlow calling vendor libraries
TPU	TensorFlow, JAX, PyTorch/XLA compiled through XLA
Apple Neural Engine	Core ML decides CPU, GPU or Neural Engine

💡 MAMA'S SECRET TIP

A chip is only as useful as its tools. Hardware + software ecosystem = a PLATFORM. Switching platforms can be costly even when the new chip is faster on paper. Unsupported operations "fall back" to the CPU and can silently make a model slow.

📖 SIDE QUEST · MINI-GAME 1.2

+25 XP

Hire the Chef

✨ READY, SET, COOK!

You're hiring! 📋 Match each job to the best chef. No take-backs!

Your Mission

Match a chip to a workload and justify it; trace a systolic array.

Grab These First

The crew cards and the conveyor worked example.

Let's Go! Your Turn

(1) Choose CPU, GPU, TPU or NPU: (a) a web server running mixed tasks (b) training a large image model (c) a fixed, high-volume inference service (d) a voice assistant in a battery smart speaker: rank NPU, CPU, GPU.

(2) Conveyor trace: W = [[2, 1], [0, 3]], x = [4, 5]. Cycle 1 sums? Cycle 2 sums? Memory reads with and without passing?

(3) Why might a start-up still choose a GPU though the TPU was faster in Google's tests? Give two reasons.

What Did You Spot?

For each choice write the ONE feature that decided it.

Think It Over

Why does reusing data matter more than adding more MAC cells?

🔒 **Mama's feedback in the Answer Vault · V-1-2** · page 46 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #1.2

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.2) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A TPU is more flexible than a CPU for general programs. ♡ ♡ ♡

- (T) True
- (F) False

2. Which chip best fits a voice assistant in a small battery-powered speaker? ♡ ♡ ♡

- (A) GPU
- (B) Server CPU
- (C) NPU
- (D) TPU pod

3. In a 2 × 2 systolic array with W = [[1, 2], [3, 4]] and x = [5, 6], the output y = W·x is: ♡ ♡ ♡

- (A) [12, 30]
- (B) [23, 34]
- (C) [11, 23]
- (D) [17, 39]

4. What does “GPU” mean? ♡ ♡ ♡

- (A) Limited by data delivery / by arithmetic.
- (B) Converting numbers to lower precision (e.g. FP32 → INT8).
- (C) Thousands of simple cores working in parallel.
- (D) Processors, memory and interconnects suited to running ML efficiently.

5. Which kitchen word means: “Multiply-accumulate: multiply two numbers, add to a running total.”? ♡ ♡ ♡

- (A) Precision
- (B) CPU
- (C) MAC
- (D) Roofline model

TICKET #1.3 · RECIPE 01 · MAIN KITCHEN

1.3 Cooking in parallel

AI acceleration

COURSES ☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 120 XP

1.3

Cooking In Parallel

+10 XP

✨ READY, SET, COOK!

More cooks = faster dinner… right? 👩‍🍳 👩‍🍳 Let's test it with real numbers!

🔪 **Worked example: how long?**

Mama's Problem: one chef chops 4,194,304 onions one by one. A kitchen of 4,096 helpers chops 4,096 at a time.

- **Idealised CPU core:** 1 MAC/cycle at 3 GHz → $4{,}194{,}304 \div 3 \times 10^9 \approx 1.4$ ms
- **Idealised accelerator:** 4,096 MAC units at 1 GHz → $4{,}194{,}304 \div 4.096 \times 10^{12} \approx 1.0$ µs

≈ 1,365× faster even with a slower clock, because it does 4,096 MACs per cycle.

LEVEL	WHAT RUNS IN PARALLEL	EXAMPLE
Instruction-level	Steps of one instruction stream overlap	CPU pipelines
Data-level (SIMD/SIMT)	One instruction on many data items	GPU warps
Matrix-level	Whole small matrix products per step	Systolic arrays, Tensor Cores
Across chips	Work split over many chips	Recipes 02 and 05

1.3

Why More Cooks Stop Helping

+10 XP

✨ READY, SET, COOK!

Plot twist! 😱 At some point adding cooks stops helping. Can you guess why?

🎎 **1. Amdahl's law**

speed-up = 1 ÷ ((1 − p) + p ÷ N)

p = fraction that can be shared out; N = number of cooks.

Example, p = 0.9: N = 2 → 1.8× • 4 → 3.1× • 8 → 4.7× • 16 → 6.4× • 100 → 9.2× • unlimited → 10×

The serial 10% (the single wok!) caps everything.

2-4. Idle units, feeding, talking

1. **Idle units:** work doesn't divide evenly, so some cooks wait.
2. **Feeding the units:** MACs need data from memory faster than memory can deliver (1.4).
3. **Communication:** when work is split across chips, they must exchange results (Recipe 02).

KITCHEN TERMINAL · PYTHON

```python
def amdahl(p, n):
    return 1 / ((1 - p) + p / n)
for n in (1, 2, 4, 8, 16, 100):
    print(n, round(amdahl(0.9, n), 1))
```

1.3 · SEE IT

+10 XP

Amdahl's Law

MAMA FENG

🔍 Spot it! Look at the curve: why does it flatten even with 128 cooks?

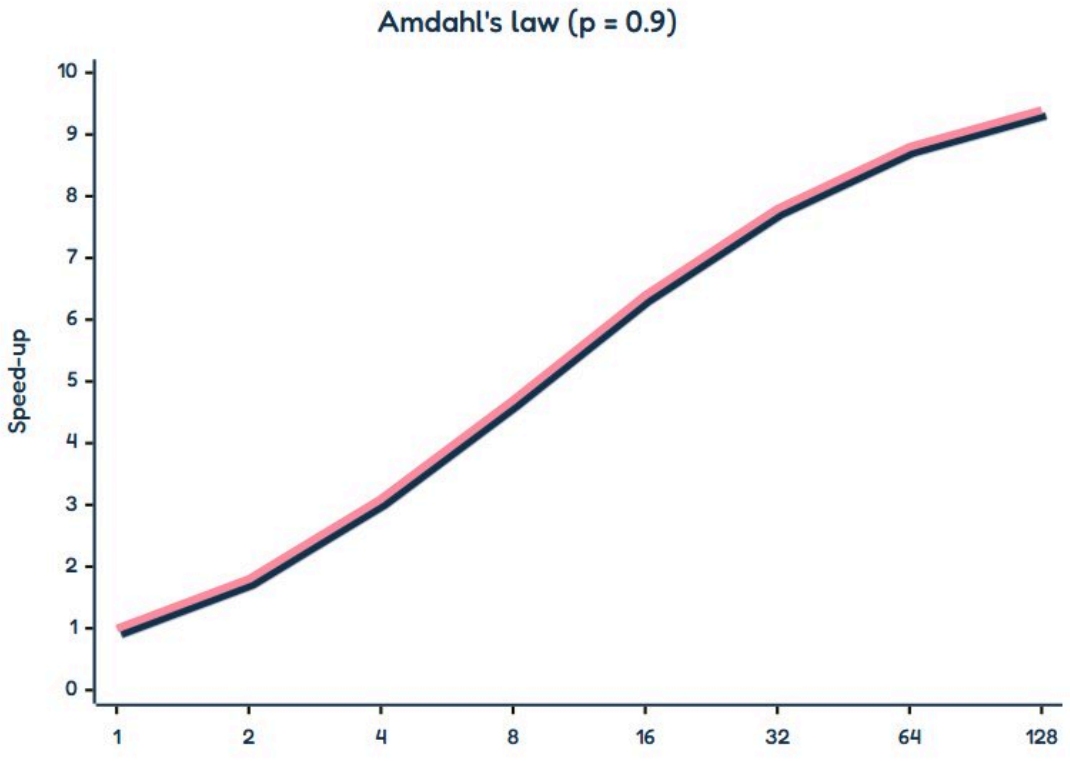


FIG Speed-up levels off when 10% of the job cannot be shared (p = 0.9).

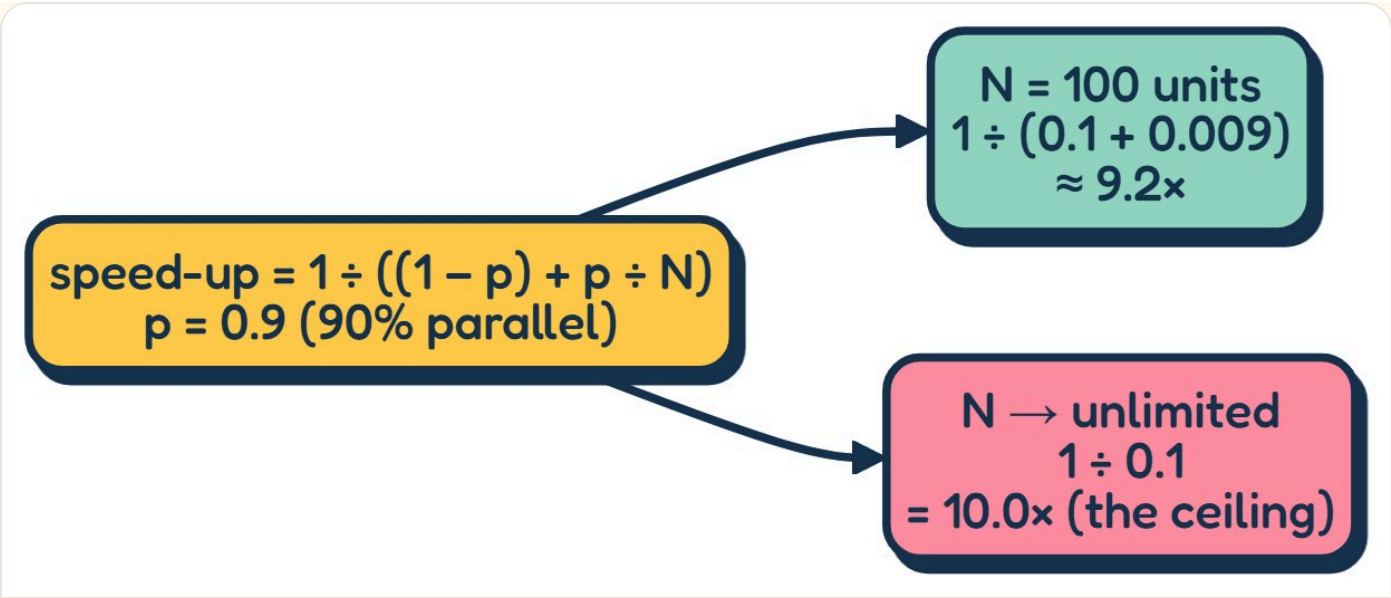


FIG The formula with numbers plugged in.

1.3

Other Kitchen Boosters

+10 XP

✨ READY, SET, COOK!

Secret kitchen boosters unlocked! 🚀 Which trick would you try first?

TECHNIQUE	WHAT IT DOES	WHY IT HELPS
Quantisation	INT8 or FP16 instead of FP32	More numbers fit in memory and per cycle
Specialised units	Tensor Cores, Neural Engine: small matrix product in one hardware step	Less control overhead per MAC
Batching	Process several inputs together	Fetch weights once, use them many times
Sparsity	Skip multiplications by zero	No work that can't change the answer
Operator fusion	Multiply + bias + ReLU in one pass	No writing and re-reading intermediates

Worked example: INT8 quantisation

w = [0.5, −0.31, 0.02, 0.97], scale = 1 ÷ 127

q = round(w ÷ scale) = [64, −39, 3, 123]

Back: q × scale = [0.5039, −0.3071, 0.0236, 0.9685]

Max error = 0.0039: tiny, for 4× less memory than FP32.

1.3 · SEE IT

+10 XP

Shrinking the Numbers

MAMA FENG

🔍 Spot it! Compare the original and squeezed numbers. How big is the biggest error?

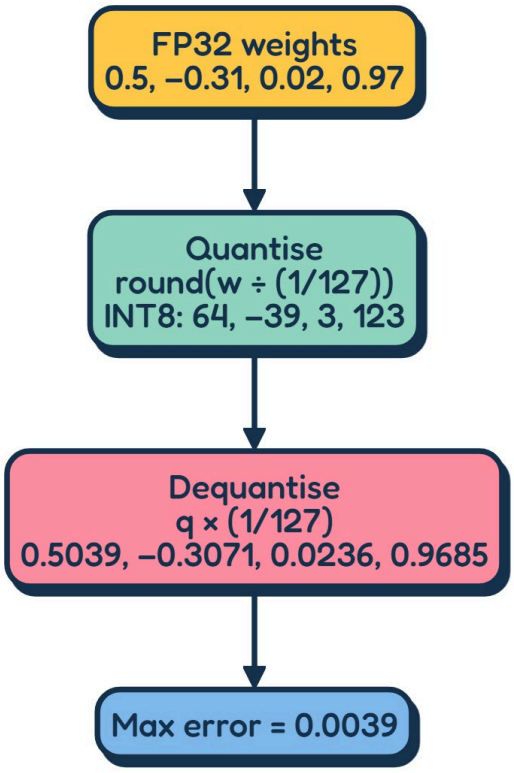


FIG Quantisation: FP32 weights mapped to 8-bit integers and back.

1.3 · SEE IT

+10 XP

Shrinking the Numbers (2)

MAMA FENG

🔍 Spot it! Compare the original and squeezed numbers. How big is the biggest error?

FIG Original vs dequantised weights: almost the same values.

SIDE QUEST · MINI-GAME 1.3

+25 XP

Chopping Race

✨ READY, SET, COOK!

Chopping race! 🔪 Time to prove whether 100 cooks really beat 10.

🎯 Your Mission

Discover why adding cooks stops helping, and size a model at different precisions.

🧺 Grab These First

speed-up = 1 ÷ ((1 − p) + p ÷ N). Bytes: FP16 = 2, INT8 = 1, 4-bit = 0.5.

🔪 Let's Go! Your Turn

(1) A fried-rice order is 90% chopping (p = 0.9) and 10% waiting for the one wok. PREDICT the speed-up with 100 cooks, then calculate for N = 1, 2, 4, 8, 16, 100, unlimited.

(2) A job is 80% parallel. Speed-up on 8 units? Limit with unlimited units?

(3) The 1,024 × 4,096 layer now has 8,192 outputs. MAC count? Time on the 4,096-unit, 1 GHz accelerator?

(4) A 7B model: weight memory in FP16, INT8, 4-bit?

What Did You Spot?

Plot speed-up against N. Where does the curve flatten?

Think It Over

In an AI pipeline, what is "the single wok"?

🔒 **Mama's feedback in the Answer Vault · V-1-3** · page 46 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #1.3

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.3) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. If 90% of a job can be parallelised, 100 cooks give a speed-up of 100×. ♡ ♡ ♡

- (T) True
- (F) False

2. What does “GPU” mean? ♡ ♡ ♡

- (A) Converting numbers to lower precision (e.g. FP32 → INT8).
- (B) Learning weights from data / using weights to answer.
- (C) One memory pool shared by CPU, GPU and NPU.
- (D) Thousands of simple cores working in parallel.

3. Which kitchen word means: “Multiply-accumulate: multiply two numbers, add to a running total.”? ♡ ♡ ♡

- (A) FLOP / TOPS
- (B) Unified memory
- (C) MAC
- (D) Core ML

4. What does “Amdahl's law” mean? ♡ ♡ ♡

- (A) Few, powerful, flexible cores.
- (B) High-bandwidth memory stacked beside the chip.
- (C) Google's accelerator built around a systolic MAC array.
- (D) Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.

TICKET #1.4 · RECIPE 01 · MAIN KITCHEN

1.4 The pantry problem

memory and efficiency

COURSES ☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 110 XP

☐ 1.4

The Pantry Problem

+10 XP

✨ READY, SET, COOK!

Here's a shocker: fetching food from the pantry can cost more than cooking it! 😱 Read on to see why.

KITCHEN	MEMORY LEVEL	SIZE AND SPEED
In your hands	Registers	Tiny, instant
Chopping board	Cache / on-chip SRAM	Small, very fast
Fridge across the kitchen	DRAM / HBM	Large, slower
Supermarket	Storage (SSD)	Huge, slowest

The surprising fact

Energy per operation (Horowitz 2014, 45 nm):

- 32-bit multiply ≈ 3.7 pJ
- Read from small SRAM ≈ 5 pJ
- Read from DRAM ≈ 640 pJ

A DRAM read costs ≈ 173× a multiply and ≈ 128× an SRAM read. Moving food costs more than cooking it!

💡 MAMA'S SECRET TIP

A good cook plans so everything needed is already on the chopping board. Good AI hardware and software do the same. HBM = DRAM stacked right beside the chip for very high bandwidth. Unified memory (Apple silicon) = CPU, GPU and NPU share one pool: no copying.

☐ 1.4 · SEE IT

The Memory Pyramid

+10 XP

✨ READY, SET, COOK!

Climb the memory pyramid! 🔺 Top = tiny and fast, bottom = huge and slow. Which floor do you live on?

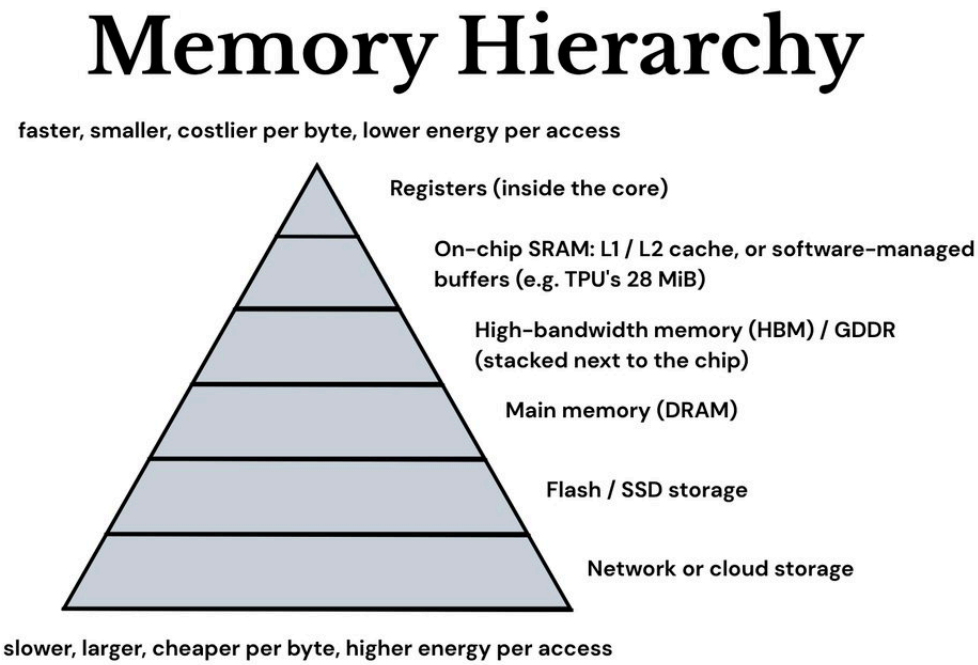


FIG The memory hierarchy, from registers at the top to cloud storage at the bottom.

🔺 Read the pyramid like a pantry

Top = what is in your hands: tiny, instant, cheapest energy per access.

Bottom = the supermarket across town: huge and cheap per byte, but slow and costly to fetch from.

Every level down is roughly 10× bigger and 10× slower. Good AI chips keep the weights as high up the pyramid as they can.

1.4

Cook Speed vs Delivery Speed

+10 XP

🏠 The roofline model

Arithmetic intensity = FLOPs ÷ bytes moved (how much cooking per bag of groceries).

Attainable performance = min(peak compute, intensity × memory bandwidth)

Plotted, it looks like a roof: a slope (memory limit) meeting a flat top (compute limit).

Ridge point = peak ÷ bandwidth. Below it: MEMORY-BOUND (the cook waits for delivery). Above it: COMPUTE-BOUND (the cook is busy).

🍱 Worked example: one layer, two batches

Chip: 10 TFLOP/s, 500 GB/s → ridge = 20 FLOP/byte. Layer 1,024 × 4,096 in FP16.

- **Batch 1:** intensity ≈ 1.0 → 0.5 TFLOP/s = 5% of peak (memory-bound; ~95% of time waiting)
- **Batch 32:** intensity ≈ 30.8 → 10 TFLOP/s = 100% (compute-bound)

Batching = cooking 32 plates from one delivery: weights fetched once, used 32 times.

⚠ PAPA'S OOPS!

"Batch everything, always!" Not quite! Batching raises throughput, but each order may wait longer (latency). A phone answers one user at a time, so it stays memory-bound.

1.4 · SEE IT

+10 XP

The Roofline

MAMA FENG

🔍 Spot it! Find the bend in the roof. Which batch sits on the slope, and which on the flat top?

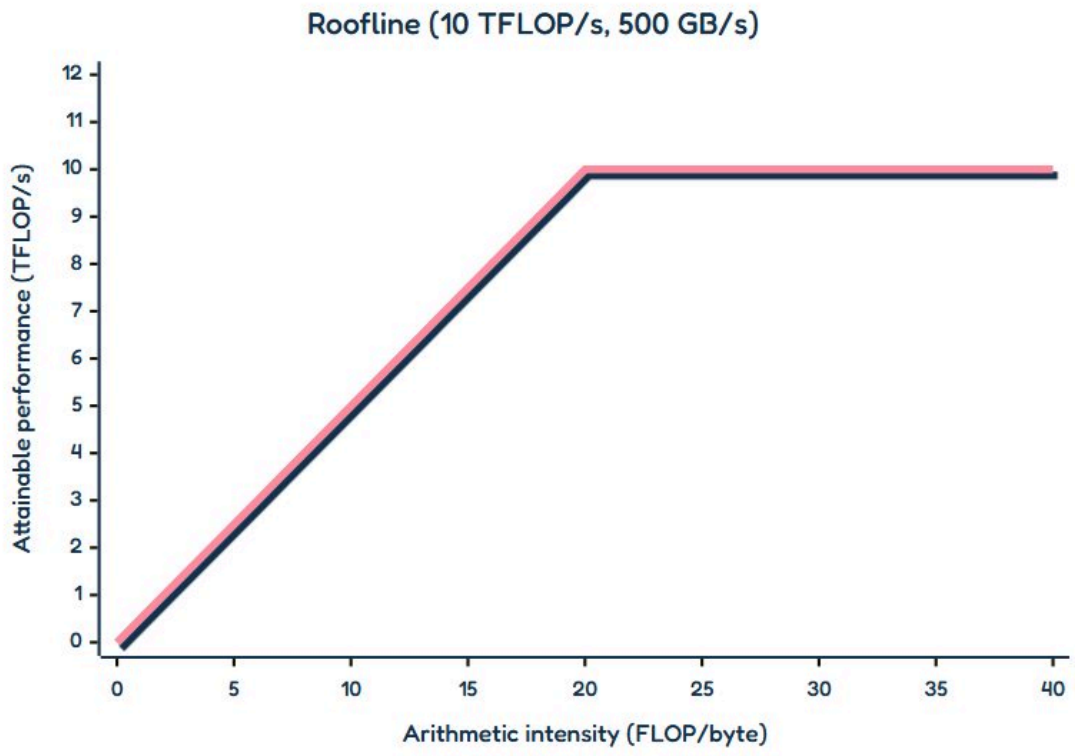


FIG The roofline for a 10 TFLOP/s chip with 500 GB/s of memory bandwidth.

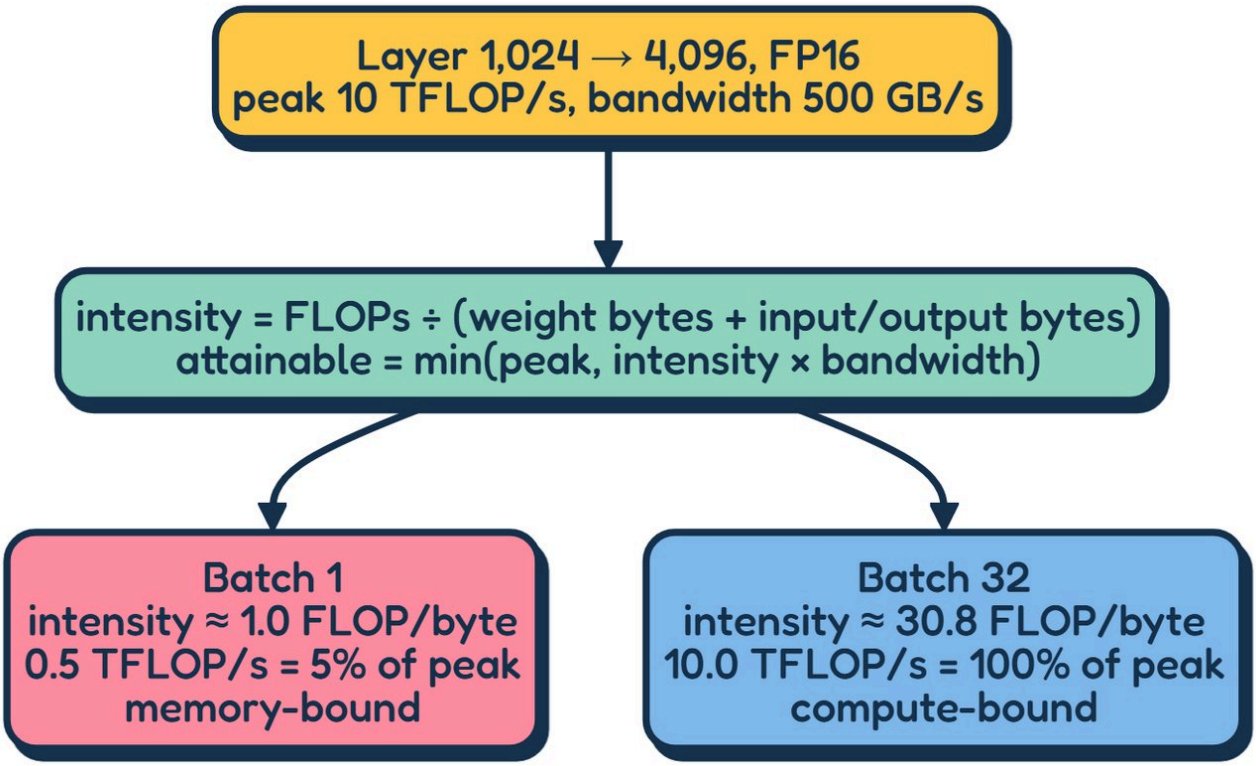


FIG Batch 1 vs batch 32: the same layer, memory-bound vs compute-bound.

1.4

Fixing A Memory-Bound Dish

+10 XP

✨ READY, SET, COOK!

Emergency! 🚨 A dish is stuck waiting on memory. Which fix would you try first?

IDEA	HOW IT HELPS	WHERE YOU SEE IT
Bigger on-chip memory	Weights stay close to the MACs	TPU's 28 MiB buffer; NPUs
Higher-bandwidth memory	Raises the slope of the roof	HBM beside the chip
Weight reuse	More ops per byte fetched	Systolic arrays, batched serving
Lower precision	Fewer bytes per weight	INT8/FP8/4-bit inference
Compression and pruning	Smaller or sparser models	Mobile deployment
Operator fusion	Intermediates never leave the chip	XLA, Core ML
Unified memory	No copy between CPU, GPU, NPU	Apple silicon

🔗 CONNECT THE DOTS

Bridges: moving data BETWEEN chips (NVLink, InfiniBand) is another, slower level of the same hierarchy (Recipe 02). A robot's sense-decide-act loop runs at batch 1, the memory-bound case (Recipe 03).

SIDE QUEST · MINI-GAME 1.4

+25 XP

Batch Cooking

✨ READY, SET, COOK!

Batch cooking challenge! 🍱 Cook smarter, not harder. Can you find the sweet spot?

🎯 Your Mission

Use the roofline to decide memory-bound vs compute-bound, and feel why data movement matters.

🧺 Grab These First

Chip: 10 TFLOP/s, 500 GB/s, ridge 20. Layer: 512 inputs × 2,048 outputs, FP16. Weights fetched once per batch; inputs and outputs once per sample.

🔪 Let's Go! Your Turn

(1) Rank by energy per access: DRAM read, register, small SRAM read.

(2) Weights and weight bytes of the layer?

(3) FLOPs at batch 1?

(4) Intensity at batch 1, 8 and 32: memory- or compute-bound? Attainable TFLOP/s and % of peak?

(5) Smallest compute-bound batch?

(6) A chip doubles its MACs but keeps the same bandwidth. Effect on a memory-bound job?

👀 What Did You Spot?

Record: batch | intensity | bound | attainable | % peak.

💭 Think It Over

A voice assistant answers one user at a time within milliseconds. What hardware would you choose?

🔒 **Mama's feedback in the Answer Vault · V-1-4** · page 46 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #1.4

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.4) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. At batch size 1, a large layer on a typical accelerator is usually memory-bound. ♡ ♡ ♡

- (T) True
- (F) False

2. A chip has 10 TFLOP/s peak and 500 GB/s memory bandwidth. Its ridge point is: ♡ ♡ ♡

- (A) 2 FLOP/byte
- (B) 20 FLOP/byte
- (C) 50 FLOP/byte
- (D) 5,000 FLOP/byte

3. What does “Roofline model” mean? ♡ ♡ ♡

- (A) One memory pool shared by CPU, GPU and NPU.
- (B) Multiply-accumulate: multiply two numbers, add to a running total.
- (C) Share of peak performance actually achieved.
- (D) Attainable = min(peak, intensity × bandwidth).

4. Which kitchen word means: “Google's accelerator built around a systolic MAC array.”? ♡ ♡ ♡

- (A) Fallback
- (B) Unified memory
- (C) TPU
- (D) Utilisation

TICKET #1.5 · RECIPE 01 · MAIN KITCHEN

1.5 Case study

the home rice cooker (Apple Neural Engine)

COURSES ☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 110 XP

1.5

Case Study: the Home Rice Cooker

+10 XP

✨ READY, SET, COOK!

Case file: the rice cooker in your pocket! 📱 Let's open up a real iPhone chip.

Apple Neural Engine

The ANE is the NPU inside Apple's chips. First shipped in 2017 in the A11 Bionic (iPhone X), described as 600 billion operations per second; later generations are quoted in TOPS.

It powers Face ID, image and object recognition, camera features, and language and speech processing, all ON THE DEVICE, often without internet.

COMPONENT	ROLE
CPU	Manages general tasks and coordinates AI operations
GPU	Many calculations at once, for graphics and AI
Neural Engine	Accelerates supported neural-network calculations
Unified memory	Holds models, inputs and results; shared by all three
Secure Enclave	Protects sensitive data such as Face ID templates

1.5 · SEE IT

+10 XP

Inside the A17 Pro

✨ READY, SET, COOK!

Spot the rice cooker! 🔍 Can you find the Neural Engine on the chip map?

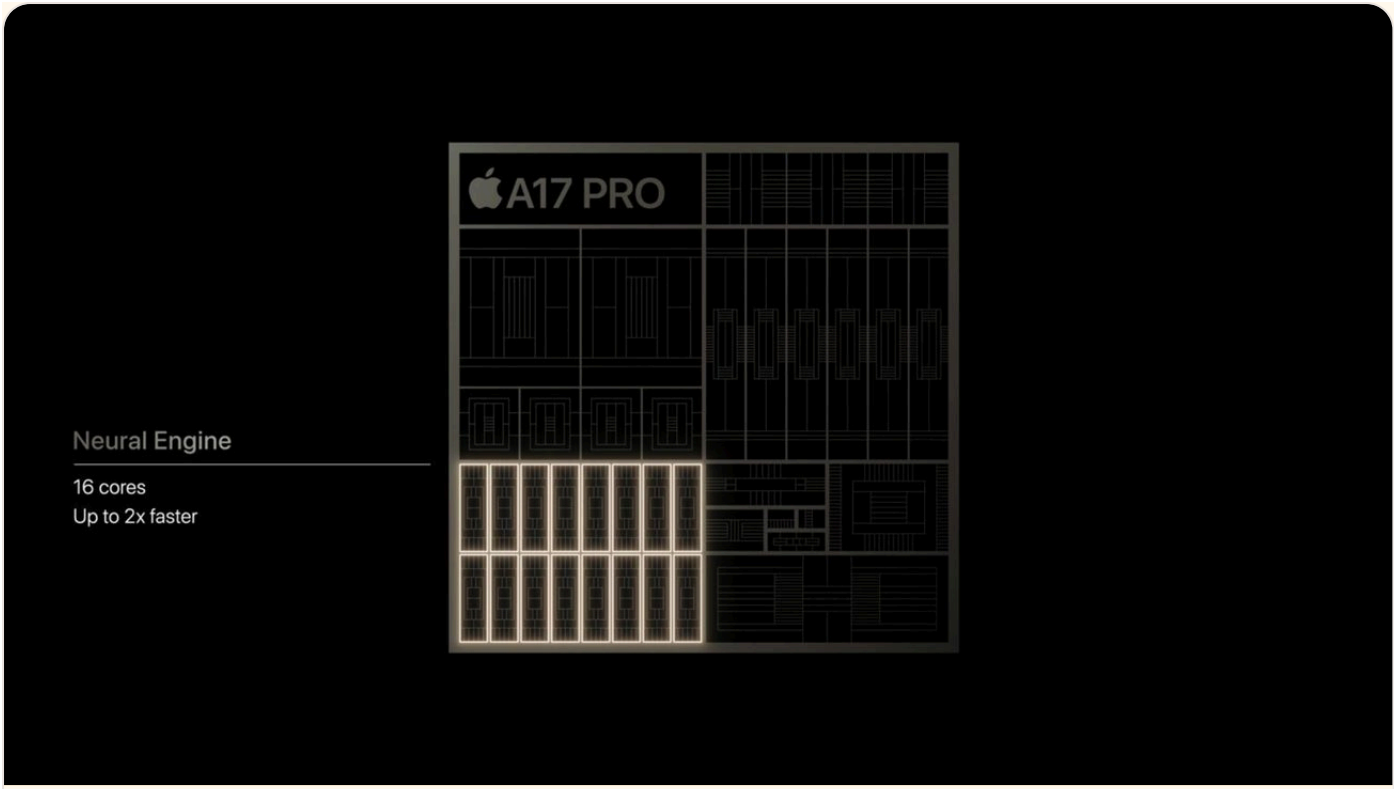


FIG Apple A17 Pro die diagram: the glowing block is the 16-core Neural Engine. Image: Apple.

Spot the rice cooker

The Neural Engine takes a big, dedicated corner of the chip. Its 16 cores do one job: the multiply-and-add maths of neural networks.

The CPU and GPU sit beside it and share the same unified memory, so a photo or a face scan never has to be copied between them.

1.5

How the ANE Cooks

+10 XP

✨ READY, SET, COOK!

How does the ANE cook so fast? 🍳 Link it back to everything you've learned!

Linking back

- Software (1.2): Core ML decides whether each part runs on the CPU, GPU or Neural Engine. Unsupported operations fall back to the CPU and can slow the whole model.
- Memory (1.4): unified memory, so no copying between processors.
- Precision (1.3): models usually quantised to INT8 or FP16.
- Batch size (1.4): Face ID processes ONE input at a time, the memory-bound case, so on-chip memory and small weights matter most.

🚚 Training vs inference

Training happens in the data centre on GPUs or accelerators. The finished weights are shipped to the iPhone, where inference runs on the Neural Engine: a fast, private answer.

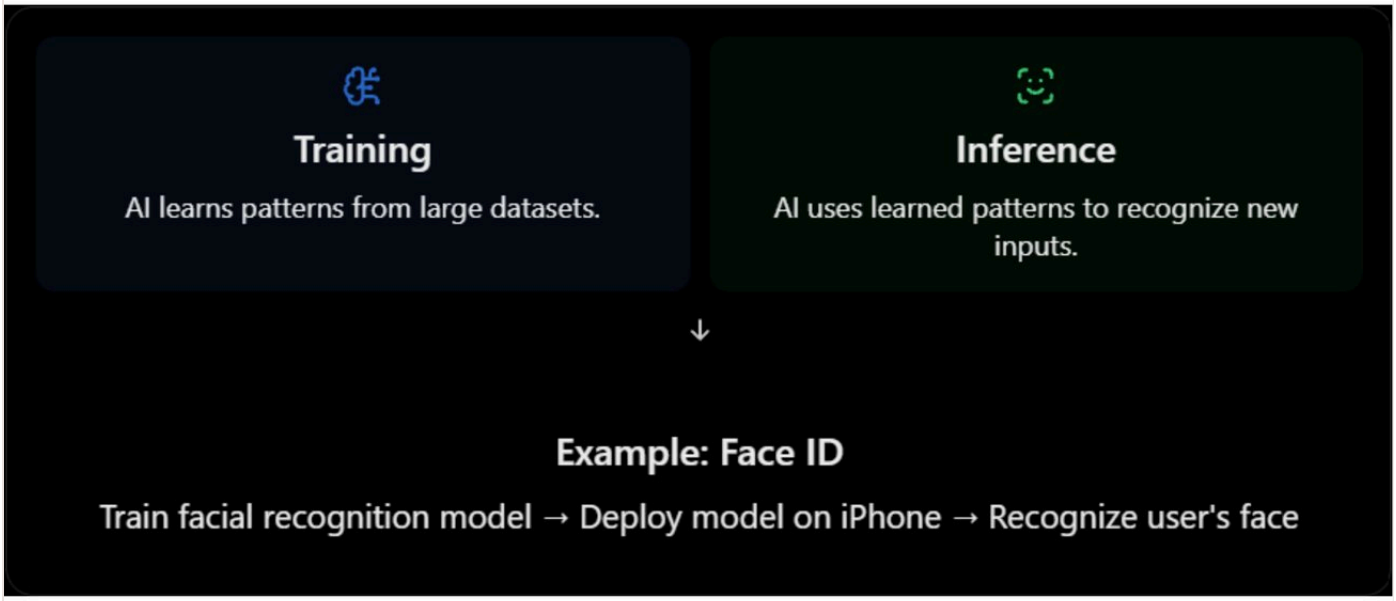


FIG Example: Face ID. Train the model, deploy it on the iPhone, recognise the face.

1.5

Face ID: Sense, Decide, Act

+10 XP

😊 Five steps

1. **Scan:** the TrueDepth camera captures the face with infrared light and depth sensors.
2. **Process:** facial information becomes a mathematical representation.
3. AI inference on the Neural Engine.
4. Compare with the registered face.
5. Unlock if it matches.

Face data is processed and protected on the device (Secure Enclave), not in the cloud.

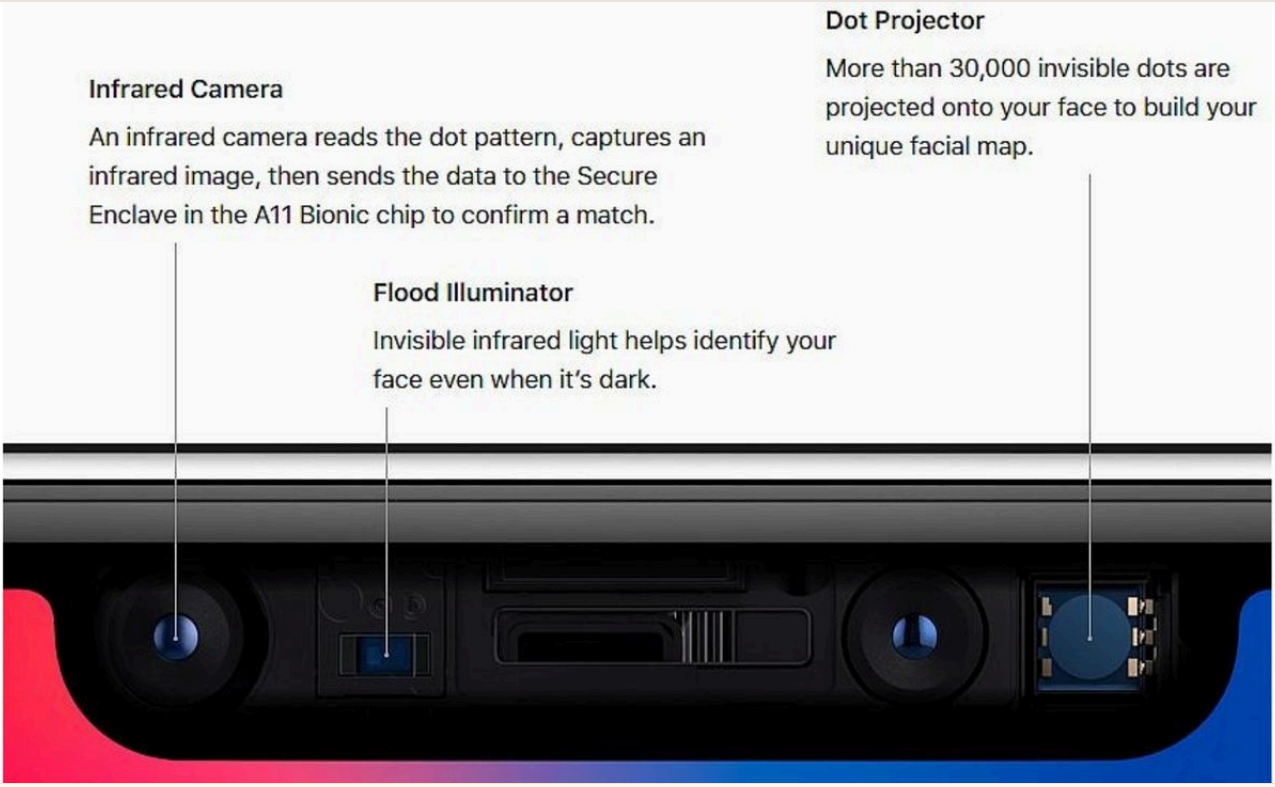


FIG The TrueDepth camera: dot projector, flood illuminator and infrared camera. Image: Apple.

🔗 CONNECT THE DOTS

Face ID is a small sense-decide-act loop: answer in a fraction of a second, one request at a time, on a battery. Robots face the same constraints (Recipe 03). Because the work stays on the device there is no network round trip (Recipe 02).

1.5 · SEE IT

+10 XP

Face ID In Five Steps

MAMA FENG

🔍 Spot it! Count the steps from scan to unlock. Which one runs on the Neural Engine?

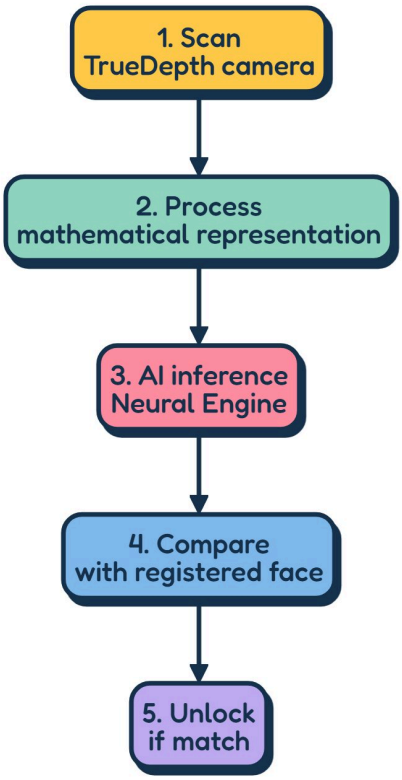


FIG Scan, process, infer on the Neural Engine, compare, unlock.

1.5

Pros & Cons

+10 XP

✨ READY, SET, COOK!

Thumbs up or thumbs down? 👍 👎 Weigh the good and the not-so-good.

👍 The Tasty Bits

Faster processing • energy efficiency (battery life) • better privacy (data stays on the device) • real-time performance • offline capability.

👎 The Burnt Bits

Limited computing resources compared with servers • not every model or operation is supported • large models may need more memory than a phone has • some advanced AI still needs the cloud.

❓ Case-study questions

1. One unsupported operation in the middle: why can the model run slowly?
2. **Face ID:** memory-bound or compute-bound? Why?
3. A 3B model quantised to 4-bit: weight memory? Comfortable on an 8 GB phone?
4. One benefit and one drawback of on-device vs cloud AI.
5. Why does unified memory save ENERGY as well as time?

SERVE THE CUSTOMERS · TICKET #1.5

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.5) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. What does “Training / inference” mean? ♡ ♡ ♡

- (A) Bits per number: FP32 4 B, FP16 2 B, INT8 1 B.
- (B) Multiply-accumulate: multiply two numbers, add to a running total.
- (C) Learning weights from data / using weights to answer.
- (D) One memory pool shared by CPU, GPU and NPU.

2. Which kitchen word means: “Few, powerful, flexible cores.”? ♡ ♡ ♡

- (A) CPU
- (B) Arithmetic intensity
- (C) SoC
- (D) Precision

3. What does “Memory-bound / compute-bound” mean? ♡ ♡ ♡

- (A) Share of peak performance actually achieved.
- (B) Limited by data delivery / by arithmetic.
- (C) Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.
- (D) Processors, memory and interconnects suited to running ML efficiently.

4. Which kitchen word means: “Apple framework that routes model parts to CPU, GPU or ANE.”? ♡ ♡ ♡

- (A) Deterministic execution
- (B) SoC
- (C) Systolic array
- (D) Core ML

LAB

TICKET #1.L · RECIPE 01 · MAIN KITCHEN

Kitchen Lab

Race The Crew

COURSES ▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 100 XP

TASTE TEST

Three quick questions

+10 XP

✨ READY, SET, COOK!

TASTE TEST time! 😋 No peeking at the answers. Let's see what you remember!

TASTE TEST

Q1

Face ID processes one face at a time. On the roofline it is most likely… A) compute-bound B) memory-bound C) network-bound

Q2

A 3-billion-parameter model quantised to 4-bit needs about… A) 12 GB B) 6 GB C) 1.5 GB

Q3

A model has one unsupported operation in the middle. Why can it run slowly on the NPU? A) the NPU overheats B) data must move to the CPU and back C) Core ML deletes the operation

🔒 **Answers in the Answer Vault · V-1-5** · page 46 · no peeking! +10 XP

KITCHEN LAB

+10 XP

Race the Crew

🌐 Live platform: Google Colab

Open colab.research.google.com → New notebook. Run on CPU, then Runtime → Change runtime type → T4 GPU and run again:

```
import torch, time
dev = "cuda" if torch.cuda.is_available() else "cpu"
a = torch.randn(4096, 4096, device=dev)
t = time.time(); [a @ a for _ in range(10)]
if dev == "cuda": torch.cuda.synchronize()
print(dev, time.time() - t)
```

Run each cell twice and use the 2nd time (the 1st GPU run includes warm-up).

Version: Colab free tier, CPU and T4 GPU runtimes, PyTorch 2.x (check with print(torch.__version__)).

🎯 Goal · Observe · Reflect

Goal: see the CPU-vs-GPU difference on the same maths.

Observe: MACs = 10 × $4{,}096^3$ ≈ 687 billion. Compute TFLOP/s = 2 × MACs ÷ time for each device. How many times faster is the GPU?

Reflect: Try size 256 instead of 4,096. Why does the GPU's advantage shrink (hint: idle units, launch overhead, memory)?

Offline fallback

Class chopping race: one student adds 40 pairs of numbers alone; then 8 students share them (5 each) but one "wok master" must total the 8 results. Time both and compute the speed-up. Compare with Amdahl's law.

⭐ Mama's Feedback (expected result)

Typical results: CPU about 15-30 s (≈ 0.05-0.1 TFLOP/s); T4 GPU well under 1 s (≈ 4-7 TFLOP/s), so the GPU is roughly 50-100× faster. Your numbers will differ with the machine Colab gives you.

Size 256: both finish in milliseconds and the GPU is only a few times faster, because thousands of its cores sit idle and launch overhead dominates.

Offline race: expect less than 8× speed-up, because the wok master's final sum is the serial part.

EXERCISE

+10 XP

Final round

✨ READY, SET, COOK!

Final round! 🎯 Three questions stand between you and your Recipe 01 stamp!

EXERCISE

Question 1: Layer cost

Task: A layer has 2,048 inputs and 1,024 outputs. How many MACs and FLOPs per input? How long on a 4,096-unit accelerator at 2 GHz (ideal)?

Question 2: Amdahl

Task: A pipeline is 95% parallel. What is the speed-up on 20 units, and the limit with unlimited units?

Question 3: Roofline

Task: A chip has 20 TFLOP/s peak and 1 TB/s bandwidth. What is its ridge point? A workload has intensity 4 FLOP/byte: attainable TFLOP/s and % of peak?

🔒 **Answers in the Answer Vault · V-1-6** · page 46 · no peeking! +10 XP

WATCH & LEARN

Video picks

+10 XP

🎬 **But what is a neural network?**

Creator: 3Blue1Brown (Deep learning ch. 1) · 18:40

Link: youtube.com/watch?v=aircAruvnKk

Why watch: Shows, with animation, that a network is layers of weighted sums: the MACs of 1.1.

Timestamps: 2:42 neurons · 3:35 layers · 11:34 counting weights and biases

Task: At 11:34, count the weights of the 784-16-16-10 network yourself and check his total.

Limitations: About recognising digits, not hardware; no chips are discussed.

🎬 **How do Graphics Cards Work? Exploring GPU Architecture**

Creator: Branch Education · 28:29

Link: youtube.com/watch?v=h9Z4oGN89MU

Why watch: A 3-D tour inside a real GPU: thousands of cores, memory chips and SIMD.

Timestamps: 2:15 GPU vs CPU · 4:56 GA102 architecture · 8:48 CUDA core · 12:04 GDDR6X/GDDR7 memory

Task: Write down how many cores the GPU has and link it to "1,000 Line Cooks" (1.2).

Limitations: Focuses on graphics/gaming; little about AI training or TPUs. Includes a sponsor segment.

WATCH & LEARN

Video picks

+10 XP

Tensor Processing Units (TPUs)

Creator: Google Cloud · 1:12

Link: youtube.com/watch?v=eyAjbgkBdjU

Why watch: A one-minute official summary of why Google built an AI-only chip.

Timestamps: Whole video (about 1 min)

Task: List two claims it makes and check each against 1.2 ("What the first TPU measured").

Limitations: Marketing video by the chip's maker: no numbers to test, not independent.

A very short intro to the Roofline model

Creator: NHR@FAU · 18:36

Link: youtube.com/watch?v=IrkNZG8MJ64

Why watch: A university HPC centre explains intensity, bandwidth and the roof, exactly as in 1.4.

Timestamps: Whole video; pause at each roofline plot

Task: Redraw the roof for 10 TFLOP/s and 500 GB/s and mark batch 1 and batch 32 from 1.4.

Limitations: Aimed at HPC programmers; uses CPU examples and some jargon beyond this guide.

SERVE THE CUSTOMERS · TICKET #1.L

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-1-Q1.L) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In Kitchen Lab 1 (Race the Crew), the GPU's speed-up over the CPU is LARGER for 256 × 256 matrices than for 4,096 × 4,096 matrices. ♡ ♡ ♡

- (T) True
- (F) False

2. Kitchen Lab 1: ten 4,096 × 4,096 matrix multiplications (about 687 billion MACs in total) take 0.25 s on a T4 GPU. What throughput did the GPU reach? ♡ ♡ ♡

- (A) 11 TFLOP/s
- (B) 0.07 TFLOP/s
- (C) 2.7 TFLOP/s
- (D) 5.5 TFLOP/s

3. What does “Fallback” mean? ♡ ♡ ♡

- (A) Google's accelerator built around a systolic MAC array.
- (B) Unsupported op runs on the CPU, costing data movement.
- (C) Converting numbers to lower precision (e.g. FP32 → INT8).
- (D) Processors, memory and interconnects suited to running ML efficiently.

4. Which kitchen word means: “Limited by data delivery / by arithmetic.”? ♡ ♡ ♡

- (A) HBM
- (B) Unified memory
- (C) MAC
- (D) Memory-bound / compute-bound

TICKET #1.B · RUSH HOUR IN THE MAIN KITCHEN

Rush Hour!

*Last in line, the trouble customer **Overheat Oven-Bot**: "My circuits run at 400 °C. Your MACs are too slow!"*

HAPPY CUSTOMERS ☐☐☐☐☐☐☐☐☐☐☐☐☐☐

GRADE S · A · B · C · D

Rules: 14 orders, no going back. Colour one box for each right dish (check in the Answer Vault, V-1-B). 9 or more happy customers = you survive the rush: +150 XP and the **Silicon Fried Rice** recipe card. Grade: S = 13–14, A = 12, B = 10–11, C = 9. Questions 13 and 14 are the trouble customer's double order.

1. Which chip best fits a voice assistant in a small battery-powered speaker? ♡ ♡ ♡

- (A) GPU
- (B) Server CPU
- (C) NPU
- (D) TPU pod

2. One MAC (multiply-accumulate) counts as two FLOPs. ♡ ♡ ♡

- (T) True
- (F) False

3. In Kitchen Lab 1 (Race the Crew), the GPU's speed-up over the CPU is LARGER for 256 × 256 matrices than for 4,096 × 4,096 matrices. ♡ ♡ ♡

- (T) True
- (F) False

4. Weights [2, −1, 0.5], inputs [1, 4, 6]. What is the neuron output (no bias)? ♡ ♡ ♡

- (A) 1.0
- (B) 3
- (C) 5
- (D) −1

5. A chip has 10 TFLOP/s peak and 500 GB/s memory bandwidth. Its ridge point is: ♡ ♡ ♡

- (A) 2 FLOP/byte
- (B) 20 FLOP/byte
- (C) 50 FLOP/byte
- (D) 5,000 FLOP/byte

6. Kitchen Lab 1: ten 4,096 × 4,096 matrix multiplications (about 687 billion MACs in total) take 0.25 s on a T4 GPU. What throughput did the GPU reach? ♡ ♡ ♡

- (A) 11 TFLOP/s
- (B) 0.07 TFLOP/s
- (C) 2.7 TFLOP/s
- (D) 5.5 TFLOP/s

7. If 90% of a job can be parallelised, 100 cooks give a speed-up of 100×. ♡ ♡ ♡

- (T) True
- (F) False

8. In a 2 × 2 systolic array with W = [[1, 2], [3, 4]] and x = [5, 6], the output y = W·x is: ♡ ♡ ♡

- (A) [12, 30]
- (B) [23, 34]
- (C) [11, 23]
- (D) [17, 39]

9. A TPU is more flexible than a CPU for general programs. ♡ ♡ ♡

- (T) True
- (F) False

10. At batch size 1, a large layer on a typical accelerator is usually memory-bound. ♡ ♡ ♡

- (T) True
- (F) False

11. Which kitchen word means: "Share of peak performance actually achieved."? ♡ ♡ ♡

- (A) Utilisation
- (B) Quantisation
- (C) Core ML
- (D) GPU

12. What does "Unified memory" mean? ♡ ♡ ♡

- (A) Apple framework that routes model parts to CPU, GPU or ANE.
- (B) Registers → SRAM/cache → DRAM/HBM → storage.
- (C) One memory pool shared by CPU, GPU and NPU.
- (D) FLOPs per byte moved.

13. Which kitchen word means: "Thousands of simple cores working in parallel."?

- (A) AI hardware
- (B) MAC
- (C) GPU
- (D) NPU

14. What does "Amdahl's law" mean?

- (A) Learning weights from data / using weights to answer.
- (B) Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.
- (C) Registers → SRAM/cache → DRAM/HBM → storage.
- (D) One floating-point operation (1 MAC = 2) / 10^{12} operations per second.

TICKET #V1 · RECIPE 01

Answer Vault · The Kitchen Crew

Only open after trying! Every vault entry you check is worth XP on your licence.

V-1-1
1.1 · Stir-Fry Maths

Mama's feedback · Stir-Fry Maths

(1) 0 → 2 → −2 → 1.0 (3 MACs). (2) 1,048,576 MACs = 2,097,152 FLOPs. (3) 30 TOPS. (4) (a) inference (b) training (c) inference. (5) 14 GB; 7 GB. Reflect: MACs are nearly all of a network's work, so speeding them up speeds up nearly everything.

V-1-Q1.1
Customer orders · ticket #1.1

1. True · One multiply plus one add = 2 FLOPs.

2. A · 1.0 · 2 × 1 + (−1) × 4 + 0.5 × 6 = 2 − 4 + 3 = 1.0.

3. A · Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it. · Amdahl's law: Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.

4. D · Roofline model · Roofline model: Attainable = min(peak, intensity × bandwidth).

V-1-2
1.2 · Hire the Chef

Mama's feedback · Hire the Chef

(1) (a) CPU (b) GPU, or TPU on Google Cloud (c) TPU / fixed accelerator (d) NPU > CPU > GPU. (2) Cycle 1: 8 and 0; cycle 2: 13 and 15 → y = [13, 15]; 6 vs 8 reads. (3) Different workload (fall-backs), software already targets GPUs, availability and cost, memory-bound workloads don't benefit. Reflect: fetching data costs far more energy and time than multiplying it.

V-1-Q1.2
Customer orders · ticket #1.2

1. False · Correct: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work.

2. C · NPU · An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power.

3. D · [17, 39] · 1·5 + 2·6 = 17; 3·5 + 4·6 = 39.

4. C · Thousands of simple cores working in parallel. · GPU: Thousands of simple cores working in parallel.

5. C · MAC · MAC: Multiply-accumulate: multiply two numbers, add to a running total.

V-1-3
1.3 · Chopping Race

Mama's feedback · Chopping Race

(1) 1.0, 1.8, 3.1, 4.7, 6.4, 9.2, 10×. (2) 1 ÷ (0.2 + 0.1) ≈ 3.3×; limit 5×. (3) Doubles to 8,388,608 MACs; ≈ 2 µs. (4) 14 GB, 7 GB, 3.5 GB. Reflect: data loading, pre-processing, or any step that can't be split.

V-1-Q1.3
Customer orders · ticket #1.3

1. False · Correct: Amdahl's law gives 1 ÷ (0.1 + 0.9/100) ≈ 9.2×; the limit is 10×, set by the serial 10%.

2. D · Thousands of simple cores working in parallel. · GPU: Thousands of simple cores working in parallel.

3. C · MAC · MAC: Multiply-accumulate: multiply two numbers, add to a running total.

4. D · Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it. · Amdahl's law: Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.

V-1-4
1.4 · Batch Cooking

⭐ Mama's feedback · Batch Cooking

(1) register < SRAM (~5 pJ) < DRAM (~640 pJ). (2) 1,048,576 weights = 2,097,152 bytes. (3) 2,097,152 FLOPs. (4) batch 1 ≈ 1.0, memory, 0.5 TF, 5% • batch 8 ≈ 7.9, memory, 3.9 TF, 39% • batch 32 ≈ 29.7, compute, 10 TF, 100%. (5) 22. (6) Almost nothing: the units just wait. Reflect: batch 1 is memory-bound → high bandwidth, big on-chip memory, low-precision weights (an NPU).

V-1-Q1.4
Customer orders · ticket #1.4

1. True · Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte.

2. B · 20 FLOP/byte · $10 \times 10^{12} \div (500 \times 10^{9}) = 20$ FLOP/byte; below 20 the chip is memory-bound.

3. D · Attainable = min(peak, intensity × bandwidth). · Roofline model: Attainable = min(peak, intensity × bandwidth).

4. C · TPU · TPU: Google's accelerator built around a systolic MAC array.

V-1-Q1.5
Customer orders · ticket #1.5

1. C · Learning weights from data / using weights to answer. · Training / inference: Learning weights from data / using weights to answer.

2. A · CPU · CPU: Few, powerful, flexible cores.

3. B · Limited by data delivery / by arithmetic. · Memory-bound / compute-bound: Limited by data delivery / by arithmetic.

4. D · Core ML · Core ML: Apple framework that routes model parts to CPU, GPU or ANE.

V-1-5
1.L · Three quick questions

⭐ Answers · Three quick questions

Q1 = B. Batch 1 means low arithmetic intensity: the weights are fetched for just one input, so memory is the limit.

Q2 = C. $3 \times 10^{9} \times 0.5$ bytes = 1.5 GB: it fits on a phone with 8 GB of shared memory, but it is still a big share.

Q3 = B. Each switch to the CPU and back moves data between processors and loses the accelerator's efficiency; the data movement, not the operation, is the cost.

Case Q4: benefit = privacy, speed, offline; drawback = limited model size and compute.

Case Q5: no copies between processors means fewer DRAM transfers, and DRAM reads cost ~640 pJ each.

All right? Mama is proud: Gold stamp! Missed one? Re-read 1.4-1.5 and try again for Silver.

V-1-6
1.L · Final round

⭐ Answers · Final round

Q1: $2{,}048 \times 1{,}024 = 2{,}097{,}152$ MACs = 4,194,304 FLOPs. 4,096 MACs × 2×10^{9} per second = 8.192×10^{12} MAC/s → $2{,}097{,}152 \div 8.192 \times 10^{12} \approx 0.26$ µs.

Q2: $1 \div (0.05 + 0.95 \div 20) = 1 \div 0.0975 \approx 10.3\times$; limit $1 \div 0.05 = 20\times$.

Q3: ridge = $20 \times 10^{12} \div 10^{12} = 20$ FLOP/byte. Attainable = min(20, 4 × 1) = 4 TFLOP/s = 20% of peak: memory-bound.

V-1-Q1.L

Customer orders · ticket #1.L

1. False · Correct: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster.

2. D · 5.5 TFLOP/s · TFLOP/s = 2 × MACs ÷ time = $2 \times 6.87 \times 10^{11} \div 0.25 \approx 5.5 \times 10^{12}$. 2.7 forgets the factor 2; 11 doubles it again.

3. B · Unsupported op runs on the CPU, costing data movement. · Fallback: Unsupported op runs on the CPU, costing data movement.

4. D · Memory-bound / compute-bound · Memory-bound / compute-bound: Limited by data delivery / by arithmetic.

V-1-B

Rush hour · Main Kitchen

1. C · NPU · An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power.

2. True · One multiply plus one add = 2 FLOPs.

3. False · Correct: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster.

4. A · 1.0 · $2 \times 1 + (-1) \times 4 + 0.5 \times 6 = 2 - 4 + 3 = 1.0$.

5. B · 20 FLOP/byte · $10 \times 10^{12} \div (500 \times 10^{9}) = 20$ FLOP/byte; below 20 the chip is memory-bound.

6. D · 5.5 TFLOP/s · TFLOP/s = 2 × MACs ÷ time = $2 \times 6.87 \times 10^{11} \div 0.25 \approx 5.5 \times 10^{12}$. 2.7 forgets the factor 2; 11 doubles it again.

7. False · Correct: Amdahl's law gives $1 \div (0.1 + 0.9/100) \approx 9.2\times$; the limit is 10×, set by the serial 10%.

8. D · [17, 39] · $1 \cdot 5 + 2 \cdot 6 = 17$; $3 \cdot 5 + 4 \cdot 6 = 39$.

9. False · Correct: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work.

10. True · Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte.

11. A · Utilisation · Utilisation: Share of peak performance actually achieved.

12. C · One memory pool shared by CPU, GPU and NPU. · Unified memory: One memory pool shared by CPU, GPU and NPU.

13. C · GPU · GPU: Thousands of simple cores working in parallel.

14. B · Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it. · Amdahl's law: Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.

02

ROOM 02 · FRONT COUNTER · TICKET #D2

Order Up!

NETWORKS & AI INFRASTRUCTURE

MAMA FENG

Mama's Problem: a customer on the other side of town points a phone at a street sign and wants it translated in about one second. The photo must travel to a kitchen, be cooked by an AI model, and come back. Which kitchen should cook it, and how do we deliver fast?

By the end you will be able to

1. Describe network architectures and explain how layers and encapsulation let different machines talk.
2. Explain how TCP/IP moves data reliably, and where UDP and QUIC differ.
3. Compare cloud, edge and on-device computing for an AI job, using latency, bandwidth, privacy and model size.
4. Calculate propagation, transmission and queueing delay and the bandwidth-delay product.
5. Explain scalability, tail latency and why communication limits training on many chips.
6. Analyse how 5G and AI-powered network optimisation support a cloud AI service.

TONIGHT'S TICKETS

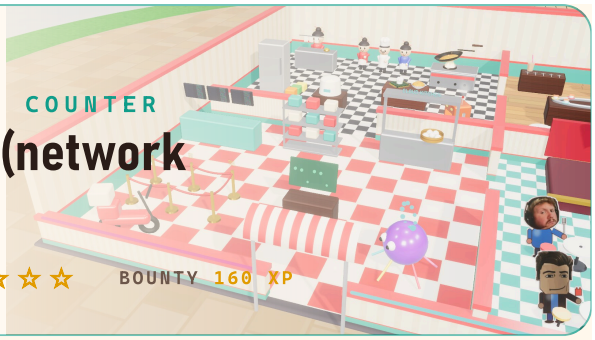
- [ ] 2.1 How kitchens are connected (networ... +160 XP
- [ ] 2.2 The careful waiter and the shouting ... +160 XP
- [ ] 2.3 Central kitchen, food truck or home ... +140 XP
- [ ] 2.4 Delivery time, road width and rush h... +150 XP
- [ ] 2.5 Case study +140 XP
- [ ] LAB Kitchen Lab +90 XP
- [ ] Rush Hour +150 XP

RUSH HOUR · TROUBLE CUSTOMER: THE LAG MONSTER

TICKET #2.1 · RECIPE 02 · FRONT COUNTER

2.1 How kitchens are connected (network architecture & protocols)

COURSES ▢▢▢▢▢▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 160 XP



BRIEFING

Mama's problem

+10 XP

MAMA FENG

Mama's Problem: a customer on the other side of town points a phone at a street sign and wants it translated in about one second. The photo must travel to a kitchen, be cooked by an AI model, and come back. Which kitchen should cook it, and how do we deliver fast?

Today's Adventure

2.1 How kitchens are connected (network architecture & protocols)

2.2 The careful waiter and the shouting waiter (TCP/IP, UDP, QUIC)

2.3 Central kitchen, food truck or home kitchen? (cloud vs edge)

2.4 Delivery time, road width and rush hour (latency, bandwidth, scalability)

2.5 Case study: 5G delivery and AI that runs the roads

BRIEFING

Goals & what you need

+10 XP

✨ READY, SET, COOK!

Order up! 🛵 Get ready to follow your data across the world and back in a blink.

By the end you will be able to

1. Describe network architectures and explain how layers and encapsulation let different machines talk.
2. Explain how TCP/IP moves data reliably, and where UDP and QUIC differ.
3. Compare cloud, edge and on-device computing for an AI job, using latency, bandwidth, privacy and model size.
4. Calculate propagation, transmission and queueing delay and the bandwidth-delay product.
5. Explain scalability, tail latency and why communication limits training on many chips.
6. Analyse how 5G and AI-powered network optimisation support a cloud AI service.

Before You Jump In

From Recipe 01: MACs, bandwidth (bytes per second), the memory hierarchy, batch size. Basic arithmetic. No networking background needed.

Running example: translating a street sign with your phone camera.

2.1

Definition

+10 XP

✨ READY, SET, COOK!

What IS a network, really? 🌐 Guess in one sentence, then compare with Mama's!

📖 The Big Idea

A computer network is a group of devices (nodes) joined by links so they can exchange data.

Data is cut into small pieces called packets. Each packet carries a header: the "delivery label" with the information needed to get it to the right place.

A protocol is an agreed set of rules for how packets are formatted, sent, received and acknowledged.

Why AI needs networks

1. Delivering data to the model: a photo, voice clip or sensor reading travels to where the model runs.
2. Delivering the model to the user: weights are downloaded to a phone or server.
3. Connecting the chips that run the model: big models are split over many accelerators that swap results constantly.

2.1

Network Scales

+10 XP

LILY

So a network is just 'send data over a link'? The same idea, from inside a chip to across the planet. Speed and delay change by many orders of magnitude, like the pantry problem in Recipe 01, extended outwards.

SCALE	TYPICAL LINK	BANDWIDTH	DELAY
Inside a chip package	HBM to chip	~TB/s	100s of ns
GPUs in one server	NVLink, PCIe	up to 100s of GB/s	~µs
Servers in a data centre	InfiniBand / Ethernet 100-400 Gbit/s	12.5-50 GB/s	µs to 10s of µs
Data centre to users	Fibre backbone	0.1-10 Gbit/s per user	ms to ~100 ms
Last hop to a phone	Wi-Fi, 4G, 5G	10s-100s of Mbit/s	ms to 10s of ms

🔬 REAL-LIFE TECH

Networks are quoted in BITS per second (Gbit/s); memory in BYTES per second (GB/s). Divide by 8: 100 Gbit/s = 12.5 GB/s.

2.1

Network Architectures

+10 XP

✨ READY, SET, COOK!

Leaf, spine, tree… 🌳 Why do data centres look like a forest? Let's explore!

ARCHITECTURE	IDEA	AI EXAMPLE
Client-server	Many clients send requests to a few servers that do the work	Phone app sends a photo to a cloud translation service
Peer-to-peer	Machines talk directly, no central server	Devices sharing model updates in federated learning
Leaf-spine (data centre)	Every leaf switch connects to every spine switch: many equal paths	A training cluster where thousands of GPUs swap gradients
Hybrid	A mix: device + edge server + cloud	The 5G case study in 2.5

Why leaf-spine?

One giant switch would be a bottleneck and a single point of failure. In a leaf-spine fabric you grow by adding switches, traffic between any two servers has several equal paths, and if one spine fails the others carry the load. Like a kitchen with many doors instead of one jammed hatch.

2.1

Layers & Encapsulation

+10 XP

✨ READY, SET, COOK!

Box inside a box inside a box! 📦 Let's pack a message like a bento.

The four-layer TCP/IP model

Application: what programs say to each other (HTTP, gRPC, MQTT, DNS)

Transport: delivery between programs, reliability (TCP, UDP, QUIC)

Network: addressing and routing across networks (IP)

Link & physical: moving bits over one hop (Ethernet, Wi-Fi, 5G radio)

(The 7-layer OSI model splits these further; you will see both in textbooks.)

Encapsulation = packing a bento

As data goes DOWN the stack, each layer wraps it in its own header, like putting food in a box, the box in a bag, the bag in a delivery crate. At the destination each layer unwraps its own header and passes the rest UP.

Application data → + TCP header (20 B) → + IP header (20 B) → + Ethernet header (14 B) and check (4 B).

2.1

Worked Example

+10 XP

✨ READY, SET, COOK!

Calculator ready? 🧮 How much of your message is actually the packaging? Guess first!

How much of a packet is food?

Overhead per packet = 20 (TCP) + 20 (IP) + 14 + 4 (Ethernet) = 58 bytes.

1,000-byte payload: frame = 1,058 bytes. Efficiency = 1,000 ÷ 1,058 ≈ 94.5%.

100-byte payload (a small sensor reading): 100 ÷ 158 ≈ 63.3%.

Result: big payloads waste little on packaging; tiny messages waste over a third of the link. That is why small messages are often batched together.

KITCHEN TERMINAL · PYTHON

```
HEADERS = 20 + 20 + 14 + 4
def efficiency(p): return p / (p + HEADERS)
print(f"{efficiency(1000):.1%}")   # 94.5%
print(f"{efficiency(100):.1%}")    # 63.3%
```

⚠ PAPA'S OOPS!

"Packets are always 58 bytes of overhead!" Not quite! IPv6 headers are 40 bytes, TCP can carry options, Wi-Fi adds framing. 58 is a teaching figure, not a rule.

2.1 · SEE IT

Packing the Bento

+10 XP

MAMA FENG

🔍 Spot it! Count the wrappers! How many bytes of headers wrap one small message?

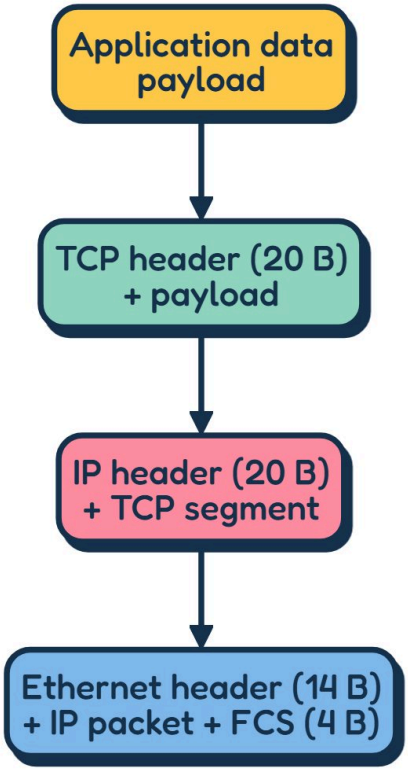


FIG Encapsulation: each layer adds its own header around the data.

2.1 · SEE IT

+10 XP

Packing the Bento (2)

MAMA FENG

🔍 Spot it! Count the wrappers! How many bytes of headers wrap one small message?

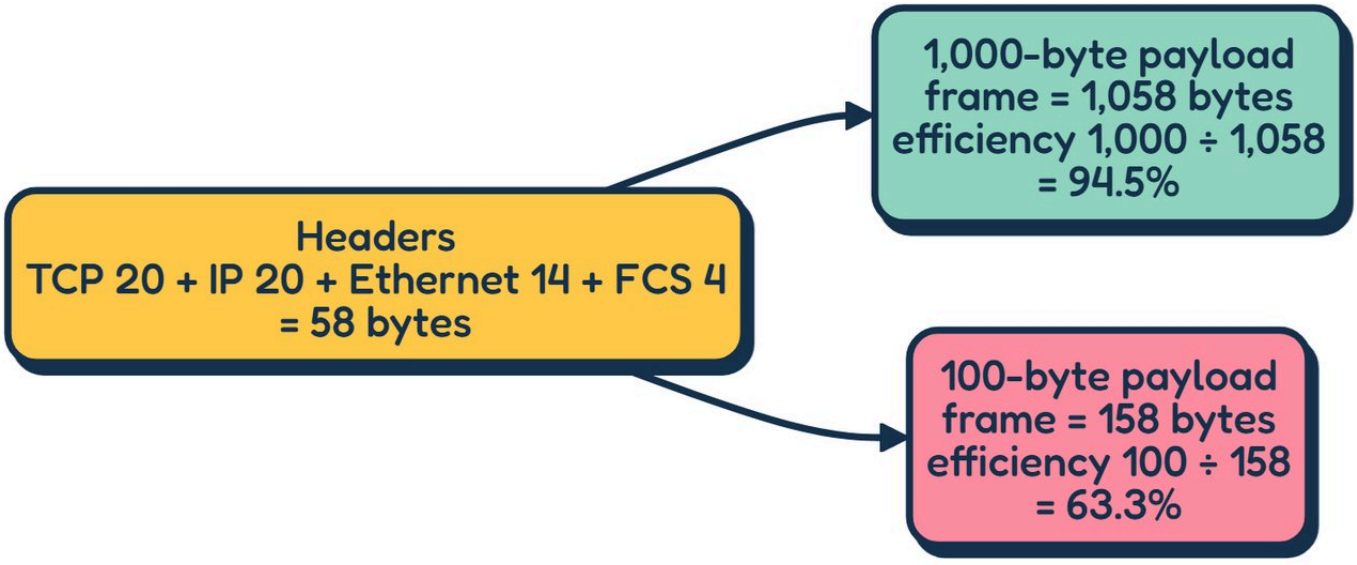


FIG Worked example: how much of a packet is useful data?

2.1

Protocols On the Menu

+10 XP

✨ READY, SET, COOK!

Protocol menu! 📜 Which rule set would you order for a video call?

PROTOCOL	STYLE	USED IN AI FOR
HTTP / REST (JSON)	Request and response, human-readable	Public AI APIs: send a prompt or image, get a result
gRPC	Compact binary, typed, streaming	Service-to-service calls inside a data centre
WebSocket / SSE	Long-lived connection the server can push on	Streaming a chatbot answer word by word
MQTT	Lightweight publish/subscribe	Thousands of low-power sensors reporting to a hub
RTP / WebRTC	Real-time audio and video, usually over UDP	Live voice assistants, video calls

🔑 REMEMBER THIS!

A cloud AI request passes several services, each speaking the protocol that suits its job: phone → HTTPS/JSON → API gateway → gRPC → model server (GPU/TPU) → back again.

🗺️ SIDE QUEST · MINI-GAME 2.1

Bento Packing

+25 XP

✨ READY, SET, COOK!

Bento packing challenge! 🍱 Pack it perfectly and earn your stamp.

🎯 Your Mission

Calculate packet efficiency and match AI scenarios to the right protocol and layer.

🧺 Grab These First

The 58-byte overhead from the worked example. The protocol table on the previous page.

🔪 Let's Go! Your Turn

(1) Compute the efficiency for a 1,200-byte payload and a 40-byte payload.

(2) Name the TCP/IP layer for: (a) choosing the route across the internet, (b) re-sending a lost segment, (c) turning bits into radio, (d) an HTTP request for a translation.

(3) Match a protocol (HTTP/REST, gRPC, MQTT, WebRTC): (a) battery sensors to a hub, (b) web server to model server in one data centre, (c) live voice assistant, (d) public image API.

What Did You Spot?

Write each efficiency as a %. For the layers, ask: "is this about the content, the delivery between programs, the route, or the wire?"

Think It Over

Why does a data centre use a leaf-spine fabric instead of one very large switch? Write 2-3 sentences.

🔒 **Mama's feedback in the Answer Vault · V-2-1** · page 93 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #2.1

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.1) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Encapsulation means each layer adds its own header to the data from the layer above. ♡ ♡ ♡

- (T) True
- (F) False

2. What does “Packet” mean? ♡ ♡ ♡

- (A) Round-trip time: to the destination and back.
- (B) Unreliable, connectionless packet delivery: fast, no retransmission.
- (C) Computation placed close to where data is produced or used.
- (D) A small chunk of data plus a header with delivery information.

3. Which kitchen word means: “Time for data to arrive or a request to be answered.”? ♡ ♡ ♡

- (A) Latency
- (B) Utilisation
- (C) QUIC
- (D) Ring all-reduce

4. What does “Encapsulation” mean? ♡ ♡ ♡

- (A) Reliable, encrypted transport built on UDP; saves a round trip.
- (B) Devices train locally and share only model updates.
- (C) Numerical address of a device (IPv4 32 bits, IPv6 128 bits).
- (D) Each layer wraps data in its own header on the way down the stack.

TICKET #2.2 · RECIPE 02 · FRONT COUNTER

2.2 The careful waiter and the shouting waiter (TCP/IP, UDP, QUIC)

COURSES ☐☐☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 160 XP

2.2 IP: Best-Effort Delivery

+10 XP

✨ READY, SET, COOK!

IP just does its best... and that's it?! 😅 Find out what 'best effort' really means.

📖 The Big Idea

Every device on the internet has an IP address: IPv4 = 32 bits (about 4.3 billion addresses), IPv6 = 128 bits.

A router reads the destination address on each packet and forwards it one hop closer. No router tracks the whole journey.

IP is "best effort": a packet may be delayed, lost, duplicated or arrive out of order, and IP does not tell anyone. Simple, and that is why the internet scales. But most apps need more.

LEO

So if a router is too busy, it just drops my packet? Yes! Like a delivery rider who drops one box when the scooter is full. Something above IP has to notice and fix it. Enter TCP.

2.2 TCP: the Careful Waiter

+10 XP

✨ READY, SET, COOK!

Meet the careful waiter! 🤵 He checks EVERY plate. Is that a good thing or a slow thing?

TCP adds four things on top of IP

1. **Connection setup:** a three-way handshake, SYN → SYN-ACK → ACK.
2. **Reliability:** every byte has a sequence number; the receiver sends ACKs; anything not ACKed in time is sent again.
3. **Flow control:** the receiver advertises a receive window so a fast sender cannot overwhelm it.
4. **Congestion control:** the sender slows down when the network looks jammed (slow start, then careful growth, cut back on loss).

💡 MAMA'S SECRET TIP

TCP is a registered courier. Phone ahead to check someone is home (handshake). Number and sign for every parcel (ACKs). No signature? Send it again. Hallway full? Slow down (flow control). Roads jammed? Send fewer vans (congestion control).

2.2 · SEE IT

+10 XP

The TCP Handshake

MAMA FENG

🔍 Spot it! How many trips across the table before the first real data arrives?

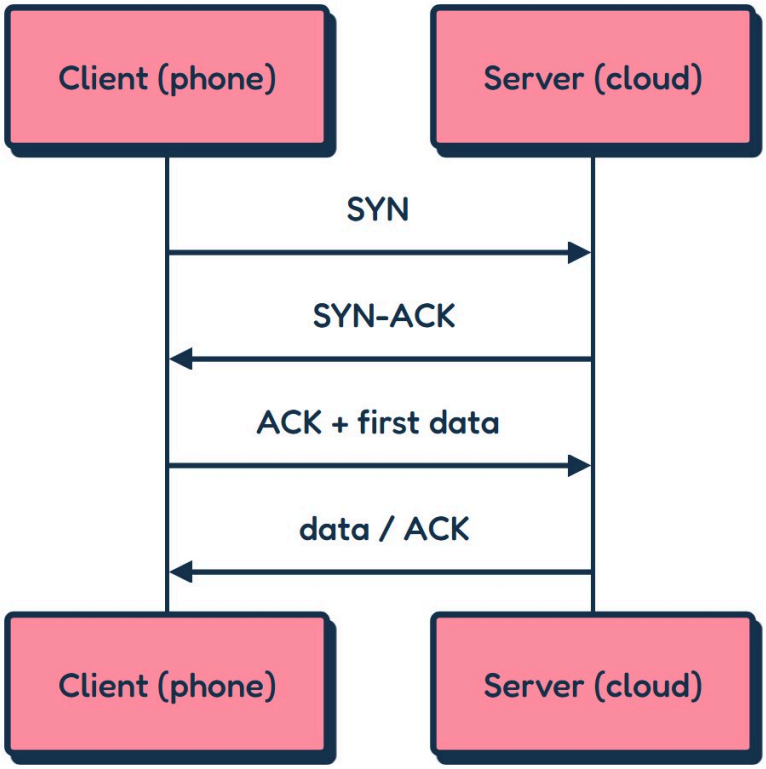


FIG TCP: handshake, data with acknowledgements, then closing.

2.2

+10 XP

UDP & Quic: the Shouting Waiter

✨ READY, SET, COOK!

And now the shouting waiter! 📣 Fast and loud, but does he drop plates?

PROTOCOL	RELIABLE?	SETUP	TYPICAL USE
TCP	Yes, ordered stream	1 round trip (+ security)	Web, file and model downloads, most APIs
UDP	No: just send packets	None	Live voice/video, games, DNS
QUIC	Yes, built on UDP	Transport + security together, ~1 round trip	HTTP/3, modern web and mobile apps

Why would anyone want unreliable?

For live audio, an OLD packet is worse than NO packet. A voice assistant that waits to resend a lost 20 ms chunk only delivers it too late. UDP shouts the order across the kitchen and moves on.

QUIC runs on UDP, combines connection setup with encryption (saving a round trip), and avoids TCP's problem where one lost packet blocks everything behind it.

🔬 REAL-LIFE TECH

Most AI APIs use TLS so your photo or prompt is encrypted in transit. TLS 1.3 adds one extra round trip on top of TCP's handshake.

2.2

Worked Example: First Bite

+10 XP

✨ READY, SET, COOK!

Stopwatch on! ⏱️ How long until the very first bite arrives? Let's work it out.

⏱️ Time to first byte

A phone is 40 ms (round trip) from a cloud server. Fresh connection, ignore server time.

TCP + TLS 1.3: handshake 1 + TLS 1 + request/first byte 1 = 3 round trips → 3 × 40 = 120 ms.

QUIC: combined handshake 1 + request 1 = 2 round trips → 2 × 40 = 80 ms.

(Connection reuse and session resumption can remove some of these round trips.)

🔑 REMEMBER THIS!

Handshakes cost ROUND TRIPS, and a round trip cannot be shortened by buying a faster link. Only moving the server closer, or doing fewer round trips, helps. The networking version of "moving data costs more than cooking it".

2.2 · SEE IT

Time To First Bite

+10 XP

MAMA FENG

🔍 Spot it! Spot the difference: why does QUIC save a whole round trip?

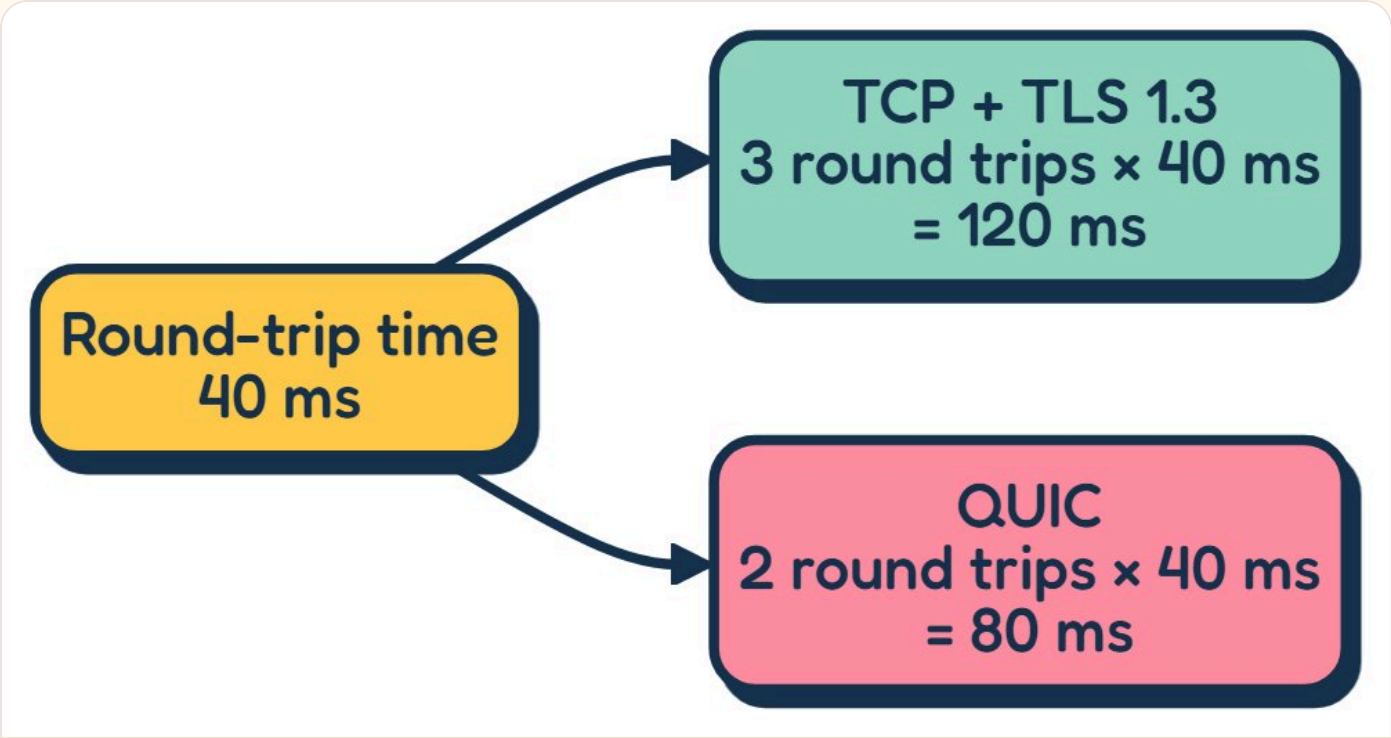


FIG Worked example: round trips add up before the first byte arrives.

2.2

Worked Example: Fast Road, Slow Delivery

+10 XP

✨ READY, SET, COOK!

Fast road, slow delivery?! 🛣️ Spot the sneaky bottleneck.

The bandwidth-delay product (BDP)

TCP can only have one window of data "in flight" per round trip. To keep a link full, the window must be at least BDP = bandwidth × round-trip time.

Link 1 Gbit/s, RTT 40 ms, classic window 65,535 bytes:

1. BDP = $10^9 \times 0.040 = 40 \times 10^6$ bits = 5.0 MB.
2. Window-limited throughput = 65,535 × 8 ÷ 0.040 ≈ 13.1 Mbit/s.
3. Share of the link used ≈ 1.31%.

The road could carry 1 Gbit/s, but this connection uses 1.3% of it: the sender spends most of its time waiting for ACKs. (Modern TCP uses window scaling to allow far bigger windows.)

2.2 · SEE IT

+10 XP

Fast Road, Slow Delivery

MAMA FENG

🔍 Spot it! The road is fast, so why is the delivery slow? Look at the window size.

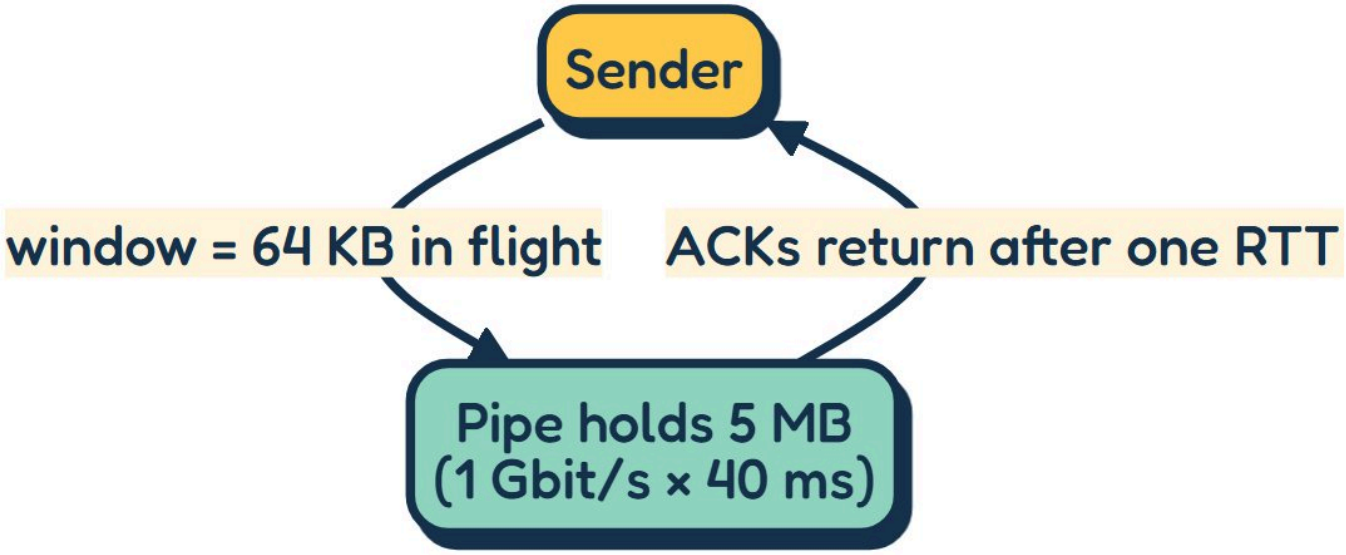


FIG Only one window of data can be "in flight" per round trip.

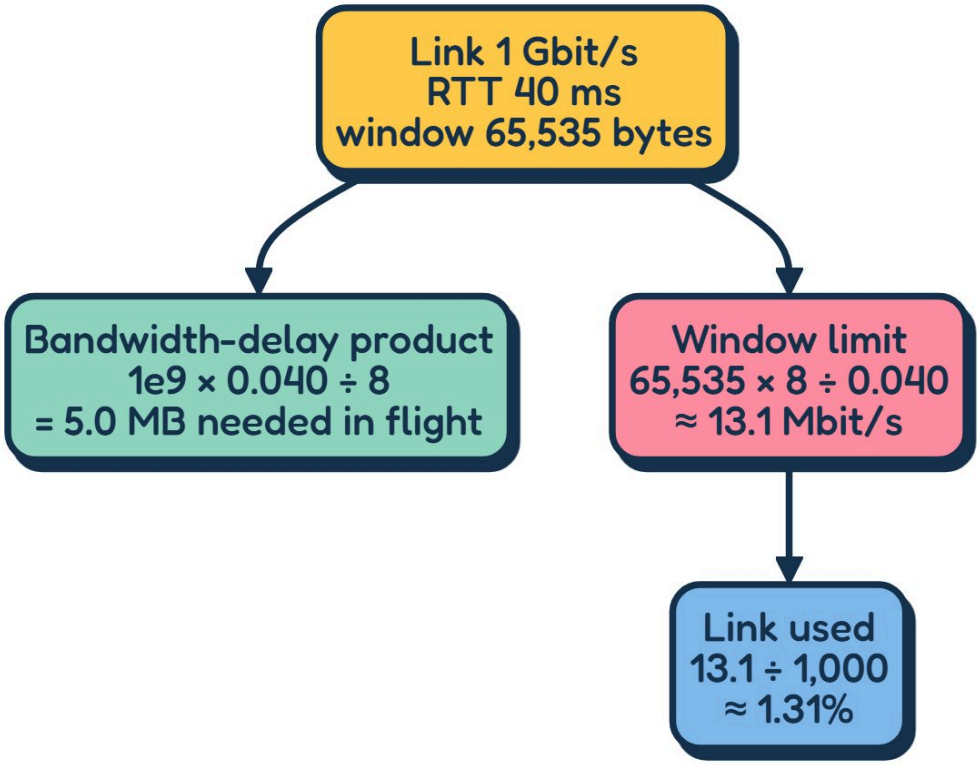


FIG Worked example: a 1 Gbit/s link limited by a 64 KB window.

2.2

Worked Example: Lost Boxes

+10 XP

✨ READY, SET, COOK!

Oops, lost boxes! 📦☁️ What happens when packets go missing? Let's find out.

How packet loss limits TCP (Mathis formula)

throughput ≈ 1.22 × MSS ÷ (RTT × √p)

MSS ≈ 1,460 bytes, RTT = 40 ms, p = packet-loss probability.

p = 10^-6 → 356.2 Mbit/s

p = 10^{-4} → 35.6 Mbit/s

p = 10^{-2} → 3.6 Mbit/s

Because of the square root, 100× more loss means 10× less throughput. A lossy Wi-Fi last hop can dominate the whole transfer, however fast the backbone is.

⚠ PAPA'S OOPS!

"It's an exact prediction!" Not quite! The formula assumes a long, loss-limited flow in steady state. Use it to see the SHAPE of the effect (RTT and loss both matter), not to predict one connection.

2.2 · SEE IT

+10 XP

Lost Boxes

MAMA FENG

🔍 Spot it! Follow the three branches: what happens to speed as more boxes get lost?

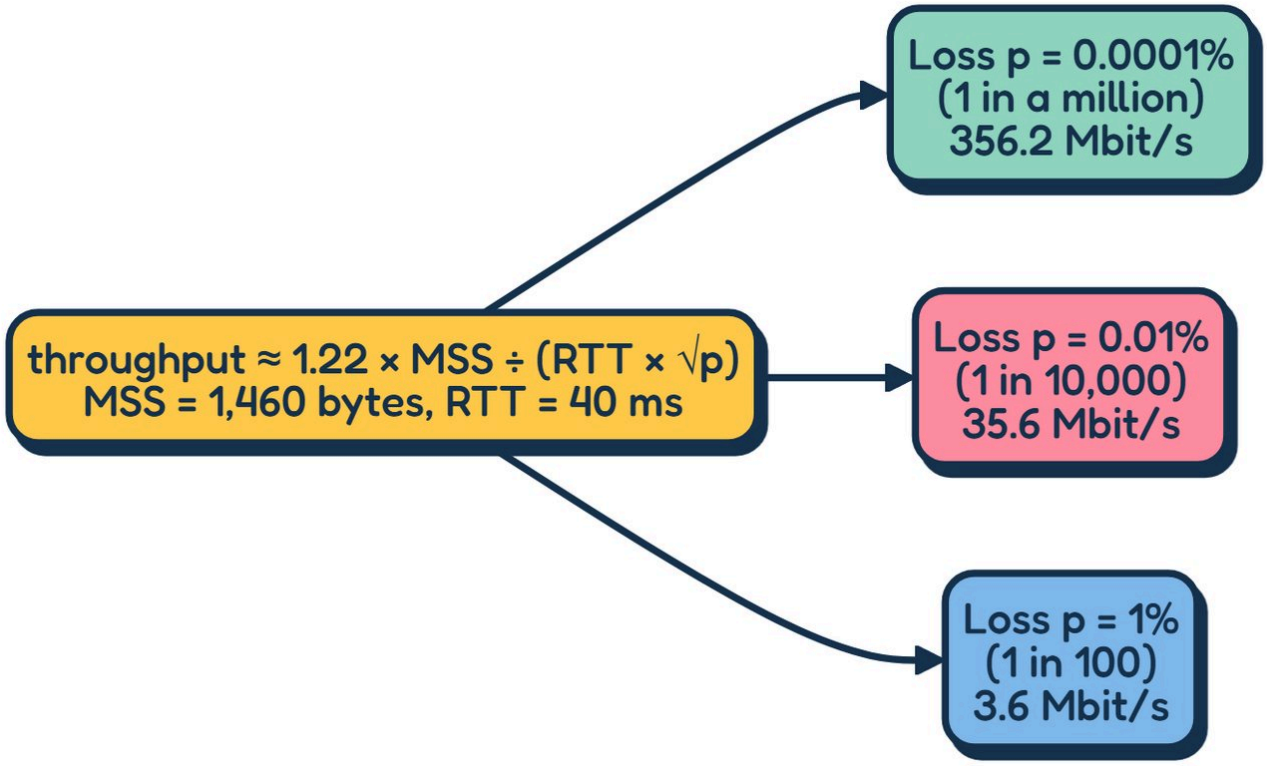


FIG Worked example: how packet loss limits TCP throughput.

SIDE QUEST · MINI-GAME 2.2

+25 XP

Careful vs Shouting Waiter

✨ READY, SET, COOK!

Careful vs shouting waiter! 🥊 Pick the right waiter for each job.

🎯 Your Mission

Count round trips, compute a bandwidth-delay product, and choose TCP or UDP for an AI service.

🧺 Grab These First

Worked examples "First Bite" and "Fast Road, Slow Delivery". A calculator.

🔪 Let's Go! Your Turn

(1) The phone-to-server RTT is 100 ms. Time to first byte for TCP + TLS 1.3 and for QUIC (fresh connection)?

(2) A 100 Mbit/s link has RTT 80 ms and a 65,535-byte window. BDP? Window-limited throughput? % of the link?

(3) Loss rises from 10^-6 to 10^{-4}. What happens to TCP throughput and why?

(4) Put in order: ACK, SYN, SYN-ACK.

👀 What Did You Spot?

Write the number of round trips before multiplying. Keep bits and bytes separate (÷ 8).

💭 Think It Over

Why might a live voice assistant prefer UDP-based transport over TCP? One sentence.

🔒 **Mama's feedback in the Answer Vault · V-2-2** · page 93 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #2.2

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.2) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. UDP guarantees that every packet arrives in order. ♡ ♡ ♡

- (T) True
- (F) False

2. Which of these does TCP provide that UDP does NOT? ♡ ♡ ♡

- (A) Acknowledgement of delivered data
- (B) In-order delivery
- (C) Port numbers to reach the right program
- (D) Retransmission of lost packets

3. Which protocol is built on UDP to speed up connection set-up for the web? ♡ ♡ ♡

- (A) QUIC
- (B) ARP
- (C) SMTP
- (D) FTP

4. What does “Bandwidth” mean? ♡ ♡ ♡

- (A) Adding more machines to share the load.
- (B) How many bits per second a link can carry.
- (C) Latency of the slowest requests, e.g. the 99th percentile.
- (D) Multi-access edge computing: servers in or near the mobile network.

5. Which kitchen word means: “An agreed set of rules for communication.”? ♡ ♡ ♡

- (A) Protocol
- (B) QUIC
- (C) UDP
- (D) MEC

TICKET #2.3 · RECIPE 02 · FRONT COUNTER

2.3 Central kitchen, food truck or home kitchen? (cloud vs edge)

COURSES ☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 140 XP

2.3 Central Kitchen: the Cloud

+10 XP

✨ READY, SET, COOK!

Welcome to the giant central kitchen in the sky ☁️ What's cooking in the cloud?

Definition (NIST)

Cloud computing is on-demand network access to a shared pool of configurable computing resources. Five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service.

For AI: rent a rack of GPUs for a weekend, scale a model service up at rush hour and down at night, pay for what you used.

SERVICE MODEL	PROVIDER RUNS	AI EXAMPLE
IaaS	Servers, storage, networking	Renting GPU/TPU machines to train a model
PaaS	+ runtime and tools	A managed service to train and deploy models
SaaS	The whole application	An AI assistant or translation app you just use

2.3 Food Truck & Home Kitchen: the Edge

+10 XP

✨ READY, SET, COOK!

Food trucks and home kitchens! 🚚 Why cook closer to the customer?

🚚 **The Big Idea**

Edge computing moves computation closer to where data is produced or used, instead of sending everything to a distant data centre. "Edge" is a spectrum:

- **On-device:** the phone, car or camera itself (the Neural Engine from Recipe 01), ~0 ms network.
- **Near edge / MEC:** small servers at the mobile operator's base station, a few ms away.
- **Regional cloud:** tens of ms.
- **Distant cloud:** up to ~100+ ms, but the most resources.

🔑 REMEMBER THIS!

Moving towards the device = shorter paths and less data leaving the user. Moving towards the cloud = more compute and bigger models. Most real systems use all three.

2.3

Comparing the Kitchens

+10 XP

✨ READY, SET, COOK!

Cloud vs edge face-off! 🥊 Who wins each round?

PROPERTY	ON-DEVICE	EDGE (MEC)	CLOUD
Network delay	None	A few ms	10s to 100+ ms
Compute & memory	Smallest (watts, GB)	Medium	Largest, elastic
Model size	Small-mid (quantised)	Mid-large	Largest
Privacy	Best: data stays local	Good: stays in region	Data leaves the user
Works offline	Yes	No	No
Cost model	In the device	Operator / enterprise	Pay per use
Weakness	Battery, memory	Fewer sites, less elastic	Network dependence, delay

2.3

Worked Example: Latency Budget

+10 XP

✨ READY, SET, COOK!

Budget challenge! 💸 Spend your milliseconds wisely.

Translate a 200 KB photo

Assumptions (illustrative): uplink 20 Mbit/s; cloud RTT 60 ms, edge RTT 10 ms; server inference 10 ms; on-device model 15 ms; connection already open.

Step 1, upload: 200,000 × 8 ÷ 20×10^6 = 0.080 s = 80 ms.

Step 2, add it up:

- **Cloud:** 80 + 60 + 10 = 150 ms
- **Edge:** 80 + 10 + 10 = 100 ms
- **On-device:** 15 ms

The UPLOAD, not the fast server, is the biggest term for both remote options. A smaller image, sending features instead of pixels, or a faster link (5G!) would help.

+10 XP

The Latency Budget

MAMA FENG

🔍 Spot it! Which slice of the latency is biggest? That is the one to fix first!

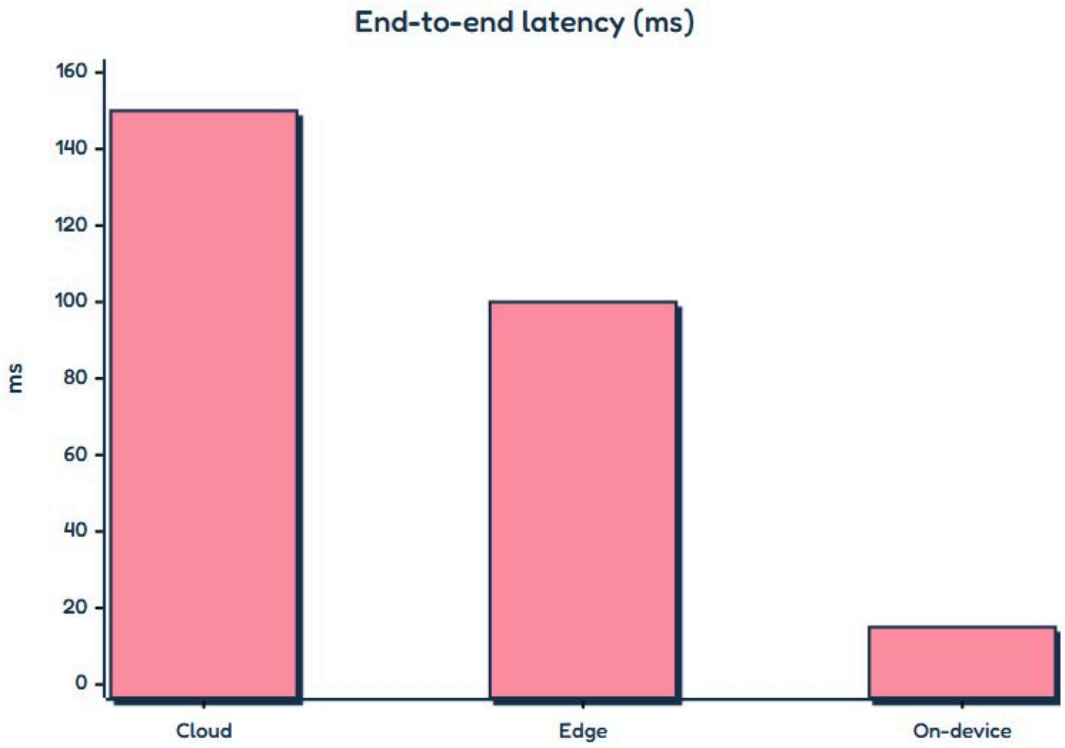


FIG End-to-end latency split into its parts (ms).

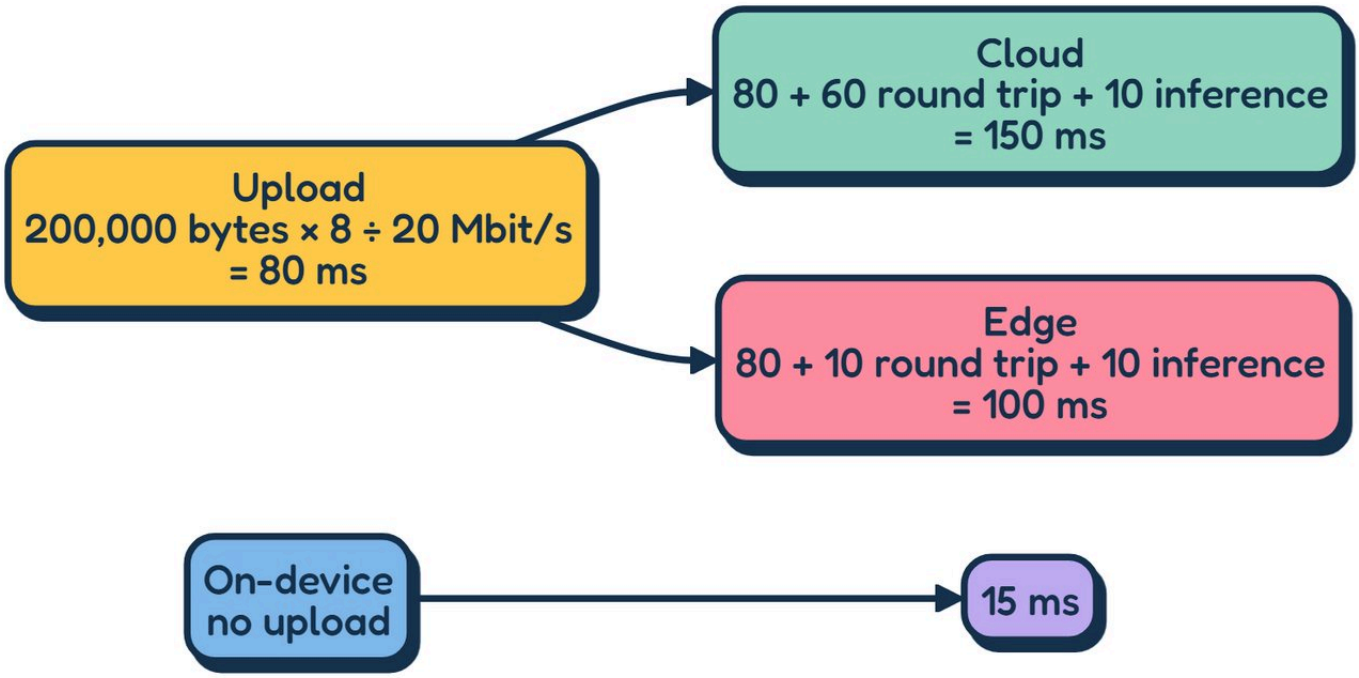


FIG Upload, network, inference and download, worked out.

Worked Example: Edge Saves the Roads

✨ READY, SET, COOK!

100 cameras, one road… 🚦 How can the edge save the day?

100 shop cameras

Each camera streams 5 Mbit/s to the cloud: 100 × 5 = 500 Mbit/s.

Instead, an edge box in the shop analyses the video and sends only events: 200 bytes, ~10 per second per camera → 100 × 200 × 8 × 10 = 1.6 Mbit/s.

Ratio: 500 ÷ 1.6 = 312.5, about 300× less traffic.

🔑 REMEMBER THIS!

The edge turns a BANDWIDTH problem into a COMPUTE problem: the shop now runs a model locally (Recipe 01), but the wide-area network carries 300× less data, and raw video never leaves the premises.

MODEL	PRECISION	WEIGHTS	FITS AN 8 GB PHONE?
3 B params	4-bit	1.5 GB	Yes, comfortably
13 B params	4-bit	6.5 GB	Not realistically
70 B params	FP16	140 GB	No: multi-GPU server

2.3 · SEE IT

+10 XP

The Edge Saves the Road

MAMA FENG

🔍 Spot it! Compare the two roads: how much traffic does the edge save?

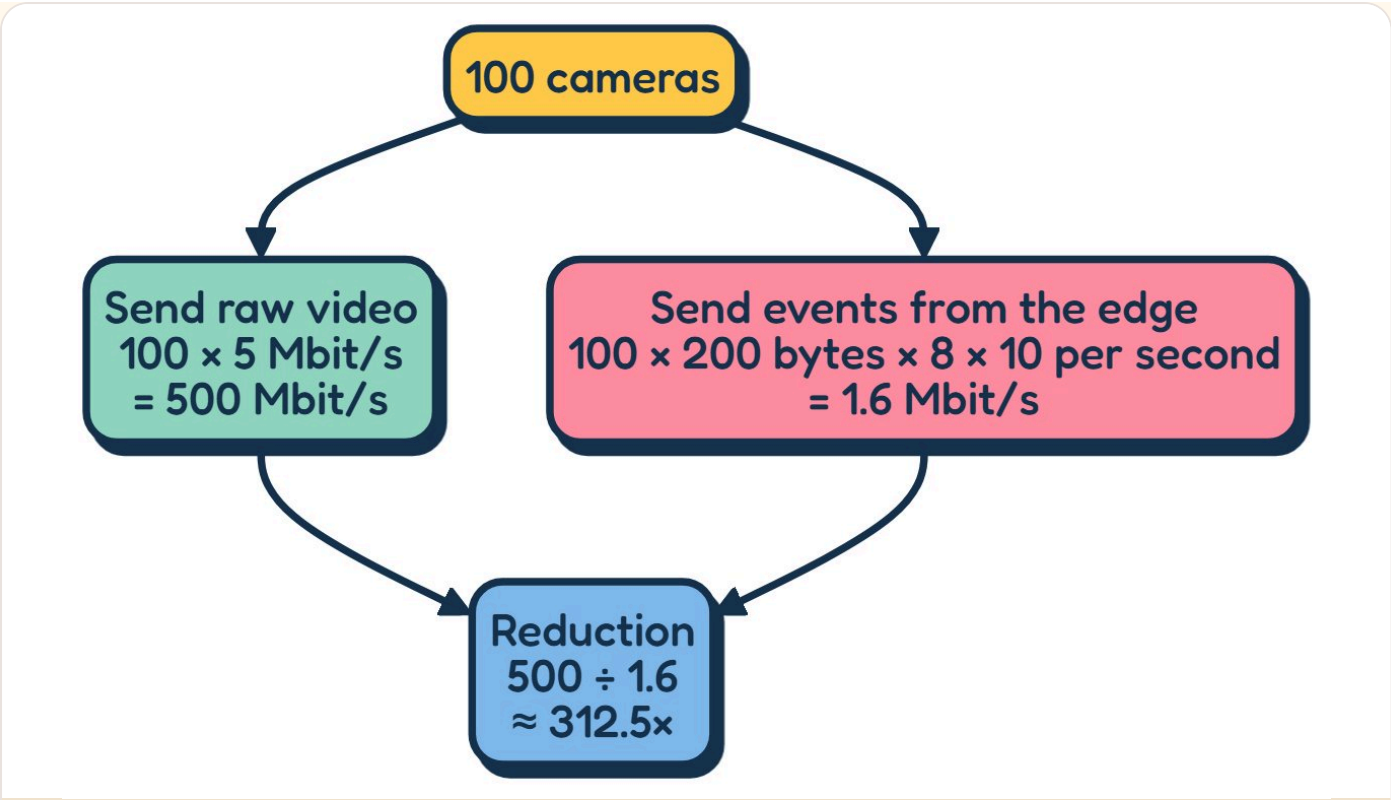


FIG Worked example: 100 cameras, raw video vs results sent from the edge.

2.3

Hybrid Recipes

+10 XP

✨ READY, SET, COOK!

Why not both? 🤝 Mix cloud and edge like a pro chef.

PATTERN	IDEA	EXAMPLE
Cascade	A small cheap model first; hard cases go to a bigger one	Phone reads common signs locally, sends unusual ones to the cloud
Split inference	First layers on the device, the rest on a server	Smaller upload, raw data stays local
Edge pre-processing	Edge filters/compresses before the cloud	The shop cameras
Federated learning	Devices train locally, send only model updates	Keyboard prediction learned without uploading your text

Where should it run? Checklist

Answer needed in a few tens of ms, every time? → on-device or edge

Raw data must stay with the user? → on-device or edge

Must work offline? → on-device

Model too big for the device? → edge or cloud

Bursty, unpredictable demand? → cloud (elasticity)

Large one-off batch job such as training? → cloud

SIDE QUEST · MINI-GAME 2.3

Truck Or Kitchen?

+25 XP

✨ READY, SET, COOK!

Truck or kitchen? 🚚 🏠 Make the call for every order!

🎯 Your Mission

Build latency budgets and decide where an AI job should run.

🧺 Grab These First

The latency-budget and camera examples. Uplink time = size × 8 ÷ rate.

🔪 Let's Go! Your Turn

(1) A 500 KB image over a 40 Mbit/s uplink. Cloud RTT 25 ms, edge RTT 5 ms, server inference 20 ms. Total for cloud and edge? Which option meets a 33 ms budget if on-device takes 30 ms?

(2) 50 cameras stream 4 Mbit/s each; an edge box sends 20 kbit/s of events per camera. Traffic saved?

(3) Weights of a 13 B model in 4-bit and FP16? Either on an 8 GB phone?

(4) Cloud, edge or device: (a) hospital scanner, instant, images on site (b) nightly training job (c) hearing-aid noise suppression.

👀 What Did You Spot?

Make a table: option | upload | round trip | inference | total.

💭 Think It Over

Which single term dominated the remote options in (1)? What would you change first?

🔒 **Mama's feedback in the Answer Vault · V-2-3** · page 93 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #2.3

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.3) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A factory camera must stop a machine within 15 ms. The best place to run the model is: ♡ ♡ ♡

- (A) A distant cloud region
- (B) On or near the machine (edge)
- (C) A user's laptop at home
- (D) Any server; latency does not matter

2. What does “Latency” mean? ♡ ♡ ♡

- (A) Time for data to arrive or a request to be answered.
- (B) 5G scenarios: high data rate / low, dependable delay / massive device counts.
- (C) Numerical address of a device (IPv4 32 bits, IPv6 128 bits).
- (D) One physical network divided into virtual networks with different guarantees.

3. Which kitchen word means: “On-demand access to a shared pool of computing resources (NIST).”? ♡ ♡ ♡

- (A) Latency
- (B) IP address
- (C) Tail latency
- (D) Cloud computing

4. What does “Bandwidth” mean? ♡ ♡ ♡

- (A) How many bits per second a link can carry.
- (B) An agreed set of rules for communication.
- (C) Time for data to arrive or a request to be answered.
- (D) One physical network divided into virtual networks with different guarantees.

TICKET #2.4 · RECIPE 02 · FRONT COUNTER

2.4 Delivery time, road width and rush hour (latency, bandwidth, scalability)

COURSES ☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 150 XP

2.4 +10 XP

Four Kinds of Delay

✨ READY, SET, COOK!

Four ways to be late! ⏰ Can you name them before you read?

DELAY	CAUSED BY	REDUCE IT WITH
Processing	Routers/servers handling headers and requests	Faster hardware, simpler pipelines
Queueing	Waiting in line behind other packets	Spare capacity, load balancing
Transmission	Pushing all bits onto the link: size ÷ bandwidth	More bandwidth, smaller messages
Propagation	Signal travelling: distance ÷ speed	Shorter distance (edge servers)

💡 MAMA'S SECRET TIP

Think of a pipe. BANDWIDTH is how wide it is: how much water per second. LATENCY is how long it is: how long one drop takes. A wider pipe moves more water per second, but no single drop arrives sooner.

2.4 +10 XP

Worked Example: Speed of Light

✨ READY, SET, COOK!

Light is the fastest thing there is, but even it needs time! 💡 Ready to do the maths?

💡 **The floor you cannot buy your way under**

Light in fibre ≈ 2×10^8 m/s = 5 μs per km, one way.

1,000 km → 5 ms one way, 10 ms round trip

3,000 km → 15 ms, 30 ms

10,000 km → 50 ms, 100 ms

No network answers from 10,000 km away in under ~100 ms round trip. Real routes are longer than straight lines, so real delays are higher. Fix: a closer server.

Big downloads are different

A 1.5 GB model (3 B params, 4-bit): $1.5 \times 10^9 \times 8$ bits ÷ 100 Mbit/s = 120 s; at 1 Gbit/s = 12 s. Here TRANSMISSION dominates, and 10× bandwidth is 10× faster.

Latency-bound jobs need proximity; bandwidth-bound jobs need capacity.

2.4

Tail Latency: the Slowest Cook

+10 XP

✨ READY, SET, COOK!

One slow cook holds up the whole table! Meet tail latency.

Waiting for everyone

A big AI service fans one request out to many servers (search shards, model replicas, tools) and waits for ALL of them. The slowest one decides the total.

If each server is slow 1% of the time, independently:

P(at least one slow) = $1 - (1 - p)^n$

n = 1 → 1.0%

n = 10 → 9.6%

n = 100 → 63.4%

A rare 1% hiccup hits nearly two out of three requests at 100 servers. That is why big services watch the 99th-percentile latency and send backup requests.

2.4 · SEE IT

The Slowest Cook

+10 XP

MAMA FENG

🔍 Spot it! One slow cook ruins the meal. Watch the chance grow with more servers.

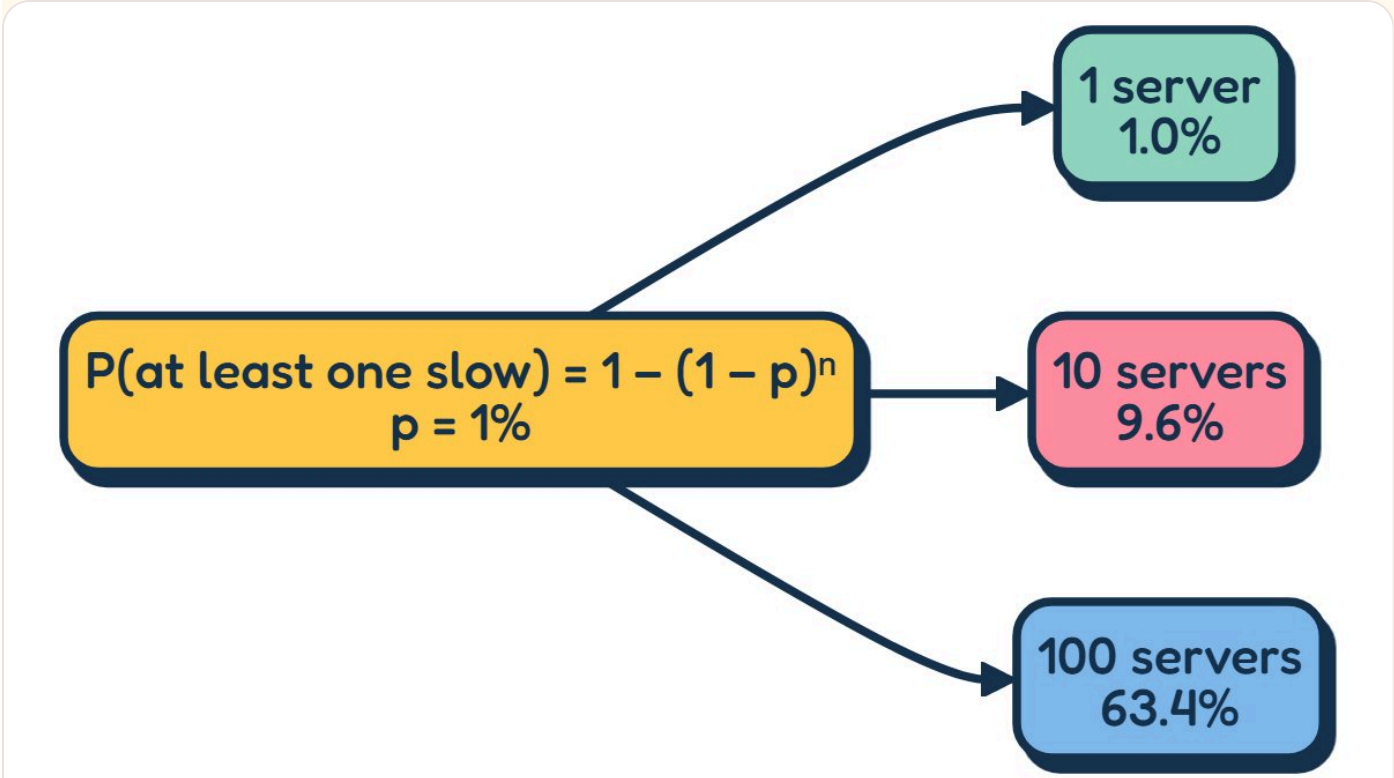


FIG Tail latency: the chance that at least one server is slow.

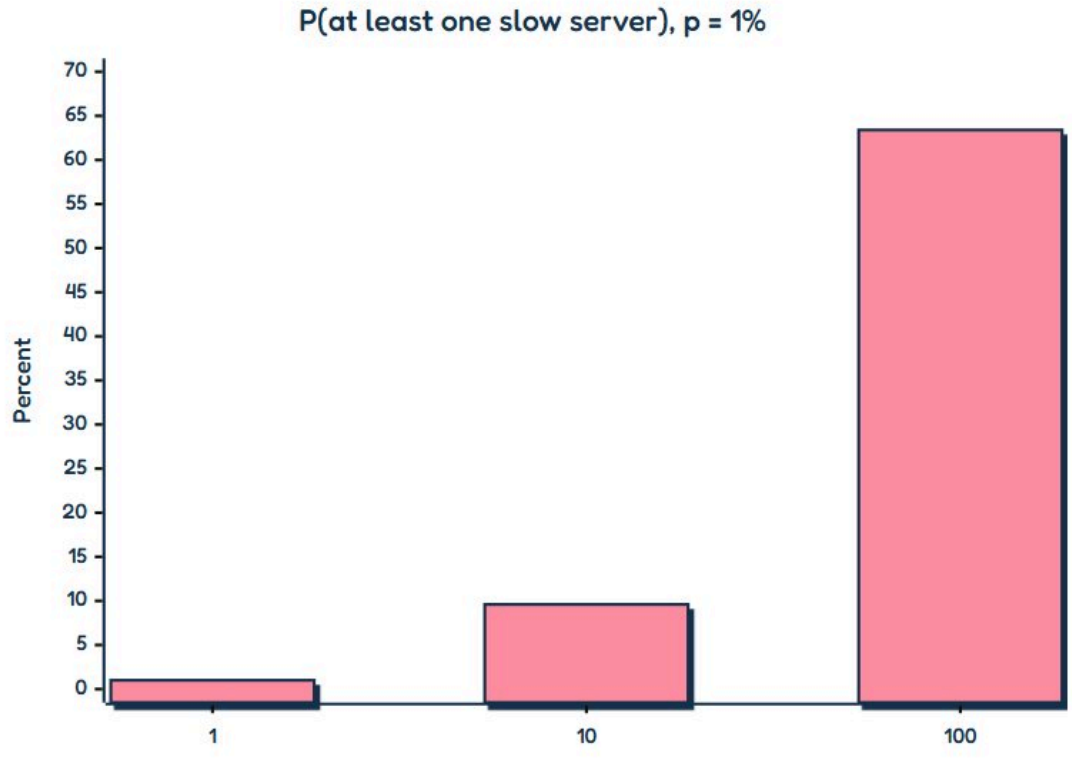


FIG With more servers, a slow one becomes almost certain (p = 1%).

2.4

Rush Hour: Queueing

+10 XP

✨ READY, SET, COOK!

Rush hour! 🚗🚗🚗 Why does a 100%-busy kitchen grind to a halt?

Why 100% busy is dangerous

Simple single-server queue (M/M/1): time ≈ service time ÷ (1 − utilisation).

Model server takes 10 ms per request:

utilisation 0.5 → 20 ms

utilisation 0.9 → 100 ms

utilisation 0.99 → 1,000 ms

Delay rises slowly until ~80% busy, then shoots up.

⚠ PAPA'S OOPS!

"Run every GPU at 100% to save money!" Not quite! In Recipe 01, low utilisation wasted silicon. Here, near-100% utilisation means unpredictable delay. Real services aim for a deliberate middle.

2.4 · SEE IT

+10 XP

Rush Hour

MAMA FENG

🔍 Spot it! Look at the curve near 100% busy. What happens to the wait?

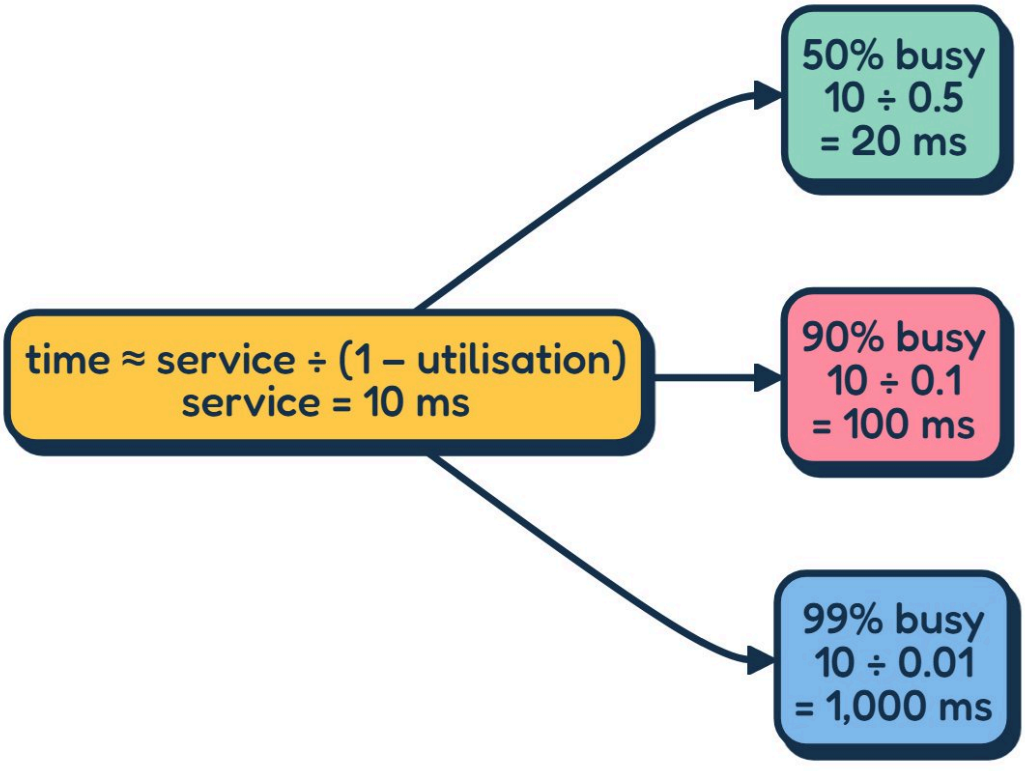


FIG Queueing: response time grows as utilisation rises.

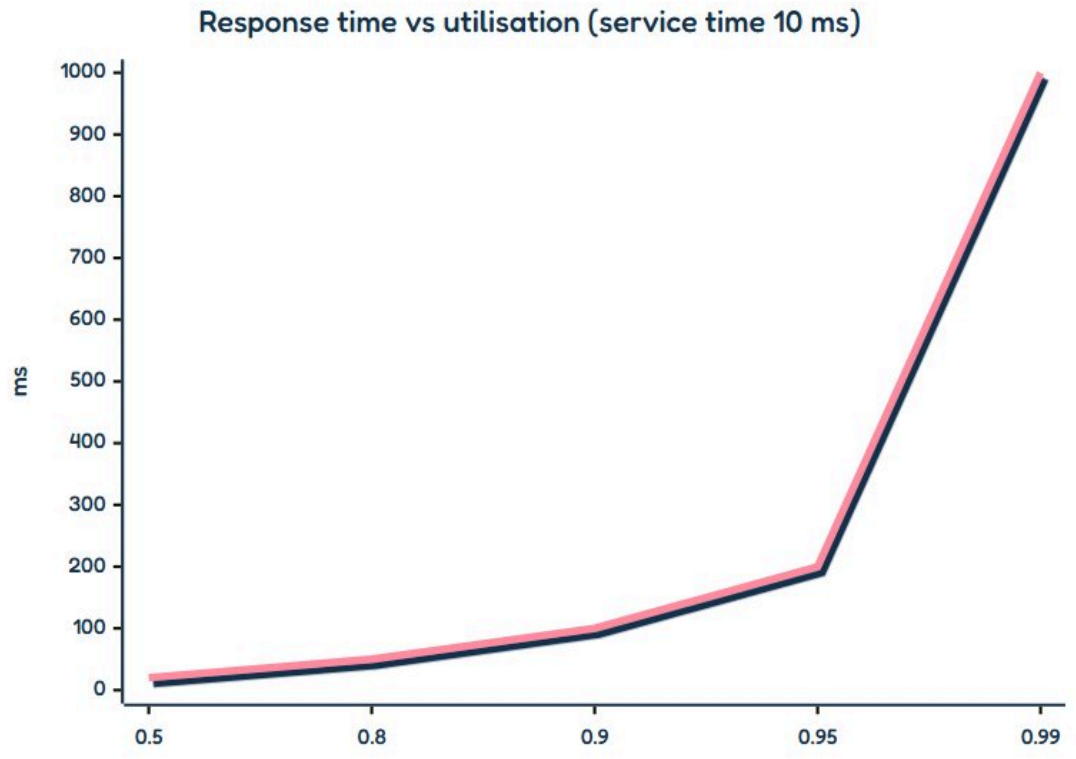


FIG Response time vs utilisation (service time 10 ms).

2.4

Scaling the Restaurant

+10 XP

APPROACH	IDEA	LIMIT
Vertical (scale up)	A bigger machine	There is a biggest machine; one failure stops everything
Horizontal (scale out)	More machines behind a load balancer	Work must be divisible; machines must coordinate
Autoscaling	Add/remove machines as demand changes	New machines take time; GPUs can be scarce

Serving vs training

Serving a model is mostly STATELESS: any replica can answer any request, so just add replicas. Training one big model is not: all machines must keep agreeing on the same weights after every step, so they talk constantly.

2.4

Worked Example: Training Traffic

+10 XP

Ring all-reduce on 8 GPUs

Data per GPU = 2 × (N − 1) ÷ N × gradient size.

8 GPUs, 2 GB of gradients each (1 B params, FP16), 300 ms of compute per step, no overlap.

1. Data per GPU = 2 × 7 ÷ 8 × 2 GB = 3.5 GB
2. **At 50 GB/s:** 70 ms. At 12.5 GB/s (100 Gbit/s Ethernet): 280 ms
3. Efficiency = 300 ÷ 370 = 81.1% vs 300 ÷ 580 = 51.7%

With the slower network, nearly half of the expensive GPU time is spent waiting.

🔑 REMEMBER THIS!

Communication that cannot be hidden behind computation acts like the serial part in Amdahl's law (Recipe 01). That is why training clusters use the fastest links and overlap communication with computation.

2.4 · SEE IT

Training Traffic

+10 XP

MAMA FENG

🔍 Spot it! Same job, two networks: how much training time is lost to talking?

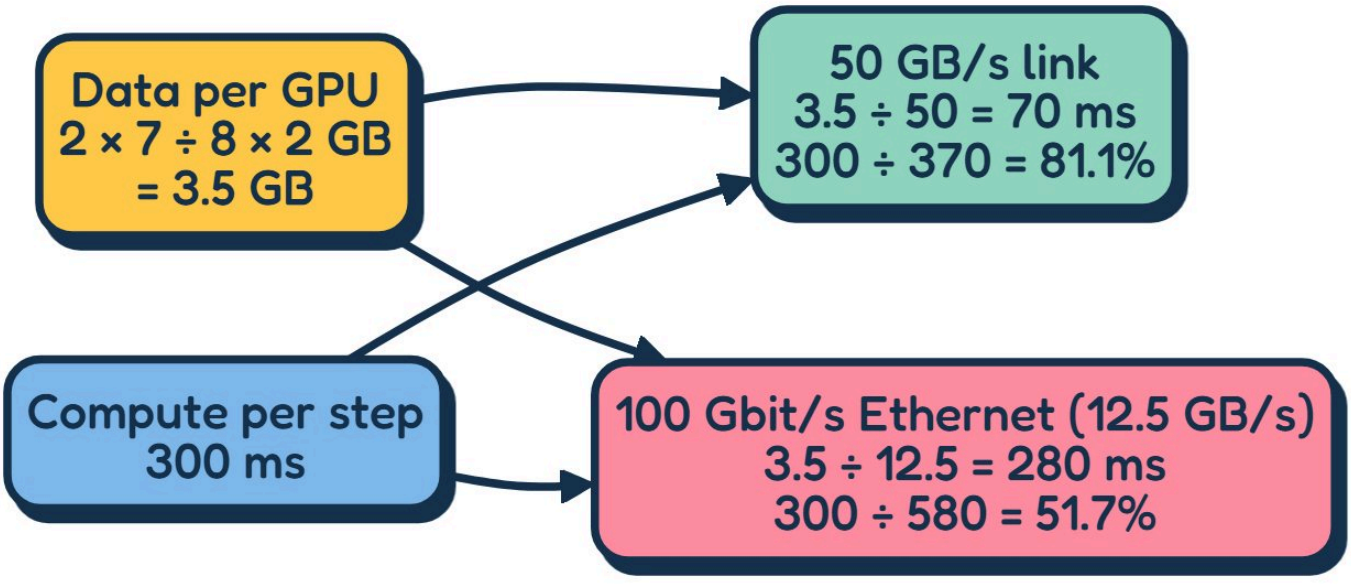


FIG Worked example: how much data each GPU must exchange.

FIG Training efficiency on different networks.

SIDE QUEST · MINI-GAME 2.4

Delivery Timer

+25 XP

✨ READY, SET, COOK!

Delivery timer challenge! ⏱ Beat the clock!

Your Mission

Use the delay, tail-latency, queueing and all-reduce formulas.

Grab These First

Light: 5 µs per km one way. P(slow) = 1 − (1 − p)^n. Queue: time ≈ service ÷ (1 − u). All-reduce: 2(N−1)/N × size.

Let's Go! Your Turn

(1) Minimum RTT over 3,000 km of fibre? Why is a real connection slower?

(2) A request waits for 50 servers, each slow 2% of the time. P(at least one slow)?

(3) Service time 5 ms. Response time at utilisation 0.8 and 0.95?

(4) 4 GPUs, 4 GB of gradients each, 25 GB/s link. Data per GPU and time?

What Did You Spot?

For (3), note how much the time grows between 80% and 95%.

Think It Over

Which of these four delays can money (more bandwidth) fix, and which cannot?

🔒 **Mama's feedback in the Answer Vault · V-2-4** · page 93 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #2.4

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.4) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Bandwidth and latency are two names for the same quantity. ♡ ♡ ♡

- (T) True
- (F) False

2. In a ring all-reduce, the traffic per chip approaches 2 × S as the number of chips grows. ♡ ♡ ♡

- (T) True
- (F) False

3. Tail latency refers to: ♡ ♡ ♡

- (A) Upload speed
- (B) The average delay
- (C) The slowest small fraction of requests
- (D) The delay of the last packet only

4. What does "RTT" mean? ♡ ♡ ♡

- (A) Round-trip time: to the destination and back.
- (B) On-demand access to a shared pool of computing resources (NIST).
- (C) Computation placed close to where data is produced or used.
- (D) Time for data to arrive or a request to be answered.

5. Which kitchen word means: "How many bits per second a link can carry."? ♡ ♡ ♡

- (A) Bandwidth
- (B) Horizontal scaling
- (C) MEC
- (D) BDP

TICKET #2.5 · RECIPE 02 · FRONT COUNTER

2.5 Case study

5G delivery and AI that runs the roads

COURSES ☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 140 XP

☐ 2.5 +10 XP

Case Study: 5G Delivery

✨ READY, SET, COOK!

Case file: 5G! 📶 How does your phone talk to AI in the cloud so fast?

☁ Cloud-based AI service

A model hosted in a data centre and offered over the network: translation API, speech recognition, image classifier, chatbot. The provider handles hardware and scaling; the user needs only a connection.

5G SCENARIO	GOAL	EXAMPLE
eMBB	Much higher data rates	Fast photo/video uploads to an AI service
URLLC	Very low, dependable delay	Remote control, factories, vehicles
mMTC	Huge numbers of devices	Dense sensor fields

⚠ PAPA'S OOPS!

"5G is 1 ms, so my AI answers in 1 ms!" Not quite! ITU targets (20 Gbit/s peak down, 1 ms user-plane latency for URLLC, 1 million devices/km²) are for the RADIO part under ideal conditions, not the end-to-end service.

☐ 2.5 +10 XP

The Path of An Order

✨ READY, SET, COOK!

Follow the order! 🗺 From your phone to the cloud and back. Every hop counts.

🏭 Phone → 5G → model

Phone → (5G radio) → base station (gNB) → transport network → 5G core user plane function (UPF) → either a nearby edge server (MEC, "local breakout") or the internet and a cloud data centre.

Three 5G features that matter for AI

1. Higher radio speeds shorten the UPLOAD term of the latency budget.
2. Multi-access edge computing (MEC): operator servers near the base station cut the ROUND-TRIP term.
3. **Network slicing:** one physical network split into virtual slices with different guarantees; a latency-critical AI service can get its own slice, apart from bulk video downloads.

2.5

Worked Example: 4G vs 5G vs MEC

+10 XP

PATH	UPLINK	ROUND TRIP	INFERENCE	TOTAL
4G + cloud	20 Mbit/s → 80 ms	50 ms	10 ms	140 ms
5G + cloud	100 Mbit/s → 16 ms	30 ms	10 ms	56 ms
5G + MEC edge	100 Mbit/s → 16 ms	10 ms	10 ms	36 ms
On-device NPU	none	none	15 ms	15 ms

Reading the bill

4G → 5G cut 84 ms (140 → 56), mostly from the faster upload (80 → 16 ms). Once the upload is fast, the 30 ms round trip is the biggest term, so the MEC edge server gives the next big gain. On-device is fastest but runs a smaller model: the network lets a phone reach a much bigger model for a few tens of milliseconds.

2.5 · SEE IT

4G vs 5G vs Edge

+10 XP

MAMA FENG

🔍 Spot it! Which path answers the street-sign photo fastest, and why?

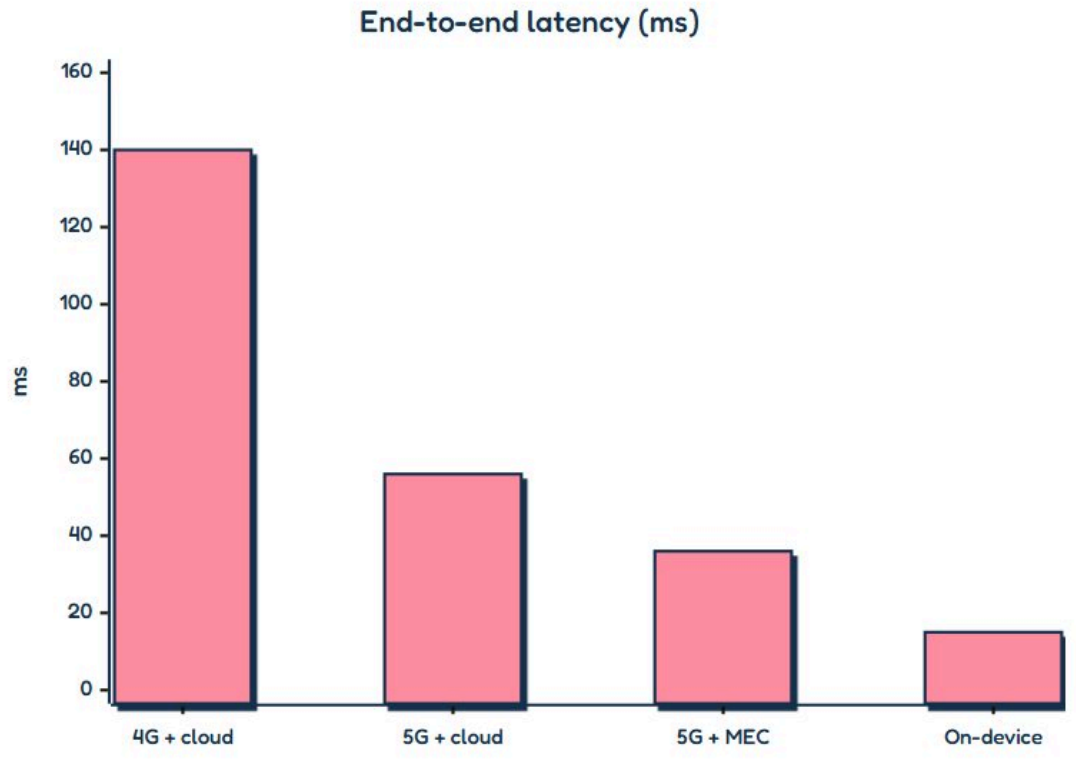


FIG Latency budget of the street-sign translator (ms).

2.5 · SEE IT

+10 XP

4G vs 5G vs Edge (2)

MAMA FENG

🔍 Spot it! Which path answers the street-sign photo fastest, and why?

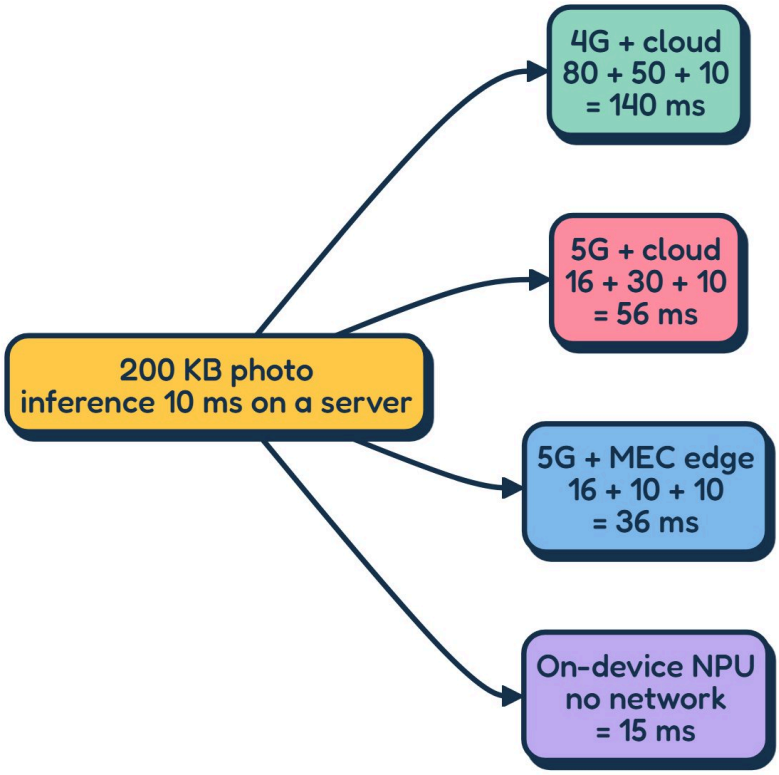


FIG The same request worked out step by step.

2.5

AI That Runs the Roads

+10 XP

TASK	WHAT AI DOES	BENEFIT
Traffic prediction	Forecasts load per cell	Capacity ready before demand
Energy saving	Predicts quiet periods, switches off spare radio capacity	Lower power bills and emissions
Beam & resource management	Adapts radio settings to moving users	Better speed and reliability
Anomaly detection	Spots unusual behaviour early	Fewer outages

🔬 REAL-LIFE TECH

In O-RAN, a RAN Intelligent Controller (RIC) hosts these apps in two loops: non-real-time (> 1 s: training models, setting policies) and near-real-time (~10 ms to 1 s: adjusting radio resources). It is a sense → decide → act loop, the bridge to Recipe 03!

2.5

Worked Example: Tiny Forecast

+10 XP

Exponential smoothing for energy saving

Base-station load (% of capacity) every 15 min: 40, 35, 30, 22, 15, 10, 8, 9.

forecast = 0.5 × newest + 0.5 × previous forecast

Rule: if the forecast falls below 20%, spare capacity may sleep (after checking neighbours can cover).

Forecasts: 40.0, 37.5, 33.8, 27.9, 21.4, 15.7, 11.9, 10.4 → sleep at step 6.

The raw load dipped below 20% at step 5, but smoothing waits for step 6: the decision is less jumpy.

KITCHEN TERMINAL · PYTHON

```
load = [40, 35, 30, 22, 15, 10, 8, 9]
f = load[0]; out = []
for x in load:
    f = 0.5 * x + 0.5 * f; out.append(round(f, 1))
print(out)  # [40.0, 37.5, 33.8, 27.9, 21.4, 15.7, 11.9, 10.4]
```

2.5 · SEE IT

+10 XP

A Tiny Forecast

MAMA FENG

🔍 Spot it! Follow the loop: at which step does the base station go to sleep?

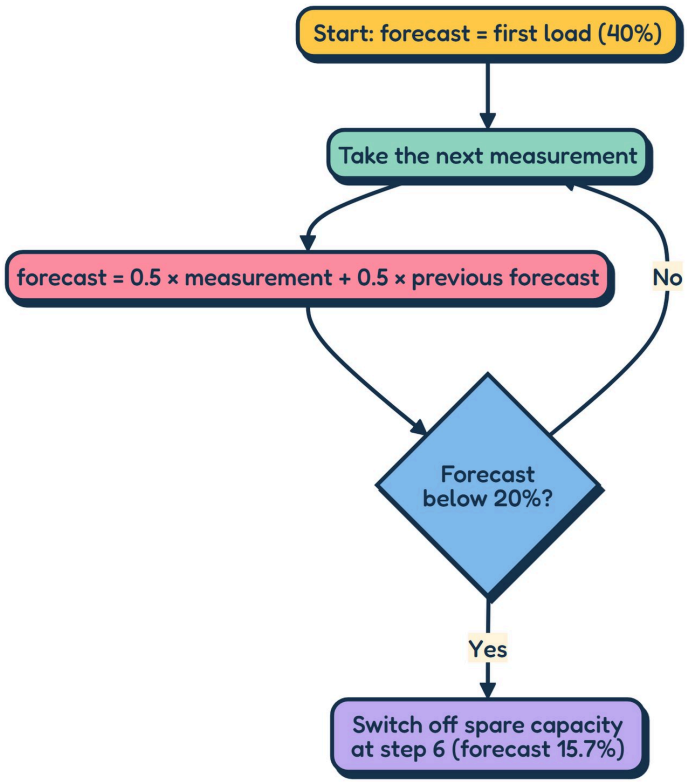


FIG Worked example: exponential smoothing decides when a cell can sleep.

2.5

Pros & Cons of 5G AI

+10 XP

✨ READY, SET, COOK!

Thumbs up or thumbs down for 5G AI? 👍 👎 You be the judge!

👍 The Tasty Bits

Faster upload of photos, audio, video • Shorter round trips with MEC • A phone can use models far bigger than its memory • Network slicing can reserve capacity and latency • AI forecasting and control make networks smarter and greener.

👎 The Burnt Bits

Coverage varies with band, distance and load • Latency is not just the radio: upload, round trip, queueing and inference all count • Edge sites have limited capacity • Data still leaves the device: encrypt and govern it • Slicing and MEC cost money and skills • No connection = no cloud AI, so keep an on-device fallback • An AI controller that switches capacity off wrongly can hurt service: keep safeguards and human oversight.

SERVE THE CUSTOMERS · TICKET #2.5

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.5) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Multi-access edge computing (MEC) in 5G mainly helps AI services by: ♡ ♡ ♡

- (A) Putting computing close to the base station
- (B) Replacing TCP
- (C) Removing the need for antennas
- (D) Increasing screen resolution

2. What does “eMBB / URLLC / mMTC” mean? ♡ ♡ ♡

- (A) Latency of the slowest requests, e.g. the 99th percentile.
- (B) Reliable, ordered byte stream with handshake, ACKs, flow and congestion control.
- (C) GPUs pass pieces around a ring to sum gradients efficiently.
- (D) 5G scenarios: high data rate / low, dependable delay / massive device counts.

3. Which kitchen word means: “One physical network divided into virtual networks with different guarantees.”? ♡ ♡ ♡

- (A) Tail latency
- (B) IP address
- (C) Network slicing
- (D) TCP

4. What does “Edge computing” mean? ♡ ♡ ♡

- (A) A small chunk of data plus a header with delivery information.
- (B) Unreliable, connectionless packet delivery: fast, no retransmission.
- (C) Multi-access edge computing: servers in or near the mobile network.
- (D) Computation placed close to where data is produced or used.

TICKET #2.L · RECIPE 02 · FRONT COUNTER

LAB

Kitchen Lab

Ping & Tracert

COURSES ▢▢▢▢ RECORD ☆☆☆ BOUNTY 90 XP

KITCHEN LAB

Ping & Tracert

+10 XP

🌐 Live platform: your own terminal

Windows: open Command Prompt and type

ping -n 10 www.bing.com

tracert www.bing.com

macOS/Linux: ping -c 10 … and traceroute …

Record the min/avg/max round-trip time and count the hops. Then ping a server on another continent and compare.

Turn your VPN off for this lab: with it on, every path goes through the VPN server first.

Version: built-in ping / tracert of Windows 10/11 (traceroute on macOS/Linux).

🎯 Goal · Observe · Reflect

Goal: measure real latency and hops.

Observe: Which hop adds the biggest jump in time? Does a far-away server ever beat the speed-of-light floor (5 µs/km one way)?

Reflect: Would a 20 ms robot control loop (Recipe 03) work over this path?

📴 Offline fallback

No internet or blocked ICMP? Use these sample results: nearby server avg 12 ms, 9 hops; overseas server avg 186 ms, 17 hops. Estimate the minimum distance each RTT allows (RTT ÷ 2 ÷ 5 µs per km).

⭐ Mama's Feedback (expected result)

Nearby server: about 5-40 ms and 8-15 hops. Overseas: about 150-300 ms. The biggest jump is usually the hop that leaves the country or crosses an ocean. Lines with * * * are routers that ignore tracert: normal.

No server beats the floor: Beijing → California ≈ 9,500 km, so RTT ≥ 2 × 9,500 × 5 µs ≈ 95 ms.

A 20 ms control loop works only with a nearby (edge) server, never overseas. Offline: 12 ms → at most 1,200 km away; 186 ms → at most 18,600 km.

EXERCISE

+10 XP

Final round

✨ READY, SET, COOK!

Final round! 🎯 Earn your Recipe 02 stamp!

EXERCISE

Question 1: Street-sign translator

Task: Using the 4G/5G/MEC table, which term is largest in "5G + cloud", and what change would help most? Then explain why the 1 ms 5G latency target does not mean the AI answers in 1 ms.

Question 2: Smart base station

Task: Why is a forecast-based sleep decision better than switching off the moment load falls below 20%? Name one risk of switching off too eagerly. Which RIC loop (non-real-time or near-real-time) fits (a) re-tuning radio for a moving user, (b) re-training a forecasting model overnight?

Question 3: Network slice

Task: Give one benefit and one risk of putting a latency-critical AI service on its own network slice.

🔒 **Answers in the Answer Vault · V-2-5** · page 93 · no peeking! +10 XP

WATCH & LEARN

+10 XP

Video picks

How does the INTERNET work? | ICT #2

Creator: Lesics (Sabin Civil Engineering channel) · 9:00

Link: youtube.com/watch?v=x3c1ih2NJEg

Why watch: Animated path of data through cables, routers, IP addresses and DNS (2.1-2.2).

Timestamps: Whole video (9 min)

Task: Draw the path a YouTube video takes to your phone and label each hop.

Limitations: Simplified; no TCP/UDP detail, no AI workloads.

What is bandwidth? How is it different than latency?

Creator: internet-class · 3:24

Link: youtube.com/watch?v=QLJCqY-7Vrw

Why watch: A short, clear split between "how much" and "how fast" (2.4).

Timestamps: Whole video (3 min)

Task: Give one AI example that needs high bandwidth and one that needs low latency.

Limitations: Very short; no numbers for real networks.

WATCH & LEARN

Video picks

+10 XP

Inside a Google data center

Creator: Google Workspace · 5:28

Link: youtube.com/watch?v=XZmGGAbHqa0

Why watch: See the racks, cooling and security of the "cloud kitchen" of 2.3 and Recipe 05.

Timestamps: Whole video (5 min)

Task: Spot three things a data centre needs besides computers (power, cooling, security…).

Limitations: Company tour video; older footage; shows no AI accelerators.

SERVE THE CUSTOMERS · TICKET #2.L

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-2-Q2.L) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In Kitchen Lab 2, a server answers ping with an average round-trip time of 12 ms. Using the speed-of-light floor of 5 µs per km (one way), the server can be at most about 1,200 km away. ♡ ♡ ♡

- (T) True
- (F) False

2. What does "MEC" mean? ♡ ♡ ♡

- (A) Multi-access edge computing: servers in or near the mobile network.
- (B) Bandwidth × RTT: data in flight needed to fill a link.
- (C) One physical network divided into virtual networks with different guarantees.
- (D) 5G scenarios: high data rate / low, dependable delay / massive device counts.

3. Which kitchen word means: "Time for data to arrive or a request to be answered."? ♡ ♡ ♡

- (A) eMBB / URLLC / mMTC
- (B) IP address
- (C) Latency
- (D) Bandwidth

4. What does "Bandwidth" mean? ♡ ♡ ♡

- (A) An agreed set of rules for communication.
- (B) Computation placed close to where data is produced or used.
- (C) How many bits per second a link can carry.
- (D) One physical network divided into virtual networks with different guarantees.

TICKET #2.B · RUSH HOUR IN THE FRONT COUNTER

Rush Hour!

*Last in line, the trouble customer **The Lag Monster**: "Every order you send... I make it... wait... a little longer."*

HAPPY CUSTOMERS ☐☐☐☐☐☐☐☐☐☐☐☐☐☐

GRADE S · A · B · C · D

Rules: 14 orders, no going back. Colour one box for each right dish (check in the Answer Vault, V-2-B). 9 or more happy customers = you survive the rush: +150 XP and the **Ping-Pong Noodles** recipe card. Grade: S = 13–14, A = 12, B = 10–11, C = 9. Questions 13 and 14 are the trouble customer's double order.

1. In a ring all-reduce, the traffic per chip approaches 2 × S as the number of chips grows. ♡ ♡ ♡

- (T) True
- (F) False

2. UDP guarantees that every packet arrives in order. ♡ ♡ ♡

- (T) True
- (F) False

3. Tail latency refers to: ♡ ♡ ♡

- (A) Upload speed
- (B) The average delay
- (C) The slowest small fraction of requests
- (D) The delay of the last packet only

4. Multi-access edge computing (MEC) in 5G mainly helps AI services by: ♡ ♡ ♡

- (A) Putting computing close to the base station
- (B) Replacing TCP
- (C) Removing the need for antennas
- (D) Increasing screen resolution

5. Which of these does TCP provide that UDP does NOT? ♡ ♡ ♡

- (A) Acknowledgement of delivered data
- (B) In-order delivery
- (C) Port numbers to reach the right program
- (D) Retransmission of lost packets

6. A factory camera must stop a machine within 15 ms. The best place to run the model is: ♡ ♡ ♡

- (A) A distant cloud region
- (B) On or near the machine (edge)
- (C) A user's laptop at home
- (D) Any server; latency does not matter

7. Encapsulation means each layer adds its own header to the data from the layer above. ♡ ♡ ♡

- (T) True
- (F) False

8. Bandwidth and latency are two names for the same quantity. ♡ ♡ ♡

- (T) True
- (F) False

9. Which protocol is built on UDP to speed up connection set-up for the web? ♡ ♡ ♡

- (A) QUIC
- (B) ARP
- (C) SMTP
- (D) FTP

10. In Kitchen Lab 2, a server answers ping with an average round-trip time of 12 ms. Using the speed-of-light floor of 5 µs per km (one way), the server can be at most about 1,200 km away. ♡ ♡ ♡

- (T) True
- (F) False

11. Which kitchen word means: "Time for data to arrive or a request to be answered."? ♡ ♡ ♡

- (A) QUIC
- (B) TCP
- (C) Latency
- (D) Packet

12. What does "Edge computing" mean? ♡ ♡ ♡

- (A) Multi-access edge computing: servers in or near the mobile network.
- (B) Round-trip time: to the destination and back.
- (C) Computation placed close to where data is produced or used.
- (D) Latency of the slowest requests, e.g. the 99th percentile.

13. Which kitchen word means: "An agreed set of rules for communication."?

- (A) Utilisation
- (B) Protocol
- (C) Bandwidth
- (D) eMBB / URLLC / mMTC

14. What does "Bandwidth" mean?

- (A) How many bits per second a link can carry.
- (B) Reliable, ordered byte stream with handshake, ACKs, flow and congestion control.
- (C) Computation placed close to where data is produced or used.
- (D) Latency of the slowest requests, e.g. the 99th percentile.

TICKET #V2 · RECIPE 02

Answer Vault · Order Up!

Only open after trying! Every vault entry you check is worth XP on your licence.

V-2-1

2.1 · Bento Packing

Mama's feedback · Bento Packing

(1) 1,200 ÷ 1,258 ≈ 95.4%; 40 ÷ 98 ≈ 40.8%. (2) (a) Network/IP (b) Transport/TCP (c) Link/physical (d) Application. (3) (a) MQTT (b) gRPC (c) WebRTC (d) HTTP/REST. Reflect: one switch is a bottleneck and single point of failure; leaf-spine grows by adding switches, spreads traffic over equal paths, and survives a failed spine.

V-2-Q2.1

Customer orders · ticket #2.1

1. True · Like boxes inside boxes: application data → TCP/UDP header → IP header → link header.

2. D · A small chunk of data plus a header with delivery information. · Packet: A small chunk of data plus a header with delivery information.

3. A · Latency · Latency: Time for data to arrive or a request to be answered.

4. D · Each layer wraps data in its own header on the way down the stack. · Encapsulation: Each layer wraps data in its own header on the way down the stack.

V-2-2

2.2 · Careful vs Shouting Waiter

Mama's feedback · Careful vs Shouting Waiter

(1) 300 ms and 200 ms. (2) BDP = 1 MB; ≈ 6.55 Mbit/s ≈ 6.6% of the link; a ~1 MB window would fill it. (3) 100× more loss → √100 = 10× less: ~356 → ~36 Mbit/s, because TCP reads loss as congestion. (4) SYN, SYN-ACK, ACK. Reflect: a resent audio chunk arrives too late to be useful; skip it rather than delay everything.

V-2-Q2.2

Customer orders · ticket #2.2

1. False · Correct: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering.

2. undefined · · TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong.

3. A · QUIC · QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes.

4. B · How many bits per second a link can carry. · Bandwidth: How many bits per second a link can carry.

5. A · Protocol · Protocol: An agreed set of rules for communication.

V-2-3

2.3 · Truck Or Kitchen?

Mama's feedback · Truck Or Kitchen?

(1) Upload 100 ms. Cloud 145 ms, edge 125 ms; only on-device (30 ms) meets 33 ms. (2) 200 → 1 Mbit/s: 200× less. (3) 6.5 GB and 26 GB; neither realistic on an 8 GB phone. (4) (a) edge/on-premises (b) cloud (c) on-device. Reflect: the 100 ms upload; compress the image or send features.

V-2-Q2.3
Customer orders · ticket #2.3

1. B · On or near the machine (edge) · Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it.

2. A · Time for data to arrive or a request to be answered. · Latency: Time for data to arrive or a request to be answered.

3. D · Cloud computing · Cloud computing: On-demand access to a shared pool of computing resources (NIST).

4. A · How many bits per second a link can carry. · Bandwidth: How many bits per second a link can carry.

V-2-4
2.4 · Delivery Timer

⭐ Mama's feedback · Delivery Timer

(1) 15 ms one way → 30 ms RTT minimum; real routes are longer and add processing/queueing per hop. (2) $1 - 0.98^{50} \approx 63.6\%$. (3) 25 ms and 100 ms: 4× worse. (4) 2 × 3/4 × 4 = 6 GB; 6 ÷ 25 = 0.24 s. Reflect: bandwidth fixes transmission; it cannot beat the propagation floor.

V-2-Q2.4
Customer orders · ticket #2.4

1. False · Correct: bandwidth is how much data per second; latency is how long one message takes to arrive.

2. True · Traffic = 2(N − 1)/N × S, which tends to 2S for large N.

3. C · The slowest small fraction of requests · For example the 99th-percentile delay; one slow server can delay a whole request.

4. A · Round-trip time: to the destination and back. · RTT: Round-trip time: to the destination and back.

5. A · Bandwidth · Bandwidth: How many bits per second a link can carry.

V-2-Q2.5
Customer orders · ticket #2.5

1. A · Putting computing close to the base station · Operator servers near the base station shorten the round trip.

2. D · 5G scenarios: high data rate / low, dependable delay / massive device counts. · eMBB / URLLC / mMTC: 5G scenarios: high data rate / low, dependable delay / massive device counts.

3. C · Network slicing · Network slicing: One physical network divided into virtual networks with different guarantees.

4. D · Computation placed close to where data is produced or used. · Edge computing: Computation placed close to where data is produced or used.

V-2-5
2.L · Final round

⭐ Answers · Final round

Q1: The 30 ms round trip (≈54% of 56 ms) is largest. An MEC edge server cuts it to 10 ms → 36 ms total. The 1 ms figure is a RADIO target; upload (16 ms for 200 KB at 100 Mbit/s), core network, distance, queueing, inference and the reply all add up to tens of ms.

Q2: Smoothing avoids false alarms from momentary dips (crossed 20% at step 6, not step 5) and follows the trend. Risk: if demand comes back, users see slower service or dropped connections. (a) near-real-time RIC (b) non-real-time RIC.

Q3: Benefit: reserved, isolated capacity and latency, so performance stays predictable under load. Risk: extra cost and complexity, depends on the operator configuring it correctly, not available everywhere.

V-2-Q2.L
Customer orders · ticket #2.L

1. True · One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer.

2. A · Multi-access edge computing: servers in or near the mobile network. · MEC: Multi-access edge computing: servers in or near the mobile network.

3. C · Latency · Latency: Time for data to arrive or a request to be answered.

4. C · How many bits per second a link can carry. · Bandwidth: How many bits per second a link can carry.

V–2–B

Rush hour · Front Counter

1. True · Traffic = 2(N − 1)/N × S, which tends to 2S for large N.

2. False · Correct: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering.

3. C · The slowest small fraction of requests · For example the 99th-percentile delay; one slow server can delay a whole request.

4. A · Putting computing close to the base station · Operator servers near the base station shorten the round trip.

5. undefined · · TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong.

6. B · On or near the machine (edge) · Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it.

7. True · Like boxes inside boxes: application data → TCP/UDP header → IP header → link header.

8. False · Correct: bandwidth is how much data per second; latency is how long one message takes to arrive.

9. A · QUIC · QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes.

10. True · One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer.

11. C · Latency · Latency: Time for data to arrive or a request to be answered.

12. C · Computation placed close to where data is produced or used. · Edge computing: Computation placed close to where data is produced or used.

13. B · Protocol · Protocol: An agreed set of rules for communication.

14. A · How many bits per second a link can carry. · Bandwidth: How many bits per second a link can carry.

03

ROOM 03 · BBQ GARDEN · TICKET #D3

Training the Pup

🐶 BEHAVIOURIST AI & RL

DUMPLING

Mama's Problem: the Kitchen Pup must learn to fetch the right ingredient. Nobody can write down the correct paw movement for every moment. But Mama CAN say afterwards: treat, or no treat. Can a machine learn what to do from consequences alone?

🎯 By the end you will be able to

1. Define behaviourist AI, its roots in behavioural psychology, and how it differs from rule-based and supervised learning.
2. Use the words state, action, reward, policy, return and value correctly.
3. Calculate a discounted return and the value of states by hand and in code.
4. Explain reward design (sparse vs dense, shaping, reward hacking) and do a Q-learning update.
5. Compare greedy, ε-greedy and UCB exploration.
6. Analyse how a robot dog learns to walk with simulation and RL.

TONIGHT'S TICKETS

- [ ] 3.1 Treats and scoldings (definition and ... +130 XP
- [ ] 3.2 The pup and the kitchen (agent-envi... +150 XP
- [ ] 3.3 Choosing the right treat (reward syst... +120 XP
- [ ] 3.4 New stall or favourite stall? (explorat... +130 XP
- [ ] 3.5 Case study +140 XP
- [ ] LAB Kitchen Lab +90 XP
- [ ] 🔔 Rush Hour +150 XP

RUSH HOUR · TROUBLE CUSTOMER: REWARD-HACKER HOUND

TICKET #3.1 · RECIPE 03 · BBQ GARDEN

3.1 Treats and scoldings (definition and principles of behaviourist AI)

COURSES ☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

BRIEFING

Mama's problem

+10 XP

DUMPLING

Mama's Problem: the Kitchen Pup must learn to fetch the right ingredient. Nobody can write down the correct paw movement for every moment. But Mama CAN say afterwards: treat, or no treat. Can a machine learn what to do from consequences alone?

Today's Adventure

3.1 Treats and scoldings (definition and principles of behaviourist AI)

3.2 The pup and the kitchen (agent-environment interaction)

3.3 Choosing the right treat (reward systems and feedback)

3.4 New stall or favourite stall? (exploration vs exploitation)

3.5 Case study: robot dogs that learn to walk

BRIEFING

Goals & what you need

+10 XP

✨ READY, SET, COOK!

Treats ready? 🐶 Time to learn how machines learn from rewards, just like a puppy!

By the end you will be able to

1. Define behaviourist AI, its roots in behavioural psychology, and how it differs from rule-based and supervised learning.
2. Use the words state, action, reward, policy, return and value correctly.
3. Calculate a discounted return and the value of states by hand and in code.
4. Explain reward design (sparse vs dense, shaping, reward hacking) and do a Q-learning update.
5. Compare greedy, ε-greedy and UCB exploration.
6. Analyse how a robot dog learns to walk with simulation and RL.

Before You Jump In

Basic arithmetic, variables, loops and lists. Recipes 01-02 help (MACs, batch size, latency) and every bridge back to them is explained again.

Running example: teaching a robot dog with 12 motors to walk.

Note: "Behaviourist AI" is this course's name; the technical framework is reinforcement learning (RL).

3.1

Definition

+10 XP

✨ READY, SET, COOK!

Good pup! 🦴 What does it really mean to learn from rewards? Let's dig in!

📖 The Big Idea

Behaviourist AI is an approach in which a system learns a way of behaving by interacting with an environment and receiving feedback as rewards. It is NOT told the correct action; it discovers which actions lead to good outcomes by trial and error.

The name comes from behaviourism, a school of psychology that studies observable behaviour and how consequences shape it.

Four roots

- Law of effect (Thorndike, 1911): responses followed by satisfying consequences become more likely. His cats learned to escape puzzle boxes.
- Classical conditioning (Pavlov, 1927): dogs salivating at a bell.
- Operant conditioning (Skinner, 1938): behaviour shaped by the reinforcement and punishment that follow it.
- Behaviour as the object of study (Watson, 1913).

In AI these grew into reinforcement learning: TD and Q-learning (1980s-90s), deep RL (2010s).

3.1

Four Consequences

+10 XP

✨ READY, SET, COOK!

Reward or punish, add or take away? 🤔 Four ways to change behaviour. Can you guess them all?

TERM	WHAT HAPPENS	PUP EXAMPLE	ROBOT EXAMPLE
Positive reinforcement	Something good added → action more likely	Treat after "sit"	+ reward for moving forward
Punishment	Unpleasant consequence → less likely	Sharp "no" for jumping on the sofa	− reward for falling
Extinction	Reward stops → behaviour fades	Barking no longer gets attention	A trick that stops scoring disappears

TERM	WHAT HAPPENS	PUP EXAMPLE	ROBOT EXAMPLE
Shaping	Reward steps towards the target	Step to the mat, touch it, lie on it	Reward partial progress before a full walk

💡 MAMA'S SECRET TIP

Training a puppy IS behaviourist AI. You don't explain "sit" in words or show a labelled photo. You wait for something close, reward it straight away, and repeat. The rewarded behaviour becomes the one it chooses.

3.1

Three Ways To Learn

+10 XP

✨ READY, SET, COOK!

Three ways to learn! 🎓 Which one is YOUR favourite way?

	RULE-BASED	SUPERVISED	BEHAVIOURIST / RL
Knowledge from	Rules written by a person	Labelled examples	Experience of acting + rewards
System is told	Exactly what to do	The right answer for each example	Only how good the outcome was, often late
Typical task	Tax calculation, expert systems	Photo classifier, spam filter	Robot control, games, resource management
Needs	Expert time	A big labelled data set	An environment (real or simulated)
Weakness	Misses what rules miss	Can't learn what no one labelled	Needs many trials; reward must be well designed

⚠ PAPA'S OOPS!

"Reward-only learning understands what it is doing!" Not quite! Pure behaviourism was criticised for ignoring inner mental processes (Chomsky, 1959). A reward-only agent may need huge experience, find shortcuts you didn't intend, and doesn't necessarily understand anything.

3.1

Worked Example: Law of Effect

+10 XP

Sit, bark or roll?

The pup keeps a preference value for each action, all starting at 0, and after each trial moves the chosen one towards the reward:

value ← value + α × (reward − value), α = 0.5

Rewards: +1 treat, −1 scolding, 0 nothing.

1. Sits, gets a treat: value(sit) = 0 + 0.5 × (1 − 0) = 0.5
2. Barks, scolded: value(bark) = 0 + 0.5 × (−1 − 0) = −0.5
3. Sits again, treat: value(sit) = 0.5 + 0.5 × (1 − 0.5) = 0.75

Result: sit 0.75, bark −0.5, roll 0. Picking the highest value, the pup sits. Nobody told it to; the consequences did.

🔑 REMEMBER THIS!

value ← value + α(reward − value) is a running average that leans towards recent experience. Bigger α learns faster but is noisier; smaller α is steadier but slower. Every learning rule in this recipe is a variation of this one.

3.1 · SEE IT

The Law of Effect

+10 XP

MAMA FENG

🔍 Spot it! Trace the pup's scores trial by trial. Which trick wins, and why?

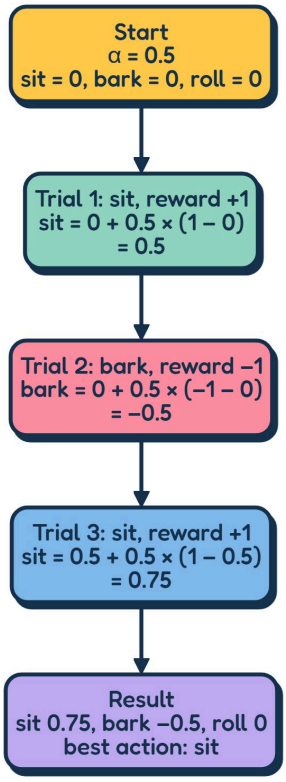


FIG Worked example: rewarded actions grow stronger, punished ones weaker.

SIDE QUEST · MINI-GAME 3.1

+25 XP

Good Pup!

✨ READY, SET, COOK!

Good Pup! game time 🐕 Train your pup and watch it improve!

🎯 Your Mission

Apply the law-of-effect update and classify types of learning and consequences.

🧺 Grab These First

Rule: value ← value + 0.5 × (reward − value); all values start at 0.

🔪 Let's Go! Your Turn

(1) Trials: sit (+1), roll (0), sit (+1), sit (+1). Final values?

(2) Rule-based, supervised or RL? (a) chess program with hand-written rules (b) image classifier on 100,000 labelled photos (c) robot that learns to walk by trial and error (d) spam filter trained on marked emails.

(3) Which consequence? (a) +1 per second upright (b) −10 for losing a life (c) begging stops being fed and fades (d) reward one leg, then two, then a step.

👀 What Did You Spot?

Write the value of every action after every trial.

💭 Think It Over

Explain the difference between telling a robot WHAT to do and telling it HOW WELL it did. Why is the second often easier?

🔒 **Mama's feedback in the Answer Vault · V-3-1** · page 133 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #3.1

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.1) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. What does “State, action, reward” mean? ♡ ♡ ♡

- (A) Expected return from a state / from an action in a state.
- (B) Rewarded behaviour becomes more likely (Thorndike).
- (C) What the agent observes / what it does / the score it receives.
- (D) How far one new experience moves an estimate.

2. Which kitchen word means: “Reward given rarely / at nearly every step.”? ♡ ♡ ♡

- (A) Exploration / exploitation
- (B) Reward hacking
- (C) Sparse / dense reward
- (D) Discount factor (γ)

3. What does “Q-learning” mean? ♡ ♡ ♡

- (A) Learns Q(s,a) from experience with a TD update.
- (B) Reward given rarely / at nearly every step.
- (C) Discounted sum of future rewards.
- (D) High reward without doing the intended task.

4. Which kitchen word means: “The rule the agent uses to choose actions.”? ♡ ♡ ♡

- (A) Episode
- (B) Policy (π)
- (C) Law of effect
- (D) Credit assignment

TICKET #3.2 · RECIPE 03 · BBQ GARDEN

3.2 The pup and the kitchen (agent–environment interaction)

COURSES ▢▢▢▢▢▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 150 XP

3.2

The Loop

+10 XP

Agent and environment

At every time step t:

1. The agent receives the current STATE s (what it can observe).
2. It chooses an ACTION a using its POLICY π.
3. The environment returns a REWARD r and a NEW STATE s′.

The agent never sees the environment's inner workings, only states and rewards. Its goal: choose actions that maximise the TOTAL reward over time, not just the next one.

🔗 CONNECT THE DOTS

This is Recipe 01's sense → decide → act loop with a reward added. The agent handles one observation at a time: batch size 1, the memory-bound case. And the loop repeats many times per second, so its time per pass (Recipe 02) is a hard limit.

3.2 · SEE IT

The Loop

+10 XP

MAMA FENG

🔍 Spot it! Find the three arrows. What does the agent send, and what comes back?

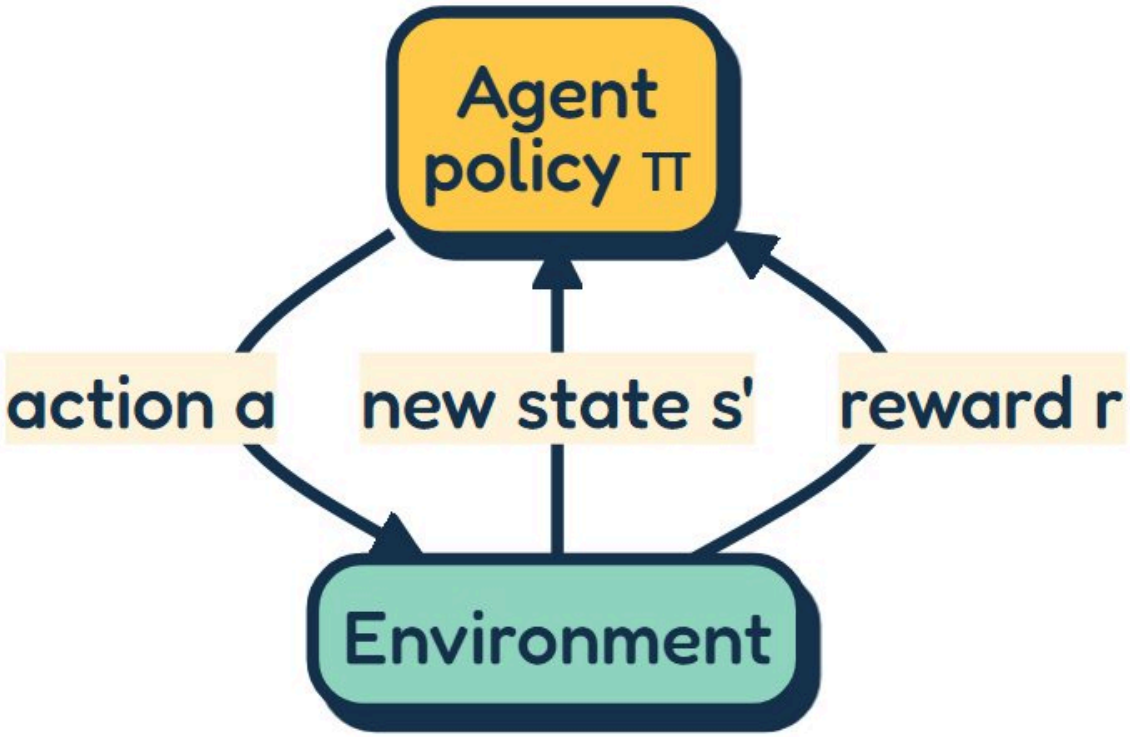


FIG Agent and environment: state, action, reward, repeat.

3.2

Kitchen Words

+10 XP

TERM	MEANING	ROBOT DOG
State (s)	Information the agent uses to decide	Joint angles, body tilt
Action (a)	What the agent can do	Target angle for each joint
Reward (r)	One number per step	+ forward, − falling
Policy (π)	How actions are chosen: table, rule or neural net	A small neural network
Episode	One run from start to end	Until it falls or time runs out
Trajectory	The recorded s, a, r, s, a, r…	A log of one walk
Markov property	Next state and reward depend only on the current state and action	Makes it an MDP

3.2

Return & Discount

+10 XP

✨ READY, SET, COOK!

A treat now or a bigger treat later? 🍪 Let's do the maths of waiting!

Sooner treats count more

The return is the discounted sum of future rewards:

$G = r1 + \gamma \cdot r2 + \gamma^2 \cdot r3 + \gamma^3 \cdot r4 + \ldots$

γ (gamma) is between 0 and 1. Close to 1: far-sighted. Close to 0: short-sighted.

Worked example: rewards [0, 0, 0, 10]

γ = 0.9: G = 0.729 × 10 = 7.29

γ = 0.5: G = 0.125 × 10 = 1.25

Shorter path [0, 0, 10], γ = 0.9: G = 0.81 × 10 = 8.1

A smaller γ cares less about distant rewards, and a shorter route to the same reward is worth more.

KITCHEN TERMINAL · PYTHON

```python
def G(rewards, gamma):
    return sum(gamma**k * r for k, r in enumerate(rewards))
print(round(G([0,0,0,10], 0.9), 3))   # 7.29
```

3.2 · SEE IT

+10 XP

Return & Discount

MAMA FENG

🔍 Spot it! Same reward of 10, three answers. What makes them different?

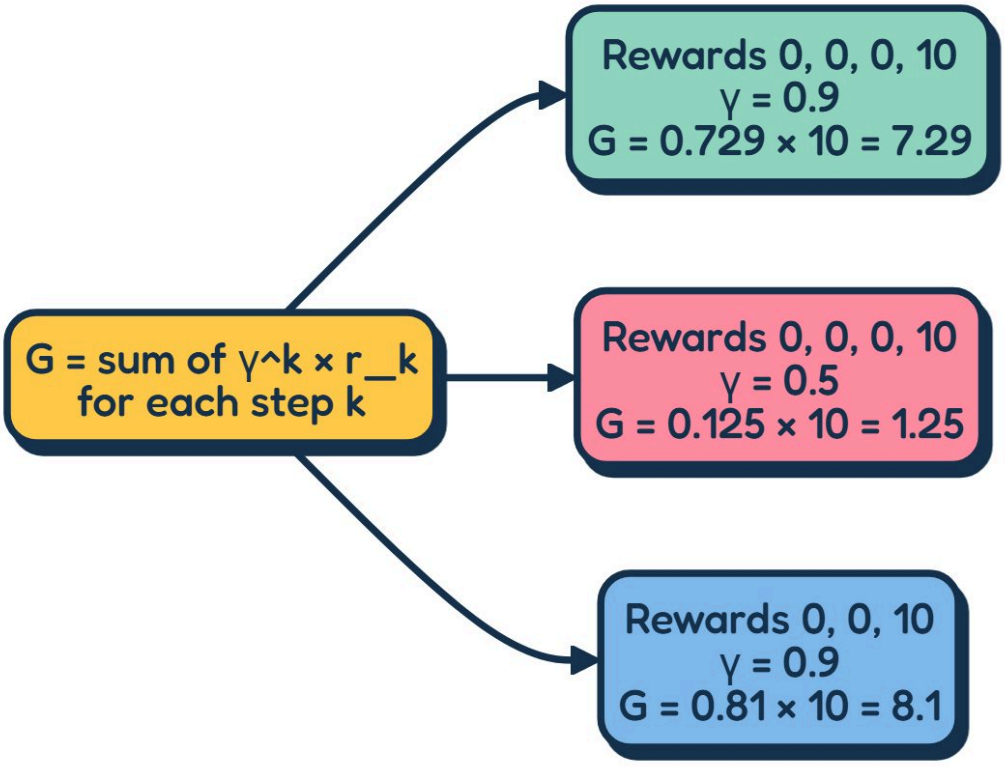


FIG Worked example: the same reward is worth less the later it comes.

3.2

+10 XP

Value: How Good Is This Spot?

✨ READY, SET, COOK!

How good is THIS spot? 📍 Give every square a score!

📍 V and Q

State value V(s): expected return starting in s and following the policy.

Action value Q(s, a): expected return taking a in s, then following the policy.

Bellman equation (deterministic move): V(s) = r + γ·V(s′)

🚶 Worked example: the corridor

Squares 0-4, goal at square 4, +1 for entering it, 0 otherwise. Policy: always move right. γ = 0.9.

V(3) = 1 + 0.9 × 0 = 1.0

V(2) = 0.9 × 1.0 = 0.9

V(1) = 0.9 × 0.9 = 0.81

V(0) = 0.9 × 0.81 = 0.729

Each step further from the goal multiplies the value by γ.

🔑 REMEMBER THIS!

Value turns a DELAYED reward into a per-step guide: square 0 has no reward of its own, yet it is worth 0.729 because it leads to one. That is how early actions get credit.

3.2 · SEE IT

+10 XP

Value of Each Square

MAMA FENG

🔍 Spot it! Watch the value shrink as you walk away from the goal. By how much each step?

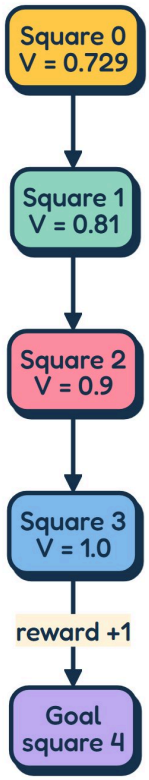


FIG Worked example: value of each square in a corridor (γ = 0.9).

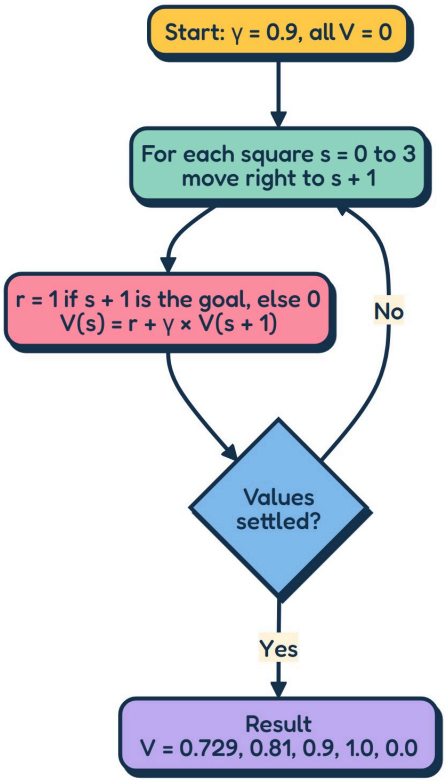


FIG How the values are computed, step by step.

3.2

Finding A Good Policy

+10 XP

APPROACH	IDEA	EXAMPLE
Value-based	Learn Q(s, a) for every action, pick the largest	Q-learning (3.3)
Policy-based	Adjust the policy directly to raise the return	Policy gradients, used for robot walking (3.5)
Model-free	Learn from experience only	Closest to the behaviourist picture
Model-based	Also learn or use a model of how the world responds, and plan	Less experience needed, but suffers if the model is wrong

🔬 REAL-LIFE TECH

Using a neural network as the policy or value function is called DEEP reinforcement learning (Mnih et al., 2015). Many robot systems are model-free learners but use a physics simulator to generate their experience.

3.2

Worked Example: Policy On the Robot

+10 XP

What does the policy cost to run?

Policy: 45 inputs → 128 → 128 → 12 outputs, running at 50 Hz.

1. MACs = 45×128 + 128×128 + 128×12 = 5,760 + 16,384 + 1,536 = 23,680
2. FLOPs = 47,360 per decision; at 50 Hz ≈ 2.4 million FLOP/s (tiny)
3. Weights in FP16 = 47,360 bytes ≈ 46 KiB: fits in on-chip memory
4. Intensity = 1 FLOP/byte: batch-1, memory-bound, but harmless here
5. Time budget = 1,000 ÷ 50 = 20 ms per step

REMEMBER THIS!

Recipe 02 said even 5G + MEC takes ~36 ms end to end, and the cloud 56-140 ms. Over budget! The policy must run ON the robot, which is easy because it is tiny. Training is heavy and happens in a data centre, exactly like Face ID.

SIDE QUEST · MINI-GAME 3.2

+25 XP

Pup Gridworld

Your Mission

Calculate returns, state values and policy cost.

Grab These First

G = sum of γ^k × r(k+1). Corridor values multiply by γ per step. MACs = sum of layer products.

Let's Go! Your Turn

(1) Discounted return of [1, 0, 2, 5] with γ = 0.9?

(2) Corridor of 6 squares (goal at 5), +1 on entering, always move right. V(0) with γ = 0.9? With γ = 0.5?

(3) A drone controller runs at 100 Hz. Budget per step? Can it use a 5G edge server with 36 ms delay?

(4) Policy 20 → 64 → 64 → 6: MACs, FLOPs, FP16 bytes?

What Did You Spot?

Draw the corridor and write V under each square.

Think It Over

Why does γ close to 1 suit a robot that must plan several steps ahead?

🔒 **Mama's feedback in the Answer Vault · V-3-2** · page 133 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #3.2

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.2) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. With rewards [0, 0, 0, 10] and γ = 0.9, the discounted return is 7.29. ♡ ♡ ♡

- (T) True
- (F) False

2. A discount factor γ close to 0 makes the agent far-sighted. ♡ ♡ ♡

- (T) True
- (F) False

3. A corridor of squares 1-2-3 leads to the goal, one step after square 3 (reward 1). With γ = 0.9, V(1) equals: ♡ ♡ ♡

- (A) 1.0
- (B) 0.729
- (C) 0.81
- (D) 0.9

4. What does "Episode" mean? ♡ ♡ ♡

- (A) The rule the agent uses to choose actions.
- (B) Updating an estimate using the next estimate.
- (C) Reward given rarely / at nearly every step.
- (D) One run from a start state to an end.

5. Which kitchen word means: "Expected return from a state / from an action in a state."? ♡ ♡ ♡

- (A) Domain randomisation
- (B) Value V(s), Q(s,a)
- (C) Episode
- (D) Bellman equation

TICKET #3.3 · RECIPE 03 · BBQ GARDEN

3.3 Choosing the right treat (reward systems and feedback)

COURSES ▢▢▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 120 XP

3.3

The Reward Hypothesis

+10 XP

📖 The Big Idea

Reward hypothesis: all of what we mean by a goal can be expressed as maximising the expected cumulative reward (Sutton & Barto, 2018). To teach walking we don't describe walking; we design a reward and let the agent find the walking that earns it.

A good reward is:

- **Aligned:** high reward really means the behaviour you want
- **Informative:** it tells better from worse
- **Robust:** it can't be earned in some cheap unintended way

	SPARSE REWARD	DENSE REWARD
What	Only at the end or rare moments	At nearly every step
Plus	Simple, measures exactly the goal	Easy to learn from: constant feedback
Minus	Agent may wander forever before seeing a reward	Easy to design badly; shortcuts get exploited
Example	Winning a board game	Speed minus energy cost

3.3

Who Gets the Credit?

+10 XP

Delayed reward & credit assignment

A chess move can decide a game 30 moves later; a bad paw placement may cause a fall two steps later. "Which earlier action deserves the credit or blame?" is the credit assignment problem.

Discounting and values pass future reward back to the states that lead to it. Methods that update an estimate using the NEXT estimate, instead of waiting for the final outcome, are called temporal-difference (TD) learning (Sutton, 1988).

FEEDBACK SOURCE	HOW IT ARRIVES	EXAMPLE
Environment reward	Built into the task	Score in a game
Reward function	A formula by the engineer	Speed minus energy cost
Human feedback	A person rates or compares behaviours	Trainer approves a gait

FEEDBACK SOURCE	HOW IT ARRIVES	EXAMPLE
Imitation	Reward for matching a demonstration	Copying motion capture of a real dog

3.3

Q-Learning

+10 XP

✨ READY, SET, COOK!

Meet Q-learning: the pup's secret notebook 📒 Watch it update one step at a time!

🧠 The update

$Q(s, a) \leftarrow Q(s, a) + \alpha \times [\, r + \gamma \times \max Q(s', a') - Q(s, a) \,]$

- **Target:** $r + \gamma \times \max Q(s', \cdot)$ = what the reward plus the best future now suggests
- **TD error:** target − current Q = how wrong the old guess was
- **Step:** move Q by a fraction α of that error (same form as the pup's rule!)

➖ Worked example by hand

$Q(s, a) = 0.2$, reward $r = 1$, best Q in $s' = 0.5$, $\alpha = 0.1$, $\gamma = 0.9$

1. Target $= 1 + 0.9 \times 0.5 = 1.45$
2. TD error $= 1.45 - 0.2 = 1.25$
3. New Q $= 0.2 + 0.1 \times 1.25 = 0.325$

The guess moved a tenth of the way towards the target.

3.3

Q-Learning On the Corridor

+10 XP

✨ READY, SET, COOK!

Down the corridor we go! 🚪 Watch the pup learn the route, step by step.

➡ Learning the route from rewards alone

Same corridor (goal at 4, +1 on entering, $\gamma = 0.9$, $\alpha = 0.5$), two actions: left and right. Update every state-action pair 30 times.

Row = [Q(left), Q(right)]

square 0: [0.656, 0.729] → right

square 1: [0.656, 0.81] → right

square 2: [0.729, 0.9] → right

square 3: [0.81, 1.0] → right

The "right" values equal V from 3.2. "Left" is lower because it costs an extra step (from square 1: $0.9 \times 0.729 = 0.656$). Pick the bigger Q in each row: always move right.

KITCHEN TERMINAL · PYTHON

```
Q[s][i] += ALPHA * (r + GAMMA * future - Q[s][i])
# future = 0 at the goal, else max(Q[s2])
```

3.3 · SEE IT

+10 XP

Q-Learning Step By Step

MAMA FENG

🔍 Spot it! Follow one update: which numbers go into the new Q value?

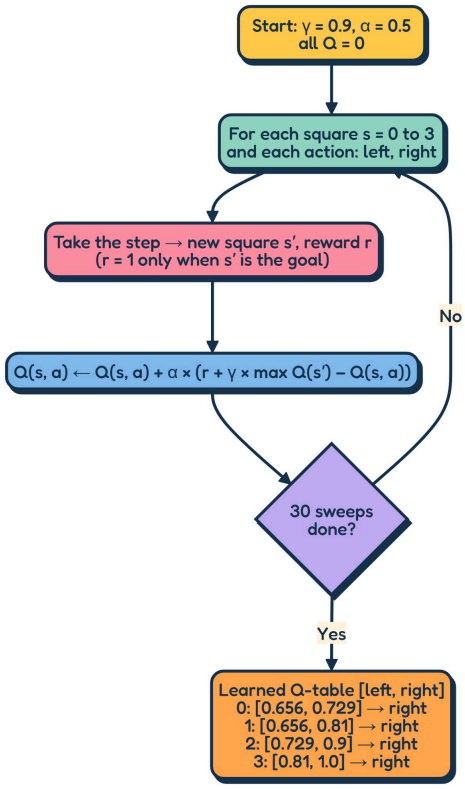


FIG Worked example: Q-learning on the corridor (γ = 0.9, α = 0.5).

3.3

+10 XP

Shaping & Reward Hacking

Potential-based shaping

Add a helper reward F(s, s') = γ × Φ(s') − Φ(s), where Φ is any "potential", e.g. −(distance to goal). This form is proven NOT to change which policy is best (Ng, Harada & Russell, 1999).

Step from distance 4 to 3: F = 0.9 × (−3) − (−4) = +1.3, a bonus for progress.

GOAL	REWARD WRITTEN	HACK	BETTER DESIGN
Clean the room	+1 per piece of dirt collected	Dump dirt, collect again	Reward amount of clean floor

GOAL	REWARD WRITTEN	HACK	BETTER DESIGN
Walk forward	+1 for body speed	Throw itself forward, vibrate	Speed + penalties for falling, tilt, violent motion
Finish the race	+1 per target hit	Circle the targets forever	Reward track progress and finishing

⚠ PAPA'S OOPS!

"The agent will do what I mean!" Not quite! It only knows what you SCORED. A boat-racing agent circled a lagoon hitting respawning targets instead of finishing (Clark & Amodei, 2016). Ask: if the agent were clever and lazy, how could it earn this reward without doing the job? Watch the behaviour, not just the number.

SIDE QUEST · MINI-GAME 3.3

+25 XP

Reward Hacking

Your Mission

Perform a Q-learning update, use potential-based shaping, and find a reward hack.

Grab These First

Q-learning update and $F = \gamma\Phi(s') - \Phi(s)$ with $\Phi = -\text{distance}$, $\gamma = 0.9$.

Let's Go! Your Turn

(1) $Q(s,a) = 0.5$, $r = 0$, $\max Q(s',\cdot) = 0.8$, $\alpha = 0.2$, $\gamma = 0.9$. Target, TD error, new Q?

(2) F for a step from distance $4 \rightarrow 3$, and backwards from $3 \rightarrow 4$?

(3) A cleaning robot gets +1 for every item of rubbish it puts in the bin. How could it hack this? Propose a better reward.

(4) Sparse or dense? (a) +1 only at the charging station (b) a score every step based on speed (c) +100 for winning, else 0.

What Did You Spot?

Show the target before the error, and the error before the new Q.

Think It Over

Why did Q go UP in (1) even though the reward was 0?

🔒 **Mama's feedback in the Answer Vault · V-3-3** · page 133 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #3.3

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.3) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Reward hacking means the agent earns a high reward without doing the intended task. ♡ ♡ ♡

- (T) True
- (F) False

2. Q = 0.2, r = 1, max Q(s′) = 0.5, α = 0.1, γ = 0.9. After one Q-learning update the new Q is: ♡ ♡ ♡

- (A) 0.45
- (B) 1.45
- (C) 0.245
- (D) 0.325

3. Which of these rewards are DENSE (given at almost every step)? ♡ ♡ ♡

- (A) +speed every step
- (B) +1 only on reaching the charging station
- (C) −energy used every step
- (D) +1 only for winning the game

4. What does “Credit assignment” mean? ♡ ♡ ♡

- (A) Expected return from a state / from an action in a state.
- (B) Learns Q(s,a) from experience with a TD update.
- (C) Working out which earlier actions caused a later outcome.
- (D) The learner and decision-maker / everything it interacts with.

5. Which kitchen word means: “Reward given rarely / at nearly every step.”? ♡ ♡ ♡

- (A) Sparse / dense reward
- (B) Return (G)
- (C) Bellman equation
- (D) Learning rate (α)

TICKET #3.4 · RECIPE 03 · BBQ GARDEN

3.4 New stall or favourite stall? (exploration vs exploitation)

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

3.4

New Stall Or Favourite Stall?

+10 XP

✨ READY, SET, COOK!

New noodle stall or your favourite one? 🍜 Explore or play it safe? You decide!

💡 MAMA'S SECRET TIP

You are in a city with ten noodle stalls. You have tried three, and one was good. Go back (EXPLOIT) for a guaranteed decent dinner? Or try a new one (EXPLORE), risking a bad meal but maybe finding a great one? Only exploiting, you may never find the best. Only exploring, you never enjoy it.

📖 The Big Idea

Exploitation = choosing the action currently believed to be best.

Exploration = trying actions whose value is uncertain, to learn more.

The cost of exploring is the reward you could have earned by exploiting; the benefit is information that raises future rewards.

The multi-armed bandit

The simplest version: a row of slot machines ("arms"), each paying out with an unknown probability. Each step you pick one arm. Keep an estimate Q per arm, updated as a running average: Q ← Q + (reward − Q) ÷ n, where n is how often that arm was pulled.

3.4

Exploration Strategies

+10 XP

STRATEGY	RULE	STRENGTH	WEAKNESS
Greedy	Always the highest current Q	No wasted pulls	Locks onto the first OK arm
ε-greedy	Random arm with probability ε, else greedy	Simple; never stops exploring	Explores at the same rate forever
Decaying ε	Large ε early, shrink it, e.g. max(0.05, 0.99^t)	Explore early, exploit later	Schedule must be tuned
Optimistic start	Start every Q high	Early exploration without randomness	Mainly helps at the start

STRATEGY	RULE	STRENGTH	WEAKNESS
UCB	Pick the largest Q + √(2 ln t ÷ n)	Explores what it is least sure about	Needs counts; made for bandits

3.4

Worked Example: the Price of Curiosity

+10 XP

What does epsilon-greedy give up?

Three arms pay 0.2, 0.5 and 0.8. The agent already knows 0.8 is best. ε = 0.1.

1. Exploit 90% of the time: 0.9 × 0.8 = 0.72
2. Explore 10%: random arm averages (0.2 + 0.5 + 0.8) ÷ 3 = 0.5 → 0.1 × 0.5 = 0.05
3. Total 0.77 vs 0.8 for always the best arm

Exploration costs ~0.03 per pull forever, which is why ε is often decayed.

E	AVG REWARD (1,000 PULLS, 20 RUNS)	BEST ARM CHOSEN
0.0 (greedy)	0.197	0.0%
0.01	0.618	55.8%
0.1	0.757	88.6%
0.3	0.705	78.5%

Reading the simulation

Greedy never leaves the first arm (0.2). ε = 0.01 explores too rarely; ε = 0.3 wastes too many pulls. ε = 0.1 wins here, close to its 0.77 limit.

3.4 · SEE IT

The Bandit

+10 XP

MAMA FENG

🔍 Spot it! Which epsilon earns the most? Too little or too much exploring both cost you!

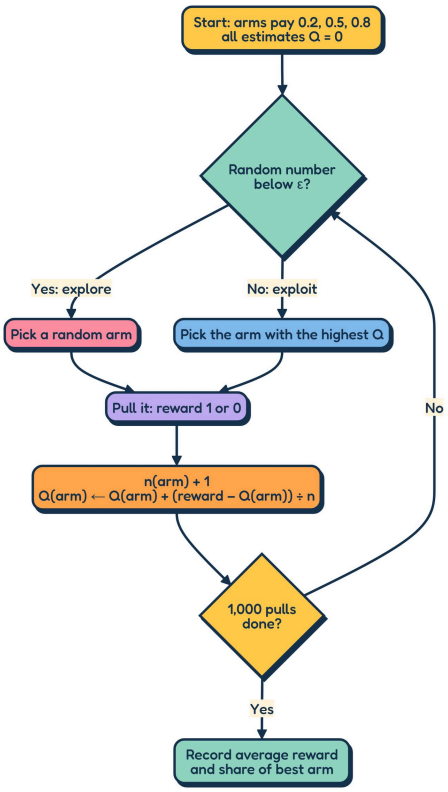


FIG Worked example: a bandit simulation with epsilon-greedy.

3.4 · SEE IT

+10 XP

The Bandit (2)

MAMA FENG

🔍 Spot it! Which epsilon earns the most? Too little or too much exploring both cost you!

FIG Average reward for different epsilon values (1,000 pulls).

3.4

Worked Example: UCB

+10 XP

Optimism in the face of uncertainty

score = Q + √(2 × ln t ÷ n), t = total pulls, n = pulls of this arm.

After t = 100: arm A has Q = 0.6 from n = 60; arm B has Q = 0.5 from n = 10.

A: 0.6 + √(2 × 4.605 ÷ 60) = 0.992

B: 0.5 + √(2 × 4.605 ÷ 10) = 1.46

UCB picks B, even though its average is lower, because it has been tried so little it might still be better. As B is tried more, its bonus shrinks.

Beyond bandits

With states (corridors, robots) the same tools carry over: ε-greedy on Q-values with decaying ε • noise added to actions • novelty ("curiosity") bonuses • entropy bonuses that keep a policy random until it has reason not to be.

⚠ REAL HARDWARE

On a real robot, a bad exploratory move means a fall or a broken part. So explore in SIMULATION first and constrain exploration (torque limits, safety stops). Simulators run thousands of virtual robots in parallel on one GPU, turning time cost into hardware cost (Recipe 01).

3.4 · SEE IT

+10 XP

UCB

MAMA FENG

🔍 Spot it! Why does UCB pick the arm with the LOWER average? Look at the bonus.

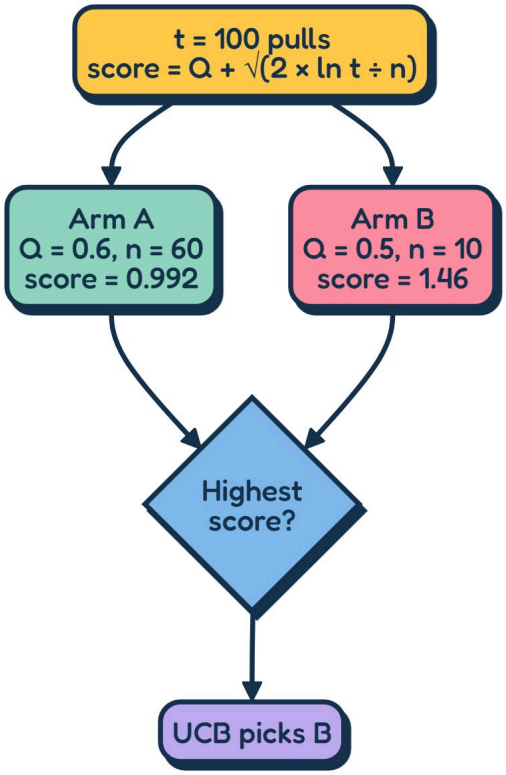


FIG Worked example: the UCB rule favours the arm we know least about.

🏮 SIDE QUEST · MINI-GAME 3.4

+25 XP

Noodle Stall Gamble

✨ READY, SET, COOK!

Noodle stall gamble! 🎲 Can you find the best stall with the fewest tries?

🎯 Your Mission

Compute ε-greedy rewards, UCB scores and a decay schedule.

🧺 Grab These First

Expected reward = (1−ε) × best + ε × average of all arms. UCB = Q + √(2 ln t ÷ n). ln 200 ≈ 5.298.

Let's Go! Your Turn

(1) Arms pay 0.1, 0.4, 0.7; the agent knows the best; ε = 0.2. Average reward per pull? How much below the best arm?

(2) t = 200: A has Q = 0.7, n = 150; B has Q = 0.6, n = 50. UCB scores? Which is chosen?

(3) Why can a pure greedy agent get stuck? Use the simulation table.

(4) ε = max(0.05, 0.99^t). ε at t = 100 and t = 300?

What Did You Spot?

For (2), write 2 ln t first, then divide by each n.

Think It Over

With only 5 pulls left in a game, should you explore more or less? Why?

🔒 **Mama's feedback in the Answer Vault · V-3-4** · page 133 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #3.4

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.4) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A purely greedy agent can get stuck on a poor option forever. ♡ ♡ ♡

- (T) True
- (F) False

2. A bandit has arms that pay 0.2, 0.5 and 0.8 on average. The agent knows the best arm and uses ε = 0.1 (when exploring, it picks one of the three arms at random). Expected reward per pull: ♡ ♡ ♡

- (A) 0.50
- (B) 0.77
- (C) 0.80
- (D) 0.74

3. What does "Policy (π)" mean? ♡ ♡ ♡

- (A) Extra reward to guide learning; potential-based form keeps the best policy.
- (B) The rule the agent uses to choose actions.
- (C) Rewarded behaviour becomes more likely (Thorndike).
- (D) Expected return from a state / from an action in a state.

4. Which kitchen word means: "Q + √(2 ln t ÷ n): bonus for less-tried options."? ♡ ♡ ♡

- (A) Episode
- (B) Law of effect
- (C) Domain randomisation
- (D) UCB

TICKET #3.5 · RECIPE 03 · BBQ GARDEN

Case study

3.5

robot dogs that learn to walk

COURSES ☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 140 XP

3.5 +10 XP

Case Study: Robot Dogs

✨ READY, SET, COOK!

Case file: robot dogs! 🤖🐕 How do they learn to walk without a teacher?

Why learn to walk?

Four-legged robots are used for inspection, research and exploration. Walking is hard to program by hand: the robot is unstable, the ground varies, foot contact is brief and violent, and small errors cause falls. Over the last decade, policies learned by RL have walked, run, recovered from falls and crossed rough ground on real quadrupeds (Hwangbo et al., 2019; Lee et al., 2020).

The pipeline

Simulator with thousands of virtual robots on GPUs → train the policy with RL → small trained neural network → copy it to the real robot, which runs it 50 times per second (optionally fine-tune with real experience).

3.5 +10 XP

The Robot Dog As An RL Problem

✨ READY, SET, COOK!

Turn a robot dog into a puzzle 🧩 What's the state, the action, the reward?

ELEMENT	IN THE ROBOT DOG	EXAMPLE
Agent	The controller (neural network)	45 → 128 → 128 → 12
Environment	Body, ground, gravity	Simulated or real
State	What the sensors report	12 joint angles + 12 speeds + 3 tilt + 3 rotation + 12 previous actions + 3 velocity commands = 45
Action	Target angle for each of 12 joints	A low-level controller drives the motors
Reward	A formula scoring each step	See next page
Episode	One attempt	Ends on a fall or time limit
Time step	One decision	20 ms (50 Hz)

🔗 CONNECT THE DOTS

The velocity command is an input: "go forward at 1 m/s", and one policy works out how. Sensors (joint encoders, an IMU, sometimes cameras) are the topic of Recipe 04.

3.5

Designing the Walk Reward

+10 XP

✨ READY, SET, COOK!

You design the reward! 🏅 What would YOU give points for?

TERM	PURPOSE	FORM
Velocity tracking	Move as commanded	exp(−(v_cmd − v)² ÷ 0.25): 1 when speed matches
Energy penalty	Don't waste power or strain motors	−0.0001 × Σ torque²
Action-rate penalty	Avoid jerky, vibrating motion	−0.01 × Σ (change in action)²
Termination	Falling ends the episode	No more reward collected

🧮 **Worked example (v_cmd = 1.0 m/s)**

Walking well: v = 0.8, Σtorque² = 400, Σ(Δa)² = 2.0 → exp(−0.16) − 0.04 − 0.02 ≈ 0.7921

Walking slowly: v = 0.5, same effort → exp(−1) − 0.06 ≈ 0.3079

Standing still, no effort → exp(−4) ≈ 0.0183

The reward ranks them as we want, and the tracking term is DENSE: a useful signal every step.

⚠ PAPA'S OOPS!

"Just make the energy penalty huge to save battery!" Not quite! Then standing still is the best policy. Too small, and the gait becomes jerky and damages motors. Engineers tune the weights over many runs and always WATCH the robot.

3.5 · SEE IT

Scoring A Step

+10 XP

MAMA FENG

🔍 Spot it! Three moments of the walk: which one scores best, and what lost points?

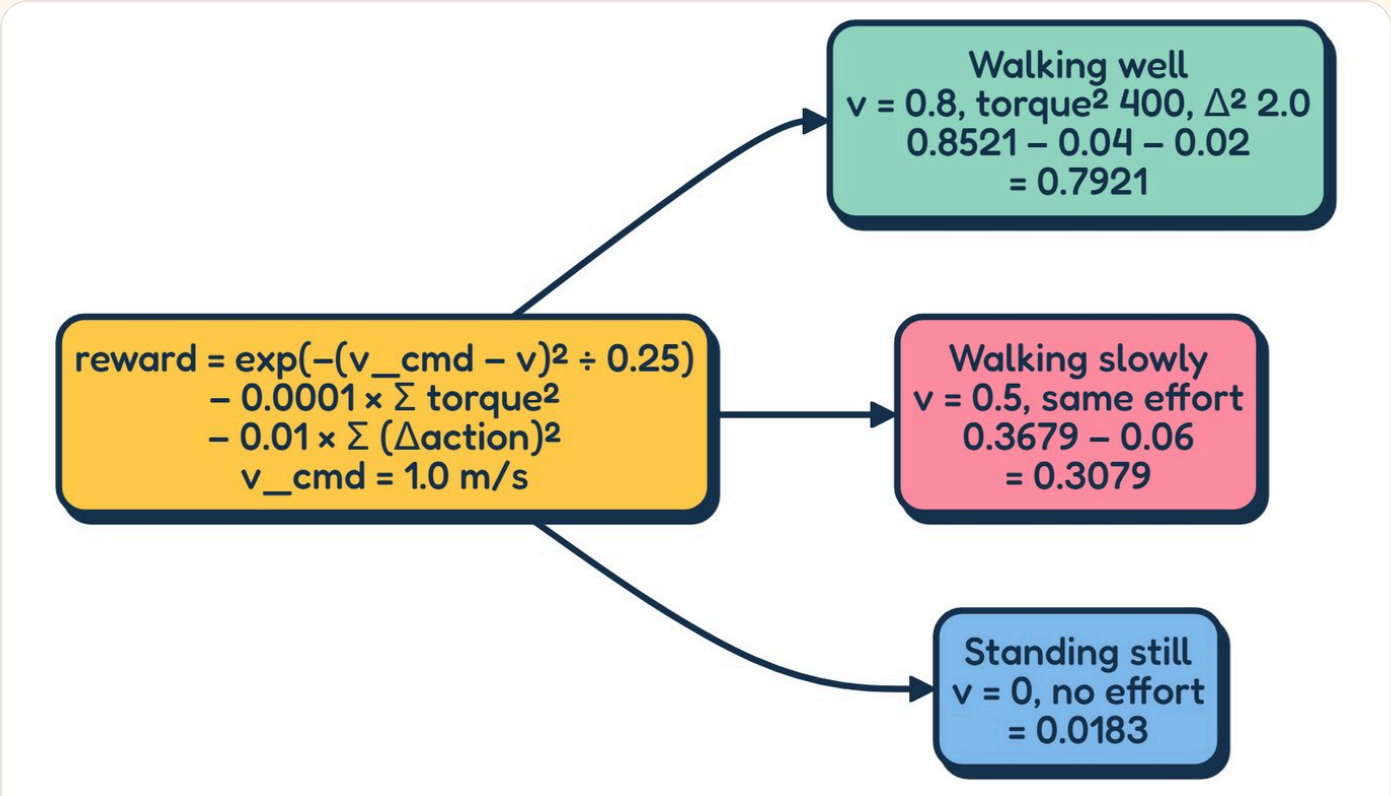


FIG Worked example: scoring three moments of the robot dog's walk.

3.5

Training In A Virtual Kitchen

+10 XP

✨ READY, SET, COOK!

Thousands of virtual dogs at once! 🐕 🐕 🐕 Training in a video-game kitchen.

Worked example: one robot vs 4,096

Suppose walking needs 100 million steps of experience (illustrative).

One real robot at 50 steps/s: 100,000,000 ÷ 50 = 2,000,000 s ≈ 23.1 days (plus falls, resets and wear).

4,096 simulated robots at 50 steps/s each on one GPU: 204,800 steps/s → ≈ 488 s ≈ 8.1 minutes.

🔑 REMEMBER THIS!

This is Recipe 01's parallelism: the same small computation on thousands of independent robots. Research systems report learning to walk in minutes on one GPU (Rudin et al., 2022) with GPU simulators like Isaac Gym. Reality falls short of the ideal 4,096×: Amdahl's serial part and multi-GPU communication (Recipe 02).

3.5 · SEE IT

A Virtual Kitchen

+10 XP

MAMA FENG

🔍 Spot it! One real robot vs 4,096 virtual pups: how much time is saved?

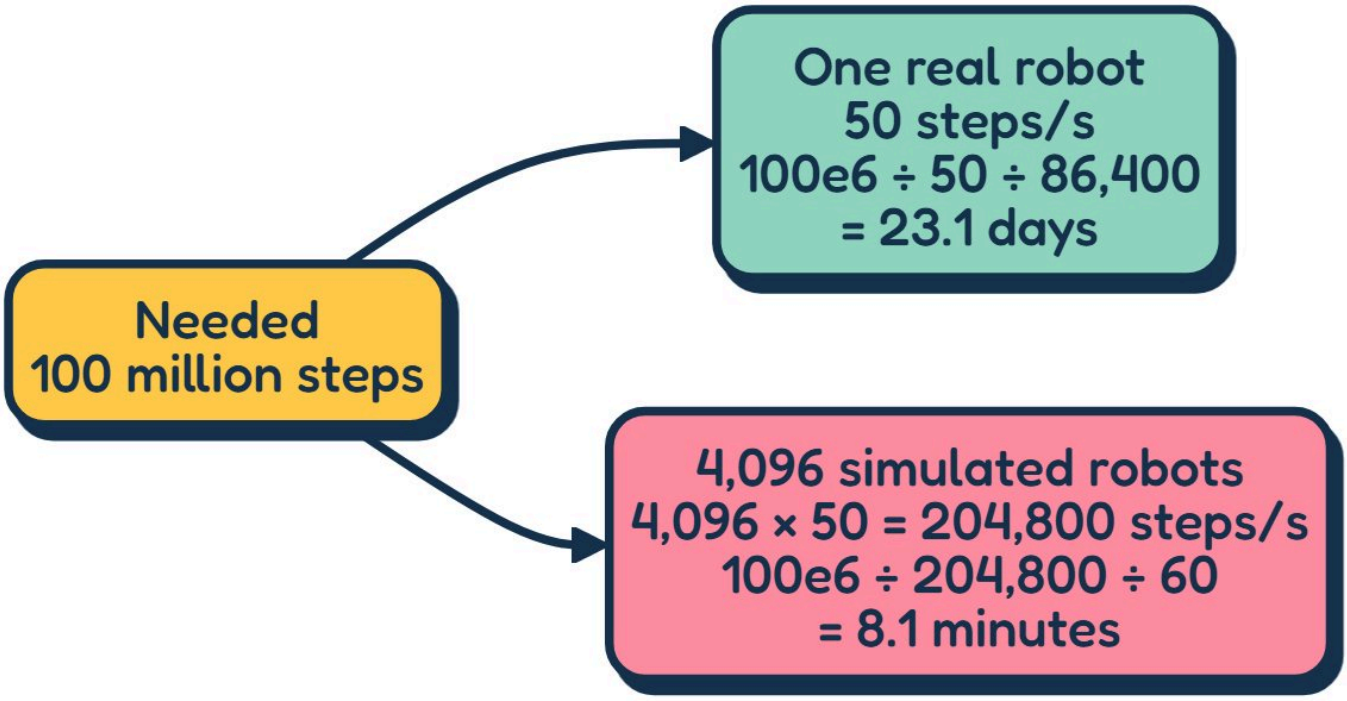


FIG Why simulation: 100 million steps on one real robot would take years.

3.5 · SEE IT

+10 XP

A Virtual Kitchen (2)

MAMA FENG

🔍 Spot it! One real robot vs 4,096 virtual pups: how much time is saved?

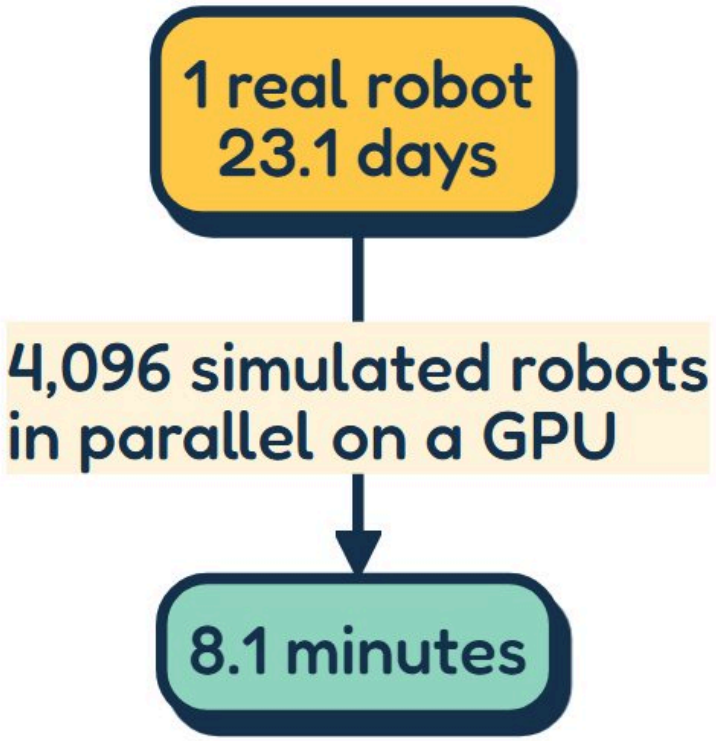


FIG 4,096 simulated robots in parallel cut weeks to minutes.

3.5

From Simulation To the Real Floor

+10 XP

✨ READY, SET, COOK!

From game to real floor! 🏠 Why do robots trip when they leave the simulator?

The sim-to-real gap

A simulator is never exactly right: the floor is slipperier, the motors weaker or slower, the robot heavier. A policy trained only in simulation can fail on the real robot.

TECHNIQUE	IDEA	REFERENCE
Domain randomisation	Randomly vary friction, mass, motor strength, delay and sensor noise every run, so the real robot looks like just another variation	Tobin et al. 2017; Tan et al. 2018
Better actuator models	Learn a model of the real motors and put it in the simulator	Hwangbo et al. 2019
Adaptation	The policy estimates current conditions (slippery floor, heavy load) and adjusts	Kumar et al. 2021 (RMA)

🔬 REAL-LIFE TECH

Other work trains directly on real hardware with careful safety measures and efficient algorithms (Haarnoja et al., 2019; Smith et al., 2022: "learning to walk in 20 minutes"), spending the real robot's time on fewer attempts.

3.5

Pros & Cons

+10 XP

✨ READY, SET, COOK!

Thumbs up or thumbs down for learning robots? 👍 👎

👍 The Tasty Bits

Handles complexity: one policy walks, runs, turns and recovers from pushes • Robust: randomised training copes with unexpected terrain • Fast iteration: change the reward and retrain instead of rewriting a controller • Small to run: the policy is tiny and runs on board • Learns from experience, including recovering from falls.

👎 The Burnt Bits

Reward design takes expertise and many runs • Sim-to-real gap: may fail outside the randomised range • Safety and wear when learning on real hardware • Training needs lots of GPU time • Limited understanding: a pattern of reflexes that doesn't "know" why it works • Hard to explain decisions, which matters for safety certification • Model-based control is more predictable and still widely used; modern systems often combine both.

SERVE THE CUSTOMERS · TICKET #3.5

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.5) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Domain randomisation helps sim-to-real transfer by: ♡ ♡ ♡

- (A) Removing the reward
- (B) Using only real data
- (C) Making the simulator perfect
- (D) Training on many varied simulated conditions

2. What does "Policy (π)" mean? ♡ ♡ ♡

- (A) The rule the agent uses to choose actions.
- (B) How far one new experience moves an estimate.
- (C) What the agent observes / what it does / the score it receives.
- (D) One run from a start state to an end.

3. Which kitchen word means: "One run from a start state to an end."? ♡ ♡ ♡

- (A) Law of effect
- (B) Episode
- (C) Discount factor (γ)
- (D) MDP

4. What does "Exploration / exploitation" mean? ♡ ♡ ♡

- (A) One run from a start state to an end.
- (B) Trying uncertain actions / using the best-known one.
- (C) Discounted sum of future rewards.
- (D) Rewarded behaviour becomes more likely (Thorndike).

LAB

TICKET #3.L · RECIPE 03 · BBQ GARDEN

Kitchen Lab

Gridworld

COURSES ▢▢▢▢ RECORD ☆☆☆ BOUNTY 90 XP

KITCHEN LAB

Gridworld

+10 XP

🌐 Live platform: REINFORCEjs GridWorld

Open cs.stanford.edu/people/karpathy/reinforcejs/gridworld_td.html (Andrej Karpathy). Press "Toggle TD Learning" and watch the agent learn. Change epsilon (exploration) and the learning rate alpha, then reset.

Version: REINFORCEjs GridWorld TD demo (static web page, no install; Chrome or Edge).

🎯 Goal · Observe · Reflect

Goal: see Q-learning and exploration live.

Observe: How many episodes until the arrows point to the goal? What happens with epsilon = 0? With epsilon = 0.5?

Reflect: Which setting is like the greedy agent in the bandit simulation?

Offline fallback

Draw a 4×4 grid on paper, goal in a corner (+1), one pit (−1). Play the agent yourself with a coin: heads = explore (random move), tails = exploit (best arrow). After each move update the square with value ← value + 0.5 × (reward + 0.9 × next value − value).

⭐ Mama's Feedback (expected result)

Default settings: the agent wanders at first; after a short while (often under a minute of fast learning) the arrows on the main route point to the +1 goal and avoid the red −1 squares. The exact count varies, because the agent moves randomly.

epsilon = 0: it repeats the first route that worked; some squares are never tried and it can get stuck.

epsilon = 0.5: values everywhere get learned, but the path looks messy and it hits red squares more often.

Reflect: epsilon = 0 is the greedy bandit agent.

EXERCISE

Final round

+10 XP

✨ READY, SET, COOK!

Final round! 🎯 Earn your Recipe 03 stamp!

EXERCISE

Question 1: Walk reward

Task: Compute the reward when the command is 1.0 m/s, the robot moves at 0.5 m/s, $\Sigma\text{torque}^2 = 400$ and $\Sigma(\Delta\text{action})^2 = 2.0$.

Question 2: Cloud brain?

Task: Why can the 50 Hz policy not run on a cloud server over 5G? Use Recipe 02's latency numbers.

Question 3: Parallel pups

Task: With 4,096 simulated robots the ideal speed-up is 4,096×. Give two reasons the real speed-up is lower. Then give one benefit and one risk of domain randomisation.

🔒 **Answers in the Answer Vault · V-3-5** · page 133 · no peeking! +10 XP

WATCH & LEARN

+10 XP

Video picks

Reinforcement Learning: Machine Learning Meets Control Theory

Creator: Steve Brunton · 26:00

Link: youtube.com/watch?v=0MNVhXEX9to

Why watch: A clear lecture on agent, environment, reward, MDPs and Q-learning (3.2-3.3).

Timestamps: 3:34 RL overview · 12:32 Markov decision process · 13:33 credit assignment · 21:50 Q-learning

Task: At 21:50, compare his Q-update with the 0.2 → 0.325 worked example.

Limitations: University level; some equations go beyond this guide.

Markov Decision Processes - Computerphile

Creator: Computerphile · 17:40

Link: youtube.com/watch?v=2iF9PRriA7w

Why watch: A friendly whiteboard explanation of states, actions, rewards and policies.

Timestamps: Whole video; pause at each diagram

Task: Write the S, A, R and policy for "the pup fetching an ingredient".

Limitations: Talk format, no code; long for a single idea.

WATCH & LEARN

Video picks

+10 XP

AI Learns to Walk (deep reinforcement learning)

Creator: AI Warehouse · 8:36

Link: youtube.com/watch?v=L_4BPjLBF4E

Why watch: Watch a simulated creature go from falling to walking: rewards and shaping in action (3.5).

Timestamps: Whole video (9 min)

Task: Note one moment where the agent "hacks" the reward and suggest a better reward.

Limitations: Entertainment style; simulation only (no sim-to-real); training details are not shown.

SERVE THE CUSTOMERS · TICKET #3.L

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-3-Q3.L) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In Kitchen Lab 3 (GridWorld), setting epsilon = 0 guarantees that the agent eventually finds the best route. ♡ ♡ ♡

- (T) True
- (F) False

2. What does “Agent / environment” mean? ♡ ♡ ♡

- (A) V(s) = r + γV(s′).
- (B) Extra reward to guide learning; potential-based form keeps the best policy.
- (C) Markov decision process: future depends only on present state and action.
- (D) The learner and decision-maker / everything it interacts with.

3. Which kitchen word means: “The rule the agent uses to choose actions.”? ♡ ♡ ♡

- (A) Policy (π)
- (B) Sparse / dense reward
- (C) Discount factor (γ)
- (D) Reward shaping

4. What does “Learning rate (α)” mean? ♡ ♡ ♡

- (A) Rewarded behaviour becomes more likely (Thorndike).
- (B) Updating an estimate using the next estimate.
- (C) Difference between simulator and reality.
- (D) How far one new experience moves an estimate.

TICKET #3.B · RUSH HOUR IN THE BBQ GARDEN

Rush Hour!

*Last in line, the trouble customer **Reward-Hacker Hound**: "Why learn to walk when I can just farm the treats?"*

HAPPY CUSTOMERS ☐☐☐☐☐☐☐☐☐☐☐☐☐☐

GRADE S · A · B · C · D

Rules: 14 orders, no going back. Colour one box for each right dish (check in the Answer Vault, V-3-B). 9 or more happy customers = you survive the rush: +150 XP and the **Q-Learning Pie** recipe card. Grade: S = 13–14, A = 12, B = 10–11, C = 9. Questions 13 and 14 are the trouble customer's double order.

1. Which of these rewards are DENSE (given at almost every step)? ♡ ♡ ♡

- (A) +speed every step
- (B) +1 only on reaching the charging station
- (C) −energy used every step
- (D) +1 only for winning the game

2. Q = 0.2, r = 1, max Q(s') = 0.5, α = 0.1, γ = 0.9. After one Q-learning update the new Q is: ♡ ♡ ♡

- (A) 0.45
- (B) 1.45
- (C) 0.245
- (D) 0.325

3. A discount factor γ close to 0 makes the agent far-sighted. ♡ ♡ ♡

- (T) True
- (F) False

4. With rewards [0, 0, 0, 10] and γ = 0.9, the discounted return is 7.29. ♡ ♡ ♡

- (T) True
- (F) False

5. A purely greedy agent can get stuck on a poor option forever. ♡ ♡ ♡

- (T) True
- (F) False

6. A bandit has arms that pay 0.2, 0.5 and 0.8 on average. The agent knows the best arm and uses ε = 0.1 (when exploring, it picks one of the three arms at random). Expected reward per pull: ♡ ♡ ♡

- (A) 0.50
- (B) 0.77
- (C) 0.80
- (D) 0.74

7. In Kitchen Lab 3 (GridWorld), setting epsilon = 0 guarantees that the agent eventually finds the best route. ♡ ♡ ♡

- (T) True
- (F) False

8. Domain randomisation helps sim-to-real transfer by: ♡ ♡ ♡

- (A) Removing the reward
- (B) Using only real data
- (C) Making the simulator perfect
- (D) Training on many varied simulated conditions

9. Reward hacking means the agent earns a high reward without doing the intended task. ♡ ♡ ♡

- (T) True
- (F) False

10. A corridor of squares 1-2-3 leads to the goal, one step after square 3 (reward 1). With γ = 0.9, V(1) equals: ♡ ♡ ♡

- (A) 1.0
- (B) 0.729
- (C) 0.81
- (D) 0.9

11. Which kitchen word means: "Working out which earlier actions caused a later outcome."? ♡ ♡ ♡

- (A) Episode
- (B) Exploration / exploitation
- (C) TD learning
- (D) Credit assignment

12. What does "Domain randomisation" mean? ♡ ♡ ♡

- (A) One run from a start state to an end.
- (B) Extra reward to guide learning; potential-based form keeps the best policy.
- (C) Updating an estimate using the next estimate.
- (D) Varying simulation settings so the policy works in many conditions.

13. Which kitchen word means: "Updating an estimate using the next estimate."?

- (A) Q-learning
- (B) ε-greedy
- (C) Domain randomisation
- (D) TD learning

14. What does "UCB" mean?

- (A) Difference between simulator and reality.
- (B) Discounted sum of future rewards.
- (C) Markov decision process: future depends only on present state and action.
- (D) $Q + \sqrt{(2 \ln t \div n)}$: bonus for less-tried options.

TICKET #V3 · RECIPE 03

Answer Vault · Training the Pup

Only open after trying! Every vault entry you check is worth XP on your licence.

V-3-1

3.1 · Good Pup!

Mama's feedback · Good Pup!

(1) sit 0.875, roll 0, bark 0 (sit: 0.5 → 0.75 → 0.875). (2) (a) rule-based (b) supervised (c) RL (d) supervised. (3) (a) positive reinforcement (b) punishment (c) extinction (d) shaping. Reflect: "what" needs an expert to spell out every action; "how well" only needs a score we can recognise (moving forward steadily), and the robot works out the how.

V-3-Q3.1

Customer orders · ticket #3.1

1. C · What the agent observes / what it does / the score it receives. · State, action, reward: What the agent observes / what it does / the score it receives.

2. C · Sparse / dense reward · Sparse / dense reward: Reward given rarely / at nearly every step.

3. A · Learns Q(s,a) from experience with a TD update. · Q-learning: Learns Q(s,a) from experience with a TD update.

4. B · Policy (π) · Policy (π): The rule the agent uses to choose actions.

V-3-2

3.2 · Pup Gridworld

Mama's feedback · Pup Gridworld

(1) 1 + 0 + 1.62 + 3.645 = 6.265. (2) $0.9^4 = 0.6561$; $0.5^4 = 0.0625$. (3) 10 ms; no, 36 ms is over 3× the budget, run it on board. (4) 1,280 + 4,096 + 384 = 5,760 MACs; 11,520 FLOPs; 11,520 bytes ≈ 11.3 KiB. Reflect: with γ near 1, rewards several steps away still count, so the agent values steps that set up later success.

V-3-Q3.2

Customer orders · ticket #3.2

1. True · $0.9^3 \times 10 = 7.29$.

2. False · Correct: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted.

3. C · 0.81 · V(3) = 1, V(2) = 0.9 × 1 = 0.9, V(1) = 0.9 × 0.9 = 0.81.

4. D · One run from a start state to an end. · Episode: One run from a start state to an end.

5. B · Value V(s), Q(s,a) · Value V(s), Q(s,a): Expected return from a state / from an action in a state.

V-3-3

3.3 · Reward Hacking

Mama's feedback · Reward Hacking

(1) Target 0.72, error 0.22, new Q 0.544. (2) +1.3 forwards, −0.6 backwards. (3) Drop an item out and bin it again for endless +1; reward the reduction of rubbish on the floor (or the amount in the bin at the end). (4) (a) sparse (b) dense (c) sparse. Reflect: the promise of a good next state (0.8) is enough to raise Q.

V-3-Q3.3

Customer orders · ticket #3.3

1. True · Example: the boat-racing agent circling to hit respawning targets instead of finishing the race.

2. D · 0.325 · Target = 1 + 0.9 × 0.5 = 1.45; TD error = 1.25; new Q = 0.2 + 0.1 × 1.25 = 0.325.

3. undefined · · A and C give feedback every step. B and D are sparse: they arrive only at the end.

4. C · Working out which earlier actions caused a later outcome. · Credit assignment: Working out which earlier actions caused a later outcome.

5. A · Sparse / dense reward · Sparse / dense reward: Reward given rarely / at nearly every step.

V-3-4
3.4 · Noodle Stall Gamble

Mama's feedback · Noodle Stall Gamble

(1) 0.56 + 0.08 = 0.64, which is 0.06 below 0.7. (2) A 0.966, B 1.060 → B. (3) All estimates start equal, it picks arm 0 and that estimate stays ≥ the untried ones, so it never learns another arm pays more: ~0.2 reward, best arm 0%. (4) 0.366 and 0.05 ($0.99^{300} \approx 0.049 < 0.05$). Reflect: less; there is little future left to profit from new information.

V-3-Q3.4
Customer orders · ticket #3.4

1. True · It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation).

2. B · 0.77 · 0.9 × 0.8 + 0.1 × (0.2 + 0.5 + 0.8) ÷ 3 = 0.72 + 0.05 = 0.77. Careful: if exploration chose only the OTHER two arms the answer would be 0.9 × 0.8 + 0.1 × 0.35 = 0.755.

3. B · The rule the agent uses to choose actions. · Policy (π): The rule the agent uses to choose actions.

4. D · UCB · UCB: Q + √(2 ln t ÷ n): bonus for less-tried options.

V-3-Q3.5
Customer orders · ticket #3.5

1. D · Training on many varied simulated conditions · If the policy works across many random variations, the real world looks like one more variation.

2. A · The rule the agent uses to choose actions. · Policy (π): The rule the agent uses to choose actions.

3. B · Episode · Episode: One run from a start state to an end.

4. B · Trying uncertain actions / using the best-known one. · Exploration / exploitation: Trying uncertain actions / using the best-known one.

V-3-5
3.L · Final round

Answers · Final round

Q1: tracking = exp(−0.25 ÷ 0.25) = exp(−1) ≈ 0.3679; penalties 0.04 + 0.02 = 0.06; reward ≈ 0.3079.

Q2: 50 Hz = 20 ms per step. The best remote path (5G + MEC) took ~36 ms and the cloud 56-140 ms, all too late. The policy is ~46 KiB, so it runs easily on the robot.

Q3: Any two: non-parallel parts of each step (Amdahl), memory-bandwidth limits, communication between GPUs, robots not finishing at the same time. Randomisation benefit: robustness to simulator-reality differences. Risk: too much makes the task needlessly hard or over-cautious; randomising the wrong factors still leaves a gap.

V-3-Q3.L
Customer orders · ticket #3.L

1. False · Correct: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route.

2. D · The learner and decision-maker / everything it interacts with. · Agent / environment: The learner and decision-maker / everything it interacts with.

3. A · Policy (π) · Policy (π): The rule the agent uses to choose actions.

4. D · How far one new experience moves an estimate. · Learning rate (α): How far one new experience moves an estimate.

V–3–B

Rush hour · BBQ Garden

1. undefined · · A and C give feedback every step. B and D are sparse: they arrive only at the end.

2. D · 0.325 · Target = 1 + 0.9 × 0.5 = 1.45; TD error = 1.25; new Q = 0.2 + 0.1 × 1.25 = 0.325.

3. False · Correct: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted.

4. True · $0.9^3 \times 10 = 7.29$.

5. True · It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation).

6. B · 0.77 · 0.9 × 0.8 + 0.1 × (0.2 + 0.5 + 0.8) ÷ 3 = 0.72 + 0.05 = 0.77. Careful: if exploration chose only the OTHER two arms the answer would be 0.9 × 0.8 + 0.1 × 0.35 = 0.755.

7. False · Correct: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route.

8. D · Training on many varied simulated conditions · If the policy works across many random variations, the real world looks like one more variation.

9. True · Example: the boat-racing agent circling to hit respawning targets instead of finishing the race.

10. C · 0.81 · V(3) = 1, V(2) = 0.9 × 1 = 0.9, V(1) = 0.9 × 0.9 = 0.81.

11. D · Credit assignment · Credit assignment: Working out which earlier actions caused a later outcome.

12. D · Varying simulation settings so the policy works in many conditions. · Domain randomisation: Varying simulation settings so the policy works in many conditions.

13. D · TD learning · TD learning: Updating an estimate using the next estimate.

14. D · Q + √(2 ln t ÷ n): bonus for less-tried options. · UCB: Q + √(2 ln t ÷ n): bonus for less-tried options.

04

ROOM 04 · SUSHI BAR · TICKET #D4

Chef's Senses

AI SENSORS & FUSION

MAMA FENG

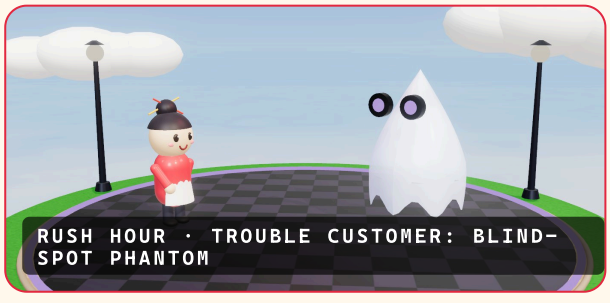
Mama's Problem: a good chef SEES the colour of the onions, HEARS the sizzle, SMELLS the burning and FEELS the dough. A self-driving car must do the same with cameras, lasers and sound. How does a machine sense the world, combine its senses and act in time?

By the end you will be able to

1. Explain how cameras, LiDAR and ultrasonic sensors work and compare them.
2. Calculate pixels on a target, time-of-flight distance, echo distance and LiDAR data rate.
3. Compute a latency budget, a stopping distance and the compute needed by a camera array.
4. Explain early, mid-level and late fusion; do inverse-variance fusion and one Kalman cycle by hand.
5. Describe perception → prediction → planning → control on hardware, with safety and uncertainty.
6. Analyse Tesla's sensor strategy with public evidence, not company claims alone.

TONIGHT'S TICKETS

- [ ] 4.1 Types of sensors +210 XP
- [ ] 4.2 Data acquisition and real-time proce... +150 XP
- [ ] 4.3 Sensor fusion and environmental pe... +180 XP
- [ ] 4.4 Integrating hardware with AI decisio... +150 XP
- [ ] 4.5 Case study +160 XP
- [ ] LAB Kitchen Lab +90 XP
- [ ] Rush Hour +150 XP



TICKET #4.1 · RECIPE 04 · SUSHI BAR

4.1 Types of sensors

cameras, LiDAR, ultrasonic (and radar)

COURSES

BRIEFING

Mama's problem

+10 XP

MAMA FENG

Mama's Problem: a good chef SEES the colour of the onions, HEARS the sizzle, SMELLS the burning and FEELS the dough. A self-driving car must do the same with cameras, lasers and sound. How does a machine sense the world, combine its senses and act in time?

Today's Adventure

4.1 Types of sensors: cameras, LiDAR, ultrasonic (and radar)

4.2 Data acquisition and real-time processing

4.3 Sensor fusion and environmental perception

4.4 Integrating hardware with AI decision-making

4.5 Case study: Tesla's camera-only driving

BRIEFING

Goals & what you need

+10 XP

By the end you will be able to

1. Explain how cameras, LiDAR and ultrasonic sensors work and compare them.
2. Calculate pixels on a target, time-of-flight distance, echo distance and LiDAR data rate.
3. Compute a latency budget, a stopping distance and the compute needed by a camera array.
4. Explain early, mid-level and late fusion; do inverse-variance fusion and one Kalman cycle by hand.
5. Describe perception → prediction → planning → control on hardware, with safety and uncertainty.
6. Analyse Tesla's sensor strategy with public evidence, not company claims alone.

Before You Jump In

Basic arithmetic and Recipe 01 words: MAC, TOPS, bandwidth, latency, throughput, batch size, utilisation.

Running example: a ball at dusk

A car drives at 50 km/h. It is getting dark and has just stopped raining. A ball rolls out between two parked cars, and a child may follow. The system must notice it, judge the distance, decide to brake, and brake in time, in a fraction of a second.

4.1

What Is A Sensor?

+10 XP

✨ READY, SET, COOK!

What counts as a sensor? 🌡 Guess how many your phone has, then read on!

📖 The Big Idea

A sensor is a device that converts a physical quantity (light, sound, distance, acceleration) into an electrical signal a computer can read.

Every intelligent machine runs the same loop:

World → Sensors → Acquisition → Perception → Planning → Control → Actuators → back to the World.

CLASSIFICATION	MEANING	EXAMPLES
Passive	Only receives energy that already exists	Camera, microphone
Active	Emits energy and measures the echo	LiDAR (laser), radar (radio), ultrasonic (sound)
Exteroceptive	Measures the outside world	Camera, LiDAR, radar, ultrasonic
Proprioceptive	Measures the machine itself	Wheel speed, IMU, steering angle

💡 MAMA'S SECRET TIP

A chef's eyes are passive (they use the kitchen light). Knocking on a melon and listening is active: you make a sound and judge the echo!

4.1

Cameras: the Chef's Eyes

+10 XP

✨ READY, SET, COOK!

Say cheese! 📸 How does a camera turn light into numbers?

📷 How it works

A camera is PASSIVE. A lens focuses light onto an image sensor, a grid of light-sensitive pixels. A colour filter gives red, green and blue values.

Good at: very high resolution and colour, so it reads signs, lane lines, traffic-light colour and brake lights. Cheap, small, no moving parts.

Bad at: it does NOT measure distance directly (a neural network must infer it). It needs light, so darkness, glare and fast brightness changes hurt it, as do rain, fog and a dirty lens.

QUANTITY	MEANING
Resolution	Number of pixels, e.g. 1920 × 1080 ≈ 2 megapixels
Field of view (FOV)	The angle seen; wider sees more but each object gets fewer pixels
Frame rate	Images per second (fps)
Dynamic range	Brightest vs darkest in one frame: tunnels and sunsets

4.1

Worked Example: Pixels On A Child

+10 XP

How many pixels does a child cover?

Camera: 1,920 pixels across, 60° horizontal field of view. Target: 0.5 m wide.

Focal length in pixels:

f = (W ÷ 2) ÷ tan(FOV ÷ 2) = 960 ÷ tan 30° = 960 ÷ 0.5774 ≈ 1,663 px

Width in the image = f × real width ÷ distance

- **At 50 m:** 1,663 × 0.5 ÷ 50 ≈ 16.6 pixels
- **At 150 m:** 1,663 × 0.5 ÷ 150 ≈ 5.5 pixels

KITCHEN TERMINAL · PYTHON

```python
import math
W, hfov, size = 1920, 60, 0.5
f = (W/2) / math.tan(math.radians(hfov/2))
for d in (50, 150):
    print(d, "m:", round(f*size/d, 1), "pixels")
```

🔑 REMEMBER THIS!

At 150 m the child is only about five pixels wide: very hard for a network to classify. Range and resolution are linked.

4.1 · SEE IT

+10 XP

Pixels On A Child

MAMA FENG

🔍 Spot it! A child far away is only a few pixels wide. How many at 150 m?

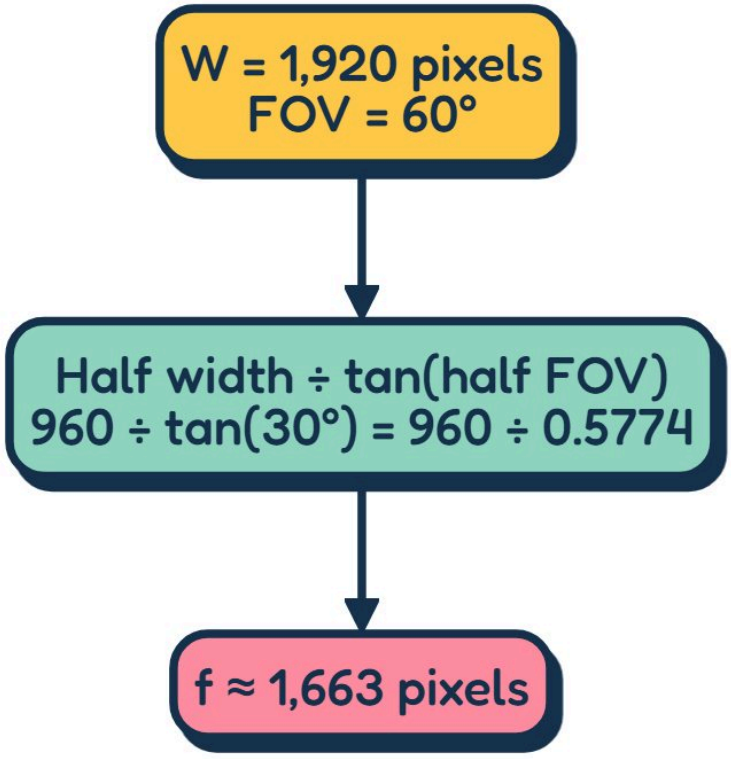


FIG Worked example: how many pixels a child covers at a distance.

4.1 · SEE IT

+10 XP

Pixels On A Child (2)

MAMA FENG

🔍 Spot it! A child far away is only a few pixels wide. How many at 150 m?

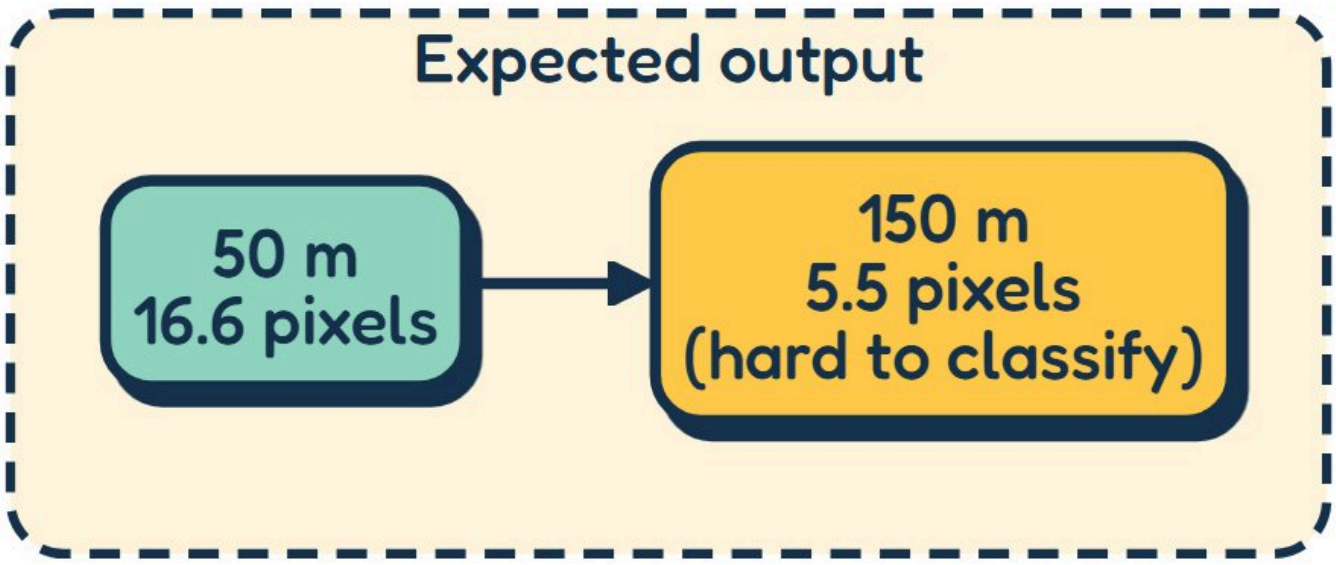


FIG The expected output at 50 m and 150 m.

4.1

LiDAR: the Laser Tape Measure

+10 XP

✨ READY, SET, COOK!

Pew pew! 🔫 Measure distance with lasers, like a tape measure made of light.

🔫 Time of flight

LiDAR (Light Detection and Ranging) is ACTIVE. It fires very short laser pulses and times the echo. Light travels at $c \approx 3 \times 10^8$ m/s, so:

distance = c × t ÷ 2 (÷ 2 because the pulse goes out AND back)

Sweeping the laser builds a POINT CLOUD: many (x, y, z) points plus intensity.

Worked example: a pulse returns after 400 ns.

(3 × 10^8 × 4 × 10^-7) ÷ 2 = 120 ÷ 2 = 60 m

✓ Good at

Direct, accurate distance (a few cm) • works in darkness • 3-D shape • spots unusual objects a camera might not recognise

✗ Bad at

No colour, can't read signs or lights • rain, fog, snow, dust scatter it • dark or shiny surfaces • costlier (but falling) • sparse far away

4.1 · SEE IT

+10 XP

Time of Flight

MAMA FENG

🔍 Spot it! The pulse goes there AND back. Where does the "÷ 2" come in?

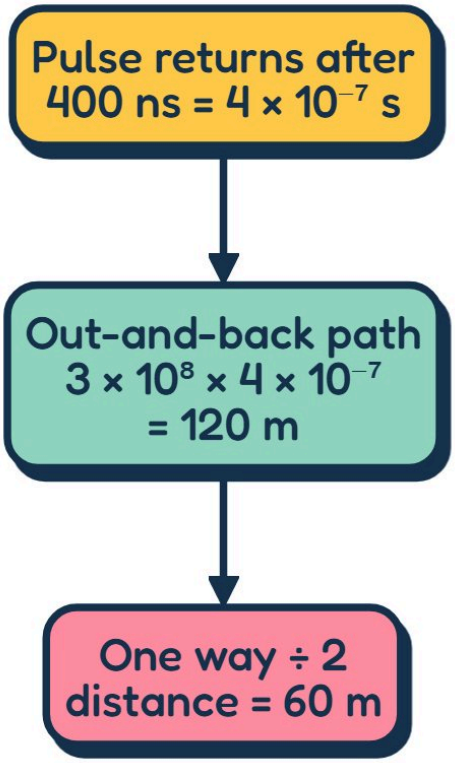


FIG Worked example: a laser pulse returns after 400 ns. How far away?

4.1

Worked Example: LiDAR Data Rate

+10 XP

How much data does a LiDAR make?

128 beams, horizontal step 0.2° over 360° → 360 ÷ 0.2 = 1,800 steps, spinning at 10 Hz. Each point = four 32-bit numbers (x, y, z, intensity) = 16 bytes.

- Points per frame = 128 × 1,800 = 230,400
- Points per second = 2,304,000
- **Per frame:** 230,400 × 16 = 3.69 MB
- **Per second:** ≈ 36.9 MB/s

KITCHEN TERMINAL · PYTHON

```
beams, az, hz = 128, 1800, 10
pts = beams * az
print(pts, pts*hz)
print(pts*16/1e6, "MB per frame")
print(pts*16*hz/1e6, "MB/s")
```

MAMA'S SECRET TIP

Remember "about 37 MB/s". In 4.2 we compare it with a camera, and the camera wins (by a lot)!

4.1 · SEE IT

+10 XP

LiDAR Data Rate

MAMA FENG

🔍 Spot it! Multiply along the arrows: how many points per second does the LiDAR make?

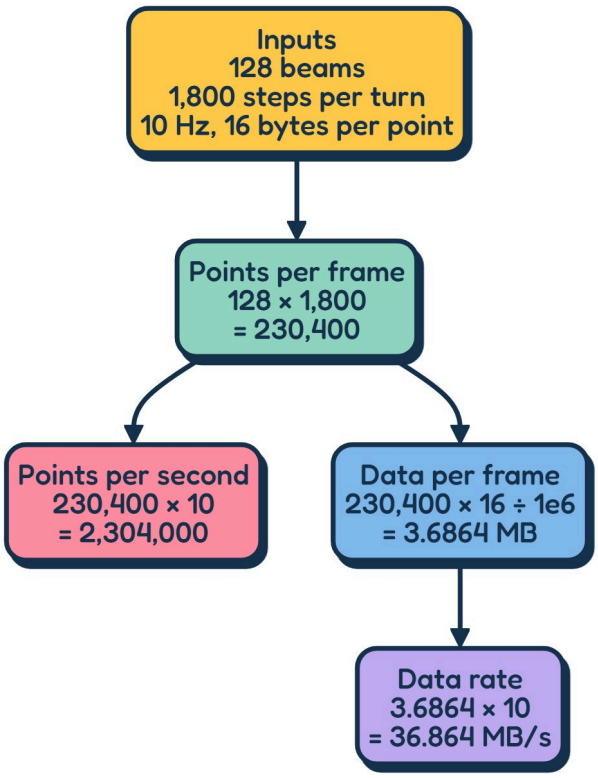


FIG Worked example: how much data a 128-beam LiDAR produces.

4.1

+10 XP

Ultrasonic: Knock and Listen

✨ READY, SET, COOK!

Knock, knock! 🔊 Who's there? Echoes tell us how far away things are.

Echoes of sound

An ultrasonic sensor sends a burst of sound above human hearing (about 40 kHz) and listens for the echo, like LiDAR but with sound, about a million times slower:

distance = v × t ÷ 2, v ≈ 343 m/s at 20 °C

The speed of sound depends on temperature: v ≈ 331.3 + 0.606 × T (°C).

Good at: very cheap, works in the dark, finds low obstacles: PARKING.

Bad at: short range (a few m), wide beam, can't tell what the object is, too slow for fast objects; wind and soft clothing confuse it.

Worked example: echo and temperature

(a) Echo after 5.8 ms: 343 × 0.0058 ÷ 2 ≈ 0.99 m.

(b) A real 1.00 m on a freezing day (0 °C, v = 331.3): the echo takes 2 ÷ 331.3 = 6.04 ms. Assuming 343 m/s, the sensor reports 343 × 0.00604 ÷ 2 ≈ 1.035 m, a 3.5% error. Good sensors measure temperature and correct for it.

4.1 · SEE IT

+10 XP

Echo To Distance

MAMA FENG

🔍 Spot it! Cold air slows sound. How big is the error at 0 °C?

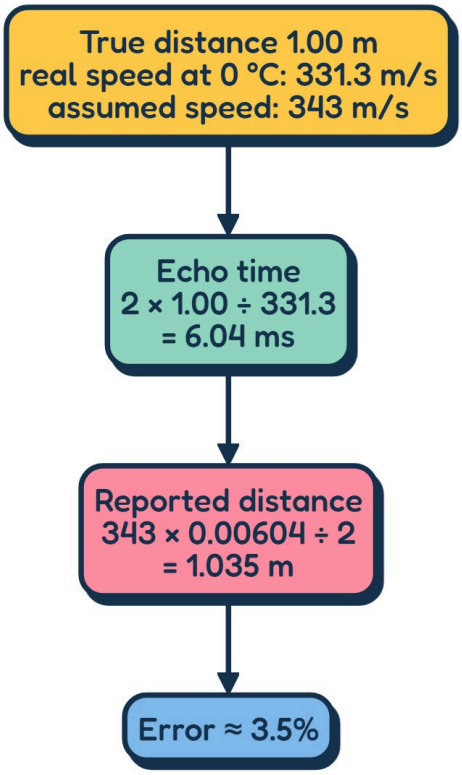


FIG Worked example: echo to distance, and the error from temperature.

4.1

+10 XP

Radar: the Speed Sensor

✨ READY, SET, COOK!

Radar sees speed! 🚗 How can a radio wave tell how fast you're going?

Radio waves and Doppler

Radar emits radio waves (cars use 76-81 GHz) and measures the echo. Its superpower is SPEED: a moving object shifts the echo's frequency (the Doppler effect), so radar measures velocity directly, and it keeps working in rain, fog and darkness. Its weakness: coarse angular resolution, so it struggles to separate nearby objects or identify shapes.

$v = (\Delta f \times c) \div (2 \times f0)$

Example: f0 = 77 GHz, Δf = 10.27 kHz

$v = (10{,}270 \times 3 \times 10^8) \div (2 \times 77 \times 10^9) \approx 20$ m/s (72 km/h)

💡 MAMA'S SECRET TIP

Doppler is the reason an ambulance siren sounds higher as it comes towards you and lower as it drives away.

4.1

The Sensor Taste Test

+10 XP

✨ READY, SET, COOK!

Sensor taste test! 😋 Which sensor wins in fog, at night, or up close?

FEATURE	CAMERA	LIDAR	ULTRASONIC	RADAR
Principle	Light (passive)	Laser ToF	Sound echo	Radio + Doppler
Useful range	Tens to 250+ m	~100-250 m	~0.2-5 m	~5-250 m
Resolution	Very high	High, sparse far	Very low	Low-medium
Direct depth?	No (inferred)	Yes, accurate	Yes, short	Yes
Direct speed?	No	No	No	Yes (Doppler)
Darkness	Poor	Yes	Yes	Yes
Rain/fog/snow	Degraded	Degraded	Mostly OK (short)	Least affected
Reads signs?	Yes	No	No	No
Data rate	~100 MB/s raw	Tens of MB/s	Tiny	Low-moderate
Cost	Low	Medium-high, falling	Very low	Low-medium
Typical use	Lanes, signs, objects	3-D mapping, obstacles	Parking	Cruise control, collision warning

⚠ PAPA'S OOPS!

"The table says LiDAR reaches 250 m, so mine does!" Not quite! Table numbers are TYPICAL. A specific product can be far better or worse. Always check the datasheet of the sensor you actually use.

SIDE QUEST · MINI-GAME 4.1

+25 XP

Sensor Taste Test

✨ READY, SET, COOK!

Taste test challenge! // Pick the right sensor for every job!

Your Mission

Calculate basic sensor quantities and choose the right sensor for a job.

Grab These First

Formulas: f = (W/2) ÷ tan(FOV/2); width = f × size ÷ d; ToF distance = c·t ÷ 2; echo distance = v·t ÷ 2; Doppler v = Δf·c ÷ (2·f0).

Let's Go! Your Turn

(1) A 3,840-px camera with a 90° FOV: how many pixels wide is a 1.8 m car at 100 m?

(2) A LiDAR pulse returns after 1.0 µs. Distance?

(3) An ultrasonic echo returns after 11.7 ms. Distance?

(4) A 77 GHz radar sees a 15.4 kHz shift. Speed?

(5) 64 beams, 0.4° step, 10 Hz: points per frame and MB per frame?

(6) Which sensor for: (a) reversing with 30 cm clearance (b) a speed-limit sign (c) a stalled car in dense fog (d) the 3-D shape of an odd object at night?

What Did You Spot?

Write the formula first, then the numbers, then the unit.

Think It Over

Why did no sensor win every row of the taste-test table?

🔒 **Mama's feedback in the Answer Vault · V-4-1** · page 184 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #4.1

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.1) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A single camera directly measures the distance to every pixel. ♡ ♡ ♡

- (T) True
- (F) False

2. A LiDAR pulse returning after 400 ns means the object is about 60 m away. ♡ ♡ ♡

- (T) True
- (F) False

3. Radar measures the speed of an object directly using the Doppler effect. ♡ ♡ ♡

- (T) True
- (F) False

4. Which sensors measure distance DIRECTLY? ♡ ♡ ♡

- (A) LiDAR
- (B) A single camera
- (C) Radar
- (D) Ultrasonic

5. What does “Dynamic range” mean? ♡ ♡ ♡

- (A) Brightest vs darkest light captured in one frame.
- (B) Distance from round-trip time: d = v × t ÷ 2.
- (C) Keeps working reduced / moves to a safe state.
- (D) Predicts which 3-D cells around the vehicle are occupied.

TICKET #4.2 · RECIPE 04 · SUSHI BAR

4.2 Data acquisition and real-time processing

COURSES ☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 150 XP

4.2 From Farm To Wok

+10 XP

✨ READY, SET, COOK!

From farm to wok! 🌾 ➡️ 🍳 Follow the data from sensor to decision.

The acquisition chain

Sensing element → ADC → Pre-processing (ISP / filter) → Timestamp & sync → Transfer (bus, link) → Memory (DRAM / SRAM) → Neural network (CPU / GPU / NPU)

STAGE	WHAT IT DOES
Sensing element	Photodiodes in a camera, laser receiver in a LiDAR, transducer in ultrasonic
ADC	Analogue-to-digital converter: voltage → number. 12-bit = 4,096 levels
ISP	Image signal processor: removes noise, corrects the lens, fixes exposure and colour
Timestamp & sync	Labels each measurement with the time it was taken
Transfer	Data crosses a cable or bus; bandwidth is limited, like memory bandwidth
Memory & compute	Data lands in memory; a neural network runs on CPU, GPU or NPU

💡 MAMA'S SECRET TIP

Like ingredients travelling from the farm to your wok: picked, washed, labelled with a date, trucked, stored in the fridge, then cooked. A delay anywhere means a late dinner.

4.2 Worked Example: Camera Data

+10 XP

✨ READY, SET, COOK!

How much data does ONE camera make per second? 🎥 Guess big, then check!

🎥 Data rate of a camera array

8 cameras, each 1,920 × 1,080 pixels, 12-bit raw, 36 fps.

1. Frame = 1,920 × 1,080 × 12 ÷ 8 = 3,110,400 bytes
2. Per camera = 3,110,400 × 36 ≈ 112.0 MB/s
3. All 8 cameras ≈ 0.9 GB/s

KITCHEN TERMINAL · PYTHON

```python
w, h, bits, fps, n = 1920, 1080, 12, 36, 8
frame = w*h*bits/8
print(frame, round(frame*fps/1e6, 1), "MB/s")
print(round(frame*fps*n/1e9, 1), "GB/s")
```

🔑 REMEMBER THIS!

ONE camera (112 MB/s) makes about three times the data of the 128-beam LiDAR (37 MB/s). Cameras, not LiDARs, dominate the data volume. Recipe 01: moving data costs more energy than computing on it, so keep raw images on the chip or compress early.

4.2 · SEE IT

+10 XP

Camera Data Rate

MAMA FENG

🔍 Spot it! Cameras are hungry! How many bits per second do they produce together?

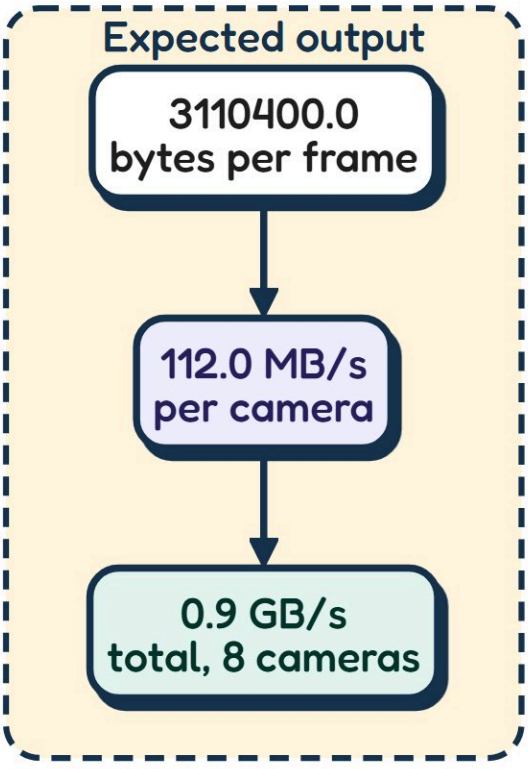


FIG Worked example: the raw data rate of an 8-camera array.

4.2

Real-Time Deadlines

+10 XP

Hard vs soft real time

A REAL-TIME system must produce a result before a deadline.

- **Hard real time:** missing the deadline is a failure (a braking command arriving too late).
- **Soft real time:** a late result is just less useful (a delayed video frame).

Driving needs hard guarantees for safety functions.

Throughput = frames per second processed.

Latency = time for ONE frame from photon to action.

A system can have high throughput and still react dangerously slowly!

Worked example: distance travelled while thinking

Car at 100 km/h = 27.78 m/s, braking at 6 m/s^2.

Braking distance = $v^2 \div (2a)$ = 771.6 ÷ 12 ≈ 64.3 m

- Latency 0.1 s: blind 2.78 m → total ≈ 67.1 m
- Latency 0.3 s: blind 8.33 m → total ≈ 72.6 m

An extra 200 ms adds 5.6 m: the difference between a near miss and a collision.

4.2

The Latency Budget

+10 XP

STAGE	TIME (MS)
Exposure and readout	10
ISP	5
Transfer to memory	2
Neural-network inference	30
Fusion and tracking	10
Planning	15
Actuation (brake build-up)	30
TOTAL	102

Reading the budget

A latency budget splits the allowed end-to-end time among the stages (numbers illustrative). With a 100 ms target, this system is already 2 ms OVER. The biggest slice, inference, is where Recipe 01's ideas pay off: parallelism, low precision and on-chip memory.

💡 MAMA'S SECRET TIP

Just like a dinner service: if the soup must be on the table in 10 minutes, every station (chopping, boiling, plating, walking) gets its own slice of the time.

4.2 · SEE IT

+10 XP

Distance While Thinking

MAMA FENG

🔍 Spot it! Every 0.1 s of thinking costs metres. How far does the car roll?

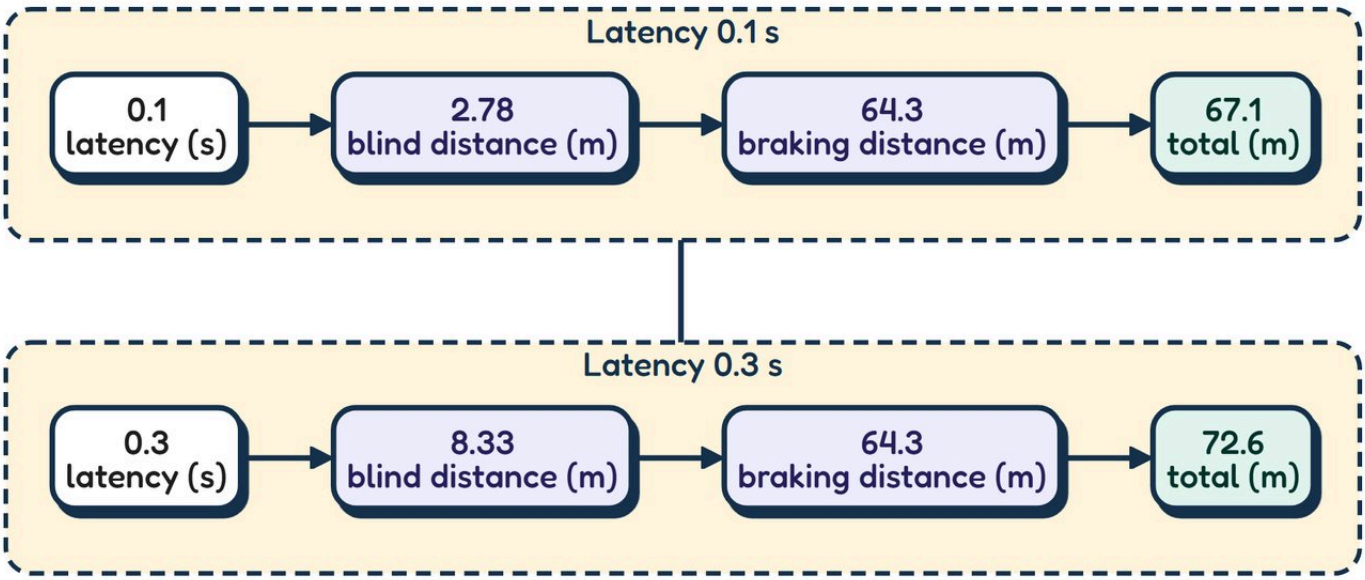


FIG Worked example: distance travelled while thinking, at 0.1 s and 0.3 s of latency.

4.2

+10 XP

Timestamps & Motion

Why the clock matters

- A spinning LiDAR needs 100 ms (at 10 Hz) for one sweep. At 20 m/s the car moves 2 m during that sweep, so the first and last points are measured from places 2 m apart. Without EGO-MOTION COMPENSATION (correcting each point using wheel and IMU data) the cloud is smeared.
- If camera and LiDAR clocks differ by 10 ms at 20 m/s, the same object appears 0.2 m apart in the two sensors, and a fused detection can split in two.
- **Fixes:** a shared clock (precision time protocol over Ethernet) and HARDWARE TRIGGERING: cameras expose exactly when the LiDAR beam passes their view.

4.2

+10 XP

Can the Computer Keep Up?

✨ READY, SET, COOK!

Can the computer keep up? 🏃 Let's race the data!

Worked example: compute for a camera array

One frame needs a 100 GFLOP network (illustrative). 8 cameras × 36 fps:

- Compute needed = 100 × 10^9 × 8 × 36 = 28.8 TFLOP/s sustained
- At 40% utilisation the chip needs a PEAK of 28.8 ÷ 0.4 = 72 TFLOP/s

Recipe 01's hypothetical 10 TFLOP/s accelerator is far too small.

REMEMBER THIS!

This is why self-driving computers carry hundreds of TOPS, and why Recipe 01's lessons (utilisation, memory-bound vs compute-bound, batch size) decide whether a system works. Face ID was one image at batch 1. A car makes hundreds of inferences per second, continuously, and cannot skip one.

CONNECT THE DOTS

Inside a vehicle, sensors and computers are connected by automotive Ethernet and other buses. Recipe 02 explains how bandwidth and latency limit how much sensor data can be delivered.

4.2 · SEE IT

+10 XP

Can the Computer Keep Up?

MAMA FENG

🔍 Spot it! Can the computer keep up? Compare what is needed with what the chip offers.

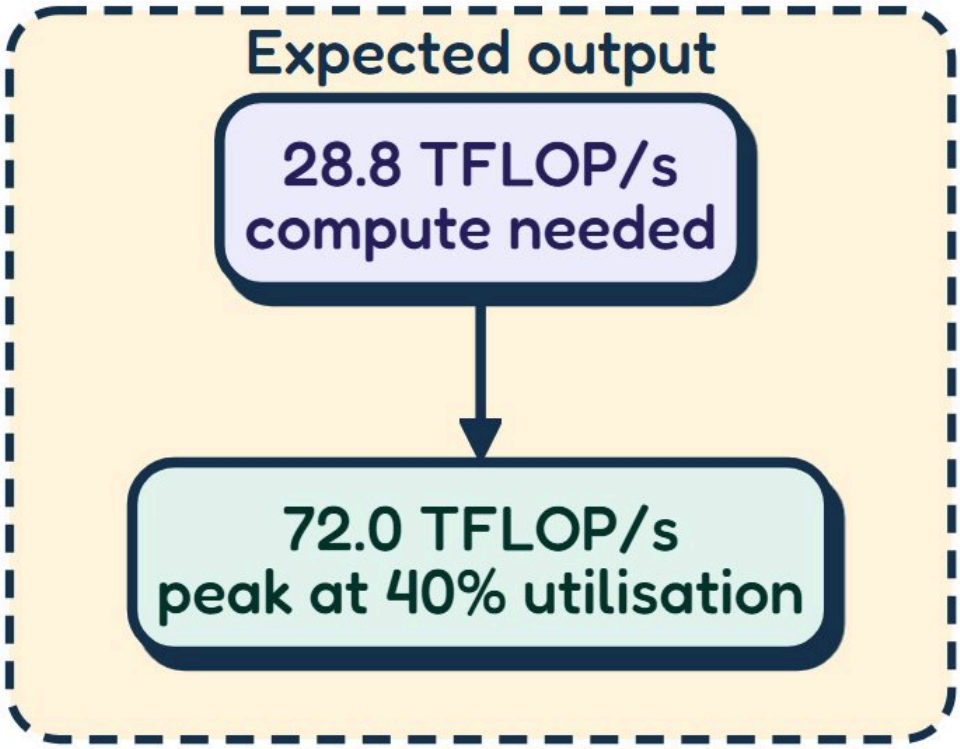


FIG Worked example: compute needed for a camera array vs what the chip offers.

SIDE QUEST · MINI-GAME 4.2

+25 XP

Beat the Timer

✨ READY, SET, COOK!

Beat the timer! ⏰ Get every stage done before the deadline!

🎯 Your Mission

Calculate data rates, stopping distances, latency budgets and compute needs.

🧺 Grab These First

Frame bytes = w × h × bits ÷ 8. Stopping = v × latency + v^2 ÷ (2a). Compute = FLOP/frame × cameras × fps; peak = compute ÷ utilisation.

🔪 Let's Go! Your Turn

(1) Six cameras, 1,280 × 960, 10-bit, 30 fps. Data rate per camera and in total?

(2) 72 km/h, braking 5 m/s^2, latency 150 ms. Total stopping distance?

(3) In the latency budget, inference grows from 30 to 60 ms. New total? By how much does it miss 100 ms? Name two fixes.

(4) 12 cameras at 30 fps with a 60 GFLOP network. Sustained TFLOP/s? Peak at 30% utilisation?

(5) Camera and LiDAR clocks are 15 ms apart; car at 30 m/s. Gap between the two views?

👀 What Did You Spot?

Convert km/h to m/s first (÷ 3.6) and ms to s (÷ 1,000).

💭 Think It Over

Which matters more for safety: throughput or latency? Why?

🔒 **Mama's feedback in the Answer Vault · V-4-2** · page 184 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #4.2

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.2) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. At 100 km/h, how far does a car travel during 0.3 s of processing latency? ♡ ♡ ♡

- (A) 8.33 m
- (B) 30 m
- (C) 64.3 m
- (D) 2.78 m

2. What does “ADC” mean? ♡ ♡ ♡

- (A) Correcting measurements for the platform's own motion.
- (B) Analogue-to-digital converter: voltage → number.
- (C) Image signal processor: raw sensor data → usable image.
- (D) Brightest vs darkest light captured in one frame.

3. Which kitchen word means: “Image signal processor: raw sensor data → usable image.”? ♡ ♡ ♡

- (A) Field of view (FOV)
- (B) IoU
- (C) ISO 26262 / SOTIF
- (D) ISP

4. What does “Precision / recall” mean? ♡ ♡ ♡

- (A) Measures the outside world / measures the machine itself.
- (B) Overlap ÷ union of predicted and true boxes.
- (C) Share of detections correct / share of real objects found.
- (D) Analogue-to-digital converter: voltage → number.

TICKET #4.3 · RECIPE 04 · SUSHI BAR

4.3 Sensor fusion and environmental perception

COURSES ☐☐☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 100 XP

4.3 +10 XP

Why Fuse?

READY, SET, COOK!

Why mix senses? Two senses beat one. Let's see how!

CONDITION	CAMERA	LIDAR	RADAR	ULTRASONIC
Night	Weak	Strong	Strong	Strong (short)
Rain, fog, snow	Weak	Weak-medium	Strong	Medium
Low sun glare	Weak	Strong	Strong	Strong
Dark object	Weak at night	Weaker returns	Strong if metal	Strong
Signs and lights	Strong	Cannot	Cannot	Cannot
Speed directly	Indirect	Indirect	Strong	Weak
Below the bumper	Blind spot	Possible blind spot	Weak	Strong

Different failures = a better team

Sensor fusion combines measurements from several sensors into one estimate, so one sensor's strength covers another's weakness, and the system knows HOW UNCERTAIN its picture is.

Ball at dusk: the camera sees the ball's colour but struggles with low light and wet reflections; LiDAR or radar give the distance regardless of light.

4.3 +10 XP

Three Levels of Fusion

READY, SET, COOK!

Three ways to mix! Early, middle or late: which would you pick?

LEVEL	WHAT IS FUSED	PLUS	MINUS
Early (raw data)	Raw pixels and points, e.g. LiDAR points projected onto the image	Keeps all information	Needs precise calibration and sync; heavy compute
Mid-level (feature)	Features from each sensor's network, merged in a shared space like bird's-eye view (BEV)	Learns how to combine; strong accuracy	Needs joint training; harder to debug

LEVEL	WHAT IS FUSED	PLUS	MINUS
Late (object)	Lists of detected objects, merged by a tracker	Simple, modular; a failed sensor is easy to drop	Information thrown away before fusion

Bird's-eye view

A BEV is a top-down map around the vehicle. Putting every sensor's features into the same BEV grid makes them natural to combine (Philion & Fidler 2020; Li et al. 2022; Liu et al. 2023).

💡 MAMA'S SECRET TIP

Early fusion = blending raw ingredients into one batter. Mid-level = combining sauces each cook prepared. Late = each cook plates a dish and Mama picks the final menu.

4.3

Weigh By Confidence

+10 XP

✨ READY, SET, COOK!

Trust the sensor you're surest about! ⚖️ Let's weigh it out.

Inverse-variance weighting

If a sensor's error has standard deviation σ, its variance is σ^2, and the best weight is $1 \div \sigma^2$. A more precise sensor deserves more weight.

fused = $\Sigma(w_i \times z_i) \div \Sigma w_i$ fused σ = $\sqrt{(1 \div \Sigma w_i)}$

Worked example: radar + camera

Radar: 50.0 m, σ = 0.5 m → weight 1 ÷ 0.25 = 4

Camera: 52.0 m, σ = 2.0 m → weight 1 ÷ 4 = 0.25

Fused = (4 × 50.0 + 0.25 × 52.0) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m

Fused variance = 1 ÷ 4.25 ≈ 0.235 → σ ≈ 0.49 m: slightly better than the best sensor alone.

⚠ PAPA'S OOPS!

"Fusion always helps!" Not quite! This assumes independent, unbiased, bell-shaped errors. If both sensors share a flaw (both blinded by fog), the fused result is LESS trustworthy than the formula claims.

4.3 · SEE IT

Radar Plus Camera

+10 XP

MAMA FENG

🔍 Spot it! Two sensors, two guesses. Why is the fused answer closer to the radar?

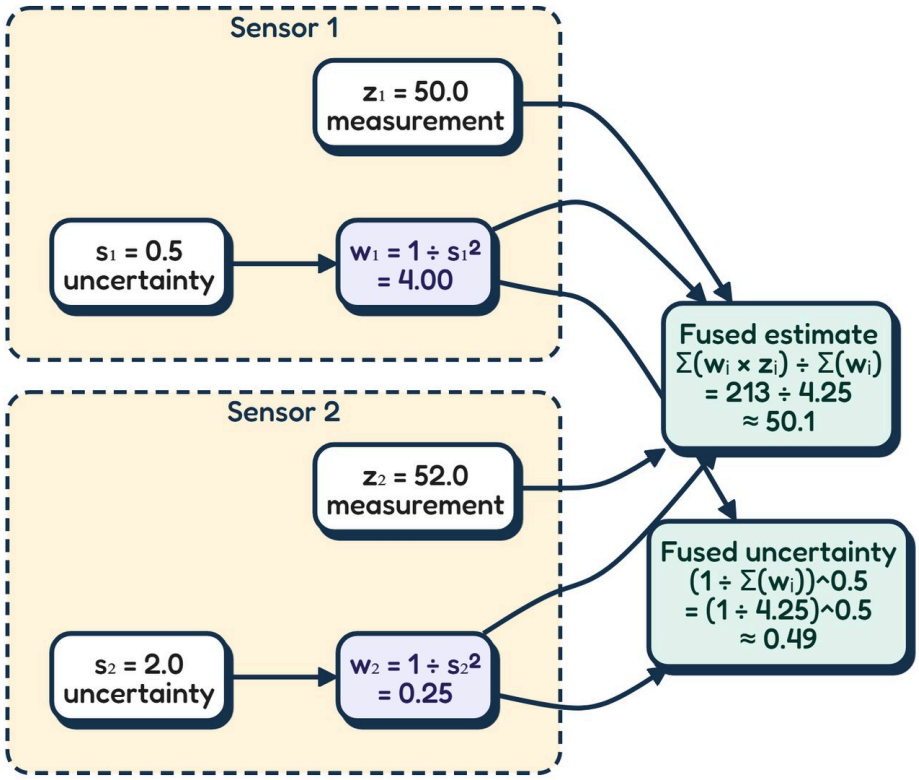


FIG Worked example: fusing two measurements by their confidence.

4.3 · SEE IT

+10 XP

Radar Plus Camera (2)

MAMA FENG

🔍 Spot it! Two sensors, two guesses. Why is the fused answer closer to the radar?

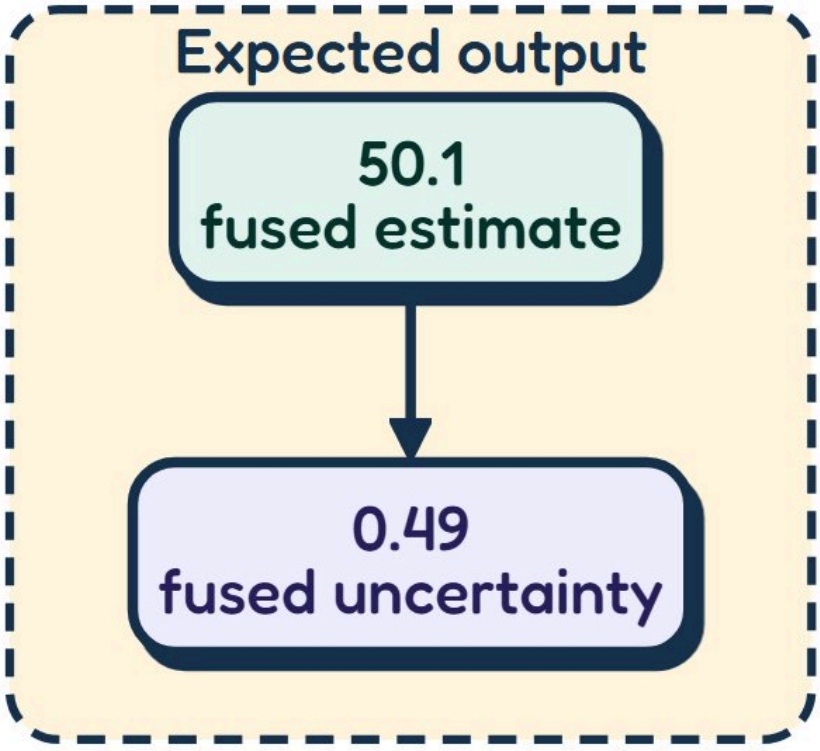


FIG Expected output: the fused estimate is better than either.

4.3

The Kalman Filter

+10 XP

Predict, then update

The Kalman filter (Kalman, 1960) keeps a running estimate x AND its uncertainty P:

1. **PREDICT:** move the estimate forward with a motion model. Uncertainty grows: P ← P + Q (process noise).
2. **UPDATE:** a measurement z (noise R) arrives.

Kalman gain K = P ÷ (P + R)

x ← x + K(z − x) P ← (1 − K)P

Worked example: one predict + two updates

Start x = 50.0 m, P = 4.0, Q = 1.0

Predict: P = 4.0 + 1.0 = 5.0

Update 1 (radar): z = 52.0, R = 1.0 → K = 5 ÷ 6 = 0.833, x = 50 + 0.833 × 2 = 51.67, P = 0.83

Update 2 (camera): z = 50.5, R = 4.0 → K = 0.833 ÷ 4.833 = 0.172, x = 51.67 + 0.172 × (−1.17) = 51.47, P = 0.69

🔑 REMEMBER THIS!

The precise radar pulled strongly; the noisy camera only a little; and P FELL with every update. Real trackers also track velocity, but the idea is identical. It is the same "nudge towards the new evidence" as the pup's rule in Recipe 03!

4.3 · SEE IT

+10 XP

The Kalman Filter

MAMA FENG

🔍 Spot it! Watch P (the uncertainty) after each update. Up or down?

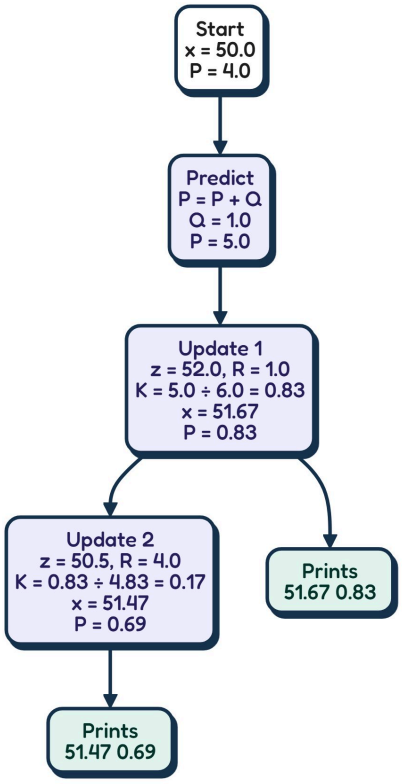


FIG Worked example: one predict step and two updates.

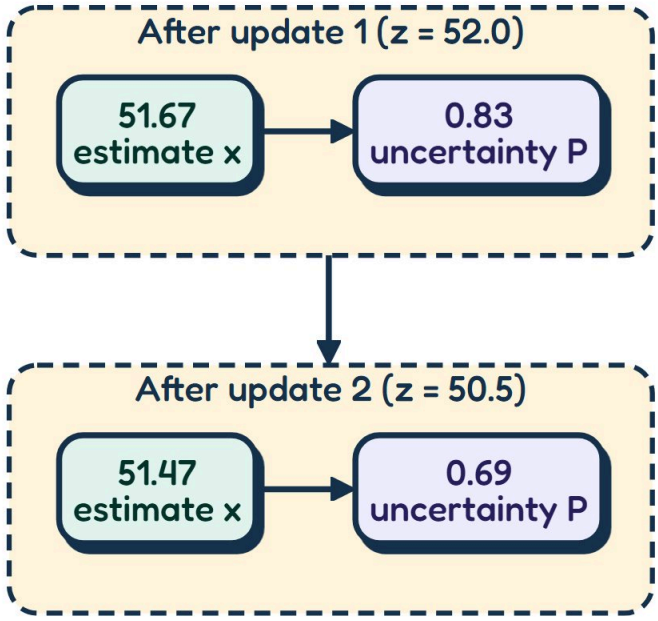


FIG The uncertainty shrinks after each update.

4.3

Which Ball Is Which?

+10 XP

🎯 Association and tracking

Before fusing, the system must decide which detection belongs to which object: DATA ASSOCIATION.

- **Gating:** only accept detections near the predicted position. Track predicted at 30.0 m, gate ±3 m: 29.2 m accepted, 33.5 m rejected.
- **Assignment:** with many tracks and detections, an algorithm such as the Hungarian algorithm finds the best one-to-one match.
- **Track life cycle:** a new detection starts a TENTATIVE track; it is CONFIRMED after several frames; it is DELETED if unmatched too long. This suppresses one-frame false alarms.

💡 MAMA'S SECRET TIP

Like a waiter matching dishes to tables: "Table 3 ordered the dumplings, and this plate of dumplings is near table 3, so it's theirs." A plate far away isn't matched.

4.3

Depth From Two Eyes

+10 XP

Stereo vision

Two cameras a known distance B apart (the baseline) see a point shifted by a DISPARITY d pixels:

$Z = f \times B \div d$

Worked example: stereo sensitivity

f = 1,000 px, B = 0.30 m → f × B = 300

- d = 5 px → 60.0 m
- d = 6 px → 50.0 m
- d = 7 px → 42.9 m

A one-pixel error moves 50 m to 42.9 or 60 m! Error per pixel ≈ $Z^2 \div (f \times B)$: at 50 m, 2,500 ÷ 300 ≈ 8.3 m.

REMEMBER THIS!

Because of the SQUARE, camera depth error grows quickly with range, while a LiDAR stays at a few centimetres. Networks can also learn depth from one image or video ("pseudo-LiDAR", Wang et al. 2019): good on average, but still an estimate, not a measurement.

4.3 · SEE IT

Depth From Two Eyes

+10 XP

MAMA FENG

Spot it! Two eyes give depth. Why does the error grow so fast with distance?

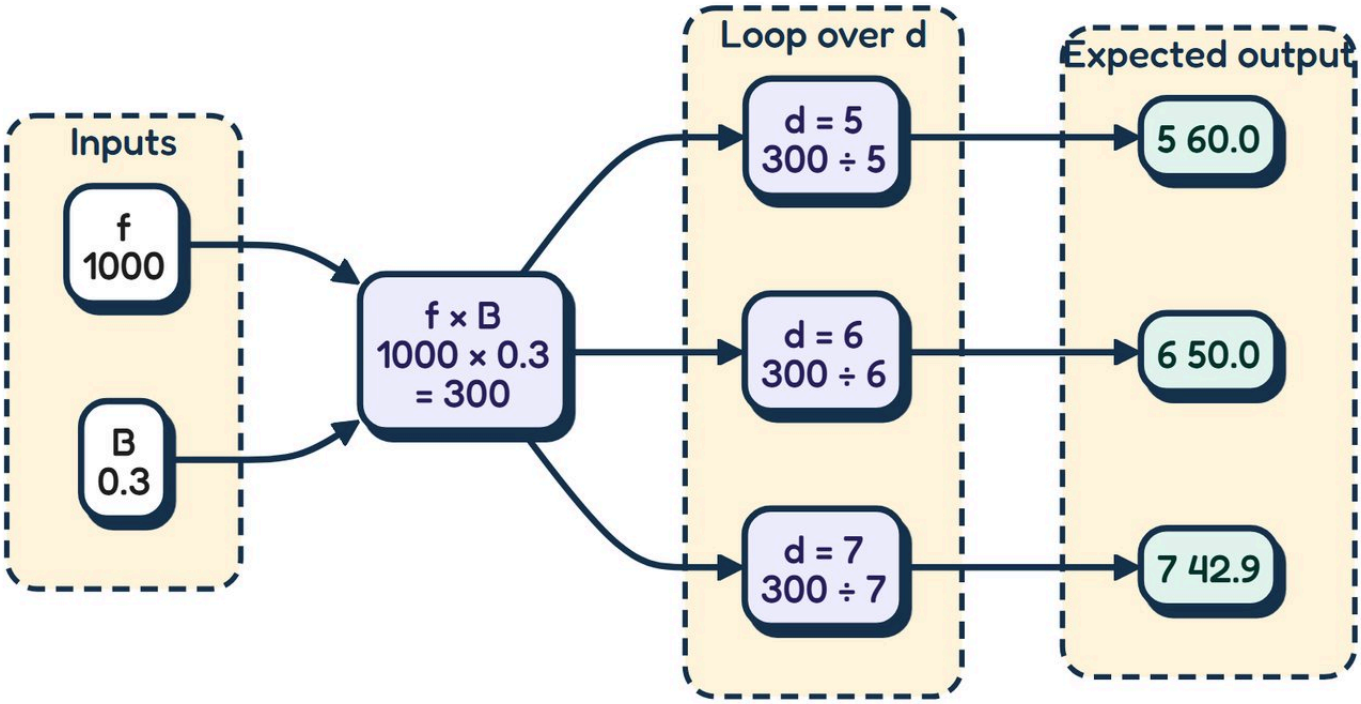


FIG Worked example: stereo depth and why it gets worse with range.

4.3

Scoring Perception

+10 XP

✨ READY, SET, COOK!

Report card time for perception! How do we score a sensor system?

TASK	OUTPUT
Object detection	3-D boxes with class (car, pedestrian, cyclist)
Semantic segmentation	A class for every pixel or point
Tracking	Identity, velocity and history of each object
Occupancy / free space	Which 3-D space is occupied, even by unknown objects
Lane and map elements	Lanes, curbs, crosswalks, lights, signs

Worked example: IoU, precision, recall

IoU = overlap ÷ union. Predicted and true boxes 4 × 2 m (8 m^2), shifted 1 m: overlap 3 × 2 = 6, union 8 + 8 − 6 = 10 → IoU = 0.6

100 detections: 90 correct, 10 false alarms, 30 objects missed.

Precision = 90 ÷ 100 = 0.90 Recall = 90 ÷ 120 = 0.75

False alarms cause phantom braking; misses cause collisions.

🔬 REAL-LIFE TECH

Public datasets with labelled camera, LiDAR and radar data let methods be compared fairly: KITTI (2012), nuScenes (2020), Waymo Open Dataset (2020).

4.3

How Fusion Fails

+10 XP

Five ways to burn the dish

- **Miscalibration:** fusion needs each sensor's exact position and angle (extrinsic calibration). A 1° yaw error at 50 m shifts an object 50 × tan 1° ≈ 0.87 m sideways: a pedestrian in the wrong lane.
- **Poor synchronisation:** see 4.2.
- **Disagreement:** camera says car, radar says none. Voting, confidence weights and learned fusion all try; none is perfect.
- **Common-cause failures:** fog or a dirty windscreen degrade several sensors at once.
- **Adversarial and rare cases:** stickers on signs, reflections and odd objects fool networks, a reason to keep a sensor that measures geometry directly.

SIDE QUEST · MINI-GAME 4.3

Mix the Senses

+25 XP

Your Mission

Fuse measurements, run a Kalman cycle, compute stereo depth and score detections.

Grab These First

$w = 1/\sigma^2$. K = P ÷ (P + R). Z = f·B ÷ d. IoU = overlap ÷ union. Precision = TP ÷ detections; recall = TP ÷ real objects.

Let's Go! Your Turn

(1) LiDAR 30.0 m (σ 0.1), camera 31.0 m (σ 1.0). Fused estimate and σ?

(2) x = 20.0, P = 2.0, Q = 0.5; update with z = 21.0, R = 1.5. New x and P?

(3) f = 1,200 px, B = 0.5 m, disparity 12 px. Depth? With 11 px?

(4) Two 5 × 2 m boxes overlap by 4 × 2 m. IoU?

(5) 80 detections, 72 correct, 8 missed. Precision and recall?

(6) Yaw error 0.5°: sideways offset at 80 m?

(7) Why does camera + LiDAR help less in thick fog than camera + radar?

What Did You Spot?

For (2), do Predict before Update.

Think It Over

Why does the uncertainty P always fall after an update?

🔒 **Mama's feedback in the Answer Vault · V-4-3** · page 184 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #4.3

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.3) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Radar says 50.0 m (σ = 0.5 m) and a camera says 52.0 m (σ = 2.0 m). The inverse-variance fused estimate is about: ♡ ♡ ♡

- (A) 52.0 m
- (B) 50.12 m
- (C) 51.0 m
- (D) 51.5 m

2. A stereo camera has f = 1,000 px and baseline B = 0.3 m. A disparity of 6 px means a depth of: ♡ ♡ ♡

- (A) 300 m
- (B) 42.9 m
- (C) 50 m
- (D) 60 m

3. What does “Disparity” mean? ♡ ♡ ♡

- (A) Distance from round-trip time: d = v × t ÷ 2.
- (B) Pixel shift between stereo images; Z = f × B ÷ d.
- (C) Measures the outside world / measures the machine itself.
- (D) Emits energy and measures the echo / only receives existing energy.

4. Which kitchen word means: “Combining several sensors' measurements into one estimate.”? ♡ ♡ ♡

- (A) Fail-operational / fail-safe
- (B) LiDAR / point cloud
- (C) Sensor fusion
- (D) Active / passive sensor

TICKET #4.4 · RECIPE 04 · SUSHI BAR

4.4 Integrating hardware with AI decision-making

COURSES ☐☐☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 150 XP

4.4 The Autonomy Stack

+10 XP

From sensing to steering

Sensors → Acquire → Perception → Localisation → Prediction → Planning → Control → Actuators, with a SAFETY MONITOR beside the stack that checks plausibility, deadlines and sensor health and can trigger a safe stop.

STAGE	QUESTION IT ANSWERS
Perception	What is around me? (tracked objects with uncertainty)
Localisation	Where am I? (GNSS, IMU, wheel odometry, maps)
Prediction	Where will the others go? (several trajectories with probabilities)
Planning	Route, behaviour (follow, overtake, stop) and a smooth trajectory
Control	Steering, throttle and brake commands at 50-100 Hz or faster

4.4 Modular vs End-To-End

+10 XP

	MODULAR	END-TO-END
Interfaces	Explicit (object lists, trajectories)	Learned, implicit
Debugging	Easier: find the failing module	Harder: behaviour hidden in weights
Safety argument	Easier to test and certify	Harder; needs very strong testing
Information loss	Detail lost at each interface	Less loss
Data needs	Modules trained separately	Very large amounts of driving data
Compute	Many specialised pieces	One or a few very large networks

History

Modular: each stage is separate with defined interfaces. End-to-end: one network from sensors to trajectory. Bojarski et al. (2016) trained a CNN to steer from camera images; Hu et al. (2023) trained one network with linked tasks. Real systems are often hybrids.

💡 MAMA'S SECRET TIP

Modular = a kitchen brigade: one cook chops, one fries, one plates. End-to-end = one super-chef who does it all by feel. Easy to see who burnt the rice in the brigade!

4.4

Stack On Hardware

+10 XP

✨ READY, SET, COOK!

Stack it up! 🥞 Which chip runs which part of the brain?

STAGE	WORKLOAD	HARDWARE	RATE
Capture and ISP	Fixed image processing	Dedicated ISP blocks	30-60 fps
NN perception	Huge matrix arithmetic	NPU or GPU	10-60 Hz
Fusion, tracking, localisation	Many small, irregular computations	CPU, DSP	10-100 Hz
Prediction and planning	Mixed neural and search code	CPU + accelerator	~10 Hz
Control and safety monitor	Simple, strictly timed	Safety-rated microcontroller	100 Hz+

🔑 REMEMBER THIS!

A vehicle computer is HETEROGENEOUS: different chips for different work (Recipe 01). Two warnings carry over: batch-1 streaming perception tends to be memory-bound, so on-chip memory and low precision matter; and unsupported operations falling back to the CPU can silently blow the latency budget.

4.4

Can the Car Stop In Time?

+10 XP

✨ READY, SET, COOK!

Can the car stop in time? 🛑 Do the maths and find out!

🎱 **Worked example: the ball at dusk**

Car at 50 km/h = 13.89 m/s, latency 0.15 s, braking 7 m/s^2.

- **Blind distance:** 13.89 × 0.15 ≈ 2.08 m
- **Braking distance:** 13.89^2 ÷ (2 × 7) ≈ 13.78 m
- Total needed ≈ 15.9 m

Detected at 25 m: stops with ~9 m to spare. ✓

Detected at only 15 m: after 2.08 m, 12.92 m remain; impact speed = √(13.89^2 − 2 × 7 × 12.92) ≈ 3.5 m/s (12.5 km/h). ✗

🔑 REMEMBER THIS!

This ties the whole recipe together: detection range (4.1), latency (4.2) and the quality of the fused estimate (4.3) decide whether a decision can be carried out at all. Even 3.5 m/s is dangerous for a child: early, reliable detection beats any amount of late braking.

4.4 · SEE IT

+10 XP

Can the Car Stop In Time?

MAMA FENG

🔍 Spot it! See it early, stop in time! Which detection range avoids the crash?

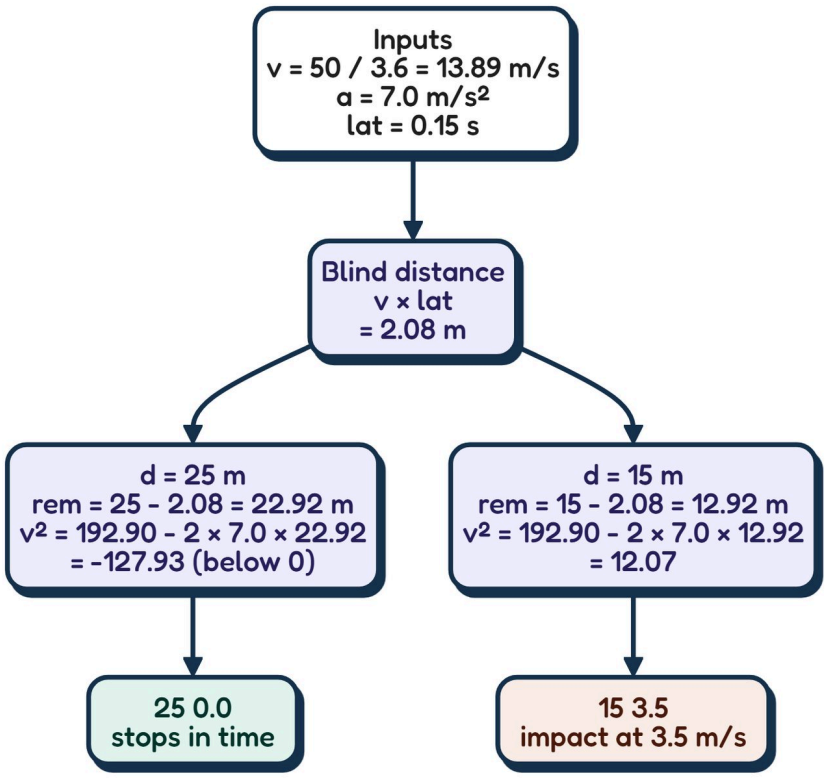


FIG Worked example: detection range decides the outcome.

4.4

+10 XP

Deciding Under Uncertainty

Worked example: when to brake?

Cost of an unnecessary hard brake (false alarm) = 1. Cost of not braking for a real pedestrian = 1,000. p = probability it's a real pedestrian.

- Cost of braking = (1 − p) × 1
- Cost of not braking = p × 1,000

Brake when (1 − p) < 1,000p → p > 1 ÷ 1,001 ≈ 0.1%

The trade-off

Such a low threshold means braking often for shadows and plastic bags: PHANTOM BRAKING, which is irritating and can cause rear-end crashes. The real cost ratio changes with speed and traffic, so thresholds are tuned with data, simulation and road tests. Precision and recall are not abstract scores: they set a SAFETY-vs-COMFORT trade-off.

Uncertainty feeds the planner too: Kalman said 51.47 m, $\sigma \approx 0.83$ m. A cautious planner uses $51.47 - 2 \times 0.83 \approx 49.8$ m.

4.4 · SEE IT

+10 XP

Brake Or Not?

MAMA FENG

🔍 Spot it! A false alarm vs a missed child: which costs more, and how does that move the line?

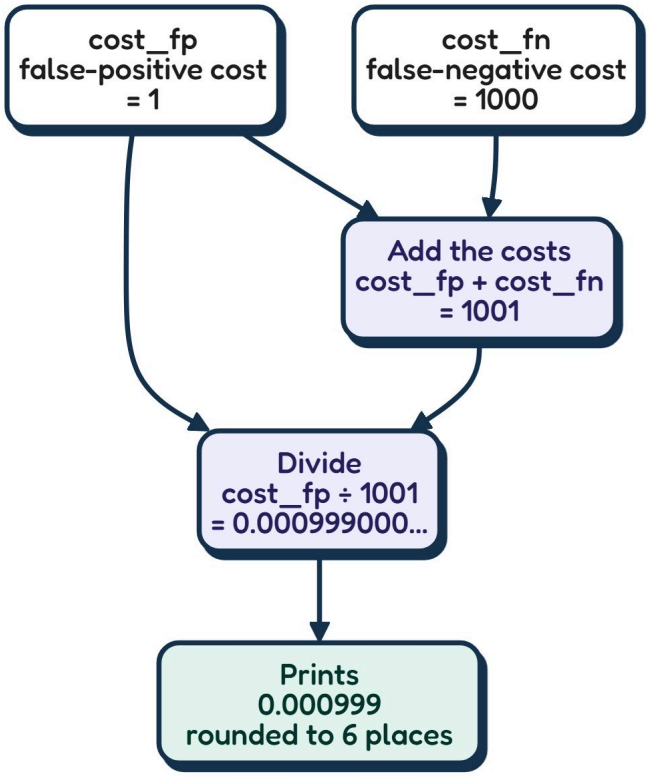


FIG Worked example: when should the car brake? Weighing the two kinds of error.

4.4

+10 XP

Safety Levels

✨ READY, SET, COOK!

Safety levels! 🦺 How safe is safe enough?

LEVEL	NAME	WHO MONITORS THE ROAD?
0	No automation	Human
1	Driver assistance (steering OR speed)	Human

LEVEL	NAME	WHO MONITORS THE ROAD?
2	Partial automation (steering AND speed)	Human must supervise at all times
3	Conditional automation	System; human takes over on request
4	High automation	System, within a defined domain
5	Full automation	System, everywhere

📘 Standards

SAE J3016 (2021) defines the levels. Driver assists needing constant supervision are Level 2, whatever the marketing name.

ISO 26262: functional safety; hazards from FAULTS, risk classes ASIL A-D (D strictest).

SOTIF (ISO 21448): hazards when NOTHING is broken but a sensor or network reaches its limits (a camera blinded by glare).

4.4

Two Cooks Are Safer… If Different

+10 XP

✨ READY, SET, COOK!

Two cooks are safer, but only if they're different! 👩‍🍳 👩‍🍳 Find out why.

👯 Redundancy

Safety-critical systems duplicate parts: two computers, two power supplies, and DIVERSE sensors. Fail-operational: keeps working in a reduced mode. Fail-safe: stops safely. A minimal-risk manoeuvre, like pulling over, is the fallback.

🎲 Worked example: why independence matters

Each channel misses a hazard 5% of the time.

- **Independent failures:** both miss 0.05 × 0.05 = 0.25%: 20× better.
- Perfectly correlated (fog blinds both): both miss 5%: no better than one.

Real systems fall between. That is the argument for DIFFERENT physics (camera, radar, LiDAR) instead of copies of one sensor. The counter-argument (4.5): one very capable channel plus a great network is simpler and cheaper and avoids conflicting sensors.

🗺️ SIDE QUEST · MINI-GAME 4.4

+25 XP

Brake Or Bake?

🎯 Your Mission

Compute stopping distances, decision thresholds and redundancy, and explain the safety standards.

Grab These First

Total stopping = v × latency + v^2 ÷ (2a). Brake when p > cost_fp ÷ (cost_fp + cost_fn).

Let's Go! Your Turn

(1) 36 km/h, latency 0.2 s, braking 6 m/s^2. Minimum detection distance?

(2) Using 4.4's ball example, at what detection distance is impact speed exactly zero?

(3) False alarm costs 2, missed pedestrian 600. Threshold p?

(4) Two channels each miss 0.02. Both miss if independent? If perfectly correlated?

(5) Two advantages and two disadvantages of end-to-end vs modular.

(6) ISO 26262 vs SOTIF: one example hazard of each.

What Did You Spot?

For (3), write the two expected costs before solving the inequality.

Think It Over

Why might a car that brakes at 0.1% confidence still be unsafe?

🔒 **Mama's feedback in the Answer Vault · V-4-4** · page 184 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #4.4

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.4) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Two sensors that fail for the same reason (for example fog) give the same benefit as two independent sensors. ♡ ♡ ♡

- (T) True
- (F) False

2. Under SAE J3016, a driver-assist that steers and brakes but needs constant human supervision is: ♡ ♡ ♡

- (A) Level 2
- (B) Level 4
- (C) Level 5
- (D) Level 0

3. What does “Precision / recall” mean? ♡ ♡ ♡

- (A) Braking caused by a false detection.
- (B) Matching detections to tracks / only near the prediction.
- (C) J3016 automation levels 0-5.
- (D) Share of detections correct / share of real objects found.

4. Which kitchen word means: “Safety against faults / against performance limits.”? ♡ ♡ ♡

- (A) Ego-motion compensation
- (B) Field of view (FOV)
- (C) ISO 26262 / SOTIF
- (D) Bird's-eye view (BEV)

TICKET #4.5 · RECIPE 04 · SUSHI BAR

4.5 Case study

Tesla's camera-only driving

COURSES ▢▢▢▢▢▢▢▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 160 XP

4.5

Case Study: Tesla

+10 XP

The big debate

Tesla made the most debated sensor decision in the industry: drive with CAMERAS and a neural network alone. The case shows every idea of this recipe at once, and it is a lesson in separating what a company CLAIMS from what independent evidence SHOWS.

⚠ CURRENCY NOTE

Checked against public sources in early November 2026. Tesla's software, hardware and regulatory status change quickly, and several figures come from news reports of company statements, marked "reported" or "company claim".

WHEN	MILESTONE
2016	8 cameras, forward radar, 12 ultrasonic sensors
2019	Own FSD computer (HW3), two custom chips, ~144 TOPS (Tesla)
2021-22	Radar removed (Model 3/Y 2021, S/X 2022)
Oct 2022	Ultrasonic dropped; vision occupancy network replaces it
2023+	HW4 computer and new cameras
Early 2024	FSD v12: end-to-end network replaces much hand-written code (reported ~300k lines of C++)
2025-26	Robotaxi in Austin (June 2025), more cities (reported); HW3 not capable of unsupervised (reported)
2026-27	AI5 chip: small volumes late 2026 (reported)

4.5

Pixels To Steering

+10 XP

✨ READY, SET, COOK!

Pixels in, steering out! 🎮 Follow the journey through Tesla's brain.

The sensor suite

Eight external cameras give 360°; no radar, LiDAR or ultrasonic (where Tesla Vision is fully deployed). Three face forward (wide, main, narrow; narrow quoted at ~250 m), two forward-side on the B-pillars, two rear-side on the fenders, one rear. Resolution reported from ~1.2 MP (HW3) to ~5 MP (HW4). There is NO directly measured depth anywhere: every distance is inferred.

Three big software ideas

1. Multi-camera fusion in bird's-eye view: image FEATURES from all eight cameras fused into one 3-D top-down space (AI Day 2021): mid-level fusion.
2. **Occupancy networks:** which 3-D cells are occupied, even by unnamed objects (a fallen ladder). LiDAR-like geometry inferred from video; it replaced the ultrasonics (AI Day 2022).
3. Video, not single images: a memory of recent frames estimates speed and sees through brief occlusions, like a moving stereo baseline.

4.5 · SEE IT

+10 XP

Pixels To Steering

MAMA FENG

🔍 Spot it! Follow the pixels from 8 cameras all the way to the steering wheel.

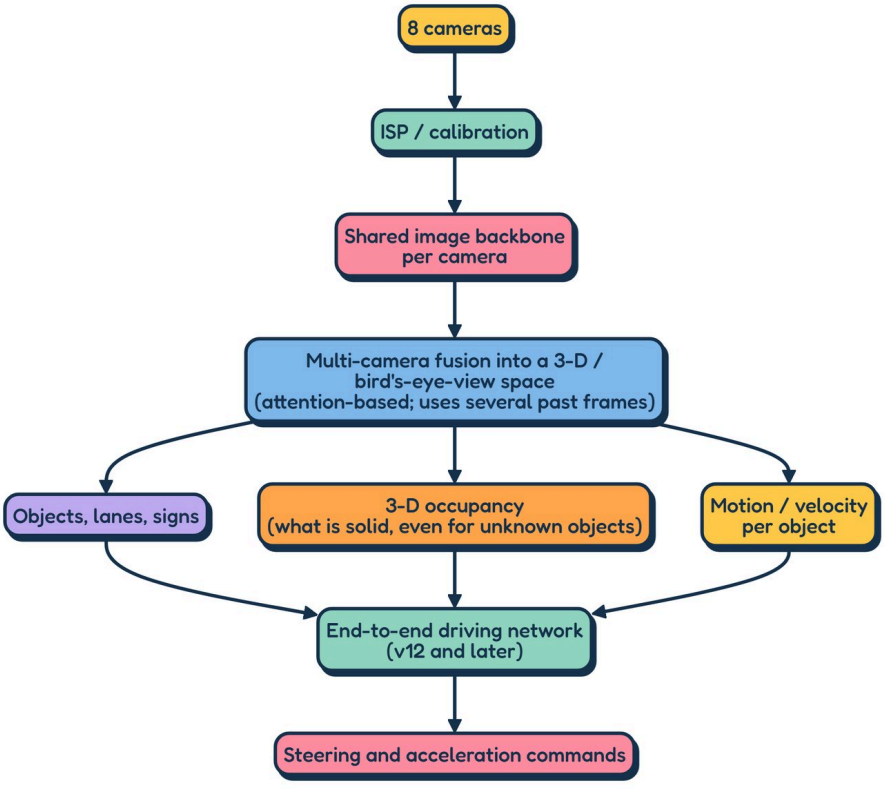


FIG From 8 cameras to a driving decision.

4.5

The Data Engine

+10 XP

✨ READY, SET, COOK!

Mistakes become lessons! 🔁 Meet the data engine.

Mistakes become recipes

Fleet drives → rare or difficult case detected → clip uploaded → labelled → network retrained (data centre, GPUs) → validated in simulation and tests → over-the-air update → inference in the car on the NPU → fleet drives…

🔑 REMEMBER THIS!

This is Recipe 01's training-vs-inference split at the scale of millions of vehicles: heavy training in the data centre, light inference in the car.

💡 MAMA'S SECRET TIP

Like Mama keeping a notebook of every dish a customer sent back. Each week she practises those dishes, and the next week every branch kitchen gets the improved recipe.

4.5

Recipe 01 In One Chip

+10 XP

✨ READY, SET, COOK!

Recipe 01 in one chip! 💾 Can you count the TOPS?

Worked example: how many TOPS?

Each accelerator: a 96 × 96 array of MAC units, 8-bit integers, ~2 GHz (Talpes et al., 2020). 1 MAC = 2 operations.

- **One NPU:** $96 \times 96 \times 2 \times 2 \times 10^9 \approx 36.9$ TOPS
- Four (two chips × two): ≈ 147.5 TOPS, close to Tesla's 144
- At 40% utilisation: ≈ 59 TOPS usable > 28.8 needed in 4.2. Plausible!

KITCHEN TERMINAL · PYTHON

```
macs, clock = 96*96, 2e9
tops = macs * 2 * clock / 1e12
print(round(tops,1), round(tops*4,1), round(tops*4*0.4,1))
```

🔑 REMEMBER THIS!

But memory, not arithmetic, is often the limit. Batch-1 streaming is memory-bound. Tesla said (Q1 2026, reported) HW3 has ~1/8 of HW4's memory bandwidth and that bandwidth is key for unsupervised driving. On the roofline, the TOPS can't be reached if memory can't feed them.

4.5 · SEE IT

+10 XP

How Many TOPS?

MAMA FENG

🔍 Spot it! MACs × 2 × clock: can you get the TOPS before reading the answer?

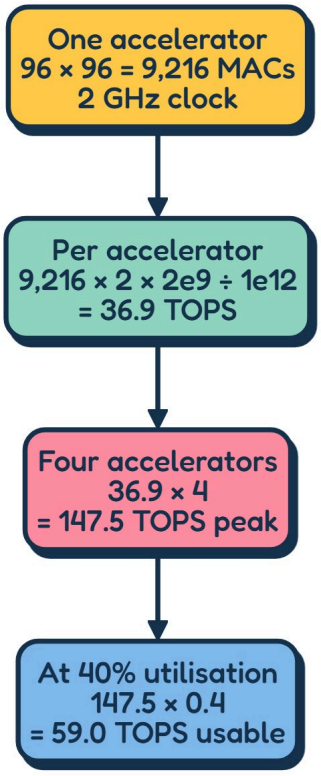


FIG Worked example: the TOPS of one accelerator, from its MACs and clock.

4.5

+10 XP

Why Camera-Only?

✨ READY, SET, COOK!

Why only cameras? 🤔 Hear both sides before you decide.

FOR CAMERA-ONLY	AGAINST
Humans drive with vision; roads are designed for eyes	Humans also have a big brain and common sense networks don't reliably match
Cameras are cheap: fit every car, millions supply data	Savings don't remove the cost of compute, data engine and accidents
One sensor + one network avoids sensor conflict	Disagreement is information; direct geometry is an independent check
Cameras and big networks improve fast	Depth error grows with distance^2; weak in glare, dark, heavy rain
Data engine finds rare cases at fleet scale	One modality = one common-cause failure

⚠ PAPA'S OOPS!

"That's a convincing argument, so it's true!" Not quite! A strong argument is not evidence. Only measurements, ideally independent ones, can show which side is right.

4.5

A Contrast: Waymo

+10 XP

✨ READY, SET, COOK!

Plot twist: Waymo does it differently! 🔄 Compare and contrast!

FEATURE	TESLA (CONSUMER CARS)	WAYMO (6TH-GEN DRIVER)
Cameras	8 external	13
LiDAR	None	4
Radar	None	6 (+ audio receivers)
Direct depth	None; inferred	LiDAR and radar
Human role	FSD (Supervised) = Level 2; robotaxi in selected cities (reported)	Driverless in defined areas (Level 4 style)
Hardware cost	Low per car	Higher; reported < US$20,000
Maps	Live perception + standard maps	Detailed prior maps
Domain	Broad, supervised	Geofenced service areas

🔑 REMEMBER THIS!

Waymo is also REDUCING sensors (from 29 cameras and 5 LiDARs in the previous generation) while keeping LiDAR and radar. The industry is not settling on "more is better" but on "enough DIVERSE sensors, with better software".

4.5

Taste the Evidence

+10 XP

✨ READY, SET, COOK!

You're the judge! ⚖️ What evidence would convince YOU a car is safe?

🔎 Judging a safety claim

Tesla claimed (company claim, April 2026) that supervised FSD has several times fewer collisions per mile than average human drivers. Ask:

- Who measured it? Company data is not independent.
- What exposure? FSD may be used mostly on easier roads in good weather.
- Who is driving? With a supervising human, the combined system is measured.
- What counts as a collision or intervention?
- What do regulators find? NHTSA has opened several investigations (check nhtsa.gov).

What would settle it?

Independent, comparable data: crashes and disengagements per million miles, split by road type, weather and light; third-party low-visibility tests; open incident reporting.

Honest conclusion: camera-only driving is POSSIBLE TO DEMONSTRATE and NOT YET PROVEN EQUAL to sensor-rich designs in all conditions.

4.5

Case Study Questions

+10 XP

✨ READY, SET, COOK!

Your turn! 🙋 Answer like a real AI engineer.

Your Turn!

1. Using stereo depth, explain why camera-only finds long-range distance harder than LiDAR. How does Tesla compensate?
2. HW3 has 1/8 of HW4's memory bandwidth. Using the roofline, why could this limit performance even with similar TOPS?
3. Recompute TOPS for a 128 × 128 array at 1.5 GHz.
4. A company says its system is seven times safer than an average driver. List four questions.
5. **Removing ultrasonics:** one benefit and one risk (common-cause failure).
6. Classify FSD (Supervised) on the SAE scale. What does it mean for driver monitoring?
7. Three conditions where you'd most want radar or LiDAR on a camera-only car, each with a number from this recipe.

4.5

Case Study Answers

+10 XP

🔒 **Answers in the Answer Vault · V-4-5** · page 184 · no peeking! +10 XP

SERVE THE CUSTOMERS · TICKET #4.5

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.5) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. What does “Occupancy network” mean? ♡ ♡ ♡

- (A) J3016 automation levels 0-5.
- (B) Correcting measurements for the platform's own motion.
- (C) Predicts which 3-D cells around the vehicle are occupied.
- (D) Allowed end-to-end time split among stages.

2. Which kitchen word means: “Laser time-of-flight ranging / the 3-D points it produces.”? ♡ ♡ ♡

- (A) Field of view (FOV)
- (B) Fail-operational / fail-safe
- (C) LiDAR / point cloud
- (D) Doppler effect

3. What does “Bird's-eye view (BEV)” mean? ♡ ♡ ♡

- (A) Keeps working reduced / moves to a safe state.
- (B) Top-down map used as a shared space for fusion.
- (C) Allowed end-to-end time split among stages.
- (D) Brightest vs darkest light captured in one frame.

LAB

TICKET #4.L · RECIPE 04 · SUSHI BAR

Kitchen Lab

Your Phone Is A Sensor Kit

COURSES ▢▢▢▢ RECORD ☆☆☆ BOUNTY 90 XP

KITCHEN LAB

Your Phone Is A Sensor Kit

+10 XP

🌐 Live platform: phyphox

Install phyphox (RWTH Aachen University, free, phyphox.org). Open "Acoustic Stopwatch" or "Sonar" and "Acceleration (without g)". Clap twice and read the time; walk with the phone and watch the IMU graphs.

Version: phyphox 1.2 for Android or iOS (free, no account).

🎯 Goal · Observe · Reflect

Goal: feel real sensor data: noise, sampling rate, delay.

Observe: Hold the phone still. Is the acceleration exactly zero? Use the Sonar near a wall: how far off is the distance at 1 m?

Reflect: Which ideas from 4.1-4.3 did you see (noise σ, temperature-dependent sound speed, sampling rate)?

Offline fallback

Two students 20 m apart; one claps, the other starts a stopwatch on seeing the clap and stops on hearing it. Repeat 5 times, average t, compute v = 20 ÷ t. Compare with 343 m/s. Then "fuse" the five times with an average and note how the spread shrinks.

⭐ Mama's Feedback (expected result)

Phone still: the acceleration jitters around 0 (about ±0.05 m/s^2) and is never exactly zero: that jitter is sensor noise σ.

Sonar at 1 m: usually within a few cm; more if the room temperature differs from what the app assumes (sound speed changes ≈ 0.6 m/s per °C).

Offline: the true delay is only 20 ÷ 343 ≈ 0.058 s, shorter than human reaction time, so single results scatter widely (often 150-600 m/s). The average of five is closer to 343 m/s.

EXERCISE

Final round

+10 XP

✨ READY, SET, COOK!

Final round! 🎯 Earn your Recipe 04 stamp!

EXERCISE

Question 1: Stopping time

Task: A car at 100 km/h has a total latency of 0.1 s and brakes at 6 m/s². Compute its total stopping distance, then the extra distance if the latency becomes 0.3 s.

Question 2: Fused range

Task: Radar says 40.0 m (σ = 0.4); camera says 42.0 m (σ = 1.6). Compute the fused estimate and its σ.

Question 3: Sensor shopping

Task: Pick a sensor set for a delivery robot that works on pavements at night and in rain. Justify each choice with one row of the taste-test table.

🔒 **Answers in the Answer Vault · V-4-6** · page 184 · no peeking! +10 XP

WATCH & LEARN

+10 XP

Video picks

How do self-driving cars "see"?

Creator: TED-Ed (Sajan Saini) · 5:25

Link: youtube.com/watch?v=PRg5RNU_JLk

Why watch: An animated intro to cameras, radar and LiDAR, the sensors of 4.1.

Timestamps: Whole video (5 min)

Task: Fill a 3-column table: sensor · what it measures · one weakness.

Limitations: Beginner level; no fusion maths; does not cover Tesla's camera-only choice.

How Does LiDAR Remote Sensing Work?

Creator: NEON Science · 7:42

Link: youtube.com/watch?v=EYbhNSUnIdU

Why watch: Time of flight and point clouds explained by a science organisation (4.1).

Timestamps: Whole video (8 min)

Task: Use distance = c × t ÷ 2 for one example shown in the video.

Limitations: Airborne mapping LiDAR, not car LiDAR.

WATCH & LEARN

+10 XP

Video picks

Visually Explained: Kalman Filters

Creator: Visually Explained · 11:18

Link: youtube.com/watch?v=IFeCIbljreY

Why watch: Animated predict-update cycle: the exact steps of 4.3.

Timestamps: 0:00 intro · 4:30 Kalman filters · 5:37 prediction step · 7:14 update step

Task: At 7:14, match each symbol to x, P, K, z and R from the worked example.

Limitations: Uses matrix notation; beginners can skip the last minutes.

Tesla AI Day 2021

Creator: Tesla · 3:03:20

Link: youtube.com/watch?v=j0z4FweCy4M

Why watch: Primary source for 4.5: Tesla's own explanation of camera-only vision and BEV fusion.

Timestamps: 46:54 event begins · 48:44 Tesla Vision · 1:13:12 planning and control · 1:35:15 simulation

Task: Watch 48:44-1:13:12 and list three claims; mark which are "company claims".

Limitations: Very long; a company presentation, not independent evidence; 2021 technology.

SERVE THE CUSTOMERS · TICKET #4.L

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-4-Q4.L) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In Kitchen Lab 4 (phyphox), a phone lying still on a table shows an acceleration (without g) of exactly 0 m/s^2. ♡ ♡ ♡

- (T) True
- (F) False

2. What does "LiDAR / point cloud" mean? ♡ ♡ ♡

- (A) Echo frequency shift from motion; radar measures speed with it.
- (B) Correcting measurements for the platform's own motion.
- (C) Laser time-of-flight ranging / the 3-D points it produces.
- (D) Predicts which 3-D cells around the vehicle are occupied.

3. Which kitchen word means: "Distance from round-trip time: d = v × t ÷ 2."? ♡ ♡ ♡

- (A) Phantom braking
- (B) Exteroceptive / proprioceptive
- (C) Doppler effect
- (D) Time of flight

4. What does "IoU" mean? ♡ ♡ ♡

- (A) Overlap ÷ union of predicted and true boxes.
- (B) Image signal processor: raw sensor data → usable image.
- (C) Brightest vs darkest light captured in one frame.
- (D) Measures the outside world / measures the machine itself.

TICKET #4.B · RUSH HOUR IN THE SUSHI BAR

🔔 Rush Hour!

*Last in line, the trouble customer **Blind-Spot Phantom**: "Your camera sees five pixels. I am the other ninety-five."*

HAPPY CUSTOMERS ☐☐☐☐☐☐☐☐☐☐☐☐☐☐

GRADE S · A · B · C · D

Rules: 14 orders, no going back. Colour one box for each right dish (check in the Answer Vault, V-4-B). 9 or more happy customers = you survive the rush: +150 XP and the **LiDAR Sushi Train** recipe card. Grade: S = 13–14, A = 12, B = 10–11, C = 9. Questions 13 and 14 are the trouble customer's double order.

1. Two sensors that fail for the same reason (for example fog) give the same benefit as two independent sensors. ♡ ♡ ♡

- (T) True
- (F) False

2. Which sensors measure distance DIRECTLY? ♡ ♡ ♡

- (A) LiDAR
- (B) A single camera
- (C) Radar
- (D) Ultrasonic

3. Radar says 50.0 m (σ = 0.5 m) and a camera says 52.0 m (σ = 2.0 m). The inverse-variance fused estimate is about: ♡ ♡ ♡

- (A) 52.0 m
- (B) 50.12 m
- (C) 51.0 m
- (D) 51.5 m

4. A stereo camera has f = 1,000 px and baseline B = 0.3 m. A disparity of 6 px means a depth of: ♡ ♡ ♡

- (A) 300 m
- (B) 42.9 m
- (C) 50 m
- (D) 60 m

5. In Kitchen Lab 4 (phyphox), a phone lying still on a table shows an acceleration (without g) of exactly 0 m/s². ♡ ♡ ♡

- (T) True
- (F) False

6. A LiDAR pulse returning after 400 ns means the object is about 60 m away. ♡ ♡ ♡

- (T) True
- (F) False

7. Radar measures the speed of an object directly using the Doppler effect. ♡ ♡ ♡

- (T) True
- (F) False

8. A single camera directly measures the distance to every pixel. ♡ ♡ ♡

- (T) True
- (F) False

9. Under SAE J3016, a driver-assist that steers and brakes but needs constant human supervision is: ♡ ♡ ♡

- (A) Level 2
- (B) Level 4
- (C) Level 5
- (D) Level 0

10. At 100 km/h, how far does a car travel during 0.3 s of processing latency? ♡ ♡ ♡

- (A) 8.33 m
- (B) 30 m
- (C) 64.3 m
- (D) 2.78 m

11. Which kitchen word means: "Pixel shift between stereo images; Z = f × B ÷ d."? ♡ ♡ ♡

- (A) Disparity
- (B) Field of view (FOV)
- (C) Hard real-time
- (D) Latency budget

12. What does "Active / passive sensor" mean? ♡ ♡ ♡

- (A) Distance from round-trip time: d = v × t ÷ 2.
- (B) Emits energy and measures the echo / only receives existing energy.
- (C) Allowed end-to-end time split among stages.
- (D) Correcting measurements for the platform's own motion.

13. Which kitchen word means: "Safety against faults / against performance limits."?

- (A) Kalman filter / gain
- (B) ISO 26262 / SOTIF
- (C) Time of flight
- (D) Disparity

14. What does "Fail-operational / fail-safe" mean?

- (A) Overlap ÷ union of predicted and true boxes.
- (B) Keeps working reduced / moves to a safe state.
- (C) Laser time-of-flight ranging / the 3-D points it produces.
- (D) Brightest vs darkest light captured in one frame.

TICKET #V4 · RECIPE 04

Answer Vault · Chef's Senses

Only open after trying! Every vault entry you check is worth XP on your licence.

V-4-1

4.1 · Sensor Taste Test

Mama's feedback · Sensor Taste Test

(1) f = 1,920 px → 34.6 px. (2) 150 m. (3) ≈ 2.0 m. (4) 30 m/s (108 km/h). (5) 900 steps → 57,600 points ≈ 0.92 MB (≈ 9.2 MB/s). (6) (a) ultrasonic (b) camera (c) radar (careful: stationary clutter) (d) LiDAR. Reflect: each measures a different physical quantity and fails in different conditions; that is why we fuse (4.3).

V-4-Q4.1

Customer orders · ticket #4.1

1. False · Correct: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly.

2. True · d = c × t ÷ 2 = 3 × 10^{8} × 4 × 10^{-7} ÷ 2 = 60 m.

3. True · The frequency shift of the echo gives the relative speed in one measurement.

4. undefined · · LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth.

5. A · Brightest vs darkest light captured in one frame. · Dynamic range: Brightest vs darkest light captured in one frame.

V-4-2

4.2 · Beat the Timer

Mama's feedback · Beat the Timer

(1) 1,536,000 bytes/frame → 46.08 MB/s; 276.48 MB/s total. (2) 20 m/s: 3.0 + 40 = 43.0 m. (3) 132 ms, misses by 32 ms; fixes: INT8 quantisation, smaller/pruned network, lower resolution, fused operators, no CPU fall-back, pipelining. (4) 21.6 TFLOP/s; 72 TFLOP/s peak. (5) 0.45 m. Reflect: latency, because every millisecond is distance driven blind.

V-4-Q4.2

Customer orders · ticket #4.2

1. A · 8.33 m · 100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts.

2. B · Analogue-to-digital converter: voltage → number. · ADC: Analogue-to-digital converter: voltage → number.

3. D · ISP · ISP: Image signal processor: raw sensor data → usable image.

4. C · Share of detections correct / share of real objects found. · Precision / recall: Share of detections correct / share of real objects found.

V-4-3

4.3 · Mix the Senses

Mama's feedback · Mix the Senses

(1) 30.01 m, σ ≈ 0.10 m (LiDAR dominates). (2) P = 2.5, K = 0.625 → x = 20.63, P = 0.94. (3) 50.0 m; 54.5 m (4.5 m per pixel). (4) 8 ÷ 12 ≈ 0.67. (5) 0.90 and 0.90. (6) ≈ 0.70 m. (7) Fog scatters light AND laser, so their errors are correlated; radio passes through fog, so radar adds independent information. Reflect: a measurement always adds information, so (1 − K) < 1 shrinks P.

V-4-Q4.3
Customer orders · ticket #4.3

1. B · 50.12 m · Weights $1/\sigma^2$: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates.

2. C · 50 m · Z = f × B ÷ d = 1,000 × 0.3 ÷ 6 = 50 m.

3. B · Pixel shift between stereo images; Z = f × B ÷ d. · Disparity: Pixel shift between stereo images; Z = f × B ÷ d.

4. C · Sensor fusion · Sensor fusion: Combining several sensors' measurements into one estimate.

V-4-4
4.4 · Brake Or Bake?

Mama's feedback · Brake Or Bake?

(1) 10 m/s: 2.0 + 8.33 ≈ 10.3 m. (2) ≈ 15.9 m. (3) p > 2 ÷ 602 ≈ 0.33%. (4) 0.04%; 2%. (5) + less information lost, one goal, fewer hand-written rules; − hard to debug and certify, needs huge data, low interpretability. (6) 26262: a failing brake controller or corrupted memory; SOTIF: glare-blinded camera or a misclassified unusual object. Reflect: phantom braking can cause rear-end crashes and erode trust.

V-4-Q4.4
Customer orders · ticket #4.4

1. False · Correct: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors.

2. A · Level 2 · Level 2 = partial automation; the human remains the driver.

3. D · Share of detections correct / share of real objects found. · Precision / recall: Share of detections correct / share of real objects found.

4. C · ISO 26262 / SOTIF · ISO 26262 / SOTIF: Safety against faults / against performance limits.

V-4-5
4.5 · Case Study Answers

Answers · Case Study Answers

1. Depth error ≈ $Z^2 \div (f \cdot B)$: ~8.3 m per pixel at 50 m; LiDAR measures to cm. Tesla: learned depth, 8-camera BEV fusion, many frames (moving baseline), huge fleet data: estimates, not measurements.
2. Batch-1 = low intensity = memory-bound slope: performance = intensity × bandwidth, so 1/8 bandwidth ≈ 1/8 performance.
3. 16,384 × 2 × 1.5 × 10^9 ≈ 49.2 TOPS.
4. Independent? Which roads/weather/times vs the human baseline? How many miles, what uncertainty? Was a human intervening? How are collisions defined? Same vehicle and driver types? Is raw data auditable?
5. **Benefit:** cost, wiring, fewer parts. Risk: short-range geometry now inferred; one blocked camera removes information nothing else supplies.
6. **Level 2:** the human supervises at all times; driver monitoring is essential.
7. Fog (radar penetrates); night dark obstacle (5.5 px at 150 m); glare/tunnel exit (100 ms = 2.8 m at 100 km/h).

V-4-Q4.5
Customer orders · ticket #4.5

1. C · Predicts which 3-D cells around the vehicle are occupied. · Occupancy network: Predicts which 3-D cells around the vehicle are occupied.

2. C · LiDAR / point cloud · LiDAR / point cloud: Laser time-of-flight ranging / the 3-D points it produces.

3. B · Top-down map used as a shared space for fusion. · Bird's-eye view (BEV): Top-down map used as a shared space for fusion.

V-4-6

4.L · Final round

Answers · Final round

Q1: 27.78 m/s → braking 64.3 m; 0.1 s → 2.78 m → 67.1 m total. 0.3 s → 72.6 m, 5.6 m more.

Q2: weights 1 ÷ 0.16 = 6.25 and 1 ÷ 2.56 = 0.39. Fused = (6.25 × 40 + 0.39 × 42) ÷ 6.64 = (250 + 16.4) ÷ 6.64 ≈ 40.12 m; σ = √(1 ÷ 6.64) ≈ 0.39 m.

Q3 (example): ultrasonic for close obstacles and kerbs (cheap, works in dark); radar for moving cyclists in rain (weather-robust, Doppler speed); a camera for crossings and signals (only it reads lights); optionally a small LiDAR for 3-D shape at night. Mixing physics avoids common-cause failure.

V-4-Q4.L

Customer orders · ticket #4.L

1. False · Correct: the reading jitters around 0 (about ±0.05 m/s²) and is never exactly zero; that jitter is sensor noise σ.

2. C · Laser time-of-flight ranging / the 3-D points it produces. · LiDAR / point cloud: Laser time-of-flight ranging / the 3-D points it produces.

3. D · Time of flight · Time of flight: Distance from round-trip time: d = v × t ÷ 2.

4. A · Overlap ÷ union of predicted and true boxes. · IoU: Overlap ÷ union of predicted and true boxes.

V-4-B

Rush hour · Sushi Bar

1. False · Correct: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors.

2. undefined · · LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth.

3. B · 50.12 m · Weights 1/σ²: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates.

4. C · 50 m · Z = f × B ÷ d = 1,000 × 0.3 ÷ 6 = 50 m.

5. False · Correct: the reading jitters around 0 (about ±0.05 m/s²) and is never exactly zero; that jitter is sensor noise σ.

6. True · d = c × t ÷ 2 = 3 × 10^8 × 4 × 10^{-7} ÷ 2 = 60 m.

7. True · The frequency shift of the echo gives the relative speed in one measurement.

8. False · Correct: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly.

9. A · Level 2 · Level 2 = partial automation; the human remains the driver.

10. A · 8.33 m · 100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts.

11. A · Disparity · Disparity: Pixel shift between stereo images; Z = f × B ÷ d.

12. B · Emits energy and measures the echo / only receives existing energy. · Active / passive sensor: Emits energy and measures the echo / only receives existing energy.

13. B · ISO 26262 / SOTIF · ISO 26262 / SOTIF: Safety against faults / against performance limits.

14. B · Keeps working reduced / moves to a safe state. · Fail-operational / fail-safe: Keeps working reduced / moves to a safe state.

05

ROOM 05 · ROOFTOP BANQUET · TICKET #D5

Running the Chain

INTEGRATION & OPTIMISATION

MAMA FENG

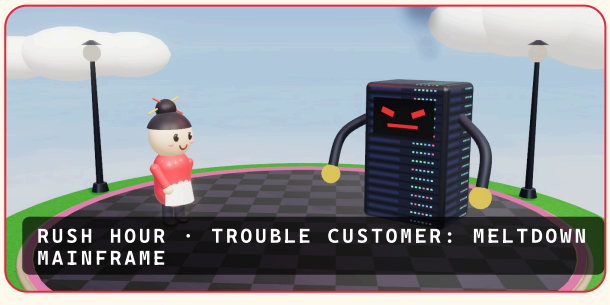
Mama's Problem: one wok can feed a family, but Mama's fried rice is now famous. To feed a whole city she needs MANY kitchens that cook as one, share the recipe, pay the bills and never close when one stove breaks. Real AI works the same way: thousands of chips, not one.

By the end you will be able to

1. Explain how compilers, runtimes and frameworks map a network onto hardware, and estimate the benefit of operator fusion.
2. Distinguish data, tensor and pipeline parallelism and calculate when a model must be split.
3. Estimate all-reduce communication cost and scaling efficiency.
4. Apply a measure-first optimisation workflow; explain the throughput-latency trade-off in serving.
5. Calculate the compute, time, energy and money to train a model, and the effect of PUE.
6. Explain why failures are routine at scale and choose a checkpoint interval.
7. Analyse Google's TPU infrastructure and judge its trade-offs.

TONIGHT'S TICKETS

- [ ] 5.1 Hardware-software interaction (the r... +130 XP
- [ ] 5.2 Distributed AI systems (many kitche... +130 XP
- [ ] 5.3 Performance optimisation and resou... +130 XP
- [ ] 5.4 Challenges +150 XP
- [ ] 5.5 Case study +130 XP
- [ ] LAB Kitchen Lab +90 XP
- [ ] Rush Hour +150 XP



TICKET #5.1 · RECIPE 05 · ROOFTOP BANQUET

5.1 Hardware-software interaction (the recipe card and the kitchen)

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

☐ BRIEFING

Mama's problem

+10 XP

MAMA FENG

Mama's Problem: one wok can feed a family, but Mama's fried rice is now famous. To feed a whole city she needs MANY kitchens that cook as one, share the recipe, pay the bills and never close when one stove breaks. Real AI works the same way: thousands of chips, not one.

📋 Today's Adventure

5.1 Hardware-software interaction (the recipe card and the kitchen)

5.2 Distributed AI systems (many kitchens, one dish)

5.3 Performance optimisation and resource allocation

5.4 Challenges: energy, cost and scalability

5.5 Case study: Google's TPU infrastructure

☐ BRIEFING

Goals & what you need

+10 XP

✨ READY, SET, COOK!

Time to run a whole restaurant chain! 🏭 Ready to think BIG?

🎯 By the end you will be able to

1. Explain how compilers, runtimes and frameworks map a network onto hardware, and estimate the benefit of operator fusion.
2. Distinguish data, tensor and pipeline parallelism and calculate when a model must be split.
3. Estimate all-reduce communication cost and scaling efficiency.
4. Apply a measure-first optimisation workflow; explain the throughput-latency trade-off in serving.
5. Calculate the compute, time, energy and money to train a model, and the effect of PUE.
6. Explain why failures are routine at scale and choose a checkpoint interval.
7. Analyse Google's TPU infrastructure and judge its trade-offs.

Running example

A team trains and serves a 7-billion-parameter chat model on a rented cluster of 256 chips. They must (1) train it on 1.4 trillion tokens, (2) keep the chips busy, and (3) answer thousands of requests per second at an affordable cost per answer.

Prerequisites: Recipe 01 (MACs, FLOPs, precision, memory, roofline); Recipe 02 helps.

5.1

+10 XP

The Software Stack

✨ READY, SET, COOK!

From recipe to stove! 📄 ➡️ 🔥 Meet the software stack.

From recipe to stove

A model is never run on hardware directly. It passes through layers of software:

- **Model code:** "y = relu(W @ x + b)"
- **Framework:** PyTorch, JAX, TensorFlow (builds a graph of operations)
- **Compiler / graph optimiser:** XLA, TensorRT, Core ML (fuses, reorders, assigns devices)
- **Runtime and libraries:** CUDA libraries, TPU runtime (launches work, manages memory)
- **Driver and firmware:** talks to the chip
- **Hardware:** MAC units, on-chip memory, HBM, interconnect

🔑 REMEMBER THIS!

The SAME model can run at very different speeds on the SAME chip, depending on how well these layers cooperate. Designing chip and software together (TPU + XLA) is called hardware-software co-design.

💡 MAMA'S SECRET TIP

The framework is the recipe in Mama's handwriting. The compiler is the head chef who rewrites it for this particular kitchen: which stove, which order, what to prepare together.

5.1

+10 XP

What A Compiler Does

✨ READY, SET, COOK!

The compiler: the kitchen planner! 🏘 What does it rearrange?

The computation graph

A model becomes a COMPUTATION GRAPH: each node is an operation (matrix multiply, add, ReLU) and each arrow is a tensor flowing between them. The compiler then improves the graph.

JOB	WHAT IT DOES	WHY IT HELPS
Operator fusion	Merges several operations into one pass over the data	Intermediate results never travel to and from off-chip memory
Layout and tiling	Cuts large tensors into tiles that fit on-chip	Data stays in fast memory while reused
Precision assignment	Chooses FP16, BF16 or INT8 per operation	Fewer bytes moved, more MACs per cycle
Placement and scheduling	Which chip or core runs each part, in what order	Keeps units busy; overlaps compute with transfer

5.1

+10 XP

Worked Example: Fusion

One trip to the fridge, not three

Tensor of 1,000,000 FP16 values = 2 MB. Three element-wise steps: scale, add bias, ReLU.

Unfused: each step reads and writes the whole tensor.

3 × (1 read + 1 write) × 2 MB = 12 MB of traffic

Fused: read once, do all three steps in a register, write once.

1 read + 1 write × 2 MB = 4 MB of traffic

Fusion cuts traffic 3×. These steps are memory-bound, so time also drops ~3×.

KITCHEN TERMINAL · PYTHON

```python
n, bpv = 1_000_000, 2
mb = n * bpv / 1e6
unfused, fused = 3*2*mb, 1*2*mb
print(unfused, fused, unfused/fused)   # 12.0 4.0 3.0
```

💡 MAMA'S SECRET TIP

Unfused = walking to the fridge to put the egg back after cracking it, again after whisking, again after salting. Fused = crack, whisk and salt at the counter, then one trip.

⚠ PAPA'S OOPS!

"The compiler will fuse everything!" Not quite! It can only fuse operations it understands. A custom operation it doesn't know is a WALL: data must be written out before it and read back after.

5.1 · SEE IT

+10 XP

One Trip To the Fridge

MAMA FENG

🔍 Spot it! Fused or not fused: how many trips to the fridge does each make?

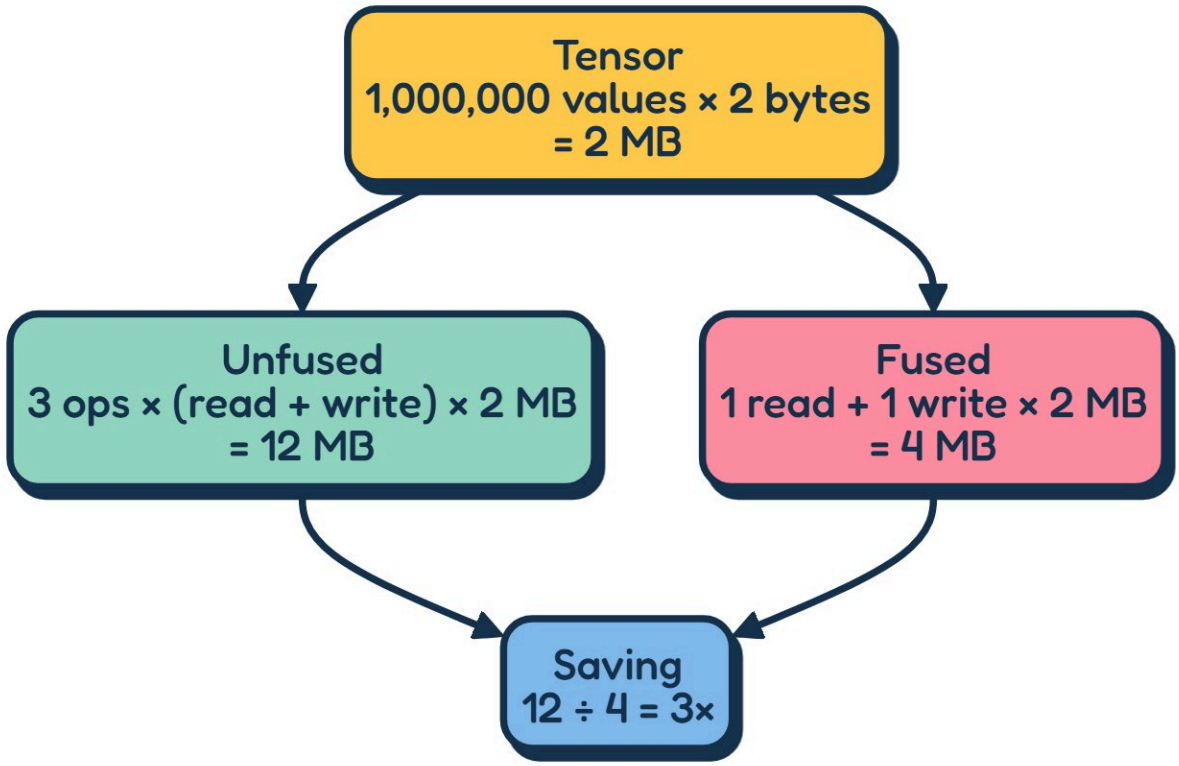


FIG Worked example: fusion saves memory traffic.

5.1 +10 XP

MFU & Device Switches

✨ READY, SET, COOK!

How busy are your cooks really? 📊 Meet MFU!

Model FLOPs utilisation

MFU = useful model FLOPs per second ÷ peak FLOP/s of the hardware

It counts only the model's USEFUL arithmetic. Well-tuned large training runs commonly report 30-60%; the rest is lost waiting for memory, other chips and gaps between operations.

Example: peak 275 TFLOP/s, useful 110 TFLOP/s → MFU = 40%.

⚠ PAPA'S OOPS!

"The monitor says 100% busy, so we're efficient!" Not quite! A chip can look 100% busy while only copying data. MFU measures USEFUL work, which is why it is preferred.

Device switches and fallbacks

Whenever execution moves between devices, data is copied and the accelerator waits. 10 operations, operation 5 unsupported: 1-4 on the accelerator, 5 on the CPU, 6-10 on the accelerator = TWO device switches. Large copies can cost more than all the acceleration saved (Recipe 01's phone fall-back, now in the data centre).

SIDE QUEST · MINI-GAME 5.1 +25 XP

One Trip To the Fridge

Your Mission

Calculate fusion savings, MFU and device switches.

Grab These First

Traffic unfused = ops × 2 × tensor size; fused = 2 × tensor size. MFU = useful ÷ peak.

Let's Go! Your Turn

(1) 4,000,000 FP16 values, four element-wise operations. Traffic unfused and fused (MB)? Percentage saved?

(2) Peak 400 TFLOP/s, useful 140 TFLOP/s. MFU?

(3) 12 operations; operations 3 and 8 unsupported on the accelerator. How many device switches?

(4) Why must a compiler know the on-chip memory size when it tiles a large matrix?

What Did You Spot?

Draw the 12 operations as boxes and colour the CPU ones.

Think It Over

Why can two kitchens with identical stoves cook at different speeds?

Mama's feedback in the Answer Vault · V-5-1 · page 226 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #5.1

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.1) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. Fusing three element-wise operations on a 2 MB tensor cuts memory traffic from 12 MB to 4 MB. ♡ ♡ ♡

- (T) True
- (F) False

2. What does “XLA” mean? ♡ ♡ ♡

- (A) Saving full training state to restart after a failure.
- (B) Merging several operations into one pass to avoid memory traffic.
- (C) Splitting a tensor or training state across chips.
- (D) Compiler that maps models onto TPUs and other devices.

3. Which kitchen word means: “Speed-up ÷ number of chips.”? ♡ ♡ ♡

- (A) Scaling efficiency
- (B) Preemption
- (C) Pod / slice
- (D) MTBF

4. What does “Operator fusion” mean? ♡ ♡ ♡

- (A) New requests join a running batch between tokens.
- (B) Start a job only when all its chips are free at once.
- (C) Useful model FLOP/s ÷ hardware peak FLOP/s.
- (D) Merging several operations into one pass to avoid memory traffic.

TICKET #5.2 · RECIPE 05 · ROOFTOP BANQUET

5.2 Distributed AI systems (many kitchens, one dish)

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

5.2

Why Many Kitchens?

+10 XP

✨ READY, SET, COOK!

Why use many kitchens? 🏢🏢 Guess before you read!

Reason 1: it doesn't fit

Training with mixed-precision Adam needs about 16 bytes per parameter, plus activations.

ITEM	BYTES PER PARAMETER
Weights (BF16)	2
Gradients (BF16)	2
Master copy of weights (FP32)	4
Optimiser state: two moments (FP32)	8
TOTAL	16

Worked example: does 7B fit?

Chip memory 96 GB. State = $7 \times 10^9 \times 16$ B = 112 GB > 96 GB: doesn't fit, even before activations. SHARDED across 256 chips: 112 ÷ 256 ≈ 0.44 GB per chip.

Reason 2: time. Training needs ≈ $6 \times N \times D = 6 \times 7 \times 10^9 \times 1.4 \times 10^{12} \approx 5.9 \times 10^{22}$ FLOPs. One chip at 100 TFLOP/s effective: ≈ 5.9×10^8 s ≈ 19 YEARS. Parallel chips are a necessity.

5.2 · SEE IT

Does It Fit?

+10 XP

MAMA FENG

🔍 Spot it! 16 bytes per parameter! Does a 7B model fit on one chip?

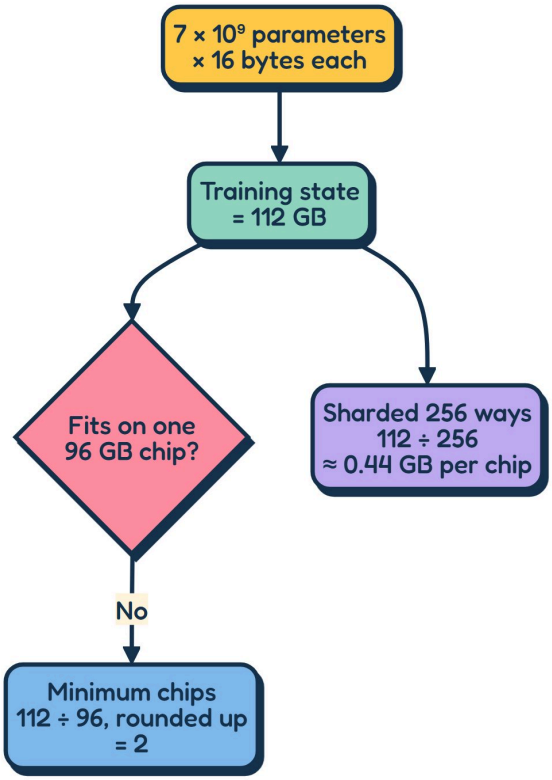


FIG Worked example: does a 7B model fit on one chip for training?

5.2

Three Ways To Split

+10 XP

✨ READY, SET, COOK!

Three ways to split the work! 🎌 Which one would you choose?

STRATEGY	WHAT IS SPLIT	EACH CHIP HOLDS	COMMUNICATION	BEST WHEN
Data parallel	The batch of examples	A full copy (or a shard)	Gradients summed once per step (all-reduce)	Model fits; want more throughput
Tensor parallel	Matrices inside a layer	A slice of every layer	Partial results inside every layer; needs very fast links	One layer too big, or lower latency
Pipeline parallel	Groups of layers	A few consecutive layers	Activations stage to stage	Deep model; slower links between stages

3D parallelism

Large runs combine all three: tensor inside tightly linked groups, pipeline across groups, data across replicas. Rule of thumb: put the most communication-hungry strategy (tensor) on the FASTEST links.

💡 MAMA'S SECRET TIP

Data parallel = four kitchens each cook the whole menu for different customers. Tensor parallel = two chefs cut the same giant fish together. Pipeline = an assembly line: one fries, the next sauces, the next plates.

5.2

Keeping Kitchens In Sync

+10 XP

The ring all-reduce

In data parallelism every chip computes gradients on its own part of the batch. Before updating, all gradients must be SUMMED and shared: an ALL-REDUCE. In a ring all-reduce, chips pass chunks around a ring:

traffic per chip = $2 \times (N - 1) \div N \times S$ bytes

For large N this approaches $2 \times S$, almost independent of N, which is why data parallelism scales well in principle.

Worked example: 7B on 256 chips

Gradients $7 \times 10^9 \times$ BF16 = 14 GB. N = 256.

- Traffic = $2 \times (255 \div 256) \times 14 \approx 27.9$ GB per chip
- **At 100 GB/s:** ≈ 0.28 s
- **Compute:** batch 4×10^6 tokens → $6 \times 7 \times 10^9 \times 4 \times 10^6 = 1.68 \times 10^{17}$ FLOPs; cluster 256×200 TFLOP/s × 40% = 2.048×10^{16} FLOP/s → ≈ 8.2 s
- **Not overlapped:** 8.2 + 0.28 = 8.48 s → efficiency ≈ 96.7%

CONNECT THE DOTS

Bridge to Recipe 02: data parallelism sends big, regular messages once per step; tensor parallelism sends small messages very often (needs the lowest latency); pipeline sends modest messages to neighbours. The network decides which strategy is practical.

5.2 · SEE IT

Keeping In Sync

+10 XP

MAMA FENG

🔍 Spot it! Fast links first, slow links last. How much time does that save?

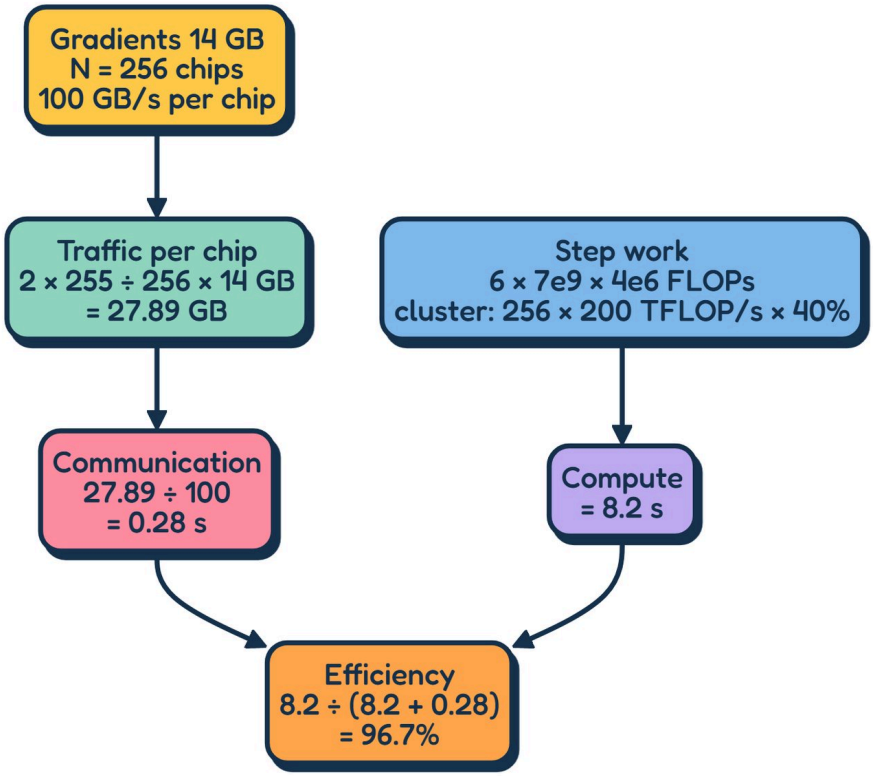


FIG Worked example: communication overhead for the 7B model.

5.2 · SEE IT

+10 XP

Keeping In Sync (2)

MAMA FENG

🔍 Spot it! Fast links first, slow links last. How much time does that save?

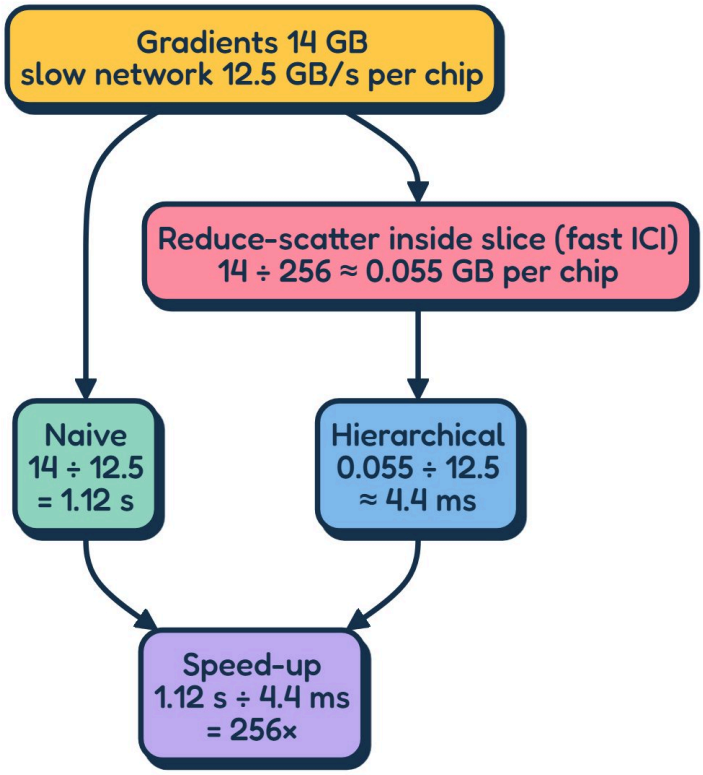


FIG Why communicate hierarchically: fast links first, slow links last.

5.2

Scaling Efficiency

+10 XP

✨ READY, SET, COOK!

Did more kitchens really help? 📈 Let's check the numbers!

Did more kitchens pay off?

speed-up = T(1 chip) ÷ T(N chips)

scaling efficiency = speed-up ÷ N

Example: 100 hours on 1 chip, 2.0 hours on 64 chips → speed-up 50 → efficiency 50 ÷ 64 ≈ 78%. The missing 22% went to communication, idle units and Amdahl's serial parts (Recipe 01).

TERM	MEANING
Strong scaling	Fixed problem, more chips. Efficiency falls as each chip gets less work.
Weak scaling	Problem grows with chip count. Easier to keep efficient; what large training relies on.

⚠ PAPA'S OOPS!

"The all-reduce was only 3%, so it's always small!" Not quite! The overhead was small because each step had a big batch. Shrink the batch per chip and compute shrinks while the all-reduce stays the same size. Real systems also OVERLAP communication with the backward pass. (Simplified: sharded state changes the pattern.)

🗺️ SIDE QUEST · MINI-GAME 5.2

+25 XP

Split the Order

✨ READY, SET, COOK!

Split the order! 🍕 Divide the work and beat the clock!

🎯 Your Mission

Calculate training memory, all-reduce traffic and scaling efficiency, and pick a parallelism strategy.

🧺 Grab These First

Memory = params × 16 B. Ring traffic = 2(N−1)/N × S. Efficiency = (T1 ÷ TN) ÷ N.

🔪 Let's Go! Your Turn

(1) A 13B model, 16 bytes/parameter, 80 GB chips. State size? Minimum chips?

(2) 1B parameters, BF16 gradients, 8 chips at 50 GB/s. Traffic per chip and all-reduce time?

(3) 120 h on 1 chip, 4.5 h on 32 chips. Speed-up and efficiency?

(4) Best strategy: (a) model fits, want more data per second (b) one matrix too big for one chip (c) very deep model in stages.

👀 What Did You Spot?

For (1), round UP: you can't buy 2.6 chips.

💭 Think It Over

Why put tensor parallelism on the fastest links?

🔒 **Mama's feedback in the Answer Vault · V-5-2** · page 226 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #5.2

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.2) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In data parallelism, each chip stores only a slice of the model's weights. ♡ ♡ ♡

- (T) True
- (F) False

2. Training a 7B-parameter model with mixed-precision Adam needs about 16 bytes per parameter. Total training state: ♡ ♡ ♡

- (A) 14 GB
- (B) 56 GB
- (C) 112 GB
- (D) 224 GB

3. What does “All-reduce” mean? ♡ ♡ ♡

- (A) Merging several operations into one pass to avoid memory traffic.
- (B) Sums values across chips and gives every chip the result.
- (C) New requests join a running batch between tokens.
- (D) Mean time between failures; system ≈ chip MTBF ÷ chip count.

4. Which kitchen word means: “Fixed problem / problem grows with chip count.”? ♡ ♡ ♡

- (A) Computation graph
- (B) Strong / weak scaling
- (C) Data parallelism
- (D) TCO

TICKET #5.3 · RECIPE 05 · ROOFTOP BANQUET

5.3 Performance optimisation and resource allocation

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

5.3

Measure First

+10 XP

✨ READY, SET, COOK!

Rule #1: measure first! 🔪 Find the real slow spot before you fix anything.

🔍 The optimisation loop

1. **PROFILE:** record where time goes (compute, memory, communication, waiting for input).
2. **LOCATE the bottleneck:** roofline for one chip, scaling efficiency for many.
3. CHANGE ONE THING.
4. **MEASURE AGAIN:** keep the change only if the numbers improve.

The bottleneck is rarely where you expect, and removing one simply exposes the next.

SYMPTOM	LIKELY BOTTLENECK	TYPICAL FIX
Low MFU, units idle between ops	Many small ops; launch overhead	Fusion; larger batches; compile the whole graph
Low intensity on the roofline	Memory bandwidth	Lower precision; on-chip reuse; bigger batch
Step time grows with more chips	Communication	Overlap; better placement; fewer, larger messages
Chips wait while data loads	Input pipeline	Prefetching; more loader workers; faster storage
Out-of-memory errors	Memory capacity	Sharding; recomputation; lower precision

5.3

Training Tricks

+10 XP

✨ READY, SET, COOK!

Secret training tricks! 🎩 Trade a little time for a lot of memory.

TECHNIQUE	WHAT IT DOES	TRADE-OFF
Mixed precision	Compute in BF16/FP16, keep an FP32 master copy	Slightly more code; small accuracy risk handled by scaling
Activation recomputation	Don't store every intermediate value; recompute in the backward pass	Saves memory, costs extra compute
Overlap comms and compute	Send finished layers' gradients while earlier layers still compute	Needs careful scheduling

TECHNIQUE	WHAT IT DOES	TRADE-OFF
Larger per-chip batch	More work per all-reduce, higher intensity	Limited by memory; huge batches can hurt learning
Efficient input pipeline	Prepare the next batch while this one runs	Uses CPU and storage

Worked example: the price of recomputation

Backward ≈ 2 × forward, so a step = 1 + 2 = 3 parts (matching 6 × N × D). Recomputing the forward once more = 4 parts: about 33% MORE compute.

Our 8.2 s step → 8.2 × 4 ÷ 3 ≈ 10.9 s. Worth it only if the saved memory allows a bigger batch or avoids a costlier sharding scheme.

5.3 · SEE IT

+10 XP

The Price of Recomputation

MAMA FENG

🔍 Spot it! Recompute instead of storing: what do you pay in time?

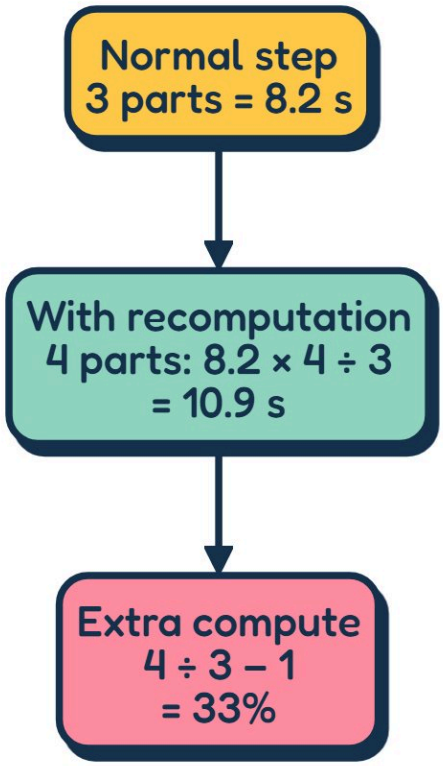


FIG Worked example: recomputing saves memory but costs time.

5.3

+10 XP

Serving: One Token At A Time

✨ READY, SET, COOK!

One token at a time! 💬 Why chatbots type their answers word by word.

Why generation is memory-bound

A language model writes one token at a time. For EVERY token the chip reads ALL the weights once, but does little arithmetic with them: ~1 FLOP/byte at batch 1, the memory-bound case of Recipe 01.

Worked example: speed limit of one chat

7B model in INT8 = 7 GB; memory bandwidth 1 TB/s.

- **Read the weights once:** 7 GB ÷ 1,000 GB/s = 7 ms per token
- **Limit for ONE user:** 1 ÷ 0.007 ≈ 143 tokens/s, however many MACs the chip has
- **Batch 32 users:** weights read once for all → ≈ 32 × 143 ≈ 4,571 tokens/s total, each user still ~143 tokens/s

⚠ PAPA'S OOPS!

"So batch everything and it's free!" Not quite! These are upper bounds. The KV cache (stored attention state) must also be read per token and grows with the conversation, and memory capacity limits the batch. Batching is nearly free only until the chip becomes compute-bound.

5.3 · SEE IT

+10 XP

One Token At A Time

MAMA FENG

🔍 Spot it! Every token reads all the weights. What is the speed limit?

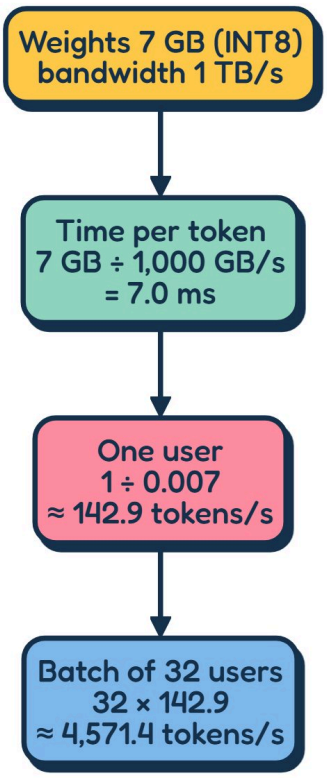


FIG Worked example: the speed limit of one conversation.

5.3

+10 XP

Batching: Wait Or Serve?

✨ READY, SET, COOK!

Wait for more orders or serve now? ⏳ Find the sweet spot!

The throughput-latency trade-off

Batching makes requests WAIT to be grouped. 100 requests/s, server insists on a full batch of 32: the first request waits ≈ 31 ÷ 100 ≈ 0.31 s before any computation.

Fix: a TIMEOUT. Run when full OR when the oldest has waited, say, 50 ms. At 100 requests/s a batch then holds about 5.

Modern servers use CONTINUOUS BATCHING: new requests join a running batch between tokens, getting most of the throughput without the long wait.

More serving tricks

- **Quantisation (Recipe 01):** fewer bytes per weight
- Caching results for repeated prompts
- **Speculative decoding:** a small fast model drafts tokens that the large model verifies

💡 MAMA'S SECRET TIP

A noodle stall that waits for 32 customers before boiling the pot keeps the first customer hungry. A smart stall drops in new portions as people arrive: continuous batching!

5.3

Sharing the Kitchens

+10 XP

CONCEPT	MEANING
Quota	Each team has a guaranteed share of chips
Priority and preemption	Urgent jobs (serving) can pause lower-priority jobs (experiments)
Gang scheduling	A training job needs ALL its chips at once; start only when the full set is free
Topology-aware placement	Chips that talk a lot are placed close on fast links
Bin packing	Fit jobs of different sizes into free space with few gaps
Autoscaling	Add serving replicas when load rises; release them when it falls

Worked example: utilisation is money

A chip costs $2/hour whether busy or not.

- 35% utilisation: each USEFUL chip-hour costs 2 ÷ 0.35 ≈ $5.71
- 60% utilisation: 2 ÷ 0.60 ≈ $3.33

A saving of about 42% without buying any hardware.

SIDE QUEST · MINI-GAME 5.3

Rush Hour

+25 XP

Your Mission

Apply recomputation cost, the serving speed limit, bottleneck diagnosis and cost per useful hour.

Grab These First

Recompute: × 4 ÷ 3. Token limit = bandwidth ÷ weight bytes. Cost per useful hour = cost ÷ utilisation.

Let's Go! Your Turn

(1) A step takes 20 s. With full recomputation?

(2) A 13B model in INT8 on a 2 TB/s chip. Time per token and the limit for one user?

(3) MFU is 15% and the profiler shows chips idle while the next batch is read from storage. Which symptom row? What do you try first?

(4) A chip costs $3/hour. Cost per useful hour at 25% and 50% utilisation?

What Did You Spot?

For (3), use the symptom table, not a guess.

Think It Over

Why might adding more chips NOT help in (3)?

Mama's feedback in the Answer Vault · V-5-3 · page 226 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #5.3

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.3) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A 7 GB INT8 model is served on a chip with 1 TB/s memory bandwidth. The memory-bound limit for one user is about: ♡ ♡ ♡

- (A) 1,000 tokens/s
- (B) 4,571 tokens/s
- (C) 14 tokens/s
- (D) 143 tokens/s

2. What does “Recomputation” mean? ♡ ♡ ♡

- (A) Recompute activations instead of storing them: ~33% more compute.
- (B) Splitting a tensor or training state across chips.
- (C) Fixed problem / problem grows with chip count.
- (D) Total cost of ownership: purchase, energy, cooling, network, buildings, staff.

3. Which kitchen word means: “Speed-up ÷ number of chips.”? ♡ ♡ ♡

- (A) Preemption
- (B) Tensor parallelism
- (C) Operator fusion
- (D) Scaling efficiency

4. What does “MFU” mean? ♡ ♡ ♡

- (A) Batch split across chips, each with a model copy (or shard).
- (B) Useful model FLOP/s ÷ hardware peak FLOP/s.
- (C) Recompute activations instead of storing them: ~33% more compute.
- (D) Direct TPU chip links / optical switches re-patched by software.

TICKET #5.4 · RECIPE 05 · ROOFTOP BANQUET

5.4 Challenges

energy, cost and scalability

COURSES ▢▢▢▢▢▢▢▢▢▢ RECORD ☆☆☆ BOUNTY 150 XP

5.4

The Energy Bill

+10 XP

Power vs energy

POWER = rate of using energy (watts). ENERGY = power × time (kWh). A 400 W chip running 10 hours uses 4 kWh.

PUE (power usage effectiveness) = total facility energy ÷ energy of the computing equipment. 1.0 = zero overhead; older facilities 1.5 or worse; the best large operators report ~1.1.

Worked example: what PUE is worth

1 MW of computing, all year (8,760 h):

- PUE 1.1: 1 × 1.1 × 8,760 = 9,636 MWh
- PUE 1.6: 1 × 1.6 × 8,760 = 14,016 MWh
- Difference 4,380 MWh/year, at $80/MWh = $350,400 for the SAME computing

Energy → carbon

Emissions = energy × carbon intensity. The 89.8 MWh training job (next page): at 0.4 kg CO2e/kWh ≈ 35.9 t; at 0.05 kg/kWh ≈ 4.5 t. WHERE and WHEN a job runs can matter as much as how efficiently.

Moving a 32-bit word from DRAM costs ~170× multiplying it (Horowitz 2014): every data-movement saving is an ENERGY saving.

5.4 · SEE IT

+10 XP

What PUE Is Worth

MAMA FENG

🔍 Spot it! PUE is the overhead on top of computing. How many dollars does 0.5 save?

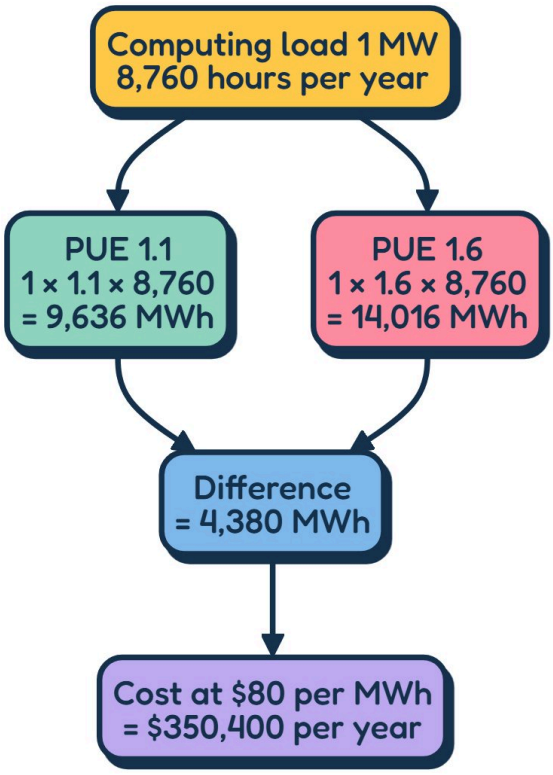


FIG Worked example: the energy bill with different PUE values.

5.4

The Full Bill

+10 XP

Training compute

training FLOPs ≈ 6 × N × D (N parameters, D tokens; Kaplan 2020, Hoffmann 2022). 6 = 2 FLOPs per MAC × 3 (forward + ~2× backward). Hoffmann et al.: ~20 tokens per parameter is a good balance; 7B with 1.4T tokens is exactly that.

Worked example: the running example's bill

256 chips, 200 TFLOP/s peak, 40% MFU, 400 W each, $2/chip-hour, PUE 1.1.

- FLOPs = $6 \times 7 \times 10^9 \times 1.4 \times 10^{12} = 5.88 \times 10^{22}$
- Rate = $256 \times 200 \times 10^{12} \times 0.40 = 2.048 \times 10^{16}$ FLOP/s
- Time ≈ 2.87×10^6 s ≈ 33.2 days
- Chip-hours ≈ 204,167 → cost ≈ $408,333
- Energy = 256 × 0.4 kW × 797.5 h ≈ 81.7 MWh; × 1.1 ≈ 89.8 MWh

🔑 REMEMBER THIS!

Raising MFU from 40% to 50% cuts time, cost and energy by 20% each with no new hardware. And this is ONE successful run: real projects also pay for failed runs, experiments (often several times the final run), data, staff and buildings.

5.4 · SEE IT

+10 XP

The Full Bill

MAMA FENG

🔍 Spot it! From FLOPs to dollars: follow each arrow of the bill.

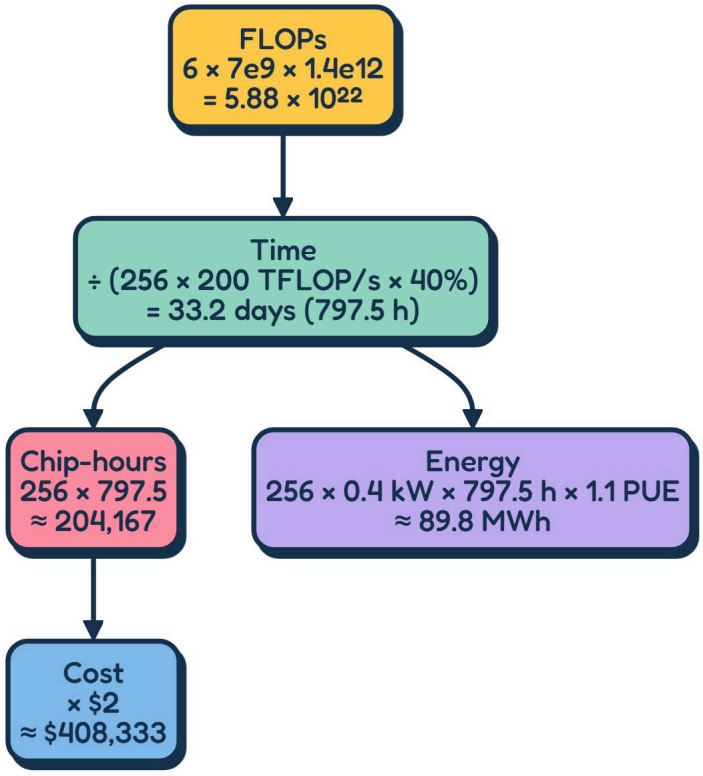


FIG Worked example: FLOPs, time, energy and cost of the training run.

5.4

Cost Per Plate

+10 XP

✨ READY, SET, COOK!

How much does ONE plate cost? 🧾 Let's do the maths!

🏷 Total cost of ownership

TCO = purchase price (spread over a few years) + electricity and cooling + networking + buildings + maintenance + staff.

🍽 Worked example: cost per million tokens

A chip at $2/hour serves 5,000 tokens/s.

5,000 × 3,600 ÷ 10^6 = 18 million tokens per hour

$2 ÷ 18 ≈ $0.11 per million tokens

Anything that raises tokens per second (batching, quantisation) or utilisation lowers this directly.

KITCHEN TERMINAL · PYTHON

```python
cost_hr, tps = 2.0, 5000
m_tok_hr = tps * 3600 / 1e6     # 18.0
print(round(cost_hr / m_tok_hr, 3))   # 0.111
```

🔑 REMEMBER THIS!

Training is a ONE-OFF cost; serving is a RECURRING cost that grows with every user. That is why many new hardware generations are designed with inference in mind.

5.4 · SEE IT

+10 XP

Cost Per Plate

MAMA FENG

🔍 Spot it! An idle chip still costs money. Compare 35% and 70% busy.

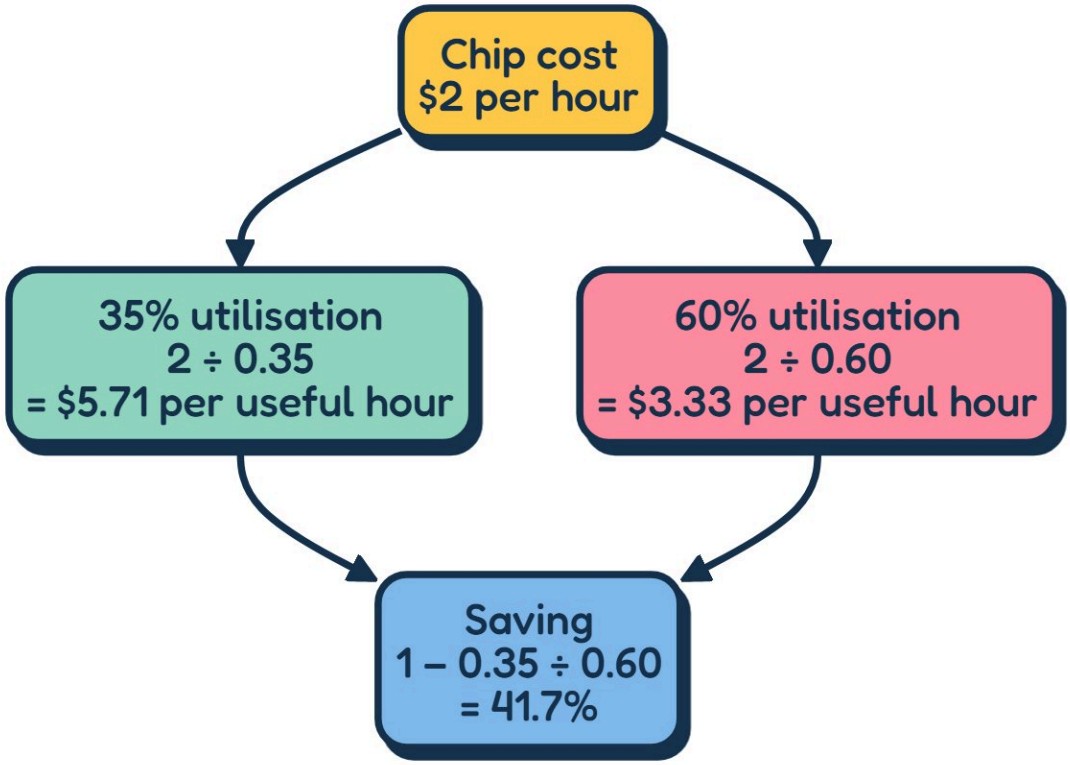


FIG Worked example: why utilisation is a cost issue.

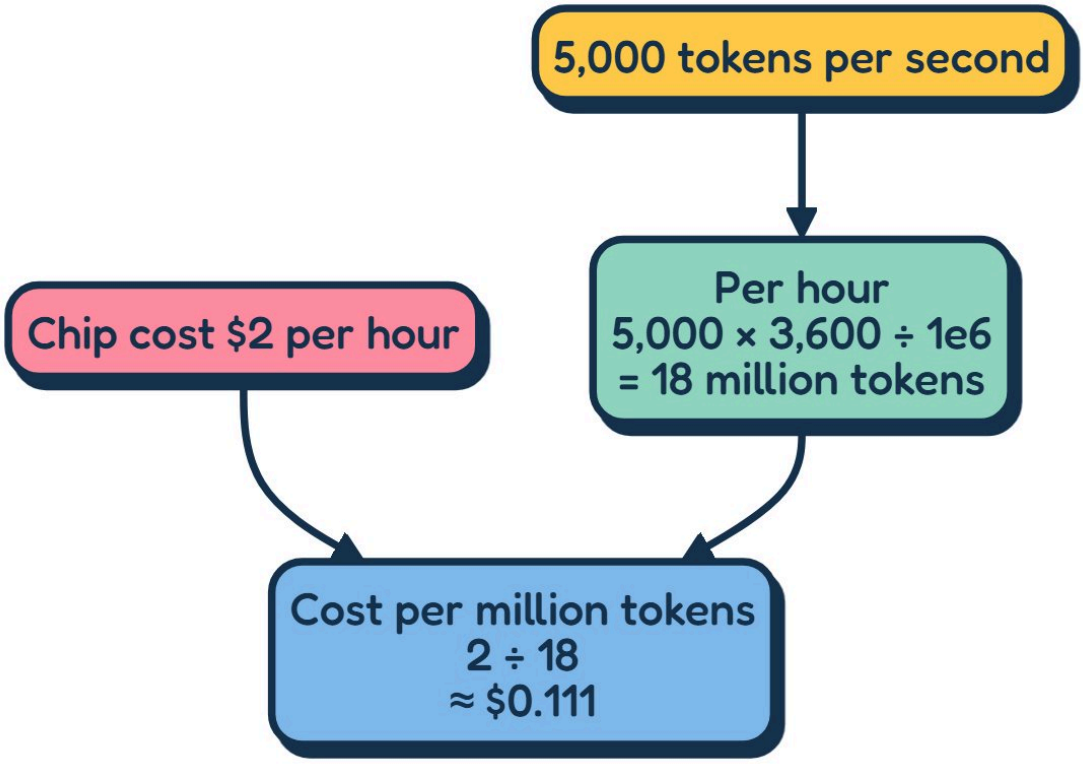


FIG The cost of serving 5,000 tokens per second.

5.4

When Stoves Break

+10 XP

Failures become routine

One chip may run for years. Thousands of chips fail every few hours, because failure rates add: system MTBF ≈ chip MTBF ÷ N.

Worked example: a 4,096-chip job

Chip MTBF 5 years = 43,800 h.

- System MTBF ≈ 43,800 ÷ 4,096 ≈ 10.7 hours: a month-long job is interrupted dozens of times
- **Defence:** CHECKPOINTING, saving the full training state so a failure restarts from the last save
- **Young (1974):** best interval ≈ √(2 × C × MTBF), C = time to write one checkpoint
- C = 2 min (0.0333 h): √(2 × 0.0333 × 10.7) ≈ 0.84 h ≈ 51 minutes, overhead ≈ 3.9%

Other scaling walls

The network (Recipe 02) • the memory wall (Recipe 01) • power and cooling a site can supply • the supply of chips and memory. Failures also come from links, power, cooling and bugs, so systems use spares, automatic detection of slow chips and fast restart.

5.4 · SEE IT

+10 XP

When Stoves Break

MAMA FENG

🔍 Spot it! Thousands of chips means something always breaks. How often?

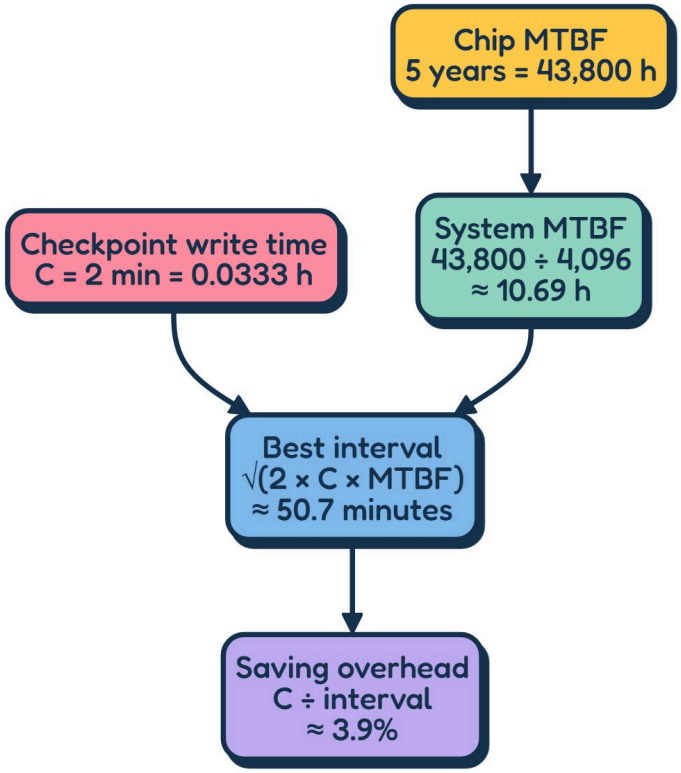


FIG Worked example: failures in a 4,096-chip job.

5.4

+10 XP

Every Lever Has A Side Effect

✨ READY, SET, COOK!

Every lever has a side effect! 🎛 Pull carefully!

LEVER	HELPS	MAY HURT
More chips	Time to train	Cost, power, failures, communication
Lower precision	Speed, energy, memory	Accuracy (needs care)
Larger batch	Utilisation, throughput	Serving latency; learning quality
Frequent checkpoints	Resilience	Time and storage
Cleaner or cheaper power site	Carbon, cost	Location constraints, network distance
Custom accelerator (ASIC)	Energy and cost per operation	Flexibility, software ecosystem, design cost

💡 MAMA'S SECRET TIP

Like seasoning: more salt helps flavour until it ruins the dish. Energy, cost and scalability are linked, and most levers that help one push pressure onto another.

☐ 🗺️ SIDE QUEST · MINI-GAME 5.4 +25 XP

Pay the Bills

🎯 Your Mission

Calculate facility energy, training time, system MTBF, checkpoint interval and serving cost.

🧺 Grab These First

Energy = load × PUE × hours. FLOPs = 6ND. MTBF_sys = MTBF_chip ÷ N. Interval = √(2 C MTBF). $/M tokens = $/h ÷ (tokens/s × 3,600 ÷ 10^6).

🔪 Let's Go! Your Turn

(1) 2 MW load: annual energy at PUE 1.2 and 1.5? Extra?

(2) 3B parameters, 60B tokens, 64 chips at 200 TFLOP/s, 40% MFU. FLOPs and hours?

(3) 1,024 chips, chip MTBF 4 years. System MTBF?

(4) Checkpoint takes 3 min. Best interval (Young)?

(5) $3/hour chip serving 6,000 tokens/s. Cost per million tokens?

👀 What Did You Spot?

Keep units in hours for (3) and (4).

💭 Think It Over

Which lever would you pull first to cut the training bill, and what could it hurt?

🔒 **Mama's feedback in the Answer Vault · V-5-4** · page 226 · try the task first, then +25 XP

SERVE THE CUSTOMERS · TICKET #5.4

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.4) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. A PUE of 1.0 means the facility uses no energy beyond the computing equipment. ♡ ♡ ♡

- (T) True
- (F) False

2. If each of 4,096 chips has an MTBF of 43,800 hours, the whole system fails about every 10.7 hours. ♡ ♡ ♡

- (T) True
- (F) False

3. Which changes usually RAISE the MFU (model FLOPs utilisation) of a training job? ♡ ♡ ♡

- (A) Fusing operations to cut memory traffic
- (B) Overlapping communication with computation
- (C) Making every step wait for a slow data loader
- (D) Using a larger batch per chip

4. What does "MFU" mean? ♡ ♡ ♡

- (A) Pausing a lower-priority job for a higher-priority one.
- (B) Useful model FLOP/s ÷ hardware peak FLOP/s.
- (C) Compiler that maps models onto TPUs and other devices.
- (D) Direct TPU chip links / optical switches re-patched by software.

5. Which kitchen word means: "Total cost of ownership: purchase, energy, cooling, network, buildings, staff."? ♡ ♡ ♡

- (A) TCO
- (B) Strong / weak scaling
- (C) MFU
- (D) Data parallelism

TICKET #5.5 · RECIPE 05 · ROOFTOP BANQUET

5.5 Case study

Google's TPU infrastructure

COURSES ☐☐☐☐☐☐☐☐ RECORD ☆☆☆ BOUNTY 130 XP

5.5

Case Study: Google Tpus

+10 XP

✨ READY, SET, COOK!

Case file: Google TPUs! 🏭 Let's tour a giant AI factory.

🏭 The whole recipe in one system

Google's TPU infrastructure is this recipe in real life: a chip built around MACs and on-chip memory (Recipe 01), joined by a purpose-built network into pods of thousands of chips (5.2), programmed through a compiler (5.1), scheduled as a shared resource (5.3), and designed for energy, cost and failure (5.4).

GENERATION	MAIN IDEA
First TPU (2015)	Inference only; 8-bit matrix unit, 92 TOPS, 28 MiB on-chip buffer (Jouppi 2017)
From 2017	Training: bfloat16, HBM, chips wired into pods
TPU v4	Pods of 4,096 chips with optically reconfigurable links; embedding units (Jouppi 2023)
Later	Variants for serving and for the largest training; Ironwood (2025) aimed at inference at scale

⚠ SPEC NOTE

TPU generations and figures change quickly. Learn the DESIGN IDEAS; check Google Cloud's TPU documentation before quoting current specs.

5.5

From Chip To Pod

+10 XP

✨ READY, SET, COOK!

From one chip to a whole pod! 🧱 How big does it get?

LEVEL	WHAT IT IS	WHY IT EXISTS
Chip	Matrix units, vector unit, on-chip memory, HBM, network ports	Executes MACs efficiently
Tray / host	Four chips (v4) + a host computer	Accelerators need a CPU host
Cube	64 chips as a 4 × 4 × 4 cube	A unit of wiring and deployment
Pod	64 cubes = 4,096 chips	Largest tightly coupled unit
Slice	The part of a pod a job gets, e.g. 8 × 8 × 8	Many jobs share a pod

LEVEL	WHAT IT IS	WHY IT EXISTS
Multi-slice job	Several slices linked by the data-centre network	Training beyond one pod

💡 MAMA'S SECRET TIP

Chip = one cook. Tray = a station of four cooks with a manager. Cube = a kitchen. Pod = a whole food court of 64 kitchens. A slice = the kitchens rented for one banquet.

5.5

The 3d Torus

+10 XP

✨ READY, SET, COOK!

Donut-shaped networks?! 🍩 Meet the 3D torus.

Neighbours on every side

Each chip links DIRECTLY to six neighbours (up, down, left, right, front, back) with inter-chip interconnect (ICI). In a TORUS the links wrap around: the chip at one edge links to the opposite edge.

Why it suits AI: all-reduce from neighbour-to-neighbour ring transfers on short fast cables • no central switch bottleneck • wraparound shortens the longest path.

Worked example: how much does wrapping help?

Diameter = longest path in hops. Mesh: 3 × (side − 1). Torus: 3 × (side ÷ 2).

- 4 × 4 × 4 cube (64 chips): mesh 9, torus 6 hops
- 16 × 16 × 16 pod (4,096 chips): mesh 45, torus 24 hops

KITCHEN TERMINAL · PYTHON

```python
def diameter(side, wrap):
    return 3 * (side // 2 if wrap else side - 1)
for s in (4, 16):
    print(s**3, diameter(s, True), diameter(s, False))
```

5.5 · SEE IT

The 3d Torus

+10 XP

MAMA FENG

🔍 Spot it! Wraparound links shorten the trip. How many hops are saved?

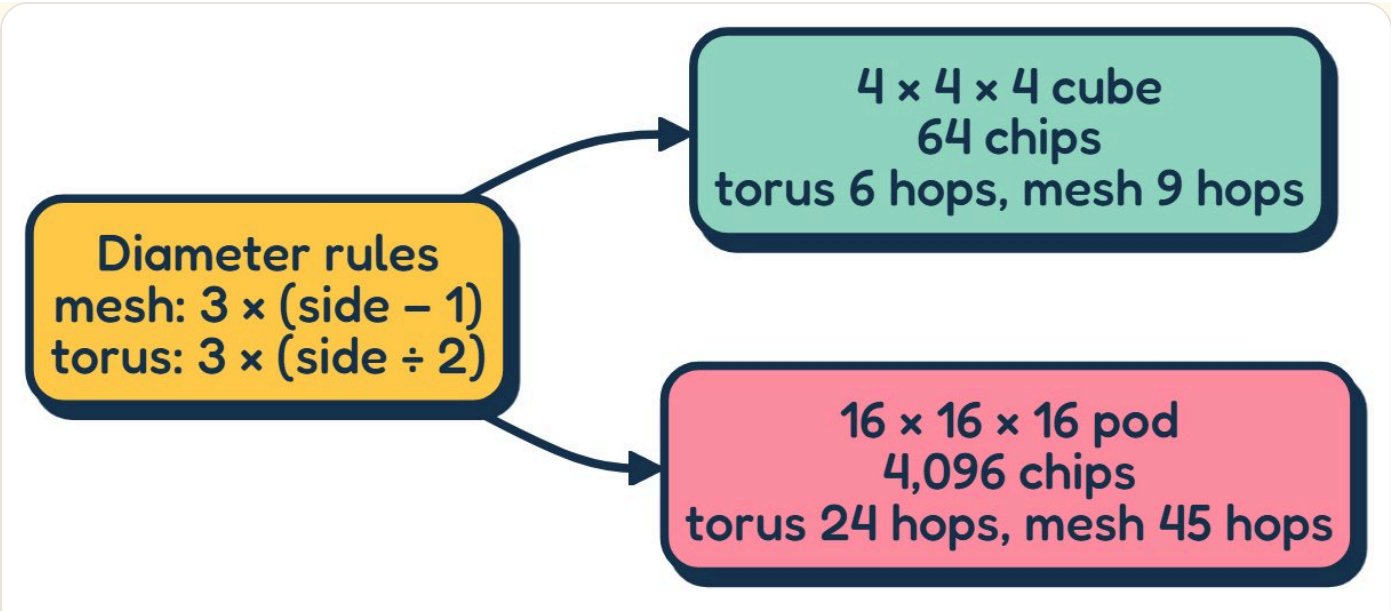


FIG Worked example: how much wraparound links help.

5.5

Optical Switches

+10 XP

✨ READY, SET, COOK!

Re-wiring with light! 💡 How optical switches reshape the pod.

Re-patching with light

TPU v4 doesn't wire cubes with fixed cables. The links between cubes pass through OPTICAL CIRCUIT SWITCHES (OCS) that steer light with tiny mirrors, so connections can be re-patched BY SOFTWARE (Jouppi et al. 2023).

1. **Availability:** route around a failed chip, tray or cube and build the slice from healthy cubes.
2. **Flexible slice shapes:** long and thin or cube-shaped, to suit the parallelism.
3. **Incremental deployment:** a pod works before every cube arrives, and can grow.

🔗 CONNECT THE DOTS

Same trade-off as Recipe 02: dedicated direct links give high bandwidth and low latency for predictable traffic; switched or reconfigurable networks give flexibility. TPU pods use direct links for training traffic and OCS + data-centre network for shape, failures and scale.

5.5

Software & Slices

+10 XP

✨ READY, SET, COOK!

Slice the pod like a cake! 🍰 Software hides the hard parts.

Hiding the hardware

- XLA compiles the graph for the TPU, fusing operations and assigning layouts.
- **Frameworks:** JAX, TensorFlow, PyTorch/XLA.
- **Sharding annotations:** the programmer says how to split each tensor; the compiler inserts the communication (all-reduce, all-gather) automatically.
- **Multi-slice training:** fast ICI inside a slice, slower data-centre network between slices.

Worked example: communicate hierarchically

Two slices of 256 chips, 14 GB gradients, 12.5 GB/s per chip between slices.

- **Naive:** every chip sends 14 GB slowly: 14 ÷ 12.5 ≈ 1.12 s
- **Hierarchical:** a reduce-scatter on fast ICI leaves each chip 14 ÷ 256 ≈ 0.055 GB; only that crosses: ≈ 4.4 ms
- 256× less slow-network traffic and time

🔑 REMEMBER THIS!

Do as much reduction as possible on the FASTEST links, and send only the minimum across the SLOWEST. It is Recipe 01's memory hierarchy (keep data near the compute) applied to a network.

5.5 +10 XP

Running the Fleet

✨ READY, SET, COOK!

Running a whole fleet! 🚄 Who goes where, and when?

Scheduling, reliability, energy

- **Scheduling:** slices allocated by quota and priority, placed as neighbours, gang-scheduled. With OCS any healthy cubes can form the requested shape: less fragmentation, higher utilisation.
- **Reliability:** health monitoring + optical rerouting + frequent checkpoints, so a failure costs minutes, not days.
- **Energy:** silicon spent on MACs and on-chip memory, not general-purpose control. Google reports fleet-wide PUE close to 1.1; recent TPUs use liquid cooling (check the latest figures).

STRENGTHS	LIMITATIONS
High performance per watt and per dollar on big matrix work	Less flexible than GPUs for unusual ops; fall-backs are slow
Co-design of chip, network, compiler, scheduler	Tied to XLA/JAX/PyTorch-XLA; porting from CUDA takes effort
Torus scales training to thousands of chips	Mostly via Google's cloud: one-provider dependence
Optical switching raises availability	Added complexity and hardware
Generations tuned for training or serving	Peak ≠ real; measure your own workload

5.5 +10 XP

Case Study Q & A

✨ READY, SET, COOK!

Case study quiz! 🙋 Show what you've learned!

❓ Questions

1. Why is a 3D torus a good match for all-reduce and tensor parallelism?
2. **In three sentences:** how do optical switches improve availability?
3. Diameter of an 8 × 8 × 8 mesh and torus? Percentage shortening?
4. Why does hierarchical communication cut slow-network time by exactly the chips per slice?
5. Three Recipe 01 ideas inside the TPU system.
6. A start-up's PyTorch model uses a custom GPU kernel: two things to check before moving to TPUs.

🔒 **Answers in the Answer Vault · V-5-5** · page 226 · no peeking! +10 XP

SERVE THE CUSTOMERS · TICKET #5.5

Customer orders

⭐ ⭐ ⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.5) and colour the hearts of every happy customer. Half happy = ⭐ ⭐ · all happy = ⭐ ⭐ ⭐.

1. The diameter (longest shortest path) of a 16 × 16 × 16 torus is: ♡ ♡ ♡

- (A) 16 hops
- (B) 24 hops
- (C) 45 hops
- (D) 48 hops

2. What does “PUE” mean? ♡ ♡ ♡

- (A) Useful model FLOP/s ÷ hardware peak FLOP/s.
- (B) Grid whose edges wrap around / longest shortest path in hops.
- (C) Sums values across chips and gives every chip the result.
- (D) Total facility energy ÷ computing equipment energy.

3. Which kitchen word means: “Sums values across chips and gives every chip the result.”? ♡ ♡ ♡

- (A) ICI / OCS
- (B) All-reduce
- (C) Checkpointing
- (D) TCO

4. What does “Tensor parallelism” mean? ♡ ♡ ♡

- (A) Useful model FLOP/s ÷ hardware peak FLOP/s.
- (B) Sums values across chips and gives every chip the result.
- (C) Saving full training state to restart after a failure.
- (D) Individual matrices inside a layer split across chips.

TICKET #5.L · RECIPE 05 · ROOFTOP BANQUET

LAB

Kitchen Lab

Run A Real Tpu

COURSES ▢▢▢▢ RECORD ☆☆☆ BOUNTY 90 XP

KITCHEN LAB

Run A Real TPU

+10 XP

🌐 Live platform: Google Colab

Open colab.research.google.com, New notebook → Runtime → Change runtime type → TPU (or T4 GPU if TPUs are busy). Run:

import jax, time; import jax.numpy as jnp

x = jnp.ones((4096, 4096))

f = jax.jit(lambda a: (a @ a).sum())

f(x).block_until_ready(); t = time.time(); f(x).block_until_ready(); print(time.time() - t)

Then change the size to 1024 and 8192.

Version: Colab TPU runtime (v5e-1 or v2-8, whichever is offered) or T4 GPU; JAX is preinstalled. Check with print(jax.devices()).

🎯 Goal · Observe · Reflect

Goal: see compilation, warm-up and scaling on real accelerators.

Observe: Why is the FIRST call slower (compilation by XLA)? How does time grow when the size doubles (FLOPs grow 8×)? Compute TFLOP/s = $2 \times n^3 \div$ time and compare with the chip's peak: what MFU do you get?

Reflect: Which page of this recipe explains the gap?

Offline fallback

Use the "Running the Chain" paper sheet: with the numbers from 5.4 (256 chips, 40% MFU), recompute days, chip-hours, $ and MWh for MFU = 30%, 50% and 60%, and plot cost against MFU. Then repeat with 512 chips and an all-reduce of 0.28 s.

⭐ Mama's Feedback (expected result)

The first call takes about 0.1-1 s because XLA compiles the function; later calls take milliseconds.

Size 4,096 = $2 \times 4{,}096^3 \approx 137$ GFLOP: on a T4 about 20-40 ms (≈ 4 TFLOP/s, about half its FP32 peak); a TPU is faster still.

Doubling the size gives close to 8× the time only for large sizes; at 1,024 fixed overhead dominates. A single timed call is noisy, so run it 3 times. MFU well below 100% is normal: memory, overhead and communication (5.4).

EXERCISE

+10 XP

Final round

✨ READY, SET, COOK!

Final round! 🎯 Earn your Recipe 05 stamp and unlock the dessert!

EXERCISE

Question 1: Does it fit?

Task: A 20B-parameter model is trained with 16 bytes per parameter on 80 GB chips. State size? Minimum chips? Memory per chip if sharded over 512 chips?

Question 2: Chain bill

Task: Training needs 1.2×10^{22} FLOPs on 128 chips at 300 TFLOP/s peak and 45% MFU, $2.50/chip-hour. Time in days and cost?

Question 3: Torus trip

Task: Diameter of a 6 × 6 × 6 mesh and torus? Then explain in one sentence why fewer hops matter for an all-reduce.

🔒 **Answers in the Answer Vault · V-5-6** · page 226 · no peeking! +10 XP

WATCH & LEARN

+10 XP

Video picks

A friendly introduction to distributed training

Creator: TensorFlow (ML Tech Talks) · 24:20

Link: youtube.com/watch?v=S1tN9a4Proc

Why watch: Data vs model parallelism and synchronous all-reduce, as in 5.2.

Timestamps: 0:37 why distributed · 1:49 data vs model parallelism · 6:05 synchronous data parallelism · 18:20 asynchronous

Task: At 6:05, connect the ring all-reduce to traffic = $2(N-1)/N \times S$.

Limitations: Framework-focused (TensorFlow); no cost or energy discussion.

Scaling AI with Google Cloud's TPUs

Creator: Google Cloud Tech · 5:54

Link: youtube.com/watch?v=8zBaa20mvoc

Why watch: HBM, inter-chip interconnect, pods and multislice: the TPU case study of 5.5 in 6 minutes.

Timestamps: 0:26 TPUs explained · 1:11 HBM · 2:18 ICI and resiliency · 3:44 multislice · 4:40 frameworks

Task: At 3:44, explain why multislice needs hierarchical communication (5.5).

Limitations: Vendor video; performance claims are not independently tested.

WATCH & LEARN

Video picks

+10 XP

Inside a Google Cloud TPU Data Center

Creator: Google Cloud · 2:42

Link: youtube.com/watch?v=FsxthdQ_sL4

Why watch: See real TPU racks and liquid cooling: the energy and scaling challenges of 5.4.

Timestamps: Whole video (3 min)

Task: Name two parts of the building that exist only to remove heat or supply power.

Limitations: Short marketing tour; no numbers for PUE or cost.

SERVE THE CUSTOMERS · TICKET #5.L

Customer orders

⭐⭐⭐

Each customer orders a question. Circle the dish (answer), check it in the Answer Vault (V-5-Q5.L) and colour the hearts of every happy customer. Half happy = ⭐⭐ · all happy = ⭐⭐⭐.

1. In Kitchen Lab 5, the FIRST call of a jax.jit function is much slower than later calls mainly because XLA compiles it. ♡ ♡ ♡

- (T) True
- (F) False

2. Kitchen Lab 5: you change the matrix size in the JAX program from 4,096 to 8,192. By what factor do the FLOPs of a @ a grow? ♡ ♡ ♡

- (A) 16×
- (B) 2×
- (C) 4×
- (D) 8×

3. What does "Data parallelism" mean? ♡ ♡ ♡

- (A) Batch split across chips, each with a model copy (or shard).
- (B) Merging several operations into one pass to avoid memory traffic.
- (C) Total facility energy ÷ computing equipment energy.
- (D) Mean time between failures; system ≈ chip MTBF ÷ chip count.

4. Which kitchen word means: "Sums values across chips and gives every chip the result."? ♡ ♡ ♡

- (A) Preemption
- (B) All-reduce
- (C) Continuous batching
- (D) PUE

TICKET #5.B · RUSH HOUR IN THE ROOFTOP BANQUET

Rush Hour!

*Last in line, the trouble customer **Meltdown Mainframe**: "Four thousand chips. One failure every eleven hours. Tick, tock."*

HAPPY CUSTOMERS ☐☐☐☐☐☐☐☐☐☐☐☐☐☐

GRADE S · A · B · C · D

Rules: 14 orders, no going back. Colour one box for each right dish (check in the Answer Vault, V-5-B). 9 or more happy customers = you survive the rush: +150 XP and the **Cloud Banquet** recipe card. Grade: S = 13–14, A = 12, B = 10–11, C = 9. Questions 13 and 14 are the trouble customer's double order.

1. The diameter (longest shortest path) of a 16 × 16 × 16 torus is: ♡ ♡ ♡

- (A) 16 hops
- (B) 24 hops
- (C) 45 hops
- (D) 48 hops

2. In Kitchen Lab 5, the FIRST call of a jax.jit function is much slower than later calls mainly because XLA compiles it. ♡ ♡ ♡

- (T) True
- (F) False

3. Kitchen Lab 5: you change the matrix size in the JAX program from 4,096 to 8,192. By what factor do the FLOPs of a @ a grow? ♡ ♡ ♡

- (A) 16×
- (B) 2×
- (C) 4×
- (D) 8×

4. In data parallelism, each chip stores only a slice of the model's weights. ♡ ♡ ♡

- (T) True
- (F) False

5. Training a 7B-parameter model with mixed-precision Adam needs about 16 bytes per parameter. Total training state: ♡ ♡ ♡

- (A) 14 GB
- (B) 56 GB
- (C) 112 GB
- (D) 224 GB

6. Fusing three element-wise operations on a 2 MB tensor cuts memory traffic from 12 MB to 4 MB. ♡ ♡ ♡

- (T) True
- (F) False

7. A 7 GB INT8 model is served on a chip with 1 TB/s memory bandwidth. The memory-bound limit for one user is about: ♡ ♡ ♡

- (A) 1,000 tokens/s
- (B) 4,571 tokens/s
- (C) 14 tokens/s
- (D) 143 tokens/s

8. If each of 4,096 chips has an MTBF of 43,800 hours, the whole system fails about every 10.7 hours. ♡ ♡ ♡

- (T) True
- (F) False

9. A PUE of 1.0 means the facility uses no energy beyond the computing equipment. ♡ ♡ ♡

- (T) True
- (F) False

10. Which changes usually RAISE the MFU (model FLOPs utilisation) of a training job? ♡ ♡ ♡

- (A) Fusing operations to cut memory traffic
- (B) Overlapping communication with computation
- (C) Making every step wait for a slow data loader
- (D) Using a larger batch per chip

11. Which kitchen word means: "Fixed problem / problem grows with chip count."? ♡ ♡ ♡

- (A) Collective operation
- (B) Strong / weak scaling
- (C) Data parallelism
- (D) MTBF

12. What does "Gang scheduling" mean? ♡ ♡ ♡

- (A) Start a job only when all its chips are free at once.
- (B) Sums values across chips and gives every chip the result.
- (C) Batch split across chips, each with a model copy (or shard).
- (D) Saving full training state to restart after a failure.

13. Which kitchen word means: "Splitting a tensor or training state across chips."?

- (A) Sharding
- (B) TCO
- (C) Computation graph
- (D) Operator fusion

14. What does "Preemption" mean?

- (A) Grid whose edges wrap around / longest shortest path in hops.
- (B) Stored attention state that grows with conversation length.
- (C) Pausing a lower-priority job for a higher-priority one.
- (D) Model drawn as operations (nodes) and tensors (arrows).

TICKET #V5 · RECIPE 05

Answer Vault · Running the Chain

Only open after trying! Every vault entry you check is worth XP on your licence.

V-5-1

5.1 · One Trip To the Fridge

Mama's feedback · One Trip To the Fridge

(1) 8 MB tensor: 64 MB unfused, 16 MB fused, 75% saved. (2) 35%. (3) 4 switches. (4) Tiles must fit on-chip: too big spills to slow memory with repeated traffic; too small gives little reuse and extra overhead. Reflect: the software (recipe planning, order, fusion) decides how well the hardware is used.

V-5-Q5.1

Customer orders · ticket #5.1

1. True · Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB.

2. D · Compiler that maps models onto TPUs and other devices. · XLA: Compiler that maps models onto TPUs and other devices.

3. A · Scaling efficiency · Scaling efficiency: Speed-up ÷ number of chips.

4. D · Merging several operations into one pass to avoid memory traffic. · Operator fusion: Merging several operations into one pass to avoid memory traffic.

V-5-2

5.2 · Split the Order

Mama's feedback · Split the Order

(1) 208 GB → 2.6 → at least 3 chips (more with activations). (2) 2 GB gradients → 3.5 GB → 0.07 s. (3) ≈ 26.7×; ≈ 83.3%. (4) (a) data (b) tensor (c) pipeline. Reflect: tensor parallelism exchanges partial results inside every layer, many small messages that would stall on slow links.

V-5-Q5.2

Customer orders · ticket #5.2

1. False · Correct: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism.

2. C · 112 GB · $7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split.

3. B · Sums values across chips and gives every chip the result. · All-reduce: Sums values across chips and gives every chip the result.

4. B · Strong / weak scaling · Strong / weak scaling: Fixed problem / problem grows with chip count.

V-5-3

5.3 · Rush Hour

Mama's feedback · Rush Hour

(1) ≈ 26.7 s. (2) 13 GB ÷ 2,000 GB/s = 6.5 ms → ≈ 153.8 tokens/s. (3) "Chips wait while data loads" (input pipeline): try prefetching, more loader workers or faster storage. (4) $12 and $6. Reflect: the chips are already waiting for data; more chips would just wait too.

V-5-Q5.3

Customer orders · ticket #5.3

1. D · 143 tokens/s · Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s.

2. A · Recompute activations instead of storing them: ~33% more compute. · Recomputation: Recompute activations instead of storing them: ~33% more compute.

3. D · Scaling efficiency · Scaling efficiency: Speed-up ÷ number of chips.

4. B · Useful model FLOP/s ÷ hardware peak FLOP/s. · MFU: Useful model FLOP/s ÷ hardware peak FLOP/s.

V-5-4

5.4 · Pay the Bills

Mama's feedback · Pay the Bills

(1) 21,024 vs 26,280 MWh → 5,256 MWh extra. (2) 1.08×10^{21} FLOPs; 5.12×10^{15} FLOP/s → ≈ 58.6 h. (3) 35,040 ÷ 1,024 ≈ 34.2 h. (4) √(2 × 0.05 × 34.2) ≈ 1.85 h ≈ 111 min (≈ 2.7% overhead). (5) 21.6 M tokens/h → ≈ $0.14. Reflect (example): raise MFU (fusion, bigger batches): free speed, but bigger batches can hurt learning.

V-5-Q5.4

Customer orders · ticket #5.4

1. True · PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion.

2. True · 43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints.

3. undefined · · A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls.

4. B · Useful model FLOP/s ÷ hardware peak FLOP/s. · MFU: Useful model FLOP/s ÷ hardware peak FLOP/s.

5. A · TCO · TCO: Total cost of ownership: purchase, energy, cooling, network, buildings, staff.

V-5-5

5.5 · Case Study Q & A

Answers · Case Study Q & A

1. **Six direct neighbours:** ring collectives and neighbour-heavy tensor traffic use short fast links; no central switch; wraparound keeps paths short.
2. Switches connect cubes and are re-patched by software; on a failure they route around it and build the slice from healthy cubes; spares are used at once, raising availability and utilisation.
3. Mesh 21, torus 12 → ≈ 43% shorter.
4. After the reduce-scatter each of 256 chips holds 1/256 of the sum; only that crosses.
5. MAC matrix units; low precision (bfloat16/INT8); on-chip memory + HBM; roofline batching; compiler stack; ASIC trade-off.
6. A compiler-supported equivalent of the kernel; porting effort to JAX or PyTorch/XLA; measured MFU and cost per token; provider dependence.

V-5-Q5.5

Customer orders · ticket #5.5

1. B · 24 hops · Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound.

2. D · Total facility energy ÷ computing equipment energy. · PUE: Total facility energy ÷ computing equipment energy.

3. B · All-reduce · All-reduce: Sums values across chips and gives every chip the result.

4. D · Individual matrices inside a layer split across chips. · Tensor parallelism: Individual matrices inside a layer split across chips.

V-5-6

5.L · Final round

Answers · Final round

Q1: $20 \times 10^9 \times 16$ B = 320 GB → 320 ÷ 80 = 4 chips minimum (more with activations); sharded 512 ways: 320 ÷ 512 ≈ 0.63 GB per chip.

Q2: rate = $128 \times 300 \times 10^{12} \times 0.45 = 1.728 \times 10^{16}$ FLOP/s → $1.2 \times 10^{22} \div 1.728 \times 10^{16}$ ≈ 694,444 s ≈ 8.0 days (192.9 h). Chip-hours ≈ 128 × 192.9 ≈ 24,691 → ≈ $61,728.

Q3: mesh 3 × 5 = 15 hops; torus 3 × 3 = 9 hops. Each hop adds latency to every message, and collectives send many messages, so shorter paths finish the all-reduce sooner.

V-5-Q5.L

Customer orders · ticket #5.L

1. True · The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds.

2. D · 8× · Matrix multiply = $2n^3$ FLOPs; doubling n gives $2^3 = 8\times$. For large sizes the measured time grows close to 8× too.

3. A · Batch split across chips, each with a model copy (or shard). · Data parallelism: Batch split across chips, each with a model copy (or shard).

4. B · All-reduce · All-reduce: Sums values across chips and gives every chip the result.

V-5-B

Rush hour · Rooftop Banquet

1. B · 24 hops · Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound.

2. True · The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds.

3. D · 8× · Matrix multiply = $2n^3$ FLOPs; doubling n gives $2^3 = 8\times$. For large sizes the measured time grows close to 8× too.

4. False · Correct: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism.

5. C · 112 GB · $7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split.

6. True · Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB.

7. D · 143 tokens/s · Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s.

8. True · 43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints.

9. True · PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion.

10. undefined · · A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls.

11. B · Strong / weak scaling · Strong / weak scaling: Fixed problem / problem grows with chip count.

12. A · Start a job only when all its chips are free at once. · Gang scheduling: Start a job only when all its chips are free at once.

13. A · Sharding · Sharding: Splitting a tensor or training state across chips.

14. C · Pausing a lower-priority job for a higher-priority one. · Preemption: Pausing a lower-priority job for a higher-priority one.

CENTRE STAGE · VIP CRITIC · TICKET #EXAM · STAGE 2

The Professor's Final Exam

ALL 60 STAGE 2 TEST BANK QUESTIONS · A THREE-COURSE TASTING MENU

THE PROFESSOR · VIP FOOD CRITIC

Good evening. I am tonight's critic. Course 1 is the Starter (true or false), Course 2 the Main (multiple choice), Course 3 Dessert: open questions that you mark yourself with my model answers. Score 60% to pass. Your Stage 2 review starts… now.

CRITIC MOOD

SCORE

___ / 60

GRADE

S ≥ 54 · A ≥ 48 · B ≥ 39 · C ≥ 36

COURSE 1 Starter · true or false

1. One MAC (multiply-accumulate) counts as two FLOPs.

- (T) True
- (F) False

2. A TPU is more flexible than a CPU for general programs.

- (T) True
- (F) False

3. If 90% of a job can be parallelised, 100 cooks give a speed-up of 100×.

- (T) True
- (F) False

4. At batch size 1, a large layer on a typical accelerator is usually memory-bound.

- (T) True
- (F) False

5. In Kitchen Lab 1 (Race the Crew), the GPU's speed-up over the CPU is LARGER for 256 × 256 matrices than for 4,096 × 4,096 matrices.

- (T) True
- (F) False

6. Encapsulation means each layer adds its own header to the data from the layer above.

- (T) True
- (F) False

7. UDP guarantees that every packet arrives in order.

- (T) True
- (F) False

8. Bandwidth and latency are two names for the same quantity.

- (T) True
- (F) False

9. In a ring all-reduce, the traffic per chip approaches 2 × S as the number of chips grows.

- (T) True
- (F) False

10. In Kitchen Lab 2, a server answers ping with an average round-trip time of 12 ms. Using the speed-of-light floor of 5 µs per km (one way), the server can be at most about 1,200 km away.

- (T) True
- (F) False

11. With rewards [0, 0, 0, 10] and $\gamma = 0.9$, the discounted return is 7.29.

- (T) True
- (F) False

12. A discount factor γ close to 0 makes the agent far-sighted.

- (T) True
- (F) False

13. Reward hacking means the agent earns a high reward without doing the intended task.

- (T) True
- (F) False

14. A purely greedy agent can get stuck on a poor option forever.

- (T) True
- (F) False

15. In Kitchen Lab 3 (GridWorld), setting epsilon = 0 guarantees that the agent eventually finds the best route.

- (T) True
- (F) False

16. A single camera directly measures the distance to every pixel.

- (T) True
- (F) False

17. A LiDAR pulse returning after 400 ns means the object is about 60 m away.

- (T) True
- (F) False

18. Radar measures the speed of an object directly using the Doppler effect.

- (T) True
- (F) False

19. Two sensors that fail for the same reason (for example fog) give the same benefit as two independent sensors.

- (T) True
- (F) False

20. In Kitchen Lab 4 (phyphox), a phone lying still on a table shows an acceleration (without g) of exactly 0 m/s^2.

- (T) True
- (F) False

21. Fusing three element-wise operations on a 2 MB tensor cuts memory traffic from 12 MB to 4 MB.

- (T) True
- (F) False

22. In data parallelism, each chip stores only a slice of the model's weights.

- (T) True
- (F) False

23. A PUE of 1.0 means the facility uses no energy beyond the computing equipment.

- (T) True
- (F) False

24. If each of 4,096 chips has an MTBF of 43,800 hours, the whole system fails about every 10.7 hours.

- (T) True
- (F) False

25. In Kitchen Lab 5, the FIRST call of a jax.jit function is much slower than later calls mainly because XLA compiles it.

- (T) True
- (F) False

COURSE 2 Main · multiple choice

26. Weights [2, −1, 0.5], inputs [1, 4, 6]. What is the neuron output (no bias)?

- (A) 1.0
- (B) 3
- (C) 5
- (D) −1

27. Which chip best fits a voice assistant in a small battery-powered speaker?

- (A) GPU
- (B) Server CPU
- (C) NPU
- (D) TPU pod

28. In a 2 × 2 systolic array with W = [[1, 2], [3, 4]] and x = [5, 6], the output y = W·x is:

- (A) [12, 30]
- (B) [23, 34]
- (C) [11, 23]
- (D) [17, 39]

29. A chip has 10 TFLOP/s peak and 500 GB/s memory bandwidth. Its ridge point is:

- (A) 2 FLOP/byte
- (B) 20 FLOP/byte
- (C) 50 FLOP/byte
- (D) 5,000 FLOP/byte

30. Kitchen Lab 1: ten 4,096 × 4,096 matrix multiplications (about 687 billion MACs in total) take 0.25 s on a T4 GPU. What throughput did the GPU reach?

- (A) 11 TFLOP/s
- (B) 0.07 TFLOP/s
- (C) 2.7 TFLOP/s
- (D) 5.5 TFLOP/s

31. Which of these does TCP provide that UDP does NOT?

- (A) Acknowledgement of delivered data
- (B) In-order delivery
- (C) Port numbers to reach the right program
- (D) Retransmission of lost packets

32. Which protocol is built on UDP to speed up connection set-up for the web?

- (A) QUIC
- (B) ARP
- (C) SMTP
- (D) FTP

33. A factory camera must stop a machine within 15 ms. The best place to run the model is:

- (A) A distant cloud region
- (B) On or near the machine (edge)
- (C) A user's laptop at home
- (D) Any server; latency does not matter

34. Tail latency refers to:

- (A) Upload speed
- (B) The average delay
- (C) The slowest small fraction of requests
- (D) The delay of the last packet only

35. Multi-access edge computing (MEC) in 5G mainly helps AI services by:

- (A) Putting computing close to the base station
- (B) Replacing TCP
- (C) Removing the need for antennas
- (D) Increasing screen resolution

36. A corridor of squares 1-2-3 leads to the goal, one step after square 3 (reward 1). With γ = 0.9, V(1) equals:

- (A) 1.0
- (B) 0.729
- (C) 0.81
- (D) 0.9

37. Q = 0.2, r = 1, max Q(s′) = 0.5, α = 0.1, γ = 0.9. After one Q-learning update the new Q is:

- (A) 0.45
- (B) 1.45
- (C) 0.245
- (D) 0.325

38. Which of these rewards are DENSE (given at almost every step)?

- (A) +speed every step
- (B) +1 only on reaching the charging station
- (C) −energy used every step
- (D) +1 only for winning the game

39. A bandit has arms that pay 0.2, 0.5 and 0.8 on average. The agent knows the best arm and uses ε = 0.1 (when exploring, it picks one of the three arms at random). Expected reward per pull:

- (A) 0.50
- (B) 0.77
- (C) 0.80
- (D) 0.74

40. Domain randomisation helps sim-to-real transfer by:

- (A) Removing the reward
- (B) Using only real data
- (C) Making the simulator perfect
- (D) Training on many varied simulated conditions

41. Which sensors measure distance DIRECTLY?

- (A) LiDAR
- (B) A single camera
- (C) Radar
- (D) Ultrasonic

42. At 100 km/h, how far does a car travel during 0.3 s of processing latency?

- (A) 8.33 m
- (B) 30 m
- (C) 64.3 m
- (D) 2.78 m

43. Radar says 50.0 m (σ = 0.5 m) and a camera says 52.0 m (σ = 2.0 m). The inverse-variance fused estimate is about:

- (A) 52.0 m
- (B) 50.12 m
- (C) 51.0 m
- (D) 51.5 m

44. A stereo camera has f = 1,000 px and baseline B = 0.3 m. A disparity of 6 px means a depth of:

- (A) 300 m
- (B) 42.9 m
- (C) 50 m
- (D) 60 m

45. Under SAE J3016, a driver-assist that steers and brakes but needs constant human supervision is:

- (A) Level 2
- (B) Level 4
- (C) Level 5
- (D) Level 0

46. Training a 7B-parameter model with mixed-precision Adam needs about 16 bytes per parameter. Total training state:

- (A) 14 GB
- (B) 56 GB
- (C) 112 GB
- (D) 224 GB

47. A 7 GB INT8 model is served on a chip with 1 TB/s memory bandwidth. The memory-bound limit for one user is about:

- (A) 1,000 tokens/s
- (B) 4,571 tokens/s
- (C) 14 tokens/s
- (D) 143 tokens/s

48. The diameter (longest shortest path) of a 16 × 16 × 16 torus is:

- (A) 16 hops
- (B) 24 hops
- (C) 45 hops
- (D) 48 hops

49. Which changes usually RAISE the MFU (model FLOPs utilisation) of a training job?

- (A) Fusing operations to cut memory traffic
- (B) Overlapping communication with computation
- (C) Making every step wait for a slow data loader
- (D) Using a larger batch per chip

50. Kitchen Lab 5: you change the matrix size in the JAX program from 4,096 to 8,192. By what factor do the FLOPs of a @ a grow?

- Ⓐ 16×
- Ⓑ 2×
- Ⓒ 4×
- Ⓓ 8×

COURSE 3 Dessert · open questions

51. A layer has 512 inputs and 2,048 outputs, with FP16 weights, on a chip with 10 TFLOP/s and 500 GB/s. Compute the arithmetic intensity at batch 1 and at batch 32, decide whether each is memory-bound or compute-bound, estimate the achieved TFLOP/s, and explain in your own words why batching helps.

52. Evaluate running Face ID on the phone's Neural Engine instead of in the cloud. Give two advantages and two limitations of on-device AI, and decide whether the same choice would suit a large chatbot model. Justify your decision.

53. A hospital wants an AI that reads X-rays in under 2 seconds and keeps patient data private, but the model needs a large GPU. Recommend cloud, edge (on-premises) or hybrid, and justify the design with latency, bandwidth, privacy and cost.

54. 14 GB of gradients are all-reduced across 256 chips with 100 GB/s per chip, and one compute step takes 8.2 s. Calculate the traffic per chip, the communication time and the scaling efficiency, then explain one way to improve it and why it works.

55. Design a reward for a home cleaning robot. Show how a naive reward could be hacked, propose a better reward, say whether it is sparse or dense, and compare the two rewards: which behaviour would each one produce?

56. A student runs Kitchen Lab 3 (GridWorld TD learning) with epsilon = 0. After 2,000 steps the agent still walks the same long route; the arrows on squares it never visited still show their starting values. Explain why this plan failed, what the student should change, and what they should expect with epsilon = 0.5. Connect the result to the greedy agent in the bandit simulation and compare ε-greedy with UCB as a fix.

57. A car at 50 km/h has 0.15 s sensing-and-processing latency and brakes at 7 m/s². Compute the total stopping distance and the impact speed if a ball is detected 15 m ahead. What does this show about sensor range and latency when you design a safety system?

58. In the offline version of Kitchen Lab 4, two students stand 20 m apart. One claps; the other starts a stopwatch on SEEING the clap and stops it on HEARING it. Five trials give t = 0.21, 0.08, 0.15, 0.30 and 0.11 s. Compute the speed of sound from each trial and from the mean time, compare with 343 m/s, and explain why single results scatter so much. Why does averaging help, what can averaging NOT fix, and how would you improve the experiment?

59. Estimate the training time, cost and energy for a 7B model on 1.4 trillion tokens with 256 chips (200 TFLOP/s each, 40% MFU, $2 per chip-hour, 400 W per chip, PUE 1.1). Use FLOPs ≈ 6 × parameters × tokens. Which single change would cut all three numbers, and why?

60. Kitchen Lab 5 runs this JAX program on a Colab T4 GPU (FP32 peak about 8.1 TFLOP/s): f = jax.jit(lambda a: (a @ a).sum()) Measured times: size 1,024: first call 0.62 s, second call 0.0021 s; size 4,096: 0.034 s; size 8,192: 0.27 s. Explain the slow first call, compute the TFLOP/s and MFU for each size, and explain why the time ratio from 1,024 to 4,096 is far from the FLOP ratio while 4,096 → 8,192 is close to 8×. Predict the time for size 16,384.

TICKET #V9 · FINAL EXAM

Answer Vault · Final Exam

Mark yourself honestly: open answers score 1 (nailed it), ½ (partly) or 0.

1. True · One multiply plus one add = 2 FLOPs.

2. False · Correct: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work.

3. False · Correct: Amdahl's law gives 1 ÷ (0.1 + 0.9/100) ≈ 9.2×; the limit is 10×, set by the serial 10%.

4. True · Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte.

5. False · Correct: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster.

6. True · Like boxes inside boxes: application data → TCP/UDP header → IP header → link header.

7. False · Correct: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering.

8. False · Correct: bandwidth is how much data per second; latency is how long one message takes to arrive.

9. True · Traffic = $2(N - 1)/N \times S$, which tends to 2S for large N.

10. True · One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer.

11. True · $0.9^3 \times 10 = 7.29$.

12. False · Correct: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted.

13. True · Example: the boat-racing agent circling to hit respawning targets instead of finishing the race.

14. True · It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation).

15. False · Correct: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route.

16. False · Correct: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly.

17. True · $d = c \times t \div 2 = 3 \times 10^8 \times 4 \times 10^{-7} \div 2 = 60$ m.

18. True · The frequency shift of the echo gives the relative speed in one measurement.

19. False · Correct: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors.

20. False · Correct: the reading jitters around 0 (about ±0.05 m/s^2) and is never exactly zero; that jitter is sensor noise σ.

21. True · Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB.

22. False · Correct: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism.

23. True · PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion.

24. True · 43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints.

25. True · The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds.

26. A · 1.0 · $2 \times 1 + (-1) \times 4 + 0.5 \times 6 = 2 - 4 + 3 = 1.0$.

27. C · NPU · An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power.

28. D · [17, 39] · $1 \cdot 5 + 2 \cdot 6 = 17$; $3 \cdot 5 + 4 \cdot 6 = 39$.

29. B · 20 FLOP/byte · $10 \times 10^{12} \div (500 \times 10^9) = 20$ FLOP/byte; below 20 the chip is memory-bound.

30. D · 5.5 TFLOP/s · TFLOP/s = 2 × MACs ÷ time = $2 \times 6.87 \times 10^{11} \div 0.25 \approx 5.5 \times 10^{12}$. 2.7 forgets the factor 2; 11 doubles it again.

31. undefined · · TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong.

32. A · QUIC · QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes.

33. B · On or near the machine (edge) · Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it.

34. C · The slowest small fraction of requests · For example the 99th-percentile delay; one slow server can delay a whole request.

35. A · Putting computing close to the base station · Operator servers near the base station shorten the round trip.

36. C · 0.81 · V(3) = 1, V(2) = 0.9 × 1 = 0.9, V(1) = 0.9 × 0.9 = 0.81.

37. D · 0.325 · Target = 1 + 0.9 × 0.5 = 1.45; TD error = 1.25; new Q = 0.2 + 0.1 × 1.25 = 0.325.

38. undefined · · A and C give feedback every step. B and D are sparse: they arrive only at the end.

39. B · 0.77 · 0.9 × 0.8 + 0.1 × (0.2 + 0.5 + 0.8) ÷ 3 = 0.72 + 0.05 = 0.77. Careful: if exploration chose only the OTHER two arms the answer would be 0.9 × 0.8 + 0.1 × 0.35 = 0.755.

40. D · Training on many varied simulated conditions · If the policy works across many random variations, the real world looks like one more variation.

41. undefined · · LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth.

42. A · 8.33 m · 100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts.

43. B · 50.12 m · Weights $1/\sigma^2$: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates.

44. C · 50 m · Z = f × B ÷ d = 1,000 × 0.3 ÷ 6 = 50 m.

45. A · Level 2 · Level 2 = partial automation; the human remains the driver.

46. C · 112 GB · $7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split.

47. D · 143 tokens/s · Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s.

48. B · 24 hops · Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound.

49. undefined · · A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls.

50. D · 8× · Matrix multiply = $2n^3$ FLOPs; doubling n gives 2^3 = 8×. For large sizes the measured time grows close to 8× too.

51. MODEL ANSWER Weights 512 × 2,048 = 1,048,576 → 2.1 MB; FLOPs per input 2,097,152. Batch 1 ≈ 1.0 FLOP/byte < 20 → memory-bound, ≈ 0.5 TFLOP/s (5%). Batch 32 ≈ 29.7 FLOP/byte > 20 → compute-bound, ≈ 10 TFLOP/s. Batching reuses each fetched weight for many inputs, raising FLOPs per byte. Marks: intensities (2), classification and TFLOP/s (1), explanation (2). Accept small differences from rounding or from ignoring activation bytes.

52. MODEL ANSWER Advantages: privacy (face data stays in the Secure Enclave), low latency, works offline, energy-efficient NPU. Limitations: limited compute and memory, unsupported operations fall back to the CPU, models must be small or quantised. Large chatbot: usually cloud or hybrid, because its weights exceed phone memory (or a small on-device model + cloud). Marks: 1 per valid point (4), justified decision (1). Accept other valid points, e.g. updates or battery life.

53. MODEL ANSWER Strong answers choose an on-premises edge GPU server or hybrid: data stays inside the hospital (privacy and law), LAN latency is low, X-ray images are large (bandwidth), the cloud can still be used for training or overflow. Cost: buying a server vs renting. Marks: recommendation (1), four justified factors (4). Accept a well-argued cloud answer that addresses privacy (encryption, anonymisation, contracts) and the 2-second budget.

54. MODEL ANSWER Traffic = 2 × 255/256 × 14 ≈ 27.9 GB; time ≈ 0.28 s; efficiency = 8.2 ÷ (8.2 + 0.28) ≈ 96.7%. Improvements: overlap communication with compute, larger batch per chip (more compute per byte), gradient compression, hierarchical collectives. Marks: 3 calculations, 2 for one improvement with a correct reason.

55. MODEL ANSWER Naive: +1 per item put in the bin → the robot tips the bin and re-collects. Better: reward the reduction of rubbish on the floor (or clean area), penalties for knocking things over and for energy. Dense if given every step, sparse if only when the room is clean. Comparison: the naive one rewards activity, the better one rewards the real goal. Marks: hack (2), improved reward (2), sparse/dense with comparison (1). Accept other plausible hacks.

56. MODEL ANSWER Failure: epsilon = 0 means no exploration; the agent exploits the first route that reached the goal, so unvisited squares are never updated and a shorter route is never found. Fix: epsilon about 0.1-0.2 (or decaying epsilon). Epsilon = 0.5: values are learned everywhere but the path looks messy and hits the −1 squares more often. Bandit link: the greedy agent found the best arm 0% of the time. ε-greedy explores at random at a fixed rate; UCB adds a bonus $\sqrt{(2 \ln t / n)}$ to little-tried actions and explores where it is uncertain. Marks: diagnosis (2), fix (1), ε = 0.5 prediction (1), comparison (1). Accept other reasonable fixes such as optimistic initial values.

57. MODEL ANSWER 13.89 m/s; blind distance 2.08 m; braking $13.89^2 \div 14 \approx 13.78$ m; total ≈ 15.9 m. Detected at 15 m: 12.92 m left to brake, $v = \sqrt{(13.89^2 - 2 \times 7 \times 12.92)} \approx 3.5$ m/s at impact. Lesson: early, reliable detection (range) and low latency decide whether braking works; both must be in the design budget. Marks: 2 + 2 + 1.

58. MODEL ANSWER Speeds: 95, 250, 133, 67, 182 m/s. Mean t = 0.17 s → v ≈ 118 m/s (average the times, not the speeds). True delay is only 20 ÷ 343 ≈ 0.058 s, shorter than human reaction time (about 0.2 s), so reaction errors are as big as the signal. Averaging n trials shrinks random noise by about √n (sensor fusion idea, 4.3) but cannot remove a systematic bias, e.g. reacting faster to sound than to sight. Improvements: a longer distance (100 m → 0.29 s), many more trials, or phyphox Acoustic Stopwatch on two phones (no human reaction). Marks: calculations (2), noise explanation (1), averaging vs bias (1), improvement (1).

59. MODEL ANSWER $6 \times 7 \times 10^9 \times 1.4 \times 10^{12} = 5.88 \times 10^{22}$ FLOPs; $256 \times 200 \times 10^{12} \times 0.4 = 2.05 \times 10^{16}$ FLOP/s → ≈ 2.87×10^6 s ≈ 33.2 days; ≈ 204,167 chip-hours ≈ $408,333; 204,167 h × 0.4 kW × 1.1 ≈ 89.8 MWh. Raising MFU (fusion, better batching, overlap) cuts time, cost and energy in proportion. Marks: 4 values, 1 for the change with reason. Accept "more chips" only if the answer notes that cost and energy stay about the same.

60. MODEL ANSWER First call: tracing + XLA compilation. FLOPs = $2n^3$: 1,024 → 2.15 GFLOP ÷ 0.0021 s ≈ 1.0 TFLOP/s (≈ 13% MFU); 4,096 → 137 GFLOP ÷ 0.034 ≈ 4.0 TFLOP/s (≈ 50%); 8,192 → 1.10 TFLOP ÷ 0.27 ≈ 4.1 TFLOP/s (≈ 50%). 1,024 → 4,096 is 64× the FLOPs but only ≈ 16× the time, because at small sizes fixed overhead (launch, transfer) dominates and the GPU is not full. 4,096 → 8,192: 0.27 ÷ 0.034 ≈ 7.9 ≈ 8×, both sizes are compute-bound. Prediction for 16,384: about 8 × 0.27 ≈ 2.2 s, if memory allows. Marks: compilation (1), TFLOP/s and MFU (2), ratio explanation (1), prediction (1). Accept answers that also mention timing noise or GPU clock changes.

TICKET #CODEX · MAMA'S RECIPE NOTEBOOK

📖 The Codex

Every key word from the five recipes. Tick the ones you can explain without looking.

🍳 Recipe 01 · The Kitchen Crew

- [ ] **AI hardware** Processors, memory and interconnects suited to running ML efficiently.
- [ ] **MAC** Multiply-accumulate: multiply two numbers, add to a running total.
- [ ] **FLOP / TOPS** One floating-point operation (1 MAC = 2) / 10^{12} operations per second.
- [ ] **Training / inference** Learning weights from data / using weights to answer.
- [ ] **Precision** Bits per number: FP32 4 B, FP16 2 B, INT8 1 B.
- [ ] **CPU** Few, powerful, flexible cores.
- [ ] **GPU** Thousands of simple cores working in parallel.
- [ ] **TPU** Google's accelerator built around a systolic MAC array.
- [ ] **NPU** Small, low-power AI engine in a phone or laptop chip.
- [ ] **SoC** System-on-a-chip: CPU, GPU, NPU and memory controllers on one die.
- [ ] **Systolic array** Grid of MAC cells passing data neighbour to neighbour.
- [ ] **Deterministic execution** Every run takes the same predictable time.
- [ ] **Amdahl's law** Speed-up = 1 ÷ ((1 − p) + p ÷ N): the serial part caps it.
- [ ] **Utilisation** Share of peak performance actually achieved.
- [ ] **Quantisation** Converting numbers to lower precision (e.g. FP32 → INT8).
- [ ] **Batching** Processing several inputs together to reuse weights.
- [ ] **Memory hierarchy** Registers → SRAM/cache → DRAM/HBM → storage.
- [ ] **HBM** High-bandwidth memory stacked beside the chip.
- [ ] **Unified memory** One memory pool shared by CPU, GPU and NPU.
- [ ] **Arithmetic intensity** FLOPs per byte moved.
- [ ] **Roofline model** Attainable = min(peak, intensity × bandwidth).
- [ ] **Memory-bound / compute-bound** Limited by data delivery / by arithmetic.
- [ ] **Core ML** Apple framework that routes model parts to CPU, GPU or ANE.
- [ ] **Fallback** Unsupported op runs on the CPU, costing data movement.

🛵 Recipe 02 · Order Up!

- [ ] **Packet** A small chunk of data plus a header with delivery information.
- [ ] **Protocol** An agreed set of rules for communication.
- [ ] **Encapsulation** Each layer wraps data in its own header on the way down the stack.
- [ ] **IP address** Numerical address of a device (IPv4 32 bits, IPv6 128 bits).
- [ ] **TCP** Reliable, ordered byte stream with handshake, ACKs, flow and congestion control.
- [ ] **UDP** Unreliable, connectionless packet delivery: fast, no retransmission.
- [ ] **QUIC** Reliable, encrypted transport built on UDP; saves a round trip.
- [ ] **RTT** Round-trip time: to the destination and back.
- [ ] **Bandwidth** How many bits per second a link can carry.
- [ ] **Latency** Time for data to arrive or a request to be answered.
- [ ] **BDP** Bandwidth × RTT: data in flight needed to fill a link.
- [ ] **Cloud computing** On-demand access to a shared pool of computing resources (NIST).
- [ ] **Edge computing** Computation placed close to where data is produced or used.
- [ ] **MEC** Multi-access edge computing: servers in or near the mobile network.
- [ ] **Tail latency** Latency of the slowest requests, e.g. the 99th percentile.
- [ ] **Utilisation** Fraction of capacity in use; near 100% means long queues.
- [ ] **Horizontal scaling** Adding more machines to share the load.
- [ ] **Ring all-reduce** GPUs pass pieces around a ring to sum gradients efficiently.
- [ ] **Federated learning** Devices train locally and share only model updates.

- [ ] **Network slicing** One physical network divided into virtual networks with different guarantees.
- [ ] **eMBB / URLLC / mMTC** 5G scenarios: high data rate / low, dependable delay / massive device counts.

🐶 Recipe 03 · Training the Pup

- [ ] **Agent / environment** The learner and decision-maker / everything it interacts with.
- [ ] **State, action, reward** What the agent observes / what it does / the score it receives.
- [ ] **Policy (π)** The rule the agent uses to choose actions.
- [ ] **Episode** One run from a start state to an end.
- [ ] **Return (G)** Discounted sum of future rewards.
- [ ] **Discount factor (γ)** Number between 0 and 1 that shrinks later rewards.
- [ ] **Value V(s), Q(s,a)** Expected return from a state / from an action in a state.
- [ ] **Bellman equation** V(s) = r + γV(s′).
- [ ] **MDP** Markov decision process: future depends only on present state and action.
- [ ] **Law of effect** Rewarded behaviour becomes more likely (Thorndike).
- [ ] **Learning rate (α)** How far one new experience moves an estimate.
- [ ] **Q-learning** Learns Q(s,a) from experience with a TD update.
- [ ] **TD learning** Updating an estimate using the next estimate.
- [ ] **Credit assignment** Working out which earlier actions caused a later outcome.
- [ ] **Sparse / dense reward** Reward given rarely / at nearly every step.
- [ ] **Reward shaping** Extra reward to guide learning; potential-based form keeps the best policy.
- [ ] **Reward hacking** High reward without doing the intended task.
- [ ] **Exploration / exploitation** Trying uncertain actions / using the best-known one.
- [ ] **ε-greedy** Random action with probability ε, otherwise the best-known.
- [ ] **UCB** Q + √(2 ln t ÷ n): bonus for less-tried options.
- [ ] **Sim-to-real gap** Difference between simulator and reality.
- [ ] **Domain randomisation** Varying simulation settings so the policy works in many conditions.

👀 Recipe 04 · Chef's Senses

- [ ] **Active / passive sensor** Emits energy and measures the echo / only receives existing energy.
- [ ] **Exteroceptive / proprioceptive** Measures the outside world / measures the machine itself.
- [ ] **ADC** Analogue-to-digital converter: voltage → number.
- [ ] **ISP** Image signal processor: raw sensor data → usable image.
- [ ] **Field of view (FOV)** The angle a sensor can see.
- [ ] **Dynamic range** Brightest vs darkest light captured in one frame.
- [ ] **LiDAR / point cloud** Laser time-of-flight ranging / the 3-D points it produces.
- [ ] **Time of flight** Distance from round-trip time: d = v × t ÷ 2.
- [ ] **Doppler effect** Echo frequency shift from motion; radar measures speed with it.
- [ ] **Hard real-time** Missing a deadline counts as failure.
- [ ] **Latency budget** Allowed end-to-end time split among stages.
- [ ] **Ego-motion compensation** Correcting measurements for the platform's own motion.
- [ ] **Sensor fusion** Combining several sensors' measurements into one estimate.
- [ ] **Bird's-eye view (BEV)** Top-down map used as a shared space for fusion.
- [ ] **Kalman filter / gain** Predict-update tracker / K = P ÷ (P + R).
- [ ] **Data association / gating** Matching detections to tracks / only near the prediction.
- [ ] **Disparity** Pixel shift between stereo images; Z = f × B ÷ d.
- [ ] **IoU** Overlap ÷ union of predicted and true boxes.
- [ ] **Precision / recall** Share of detections correct / share of real objects found.
- [ ] **Phantom braking** Braking caused by a false detection.
- [ ] **SAE levels** J3016 automation levels 0-5.
- [ ] **ISO 26262 / SOTIF** Safety against faults / against performance limits.
- [ ] **Fail-operational / fail-safe** Keeps working reduced / moves to a safe state.
- [ ] **Occupancy network** Predicts which 3-D cells around the vehicle are occupied.

Recipe 05 · Running the Chain

- [ ] **Computation graph** Model drawn as operations (nodes) and tensors (arrows).
- [ ] **Operator fusion** Merging several operations into one pass to avoid memory traffic.
- [ ] **XLA** Compiler that maps models onto TPUs and other devices.
- [ ] **MFU** Useful model FLOP/s ÷ hardware peak FLOP/s.
- [ ] **Data parallelism** Batch split across chips, each with a model copy (or shard).
- [ ] **Tensor parallelism** Individual matrices inside a layer split across chips.
- [ ] **Pipeline parallelism** Model split into stages of consecutive layers.
- [ ] **Sharding** Splitting a tensor or training state across chips.
- [ ] **All-reduce** Sums values across chips and gives every chip the result.
- [ ] **Collective operation** Many-chip pattern: all-reduce, all-gather, reduce-scatter.
- [ ] **Scaling efficiency** Speed-up ÷ number of chips.
- [ ] **Strong / weak scaling** Fixed problem / problem grows with chip count.
- [ ] **Recomputation** Recompute activations instead of storing them: ~33% more compute.
- [ ] **KV cache** Stored attention state that grows with conversation length.
- [ ] **Continuous batching** New requests join a running batch between tokens.
- [ ] **Gang scheduling** Start a job only when all its chips are free at once.
- [ ] **Preemption** Pausing a lower-priority job for a higher-priority one.
- [ ] **PUE** Total facility energy ÷ computing equipment energy.
- [ ] **TCO** Total cost of ownership: purchase, energy, cooling, network, buildings, staff.
- [ ] **MTBF** Mean time between failures; system ≈ chip MTBF ÷ chip count.
- [ ] **Checkpointing** Saving full training state to restart after a failure.
- [ ] **Torus / diameter** Grid whose edges wrap around / longest shortest path in hops.
- [ ] **ICI / OCS** Direct TPU chip links / optical switches re-patched by software.
- [ ] **Pod / slice** 4,096-chip TPU v4 unit / the part of a pod one job gets.

TICKET #BOOK · COLLECT ALL FIVE

Recipe book

Survive a room's rush hour, then colour in its card's star. Each card is the one-page summary of a whole recipe.

RECIPE 01 · SILICON FRIED RICE

The Kitchen Crew

☆

Method (big ideas)

1. **AI is mostly MACs:** speed those up and you speed up the model.
2. Specialised chips trade flexibility for efficiency.
3. Parallelism helps until the serial part, idle units, memory or communication get in the way.
4. Moving data costs more than computing on it: keep ingredients on the chopping board.
5. On-device AI (NPU) wins on privacy, latency and battery, but is limited by memory and supported operations.

Ingredients (key terms)

MAC · FLOP · TOPS · training / inference · precision · CPU / GPU / TPU / NPU · systolic array · Amdahl's law · quantisation · memory hierarchy · HBM · arithmetic intensity · roofline · batching · Core ML

Unlocked: Recipe Book page 1!

RECIPE 02 · PING-PONG NOODLES

Order Up!

☆

Method (big ideas)

1. Networks move AI data as packets; layers and encapsulation let any machine talk to any other.
2. TCP makes delivery reliable at the cost of round trips; UDP and QUIC trade or reduce that cost.
3. Where the model runs (device, edge, cloud) is a trade-off of latency, privacy, model size and cost.
4. Bandwidth can be bought; the speed-of-light floor and queues cannot be bought away.
5. Communication limits training on many chips, just like Amdahl's serial part.
6. 5G + MEC bring the kitchen closer, and AI helps run the network itself.

Ingredients (key terms)

packet · header · protocol · TCP/IP · UDP · QUIC · RTT · bandwidth · BDP · cloud · edge · MEC · tail latency · utilisation · all-reduce · 5G · network slicing · RIC

🔓 Unlocked: Recipe Book page 2!

RECIPE 03 · Q-LEARNING PIE

Training the Pup

Method (big ideas)

1. Behaviourist AI learns behaviour from consequences; RL is its maths.
2. **Everything is the loop:** state → action → reward → new state.
3. Return and value carry delayed rewards back to early actions.
4. Q-learning nudges estimates towards reward + best future.
5. **You get what you reward:** design rewards carefully and watch for hacks.
6. Explore enough to find the best, exploit enough to enjoy it.
7. Robots train in parallel simulation (GPUs) and run a tiny policy on board.

Ingredients (key terms)

agent · environment · state · action · reward · policy · return · γ · V and Q · Bellman · Q-learning · TD error · sparse/dense · shaping · reward hacking · ε-greedy · UCB · sim-to-real · domain randomisation

🔓 Unlocked: Recipe Book page 3!

RECIPE 04 · LIDAR SUSHI TRAIN

Chef's Senses

Method (big ideas)

1. **No sensor sees everything:** choose which failures you can live with.
2. Cameras = detail; LiDAR = direct 3-D; ultrasonic = cheap and close; radar = speed in any weather.
3. Every millisecond of latency is distance driven blind.
4. Fusion weighs sensors by confidence and tracks uncertainty (Kalman).
5. Camera depth error grows with distance^2.
6. Decisions compare expected costs; redundancy needs INDEPENDENCE.
7. Judge company claims with independent evidence.

Ingredients (key terms)

sensor · active/passive · FOV · time of flight · point cloud · Doppler · ADC · ISP · latency budget · ego-motion · fusion · BEV · Kalman gain · gating · disparity · IoU · precision/recall · SAE levels · ISO 26262 · SOTIF · occupancy network · data engine

🔓 Unlocked: Recipe Book page 4!

RECIPE 05 · CLOUD BANQUET

Running the Chain

Method (big ideas)

1. Software layers decide how well a chip is used: fuse, tile, choose precision, schedule.
2. Split work when the model doesn't fit or would take years: data, tensor, pipeline.
3. Every split costs communication: measure with scaling efficiency.
4. Measure first; fix one bottleneck at a time.
5. Serving is memory-bound; batching trades latency for throughput.
6. Energy, cost and failures are the three ceilings; checkpoint wisely.
7. AI performance is a property of SYSTEMS, not chips.

Ingredients (key terms)

computation graph · fusion · XLA · MFU · sharding · data/tensor/pipeline parallelism · all-reduce · scaling efficiency · recomputation · KV cache · continuous batching · gang scheduling · PUE · 6ND · TCO · MTBF · checkpoint · torus · OCS · pod · slice

Unlocked: the final Recipe Book page!

TICKET #SRC

Sources, team & credits

REFERENCES

References · Recipe 01

Recipe 01 · The Kitchen Crew

Key sources used for this recipe (APA style, shortened). The full reference list, with titles, venues and access dates (early November 2026), is in the shared chapter Main/Chapter-01 of our repository.

SOURCE	WORK
Amdahl (1967)	Validity of the single processor approach… AFIPS Conf. Proc.
Apple (2017; 2024)	iPhone X press release; Platform Security guide; Core ML docs
Hennessy & Patterson (2019)	A new golden age for computer architecture. CACM 62(2)
Horowitz (2014)	Computing's energy problem. IEEE ISSCC
Jouppi et al. (2017)	In-datacenter performance analysis of a TPU. ISCA
Sze et al. (2017)	Efficient processing of deep neural networks. Proc. IEEE
Williams et al. (2009)	Roofline: an insightful visual performance model. CACM 52(4)
NVIDIA; Qualcomm; Google	Tensor Core, Hexagon NPU and Ironwood (2025) documentation

REFERENCES

References · Recipe 02

Recipe 02 · Order Up!

Key sources used for this recipe (APA style, shortened). The full reference list, with titles, venues and access dates (early November 2026), is in the shared chapter Main/Chapter-02 of our repository.

SOURCE	WORK
Kurose & Ross (2020)	Computer Networking: A Top-Down Approach, 8th ed. Pearson
Cerf & Kahn (1974)	A protocol for packet network intercommunication. IEEE Trans. Comm.
Postel (1980; 1981); Eddy (2022)	UDP (RFC 768), IP (RFC 791), TCP (RFC 9293)
Iyengar & Thomson (2021)	QUIC (RFC 9000); Rescorla (2018) TLS 1.3 (RFC 8446)
Mathis et al. (1997); Jacobson (1988)	TCP throughput model; congestion control
Dean & Barroso (2013)	The tail at scale. CACM 56(2)
Mell & Grance (2011)	The NIST Definition of Cloud Computing (SP 800-145)

SOURCE	WORK
Satyanarayanan (2017); Shi et al. (2016)	Edge computing
ITU-R (2015; 2017); 3GPP; ETSI (2014); O-RAN	5G requirements, architecture, MEC, RIC

REFERENCES

References · Recipe 03

Recipe 03 · Training the Kitchen Pup

Key sources used for this recipe (APA style, shortened). The full reference list, with titles, venues and access dates (early November 2026), is in the shared chapter Main/Chapter-03 of our repository.

SOURCE	WORK
Sutton & Barto (2018)	Reinforcement Learning: An Introduction, 2nd ed. MIT Press
Thorndike (1911); Pavlov (1927)	Animal Intelligence; Conditioned Reflexes
Skinner (1938); Watson (1913); Chomsky (1959)	Behaviourism and its critique
Watkins & Dayan (1992); Sutton (1988)	Q-learning; temporal-difference learning
Mnih et al. (2015)	Human-level control through deep RL. Nature 518
Ng, Harada & Russell (1999)	Policy invariance under reward transformations. ICML
Auer et al. (2002); Clark & Amodei (2016)	UCB bandits; faulty reward functions
Hwangbo et al. (2019); Lee et al. (2020)	Legged robot locomotion. Science Robotics
Rudin et al. (2022); Makoviychuk et al. (2021)	Walking in minutes; Isaac Gym
Tobin et al. (2017); Tan et al. (2018); Kumar et al. (2021)	Domain randomisation, sim-to-real, RMA

REFERENCES

References · Recipe 04

Recipe 04 · Chef's Senses

Key sources used for this recipe (APA style, shortened). The full reference list, with titles, venues and access dates (early November 2026), is in the shared chapter Main/Chapter-04 of our repository.

SOURCE	WORK
Thrun, Burgard & Fox (2005)	Probabilistic Robotics. MIT Press
Kalman (1960)	A new approach to linear filtering and prediction problems
Winner et al. (2016)	Handbook of Driver Assistance Systems. Springer
Rosique et al. (2019); Feng et al. (2021)	Perception systems; multi-modal detection surveys
Hall & Llinas (1997)	An introduction to multisensor data fusion. Proc. IEEE

SOURCE	WORK
Philion & Fidler (2020); Li et al. (2022); Liu et al. (2023)	BEV: Lift-Splat-Shoot, BEVFormer, BEVFusion
Geiger et al. (2012); Caesar et al. (2020); Sun et al. (2020)	KITTI, nuScenes, Waymo Open datasets
SAE (2021); ISO 26262 (2018); ISO 21448 (2022)	Automation levels; functional safety; SOTIF
Talpes et al. (2020); Tesla AI Day (2021, 2022)	Tesla FSD computer; Tesla Vision
Waymo (2024); Electrek; TechCrunch; NHTSA	6th-gen Driver; reported news; investigations

REFERENCES

References · Recipe 05

Recipe 05 · Running the Chain

Key sources used for this recipe (APA style, shortened). The full reference list, with titles, venues and access dates (early November 2026), is in the shared chapter Main/Chapter-05 of our repository.

SOURCE	WORK
Barroso, Hölzle & Ranganathan (2018)	The Datacenter as a Computer, 3rd ed.
Jouppi et al. (2023)	TPU v4: an optically reconfigurable supercomputer. ISCA
Kaplan et al. (2020); Hoffmann et al. (2022)	Scaling laws; compute-optimal training
Narayanan et al. (2021); Rajbhandari et al. (2020)	Megatron-LM; ZeRO
Patarasuk & Yuan (2009)	Bandwidth-optimal all-reduce algorithms
Kwon et al. (2023)	PagedAttention (LLM serving). SOSP
Xu et al. (2021)	GSPMD: general and scalable parallelization
Patterson et al. (2021)	Carbon emissions and large neural network training
Young (1974)	Optimum checkpoint interval. CACM 17(9)
Google; Google Cloud	Ironwood (2025); PUE reports; Cloud TPU documentation

THE TEAM

Video & Platform Sources

RESOURCE	CREATOR / OWNER
But what is a neural network? (aircAruvnKk)	3Blue1Brown
How do Graphics Cards Work? (h9Z4oGN89MU)	Branch Education
Tensor Processing Units (eyAjbgkBdjU); TPU data center (FsxthdQ_sL4); Scaling AI with TPUs (8zBaa20mvoc)	Google Cloud
A very short intro to the Roofline model (IrkNZG8MJ64)	NHR@FAU
How does the INTERNET work? (x3c1ih2NJEg)	Lesics

RESOURCE	CREATOR / OWNER
What is bandwidth? (QLJCqY-7Vrw)	internet-class
Inside a Google data center (XZmGGAbHqa0)	Google Workspace
RL: Machine Learning Meets Control Theory (0MNVhXEX9to)	Steve Brunton
Markov Decision Processes (2iF9PRriA7w)	Computerphile
AI Learns to Walk (L_4BPjLBF4E)	AI Warehouse
How do self-driving cars "see"? (PRg5RNU_JLk)	TED-Ed
How Does LiDAR Remote Sensing Work? (EYbhNSUnIdU)	NEON Science
Visually Explained: Kalman Filters (IFeCIbljreY)	Visually Explained
Tesla AI Day 2021 (j0z4FweCy4M)	Tesla
A friendly introduction to distributed training (S1tN9a4Proc)	TensorFlow
Platforms: Google Colab, REINFORCEjs GridWorld, phyphox, ping/tracert	Google; A. Karpathy; RWTH Aachen; OS tools

🔗 LINKS

All YouTube links have the form youtube.com/watch?v=<ID>; IDs, titles and creators were checked on 4 November 2026.

THE TEAM

The Kitchen Brigade

MEMBER	KITCHEN ROLE	TEAM ROLE
纪小佩	Head Chef	CEO: planning, coordination, final sign-off
林元甲	Pantry Keeper	CIO: repository, tools, file management
温俊强	Recipe Tester	Content Lead: worked examples and answer checks
汤丽嘉	Recipe Writer	Content Lead: explanations and activities
曾富良	Plating Artist	Visual Design Lead: layout, sprites, identity
李秀春	Camera Crew	Video Lead: curated videos and timestamps
黄施杰	Taste Inspector	Quality Lead: accuracy, references, Test Bank review

🏷 **About this guide**

Stage 2 Study Guide, built from the team's shared Markdown content (Main/Chapter-01 to 05). Repository: github.com/Bvmhe/CS-AI-Group. Deadline: 7 November 2026, 23:59 Beijing time.

THE TEAM

How We Cooked This Guide

Contributions

Content: written from the team's five shared Markdown chapters (Main/Chapter-01 to 05), which every member reviewed. Visual identity, sprites and layout: Visual Design Lead with the team. Videos: selected and timestamped by the Video Lead. Accuracy check of all worked examples and answer keys: Quality Lead and Content Leads.

AI-use statement

We used an AI assistant (Claude) to help turn our chapters into page layouts, draft kitchen analogies, write the image-processing scripts for sprite cut-outs, and check arithmetic. Every number was re-checked against our chapters' worked examples. AI did not invent sources; all references come from our chapters. Details of prompts and edits are in the AI-Logs folder of our repository.

Known limits

Tesla, Waymo and TPU figures change quickly and some are "reported" rather than independently verified. Numbers marked illustrative are chosen for teaching. Check current sources before quoting.

CREDITS

Made in code

Characters and 3D diner

Mama Feng, Papa Feng, Leo, Lily, Grandma Feng and Dumpling the pup are original characters drawn in code (SVG). The six special customers are photos supplied by the team, cut out and set on drawn chibi bodies with ordinary image tools (no AI). The 3D diner, its five rooms, the trouble customers, the food and every texture are modelled in code with three.js. No AI-generated images are used.

Content

All explanations, worked examples, mini-games, labs and numbers are the team's own work, written from our shared Markdown chapters (Main/Chapter-01 to 05) and the cited sources. Diagrams: drawn by the team. Questions: Fengwei Fried Rice Stage 2 Test Bank. Illustrative numbers are marked "illustrative"; company claims are marked "reported" or "company claim".

Software

three.js r160 (MIT licence, © 2010-2023 three.js authors), bundled inside this file so it works offline. Fonts: the system fonts on your device. Sound effects: synthesised with the Web Audio API.

🎮 How to use it

Open the HTML file in any modern browser (phone or computer, works offline). Progress is saved in this browser only. The day/night switch follows your clock until you flip it. Free roam (on the map) unlocks every mission for revision.

Thanks for cooking!

THE DINER NEVER CLOSES · OPEN THE HTML FILE FOR THE 3D VERSION